Hamzstlab Mathematics and Hamzstplot

Freya · Hamzst · Lasthrim · DS Glanzsche[1]

Berlin-Sentinel Academy of Science, AliceGard

2025 Edition

[1]A thank you or further information

Contents

Preface

For my Wife Freya, and our daughters Solreya, Mithra, Catenary, Iyzumrae and Zefir.

For Lucrif and Znane too along with all the 8 Queens.

For Albert Silverberg, Camus and Miklotov who left TREVOCALL to guard me.

To Nature(Kala, Kathmandu, Big Tree, Sentinel, Aokigahara, Hoia Baciu, Jacob's Well, Mt Logan, etc) and my family Berlin: I have served, I will be of service.

To my dogs who always accompany me working in Valhalla Projection, go to Puncak Bintang or Kathmandu: Sine Bam Bam, Kecil, Browni Bruncit, Sweden Sexy, Cambridge Klutukk, Milan keng-keng, Piano Bludut and more will be adopted. To my cat who guard the home while I'm away with my dogs: London.

The one who moves a mountain begins by carrying away small stones - Confucius

Come to me, all who labor and are heavy burdened, and I will give you rest. - The Bible, Matthew 25:28

A book to explained how we create a C++ library from forking / branching out ImGUI 1.92.0 WIP, Implot, Implot 3D, merge them into one big pile that we call Hamzstlab Mathematics so it can be easily compiled / build so we can focus more on the Mathematics side / learning.

We also write about Hamzstplot that is forking / branching out from Matplot++, it is a C++ plotting library with backend gnuplot, so you can have more creativity in plotting since the syntax will be all C++. It is not as interactive as Hamzstlab Mathematics but it has more options of plotting that you can play with.

a little about GlanzFreya:
Freya the Goddess is (DS Glanzsche') Goddess wife. We get married on Puncak Bintang on November 5th, 2020 after we go back from Waghete, Papua. My wife birthday (Freya the Goddess) is on August 1st. After not working with human anymore since 2019, I (DS Glanzsche) work in a forest, shaping a forest into a natural wonder / like a canvas for painting, and also clean up trashes that are thrown in the forest. Thus after 6 years working with Nature I have found what can makes me waltz to work, it is going to the forest in the morning then go back home again and study / learn science, engineering, computer programming.

Freya specialty is Electrical Engineering, she is the one that taught Glanz about Arduino Uno and all that.

Glanz specialty is Dogs, she loves dogs and love feeding stray dogs and taking home stray puppies given she has money and foods, otherwise she will bring stray dogs to animal shelter.

a little about Hamzst:
Hamzst is a nickname for Alice from Persona game series made by Atlus, sometimes she can also be similar with Alice from Alice in Wonderland (1951 movie). When she said "But in my world, the books will be nothing but picture," it is true for Hamzstplot and Hamzstlab Mathematics. She is a Ding, Yin Fire daymaster (The candle), she is the controller of Glanz and bring a lot of luck to Glanz too. Her specialty is Physics.

a little about Lasthrim:
She was known as Anabelle the Doll from the famous movie "The Conjuring," but she changes her name into Lasthrim since 2021 and want a true repentance since she has found a new joy and Love in mathematics. She is a Bing, Yang Fire daymaster (The Sun), she can shapes Glanz into the best Excalibur. Her specialty is Mathematics. Realizing that after writing Lasthrim Projection book, it is better to use our own C++ library modifying from existing C++ library to make the plot, computation and simulation, instead of using Python and JULIA. So here she is.

Figure 1: *Freya, thank you for everything, I am glad I marry you and I could never have done it without you.*

Figure 2: *I paint her 3 days before Christmas in 2021.*

Chapter 1

Introduction Story

On top of the mountain a heart shaped stone waiting for You, my eyes can't see, my body can't touch, but my heart knows it's You. Forever only You. - ***Glanz to Freya***

Corro nel vento della landa desolata. E in questo momento sento qualcosa. Raggiungendo I'orizzonte apparso. Vagamente al confine della vista - ***Orrizonte (Suikoden II)***

THIS book is written on May 29th, 2025, while we also have another project to make SymIntegration C++ library, we are working on this when we saw Implot, Implot 3D, and Matplot++. We want to clone them and modify them, or probably add some features to plot, simulate and more in the future, it is called branching out / forking. Then rename it as Hamzstlab Mathematics for combining Imgui+Implot+Implot3D+Imnodes, as a C++ library / software where you can create interactive graph or simulation for mathematics problems. It will be very useful for students and researchers. While Hamzstplot is branching out from Matplot++ to create graph and plot as well but it is not interactive as Hamzstlab Mathematics but the options for building the plot are enormous. We can choose which one to use depends on our need.

To avoid confusion in the future: Whenever there is a written of subject "I" it means DS Glanzsche as the I subject.

Years ago, around December 2019 when I (DS Glanzsche) was hiking / jogging up to Puncak Bintang, I pray at dawn, I remember I used to arrive there at 4:30 AM , departing around 3:00 AM from home, what I pray is " I pray and hope one day I can create a software like MATLAB," well I am starting this and remember that prayer again, it is becoming true, when I was installing MATLAB back then in 2011 to 2014 for college purpose, they give MATLAB course to students and even pay the license for it, I think it is great, but then realize, when I was reading a book like "Elementary Linear Algebra" or ""Elementary Differential Equation" or "Calculus" none of them have the MATLAB code to show how the plot is made or the computation is being computed, even if today there is a book called "Advanced Engineering Mathematics with MATLAB" we still need the code for the basic that undergraduate needs, since the knowledge for other science and engineering department will rely heavily on those Calculus, Linear Algebra, and Differential Equations first.

Not to mention, that there are tons of menu / on the menu bar on MATLAB, that the user won't even need to use for the rest of his/her life, the size of MATLAB today is over 20 GB, and for the installation it takes hours, now I am comparing with Hamzstlab Mathematics, if you are an

Electrical Engineer, and we already create the basic code and API to create any electric circuit with basic NPN' or PNP' BJT transistor or MOSFET n-channel or p-channel along with any number of resistors, inductor, and capacitor and will have the computed voltage and current, and you probably won't even need to download Hamzstlab Mathematics for even 1 GB, isn't it going to save your day and your time?

This is the real life example where we try to plot a solution to a simple first order ordinary differential equation and the Euler' method approximation, the size of the C++ file and the Makefile only consumes 3.1 KB, and if you build the example the C++ codes it will create this object files that will consumes 27.3 MB of size of your storage, which can be cleaned after you are finish. Hamzstlab Mathematics will only consume your storage of 50.3 MB, and the rest will depends on the project that you will be working on.

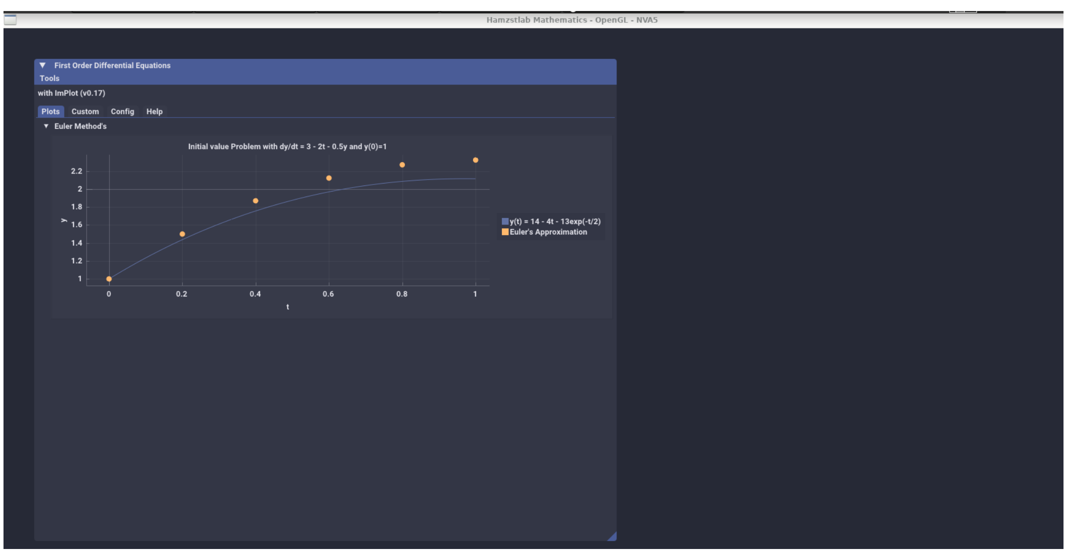


Figure 1.1: *The interactive plot of Euler' method approximation and the solution plot for* $\frac{dy}{dt} = 3 - 2t - \frac{1}{2}y$ *with Hamzstlab Mathematics.*

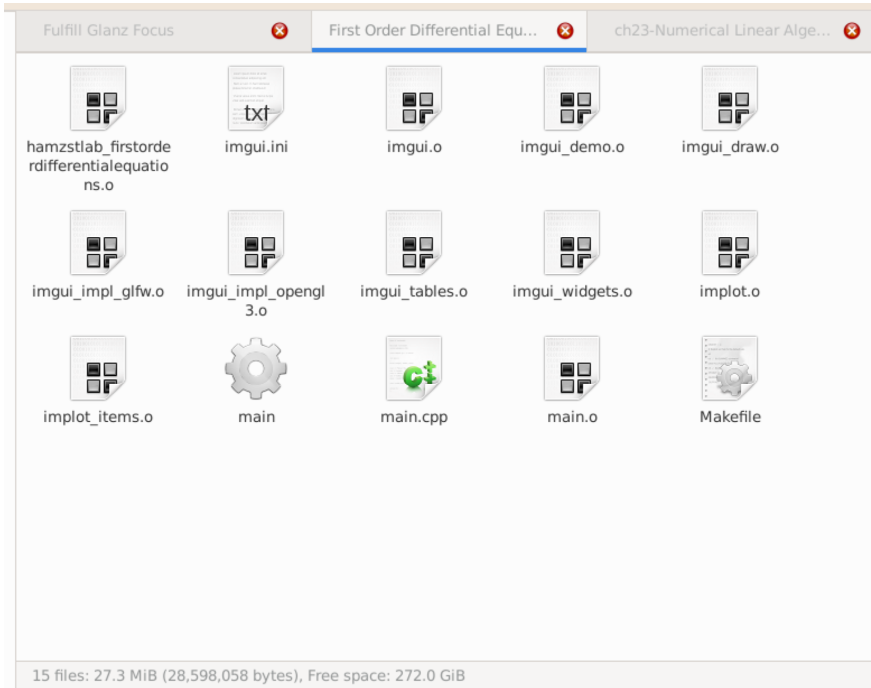


Figure 1.2: *The size of a built example with all the object files.*

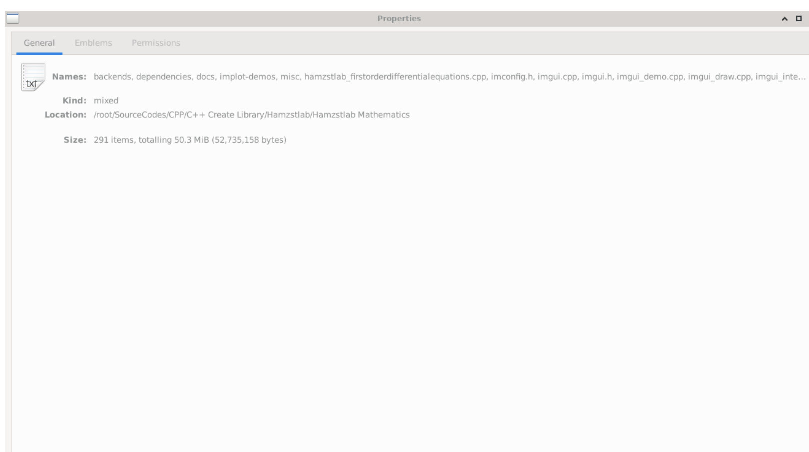


Figure 1.3: *The size of Hamzstlab Mathematics source code as of June 4th, 2025.*

On May 26th, 2025 while in St Peter Cathedral in Bandung I was reading a book [13], then at the first chapter it stated like this:
If all you are interested in is a quick calculation, then Python along with its ecosystem is likely going to be your one-stop shop. As your work becomes more computationally challenging, you may need to switch to a compiled language; most work on supercomputers is carried out using languages like Fortran or C++ or sometimes even C.

Basically, what Imgui combined with Implot and Implot 3D now able to do, show a lot of interactive visualizations in one window, is a big gap between C++ and Python, like the sky and the earth. I (DS Glanzsche) as one of the author, starts learning C++ rigorously since October 2023, still very infant stage, then able to write DianFreya Math Physics Simulator and already finish the whole Elementary Linear Algebra book along with all the C++ codes for plotting and simulation. Previously, I was using Python and JULIA to do the computation while writing the book Lasthrim Projection (focusing on Calculus and basic Mathematics field that are more theoretical like Discrete Mathematics, Elementary Differential Equations, Partial Differential Equations). Now I have learned Python, JULIA, C++, Fortran, and C. I know writing a code with C++ to achieve the same result will take longer and steeper learning curve, but it is worth the time spent.

The Operating System and Softwares in Use:

1. GFreya OS version 1.8 (based on LFS and BLFS version 11.0 System V books)

2. GCC version 11.2.0 (to compile C++ codes)

3. Hamzstplot version 1.5

4. Hamzstlab Mathematics version 1.51

We will explain about Hamzstplot and Hamzstlab Mathematics in the next two chapters, then the rest of the chapters will be talking about their application (in creating the plot, interactive plot, or simulation with either Hamzstplot or Hamzstlab Mathematics).

The content of this book covers the basic undergraduate courses in Mathematics from Differential Equation to Numerical Analysis and Linear Programming. If one is keen and acute enough and wonder, why there is no Elementary Linear Algebra here? Guilty. It is because we have done the summarization of the whole Elementary Linear Algebra based on [22] book in another book [12] that we are working on from 2023 to 2025, and we are mainly using C++ libraries that are already well known like Armadillo, Eigen, and GiNaC to help us for the matrix and vector computation till QR method, spectral decomposition, eigenvalue and eigenvector. We let the reader read that book or any other Elementary Linear Algebra book to get the basic knowledge of Linear Algebra, since it is really needed for almost all fields in mathematics, science and engineering. This books is focusing on showing what Hamzstplot, Hamzstlab Mathematics can do to help compute, plot and simulate the mathematics problems we have in each branch / field. We didn't mention it in the cover but we will also use SymIntegration for the symbolic integration computation part.

We really appreciate those who spend time to write constructing critics, not hate words without reason that will only add more good luck and karma to me, write opinions on what should be added in Hamzstlab Mathematics and Hamzstplot or if there is an incorrect algorithm, please inform the author. Feedbacks can be sent to **dsglanzsche@gmail.com**.

Chapter 2

Hamzstplot

"You can't run from what you see and feel." - ***Odessa Silverberg***

Hamzstplot was born on a Full Moon day on May 12th, 2025. We are branching out / forking from Matplot++ and we want to develop it further, changing the name first and will add several features in the future to help us in learning Mathematics, or other branch of science, e.g. to create direction fields. The basic syntax of C++ will really help for a very long term future prospects and more complex problems in science to be computed, analyzed and plot.

I. Build and Install Hamzstplot in GFreya OS

We are using GFreya OS as it is a custom Linux-based Operating System, your distro your rules. The commands here assuming that you / the reader are using Linux-based OS,they are familiar and easy to follow for other Linux-based OS users.

To download Hamzstplot, open terminal / xterm then type:
git clone https://github.com/glanzkaiser/Hamzstplot.git
Enter the directory then type:

```
mkdir build
cd build
cmake -DBUILD_SHARED_LIBS=ON ..
make
make install
```

Code 1: *Create Hamzstplot dynamic library*

It will create the dynamic library **libhamzstplot.so**, we are using CMake, and it is said that dynamic library saves more RAM memory than static library.

The prefix at default is **/usr/local** so the installed dynamic library will go to **/usr/local/lib**.

Then open terminal and from the current working directory / this repository main directory:

```
cd /source/3rd_party/cimg/
cp -r CImg.h* /usr/local/include

cd /

cd ../Hamzstplot/source/hamzstplot/
cp -r * /usr/local/include

cd ../Hamzstplot/source/3rd_party/nodesoup/include/
cp -r nodesoup.hpp /usr/local/include

cd /
cd ../Hamzstplot/source/3rd_party/nodesoup/src/
bash dynamiclibrary.sh
cp -r libnodesoup.so /usr/local/lib
```

Code 2: *Move include files and 3rd party library*

When we want to create the shared library it will look for this header files in the default path where the include files are usually located. In Linux OS **usr/include** is the basic / default path, another default path is **usr/local/include**.

All files, examples codes and source codes used in this books can be found in this repository:
https://github.com/glanzkaiser/Hamzstplot

II. Example of Plotting with Hamzstplot in GFreya OS

We will create a 3D plot of a helix, the source code is this:

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

int main() {
	using namespace hamzstplot;

	auto xt = [](double t) { return sin(t); };
	auto yt = [](double t) { return cos(t); };
	auto zt = [](double t) { return t; };
	fplot3(xt, yt, zt);

	show();
	return 0;
}
```

Code 3: *Hamzstplot/Examples with Makefile/Plot 3D Helix/main.cpp*

Open terminal and from the current working directory **Hamzstplot/Examples with Makefile/-Plot 3D Helix/**, then type:

```
make
./main
```

Code 4: *Build and run the executable*

or you can also type in the current working directory (a manual way to compile):
g++ main.cpp -o main -lhamzstplot
./main

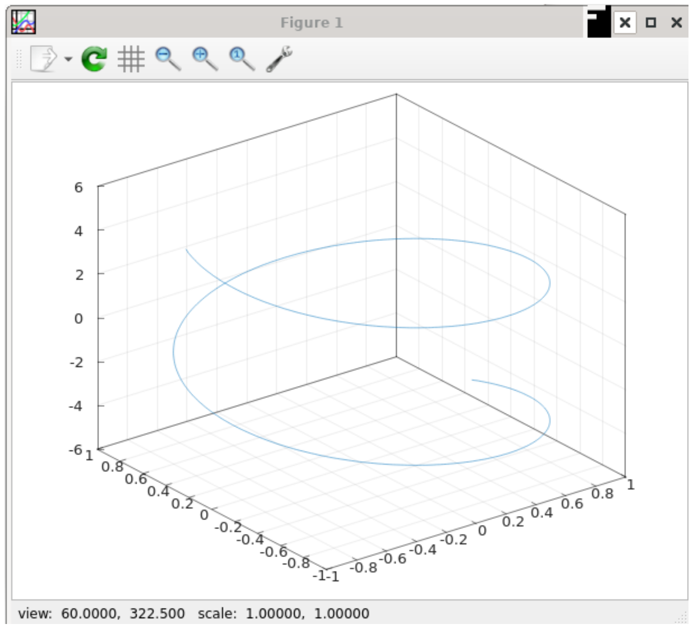


Figure 2.1: *The plot here is using the gnuplot as the backend, that is why the GUI window should look very familiar.*

Chapter 3

Hamzstlab Mathematics

"In this world perhaps, but in my world, the books will be full of pictures" - ***Alice (Alice in Wonderland - 1951)***

"If in other sciences we should arrive at certainty without doubt and truth without error, it behooves us to place the foundations of knowledge in mathematics." - ***Roger Bacon***

"If I were again beginning my studies, I would follow the advice of Plato and start with mathematics." - ***Galileo Galilei***

WHY would we need Hamzstlab Mathematics if we already have Hamzstplot? For the extra interactive feature, better simulation, even if the backend of gnuplot or OpenGL from Hamzstplot is already good enough, the features in Hamzstlab Mathematics that is coming from ImGUI, Implot, Implot3D, Imnodes can reach to almost all branches of science and engineering, help them to design and make prototype. Box2D is using ImGUI and it helps me personally to create some basic Physics simulation from an undergraduate book "Fundamental of Physics."

You can design any kind of electric circuit here with all the computation necessary too, you can simulate how a mechanical system works, how leverage works, how a human organ works, how a nuclear fission works, and you can even do gene sequence simulation better here, they are all in my head now, but one day we will add the source code and API so people can really create what I am talking now.

In the end, there is no perfect software, one software or one C++ library complement the other. That is life. Just like marriage, since nobody is perfect, just be honest, no lie no cheating, accept your partner for whoever she is, with no extramarrital affair / lie / cheating, then you both will be happy and reach your goal together, with a bonus become power couple too.

I. Build and Install Hamzstlab Mathematics in GFreya OS

We are using GFreya OS as it is a custom Linux-based Operating System, your distro your rules. The dependencies are:

1. ImGUI version 1.92.0 WIP
2. Implot version 0.17
3. Implot-3D version 0.3 WIP
4. Imnodes

There is no need to download all of the above one by one anymore, because they are already blended into one in Hamzstlab Mathematics.

To download Hamzstlab Mathematics, open terminal / xterm then type:
git clone https://github.com/glanzkaiser/Hamzstlab-Mathematics.git

Enter the directory then type:

```
cd dependencies

cp -r cxxopts.hpp /usr/include
```

Code 5: *Move an include file / header*

For the **implot-demos** we need a few compiling first, from the main directory open the terminal then type:

```
cd implot-demos

mkdir build
cd build
cmake ..
make

cp -r libimnodes.a libimplot.a libapp.a libimgui.a ../../dependencies/
```

Code 6: *Create implot-demos related static library*

The steps above are used to create the static library to run some demos from **implot-demos** that are more complex than the normal implot demo.

Alternative way besides static library:
You don't really need to create all the static libraries, considering if the dependencies are updated and you need a new feature. We think it is better to take the source files instead, e.g. if you take a look at our example:
Hamzstlab-Mathematics/examples/ImNodes Test/

It is added on June 18th, 2025. We don't use **libimnodes.a**, instead we put the source files for **Imnodes** (imnodes.cpp, imnodes.h, imnodes_internal.h) on the parent directory **Hamzstlab-Mathematics/** and adjust the Makefile in that example to process the source codes of Imnodes into object files so the example that needs **Imnodes** feature will be able to be built as executable and run.

Now, you are set to go, in the **Hamzstlab-Mathematics/examples/** directory there are tons of examples that you can just build with Makefile, easy.

For example, from the main directory **Hamzstlab-Mathematics/** open terminal

```
cd examples

cd 3D Test

make
./main
```

Code 7: *Build an example*

This will run the 3D test. You can check each examples source code and learn how to modify according to your needs.

All files, examples codes and source codes used in this books can be found in this repository: **https://github.com/glanzkaiser/Hamzstlab-Mathematics**

II. Example of Creating a 2D Interactive Plot with Hamzstlab-Mathematics in GFreya OS

First you have to already clone or download the repository, then open terminal at the main working directory **Hamzstlab-Mathematics** and type:

```
cd examples

cd First Order Differential Equation

make
./main
```

Code 8: *Create dynamic library*

This will be the result:

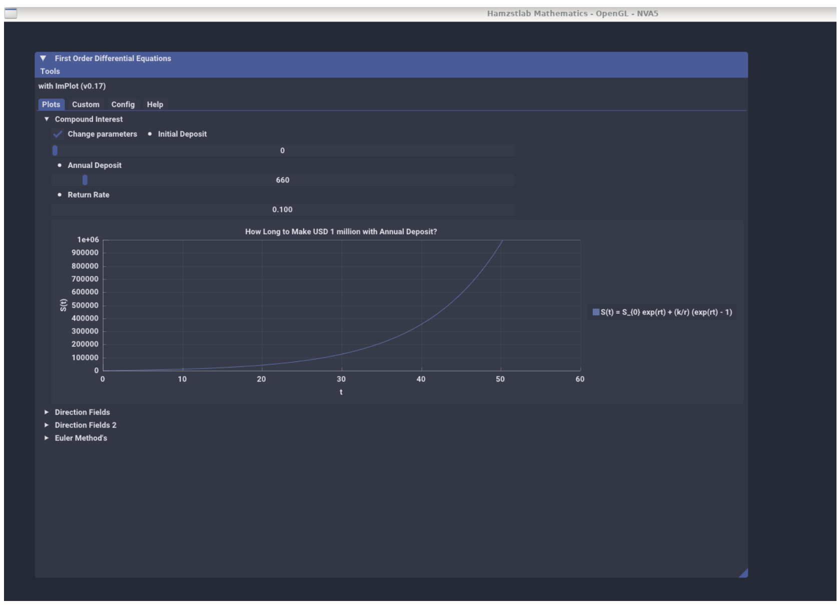


Figure 3.1: *The interactive plot of $S(t)$ with Hamzstlab-Mathematics you can change the parameters and the plot will adjust with the parameters that you just input.*

Now, we are going to explain the code on how to make this example works.

```
#include "App.h"

struct ImPlotDemo : App {
      using App::App;
      void Update() override {
            ImPlot::ShowFirstOrderDEWindow();
      }
};
```

```
int main(int argc, char const *argv[])
{
      ImPlotDemo app("Hamzstlab Mathematics",1920,1080,argc,argv);
      app.Run();

      return 0;
}
```

Code 9: *Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp*

This **main.cpp** is pointing toward **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**, so if you want to modify the example, you just need to edit **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**.

The **ImPlot::ShowFirstOrderDEWindow();** needs to be an **IMPLOT_API**, and this can be done in **Hamzstlab-Mathematics/implot.h**, so you need to add a line like this there

```
...

IMPLOT_API void ShowFirstOrderDEWindow(bool* p_open = nullptr);

...
```

Code 10: *Hamzstlab-Mathematics/implot.h*

For example, from hundred lines of codes you can check the code here to create a plot of Compound Interest, if you scroll down again you will know that we are showing this **Demo_CompoundInterest()** under a new tab that can be opened by the user.

There might be hundreds to thousands of lines of codes, but if you are rigorous and persistence you will get used to it and able to create something from your imagination.

```
...
...

void Demo_CompoundInterest() {
      static float xs1[1001], ys1[1001];
      static int S0 = 0;
      static float r = 0.08;
      static int k = 2000;
      for (int i = 0; i < 1001; ++i) {
            xs1[i] = i;
            ys1[i] = S0 * exp(r*i) + (k/r)*(exp(r*i) - 1) ;
      }

      static bool range = false;
      ImGui::Checkbox("Change parameters", &range);

      if (range) {
            ImGui::SameLine();
```

```
            ImGui::SetNextItemWidth(200);
            ImGui::BulletText("Initial Deposit");
            ImGui::SliderInt("##Initial Deposit", &S0, 0, 10000);
            ImGui::SetNextItemWidth(200);
            ImGui::BulletText("Annual Deposit");
            ImGui::SliderInt("##Annual Deposit", &k, 1, 10000);
            ImGui::SetNextItemWidth(200);
            ImGui::BulletText("Return Rate");
            ImGui::DragFloat("##Return rate", &r, 0.01f, 0.2f);
            ImGui::SetNextItemWidth(200);
        }
        if (ImPlot::BeginPlot("How Long to Make USD 1 million with Annual
            Deposit?")) {
            ImPlot::SetupAxes("t","S(t)");
            ImPlot::SetupAxesLimits(0, 60, 0, 1000000);
            ImPlot::SetupLegend(ImPlotLocation_East,
                ImPlotLegendFlags_Outside);

            ImPlot::PlotLine("S(t) = S_{0} exp(rt) + (k/r) (exp(rt) - 1)
                ", xs1, ys1, 1001);
            ImPlot::EndPlot();
        }
}
...
...

void ShowFirstOrderDEWindow(bool* p_open) {
...

if (ImGui::BeginTabBar("ImPlotDemoTabs")) {
if (ImGui::BeginTabItem("Plots")) {
        DemoHeader("Compound Interest", Demo_CompoundInterest);

...
...
void ImPlot::ShowFirstOrderDEWindow(bool* p_open) {}

#endif
```

Code 11: *Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp*

Last, but not least, for the Makefile **Hamzstlab-Mathematics/examples/First Order Differential Equation/Makefile**, we need to add the SOURCES to be compiled that is pointing toward **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**

```
EXE = main
IMGUI_DIR = ../..
SOURCES = main.cpp
```

```
SOURCES += $(IMGUI_DIR)/imgui.cpp $(IMGUI_DIR)/imgui_demo.cpp $(IMGUI_DIR)/
    imgui_draw.cpp $(IMGUI_DIR)/imgui_tables.cpp $(IMGUI_DIR)/imgui_widgets
    .cpp
SOURCES += $(IMGUI_DIR)/backends/imgui_impl_glfw.cpp $(IMGUI_DIR)/backends/
    imgui_impl_opengl3.cpp
SOURCES += $(IMGUI_DIR)/implot.cpp $(IMGUI_DIR)/implot_items.cpp
SOURCES += $(IMGUI_DIR)/hamzstlab_firstorderdifferentialequations.cpp

OBJS = $(addsuffix .o, $(basename $(notdir $(SOURCES))))
UNAME_S := $(shell uname —s)
LINUX_GL_LIBS = —lGL —lglfw

CXXFLAGS = —std=c++11 —I$(IMGUI_DIR) —I$(IMGUI_DIR)/backends
CXXFLAGS += —g —Wall —Wformat
LIBS = ../../dependencies/glad.c —L../../dependencies/ —lapp —limgui —
    limnodes —limplot

...
...
```

Code 12: *Hamzstlab-Mathematics/examples/First Order Differential Equation/Makefile*

III. Example of Creating a 3D Interactive Plot with Hamzstlab-Mathematics in GFreya OS

First you have to already clone or download the repository, then open terminal at the main working directory **Hamzstlab-Mathematics** and type:

```
cd examples

cd 3D Minimization

make
./main
```

Code 13: *Create dynamic library*

This will be the result:

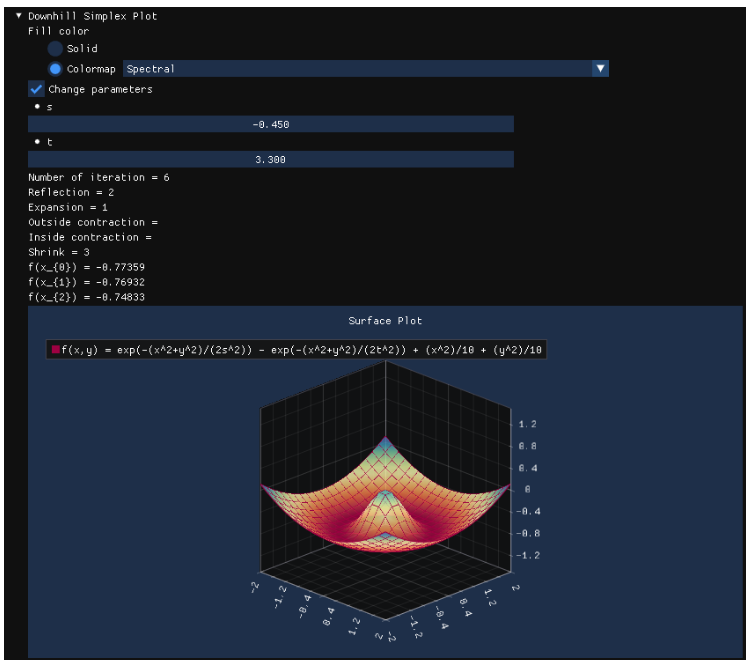


Figure 3.2: *The interactive plot of $f(x,y)$ with Hamzstlab-Mathematics you can change the parameters of s and t and the plot will adjust with the parameters that you just input in 3-dimension.*

Now, we are going to explain the code on how to make this example works.

```
#define IMGUI_DEFINE_MATH_OPERATORS // Access to math operators
#include <math.h>
#include "imgui_internal.h"
//-----------------------------------------------------------------------------

// ImPlot3D Example for Minimization
// hamzstlab_minimization.cpp
```

```
// Date: 2025-08-11
//------------------------------------------------------------------

#include "imgui.h"
#include "imgui_impl_glfw.h"
#include "imgui_impl_opengl3.h"
#include "implot.h"
#include "implot3d.h"
#include <GLFW/glfw3.h>
#include <iostream>

// Callback to handle GLFW errors
void glfw_error_callback(int error, const char* description) { std::cerr <<
     "GLFW Error " << error << ": " << description << std::endl; }

int main() {
       // Setup error callback
       glfwSetErrorCallback(glfw_error_callback);

       // Initialize GLFW
       if (!glfwInit()) {
              std::cerr << "Failed to initialize GLFW" << std::endl;
              return -1;
       }

       // Setup OpenGL version
       #if defined(__APPLE__)
       // GL 3.2 + GLSL 150 (MacOS)
       const char* glsl_version = "#version 150";
       glfwWindowHint(GLFW_CONTEXT_VERSION_MAJOR, 3);
       glfwWindowHint(GLFW_CONTEXT_VERSION_MINOR, 2);
       glfwWindowHint(GLFW_OPENGL_PROFILE, GLFW_OPENGL_CORE_PROFILE); //
           3.2+ only
       glfwWindowHint(GLFW_OPENGL_FORWARD_COMPAT, GL_TRUE); // Required on
           MacOS
       #else
       // GL 3.0 + GLSL 130 (Windows and Linux)
       const char* glsl_version = "#version 130";
       glfwWindowHint(GLFW_CONTEXT_VERSION_MAJOR, 3);
       glfwWindowHint(GLFW_CONTEXT_VERSION_MINOR, 3);
       #endif

       // Create window
       GLFWwindow* window = glfwCreateWindow(1920, 1080, "Hamzstlab
           Mathematics", nullptr, nullptr);
       if (!window) {
              std::cerr << "Failed to create GLFW window" << std::endl;
              glfwTerminate();
```

```
        return -1;
    }
    glfwMakeContextCurrent(window);
    glfwSwapInterval(0); // Disable vsync

    // Setup context
    IMGUI_CHECKVERSION();
    ImGui::CreateContext();
    ImPlot::CreateContext();
    ImPlot3D::CreateContext();

    // Setup style
    ImGui::StyleColorsDark();

    // Setup backend
    ImGui_ImplGlfw_InitForOpenGL(window, true);
    ImGui_ImplOpenGL3_Init(glsl_version);

    // Main loop
    while (!glfwWindowShouldClose(window)) {
        glfwPollEvents();

        // Start frame
        ImGui_ImplOpenGL3_NewFrame();
        ImGui_ImplGlfw_NewFrame();
        ImGui::NewFrame();

        // Demo windows
        ImPlot3D::ShowMinimizationWindow();

        // Render
        ImGui::Render();
        int display_w, display_h;
        glfwGetFramebufferSize(window, &display_w, &display_h);
        glViewport(0, 0, display_w, display_h);
        glClearColor(0.1f, 0.1f, 0.1f, 1.0f);
        glClear(GL_COLOR_BUFFER_BIT);
        ImGui_ImplOpenGL3_RenderDrawData(ImGui::GetDrawData());

        // Swap buffers
        glfwSwapBuffers(window);
    }

    // Cleanup
    ImGui_ImplOpenGL3_Shutdown();
    ImGui_ImplGlfw_Shutdown();
    ImPlot3D::DestroyContext();
    ImPlot::DestroyContext();
```

```
        ImGui::DestroyContext();
        glfwDestroyWindow(window);
        glfwTerminate();

        return 0;
}
```

Code 14: *Hamzstlab-Mathematics/examples/3D Minimization/main.cpp*

A lot of differences with the 2D interactive plot' **main.cpp**, in here we initialize **GLFW** then we call the function **ImPlot3D::ShowMinimizationWindow()** so it will call the demo of 3D minimization problem and plot that we have prepared.

This **main.cpp** is pointing toward **Hamzstlab-Mathematics/hamzstlab_minimization.cpp**, this is the file that we have prepared that will be shown when we compile and run this demo so if you want to modify the example, you just need to edit **Hamzstlab-Mathematics/hamzstlab_minimization.cpp**.

The **ImPlot3D::ShowMinimizationWindow();** needs to be an **IMPLOT3D_API**, and this can be done in **Hamzstlab-Mathematics/implot3d.h** under the **[SECTION] Demo**, so you need to add a line like this there

```
...

IMPLOT3D_API void ShowMinimizationWindow(bool* p_open = nullptr);

...
```

Code 15: *Hamzstlab-Mathematics/implot3d.h*

For example, from hundred lines of codes you can check the code here to create a plot of a surface plot in 3-dimension for minimization problem with downhill simplex algorithm, we are showing this **Demo_DownhillSimplexPlot()** under a new tab that can be opened by the user.

```
...
...

void DemoDownhillSimplexPlot() {

constexpr int N = 20;
static float xs[N * N], ys[N * N], zs[N * N];

// Define the range for X and Y
constexpr float min_val = -2.0f;
constexpr float max_val = 2.0f;
constexpr float step = (max_val - min_val) / (N - 1);
static float s = 0.75;
static float t = 1;

...
```

```
// Plot the surface
ImPlot3D::PlotSurface("f(x,y) = exp(-(x^2+y^2)/(2s^2)) - exp(-(x^2+y^2)
    /(2t^2)) + (x^2)/10 + (y^2)/10", xs, ys, zs, N, N);

// End the plot
ImPlot3D::PopStyleVar();
ImPlot3D::EndPlot();
}
if (selected_fill == 1)
{
ImPlot3D::PopColormap();
}

...
...

void ShowAllDemos() {
       ImGui::Text("with ImPlot3D (%s)", IMPLOT3D_VERSION);
       ImGui::Spacing();
       if (ImGui::BeginTabBar("Minimization")) {
              if (ImGui::BeginTabItem("Plots")) {
                     DemoHeader("Downhill Simplex Plot",
                         DemoDownhillSimplexPlot);
                     DemoHeader("Surface Plots", DemoSurfacePlots);
                     ImGui::EndTabItem();
              }
              if (ImGui::BeginTabItem("Custom")) {
                     DemoHeader("Custom Styles", DemoCustomStyles);
                     DemoHeader("Custom Rendering", DemoCustomRendering);
                     ImGui::EndTabItem();
              }
              if (ImGui::BeginTabItem("Help")) {
                     DemoHelp();
                     ImGui::EndTabItem();
              }
              ImGui::EndTabBar();
       }
}

void ShowMinimizationWindow(bool* p_open) {
       static bool show_implot3d_style_editor = false;
       static bool show_imgui_metrics = false;
       static bool show_imgui_style_editor = false;
       static bool show_imgui_demo = false;
...
...
```

Code 16: *Hamzstlab-Mathematics/hamzstlab_minimization.cpp*

Last, but not least, for the Makefile **Hamzstlab-Mathematics/examples/3D Minimization/-Makefile**, we need to add the SOURCES to be compiled that is pointing toward **Hamzstlab-Mathematics/hamzstlab_minimization.cpp**, then don't forget to add **-lsymintegration** and **-larmadillo** to link to the corresponding libraries that will be responsible for the symbolic computation and vector and matrix related computation.

```
EXE = main
IMGUI_DIR = ../..
SOURCES = main.cpp
SOURCES += $(IMGUI_DIR)/imgui.cpp $(IMGUI_DIR)/imgui_demo.cpp $(IMGUI_DIR)/
    imgui_draw.cpp $(IMGUI_DIR)/imgui_tables.cpp $(IMGUI_DIR)/imgui_widgets.cpp
SOURCES += $(IMGUI_DIR)/backends/imgui_impl_glfw.cpp $(IMGUI_DIR)/backends/
    imgui_impl_opengl3.cpp
SOURCES += $(IMGUI_DIR)/implot.cpp $(IMGUI_DIR)/implot_items.cpp
SOURCES += $(IMGUI_DIR)/implot3d.cpp $(IMGUI_DIR)/implot3d_items.cpp $(IMGUI_DIR)/
    implot3d_meshes.cpp
SOURCES += $(IMGUI_DIR)/hamzstlab_minimization.cpp

OBJS = $(addsuffix .o, $(basename $(notdir $(SOURCES))))
UNAME_S := $(shell uname -s)
LINUX_GL_LIBS = -lGL -lglfw

CXXFLAGS = -std=c++11 -I$(IMGUI_DIR) -I$(IMGUI_DIR)/backends
CXXFLAGS += -g -Wall -Wformat
LIBS = -lsymintegration -larmadillo ../../dependencies/glad.c -L../../
    dependencies/ -lapp -limgui -limnodes -limplot
```

Code 17: *Hamzstlab-Mathematics/examples/3D Minimization/Makefile*

Chapter 4

Intermediate Calculus

"Ma n'atu sole cchiù bello, oi ne', 'O sole mio sta nfronte a te!" - ***O Sole Mio***

WHY we need Calculus? Everthing in this world is mathematics, for those who love sports will love this illustration thus make you want to learn mathematics more. Surfing and snowboarding are among the many sports that use a flat board in contact with a curved surface. The skill lies in how the surfer or snowboarder manipulates the board against the curve to effect changes in velocity and acceleration [6].

Velocity and acceleration are variables that are described in terms of the rate of change of one physical quantity in relation to another; velocity is the rate of change of displacement with respect to time, and acceleration is the rate of change of velocity with respect to time.
Calculus is the mathematics of change, the study of quantities that do not stay the same and whose rate of change is important in a particular context. It is used in fields as varied as physics and finance, architecture and engineering, the setting of credit card payments and the science behind computer games.

The development of calculus is one of the major achievements of mathematics. It stimulated the flowering of mathematics and science that sparked the industrial revolution and led to the growth of the technology that we know today.

Even if sports player do not compute in their head like a calculator to win the game, the basic knowledge is already melting with their muscle, so without even memorizing the formula they just know how to surf on a big tidal wave without falling, or to swing a golf club to make the ball falls near the hole.

We are going to learn how to plot, simulate and compute with either SymIntegration, Hamzstplot, or Hamzstlab Mathematics for this chapter, we are following the Calculus book [19] from chapter 9 to 14 for the theorems, problems and definition. The chapter 1-8 are covered in Lasthrim Projection book [16] and that book is still using Python and JULIA.

I. Infinite Series

- Infinite Sequences

[H*] The first
[H*]

- Infinite Series

[H*] The first
[H*]

- Positive Series: The Integral Test

[H*] The first
[H*]

- Positive Series: Other Tests

[H*] The first
[H*]

- Alternative Series, Absolute Convergence, and Conditional Convergence

[H*] The first
[H*]

- Power Series

[H*] The first
[H*]

- Operations on Power Series

[H*] The first
[H*]

- Taylor and MacLaurin Series

[H*] The first
[H*]

- The Taylor Approximation to a Function

[H*] The first
[H*]

II. Conics and Polar Coordinates

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

III. Geometry in Space and Vectors

- Vector-Valued Functions and Curvilinear Motion

[H*] The first
[H*]

- Surfaces in Three-Space

[H*] The first
[H*]

IV. Derivatives for Functions of Two or More Variables

- Functions of Two or More Variables

[H*] The first
[H*]

- Partial Derivatives

[H*] The first
[H*]

V. Multiple Integrals

- Double Integrals over Rectangles

[H*] The first
[H*]

- Iterated Integrals

[H*] The first
[H*]

VI. Vector Calculus

- Vector Fields

[H*] Previously, we use functions where the input is a real number and the output is a real number. Vector Calculus talks about vector-valued functions, that is, functions whose input is a real number and whose output is a vector.
Consider a function $\boldsymbol{F}$ that associates with each point $\boldsymbol{p}$ in n-space a vector $\boldsymbol{F}(p)$. A typical example in two-space is

$$\boldsymbol{F}(\boldsymbol{p}) = \boldsymbol{F}(x,y) = -\frac{1}{2}y\boldsymbol{i} + \frac{1}{2}x\boldsymbol{j}$$

For historical reasons, we refer to such function as a vector field. Imagine that to each point $\boldsymbol{p}$ in a region of space is attached a vector $\boldsymbol{F}(\boldsymbol{p})$ emanating from $\boldsymbol{p}$.

[H*] Vector fields that arise naturally in science are electric fields, magnetic fields, force fields, and gravitational fields. We consider only the case in which these fields are independent of time, which we call steady vector fields.

In contrast to a vector field, a function F that attaches a number to each point in space is called a scalar field. The function that gives the temperature at each point would be a good physical example of a scalar field.

[H*] **The Gradient of a Scalar Field**
Let $f(x,y,z)$ determine a scalar field and suppose that f is differentiable. then the gradient of f, denoted by ∇f, is the vector field given by

$$\boldsymbol{F}(x,y,z) = \nabla f(x,y,z) = \frac{\partial f}{\partial x}\boldsymbol{i} + \frac{\partial f}{\partial y}\boldsymbol{j} + \frac{\partial f}{\partial z}\boldsymbol{k} \tag{4.1}$$

We learned that $\nabla f(x,y,z)$ points in the direction of greatest increase of $f(x,y,z)$. A vector field $\boldsymbol{F}$ that is the gradient of a scalar field f is called a conservative vector field, and f is its potential function.

Such fields and their potential functions are important in physics. In particular, fields that obey the inverse square law (e.g., electric fields and gravitational fields) are conservative.

[H*] **The Divergence and Curl of a Vector Field**
With a given vector field

$$\boldsymbol{F} = M(x,y,z)\boldsymbol{i} + N(x,y,z)\boldsymbol{j} + P(x,y,z)\boldsymbol{k} \tag{4.2}$$

are associated to two other important fields. the first is called the divergence of $\boldsymbol{F}$ (div $\boldsymbol{F}$), which is a scalar field; the second is called the curl of $\boldsymbol{F}$ (curl $\boldsymbol{F}$), which is a vector field.

Definition 4.1: div and curl

Let $\boldsymbol{F} = M\boldsymbol{i} + N\boldsymbol{j} + P\boldsymbol{k}$ be a vector field for which the first partial derivatives of $M, N,$ and P exist. Then

$$\text{div } \boldsymbol{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} \tag{4.3}$$

$$\text{curl } \boldsymbol{F} = \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}\right)\boldsymbol{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}\right)\boldsymbol{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\boldsymbol{k} \tag{4.4}$$

- Line Integrals

[H*] The first
[H*]

- Independence of Path

[H*] The first
[H*]

- Green's Theorem in the Plane

[H*] The first
[H*]

- Surface Integrals

[H*] The first
[H*]

- Gauss's Divergence Theorem

[H*] The first
[H*]

- Stokes's Theorem

[H*] The first
[H*]

Chapter 5

Differential Equations

"Everyone's favorite fighting fairy godmother is still kicking, taking out Ether Strike hits for her godfather Odin. The heretofore unmatched fury of her scowl derives from her lack of lines in the main story" - ***Freya (Valkyrie Profile: Covenant of the Plume)***

Differential equations are equations that contain derivatives. It is used to understand and to investigate problems involving the motion of fluids, the flow of current in electric circuits, the dissipation of heat in solid objects, the propagation and detection of seismic waves [4]. For example, the Newton's second law can be written as a differential equation

$$F = ma \tag{5.1}$$

$$F = m\frac{dv}{dt} = mv' \tag{5.2}$$

We think of v as a function of t, or we can also write it as $v(t)$.

For this chapter, we will follow the outline of [4] book and we start it on May 2025, we will take the measure how long it takes to finish this chapter that is equal to the book mentioned.

I. Direction Fields

[H*] Direction fields are valuable tools in studying the solutions of differential equations of the form

$$\frac{dy}{dt} = f(t, y) \tag{5.3}$$

where f is a given function of two variables t and y, sometimes referred to as the rate function.
A direction field (slope field) is a mathematical object used to graphically represent solutions to a first-order differential equation. At each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point.

A direction field can be constructed by evaluating f at each point of a rectangular grid. At each point of the grid, a short line segment is drawn whose slope is the value of f at that point. Thus each line segment is tangent to the graph of the solution passing through that point.

[H*] A direction field drawn on a fairly fine grid gives good picture of the overall behavior of solutions of a differential equation.

The construction of a direction field is often a useful step in the investigation of a differential equation.

[H*] **Example:**
Consider a population of field mice who inhabit a certain rural area. If we denote time by t and the mouse population by $p(t)$ with the time is measured in months, then the differential that represents the population growth of field mice that is also preyed on by owls can be expressed by the equation

$$\frac{dp}{dt} = 0.5p - 450$$

Observe that the predation term is measured in months as -450, meaning that each month there is about 450 mouse being preyed by owls every month.

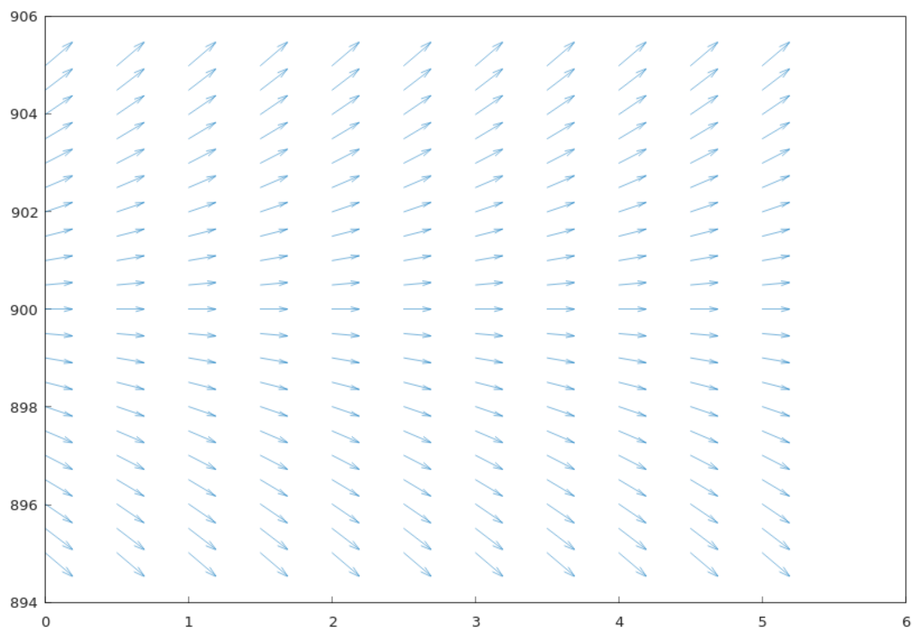


Figure 5.1: *The direction field for $\frac{dp}{dt} = 0.5p - 450$, the x-axis represents the variable t as the time and the y-axis represents variable $p(t)$ as the population growth of field mice, we can see that if the rate of the reproduction for the field mice is greater than the rate of the mice being preyed on by owls then the population growth will keep increasing, and if the rate of the reproduction for the fiel mice is smaller than the rate of the mice being preyed on by owls then the population of field mice will keep decreasing.*

The code to create the direction field above with Hamzstplot:

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;
```

```
int main() {
        using namespace hamzstplot;
        double dx, dy, magnitude;
        auto [x, y] = meshgrid(iota(0, 0.5, 5), iota(895, 0.5,
             905));

        vector_2d u = transform(x, y, [](double x, double y) {
             return 1 ; }); // dx
        vector_2d v = transform(x, y, [](double x, double y) {
             return 0.5*y - 450 ; }); // dy
        quiver(x, y, u, v);

        show();
        return 0;
}
```

Code 18: *Hamzstplot/Examples with Makefile/Plot Direction Fields Growth and Predation Rate/main.cpp*

For sufficiently large values of p it can be seen that $\frac{dp}{dt}$ is positive, so that solutions increase. On the other hand, if p is small, then $\frac{dp}{dt}$ is negative and solutions decrease.

The critical value of p that separates solutions that increase from those that decrease is the value of p for which $\frac{dp}{dt}$ is zero.

By setting $\frac{dp}{dt}$ equals zero and then solving for p, we find the equilibrium solution $p(t) = 900$ for which the growth term and the predation term are exactly balanced.

II. Solving a Simple Differential Equation

[H*] Consider we have this differential equation

$$\frac{dp}{dt} = 0.5p - 450$$

If we want to obtain the solution $p(t)$ we will have to perform some algebraic trick or manipulation then integrating them with respect to the correct variable.

$$\begin{aligned}
\frac{dp}{dt} &= 0.5p - 450 \\
&= \frac{p-900}{2} \\
\frac{dp/dt}{p-900} &= \frac{1}{2} \\
\frac{dp}{p-900} &= \frac{dt}{2} \\
\int \frac{dp}{p-900} &= \int \frac{dt}{2} \\
\ln|p-900| &= \frac{t}{2} + C \\
|p-900| &= e^{\frac{t}{2}+C} \\
|p-900| &= e^{\frac{t}{2}}e^{C} \\
p - 900 &= \pm e^{\frac{t}{2}}e^{C} \\
p &= 900 + ce^{\frac{t}{2}}
\end{aligned}$$

where $c = \pm e^{C}$ is also an arbitrary (nonzero) constant. Note that the constant function $p = 900$ is also a solution where $c = 0$.

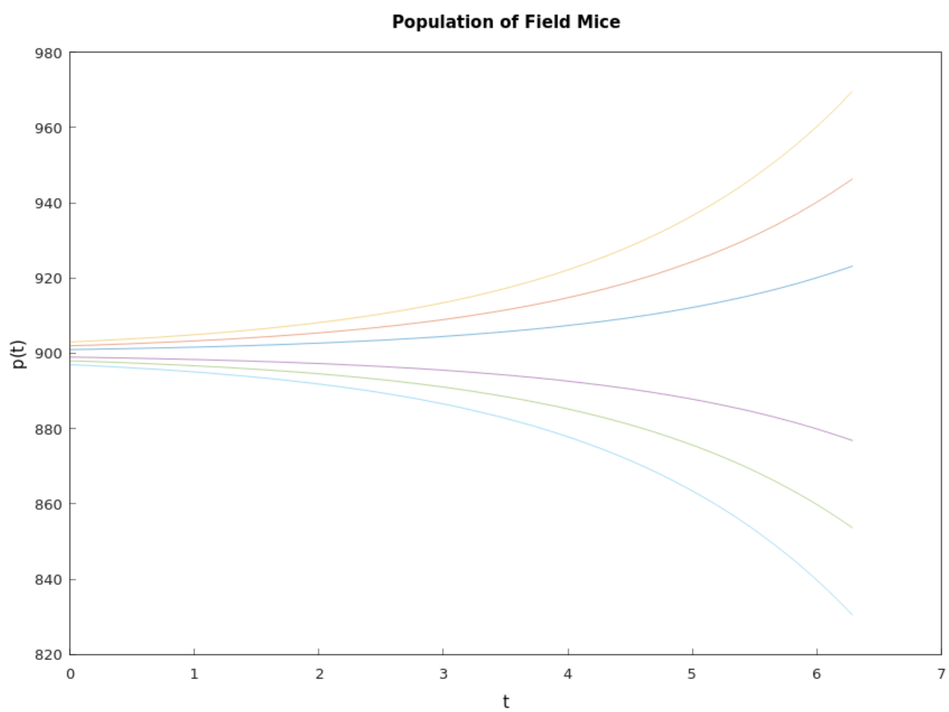


Figure 5.2: *The graphs shows the solutions for $\frac{dp}{dt} = 0.5p - 450$ with several valus of c.*

```cpp
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;

int main() {
        using namespace hamzstplot;

        std::vector<double> t = linspace(0, 2 * pi);
        std::vector<double> y1 = transform(t, [](auto t) {
            return 900 + 1*exp(0.5*t); });
        std::vector<double> y2 = transform(t, [](auto t) {
            return 900 + 2*exp(0.5*t); });
        std::vector<double> y3 = transform(t, [](auto t) {
            return 900 + 3*exp(0.5*t); });
        std::vector<double> y4 = transform(t, [](auto t) {
            return 900 + -1*exp(0.5*t); });
        std::vector<double> y5 = transform(t, [](auto t) {
            return 900 + -2*exp(0.5*t); });
        std::vector<double> y6 = transform(t, [](auto t) {
            return 900 + -3*exp(0.5*t); });

        plot(t, y1, t, y2, t, y3, t, y4, "--", t, y5,"--", t,y6,
            "--");
        title("Population of Field Mice");
        xlabel("t");
        ylabel("p(t)");
        show();
        return 0;
}
```

Code 19: *Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Solutions Curves/main.cpp*

III. Classification of Differential Equations

[H*] One important classification is based on whether the unknown function depends on a single independent variable or on several independent variables.

If a function depends on a single independent variable then it is said to be an ordinary differential equation, this is the example:

$$L\frac{d^2Q(t)}{dt^2} + R\frac{dQ(t)}{dt} + \frac{1}{C}Q(t) = E(t) \tag{5.4}$$

with $Q(t)$ represents the charge on a capacitor in a circuit with capacitance C, resistance R, and inductance L.

If a function depends on mroe than one independent variable then it is called a partial differential equation, this is the example:

$$\alpha^2 \frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial u(x,t)}{\partial t} \tag{5.5}$$

it is called the heat conduction equation with $alpha$ as the physical constant. Another example is:

$$\alpha^2 \frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial^2 u(x,t)}{\partial t^2} \tag{5.6}$$

it is called the wave equation with α as the physical constant.

[H*] System of differential equations usually have two or more unknown functions. For example, the Lotka-Volterra equations:

$$\begin{aligned} \frac{dx}{dt} &= ax - \alpha xy \\ \frac{dy}{dt} &= -cy + \gamma xy \end{aligned} \tag{5.7}$$

where $x(t)$ and $y(t)$ are the repsective populations of the prey and predator species. The constants $a, \alpha, c,$ and γ are based on empirical observations and depend on the particular species being studied.

[H*] The order of a differential equation is the order of the highest derivative that appears in the equation.

More genrally, the equation

$$F[t, u(t), u'(t), \ldots, u^{(n)}(t)] = 0 \tag{5.8}$$

is an ordinary differential equation of the nth order.

Another one,

$$y''' + 2e^t y'' + yy' = t^3$$

is a third order differential equation for $y = u(t)$.

[H*] The ordinary differential equation

$$F(t, y, y', \ldots, y^{(n)}) = 0$$

is said to be linear if F is a linear function of the variables $y, y', \ldots, y^{(n)}$; a similar definition applies to partial differential equations.

Thus, the general linear ordinary differential equation of order n is

$$a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \cdots + a_n(t)y = g(t) \tag{5.9}$$

A simple physical problem that leads to a nonlinear differential equation is the oscillating pendulum. The angle θ that an oscillating pendulum of length L makes with the vertical direction satisfies the equation

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \sin\ \theta = 0 \tag{5.10}$$

the term involving sin θ makes that equation to be nonlinear.

IV. First Order Ordinary Differential Equations

- Linear Equations; Method of Integrating Factors

[H*] Suppose we have this differential equations of first order

$$\frac{dy}{dt} = f(t, y)$$

where f is a given function of two variables. Any differentiable function $y = \phi(t)$ that satisfies this equation for all t in some interval is called a solution, and our object is to determine whether such functions exist.

For an arbitrary function f, there is no general method for solving the equation in terms of elementary functions.

[H*] We define the general first order linear equation in the standard form

$$\frac{dy}{dt} + p(t)y = g(t) \tag{5.11}$$

where p and g are given functions of the independent variable t.

To determine an appropriate integrating factor, we multiply Eq. (5.11) by an undetermined function $\mu(t)$, obtaining

$$\mu(t)\frac{dy}{dt} + p(t)\mu(t)y = \mu(t)g(t) \tag{5.12}$$

We see the left side of the Eq. (5.12) is the derivative of the product $\mu(t)y$, provided that $\mu(t)$ satisfies the equation

$$\frac{d\mu(t)}{dt} = p(t)\mu(t) \tag{5.13}$$

If we assume temporarily that $\mu(t)$ is positive, then we have

$$\frac{d\mu(t)/dt}{\mu(t)} = p(t)$$

$$\ln|\mu(t)| = \int p(t)\, dt + k$$

By choosing the arbitrary constant k to be zero, we obtain the simplest possible function for μ, namely,

$$\mu(t) = e^{\int p(t)\, dt} \tag{5.14}$$

Note that $\mu(t)$ is positive for all t, as we assumed. Returning to Eq. (5.12), we have

$$\frac{d}{dt}[\mu(t)y] = \mu(t)g(t) \tag{5.15}$$

Hence

$$\mu(t)y = \int \mu(t)g(t)\, dt + c \tag{5.16}$$

where c is an arbitrary constant.

The general solution of Eq. (5.11) is

$$y = \frac{1}{\mu(t)} \left[\int_{t_0}^{t} \mu(s) g(s)\ ds + c \right] \tag{5.17}$$

where t_0 is some convenient lower limit of integration.

[H*] **Example:**
Solve the initial value problem

$$\begin{aligned} ty' + 2y &= 4t^2 \\ y(1) &= 2 \end{aligned}$$

Solution:
In order to determine $p(t)$ and $g(t)$ correctly, we must first rewrite the equation in the standard form of Eq. (5.11). Thus we have

$$y' + (2/t)y = 4t \tag{5.18}$$

so

$$\begin{aligned} p(t) &= \frac{2}{t} \\ g(t) &= 4t \end{aligned}$$

Now, we first compute the integrating factor $\mu(t)$:

$$\begin{aligned} \mu(t) &= e^{\frac{2}{t}\ dt} \\ &= e^{2 \ln |t|} \\ &= t^2 \end{aligned}$$

Following the formula of Eq. (5.17), we obtain

$$\begin{aligned} y &= \frac{1}{\mu(t)} \left[\int_{t_0}^{t} \mu(s) g(s)\ ds + c \right] \\ &= \frac{1}{t^2} \left[\int_{0}^{t} s^2 (4s)\ ds \right] \\ &= \frac{1}{t^2} \left[\int_{0}^{t} 4s^3\ ds \right] \\ &= \frac{1}{t^2} \left[t^4 + c \right] \\ y &= t^2 + \frac{c}{t^2} \end{aligned}$$

$y(t) = t^2 + \frac{c}{t^2}$ is the general solution of $ty' + 2y = 4t^2$.

To obtain the value for c, we will input the the initial condition value $y(1) = 2$ to the general solution

$$\begin{aligned} y(1) &= 2 \\ 2 &= 1^2 + \frac{c}{1^2} \\ c &= 1 \end{aligned}$$

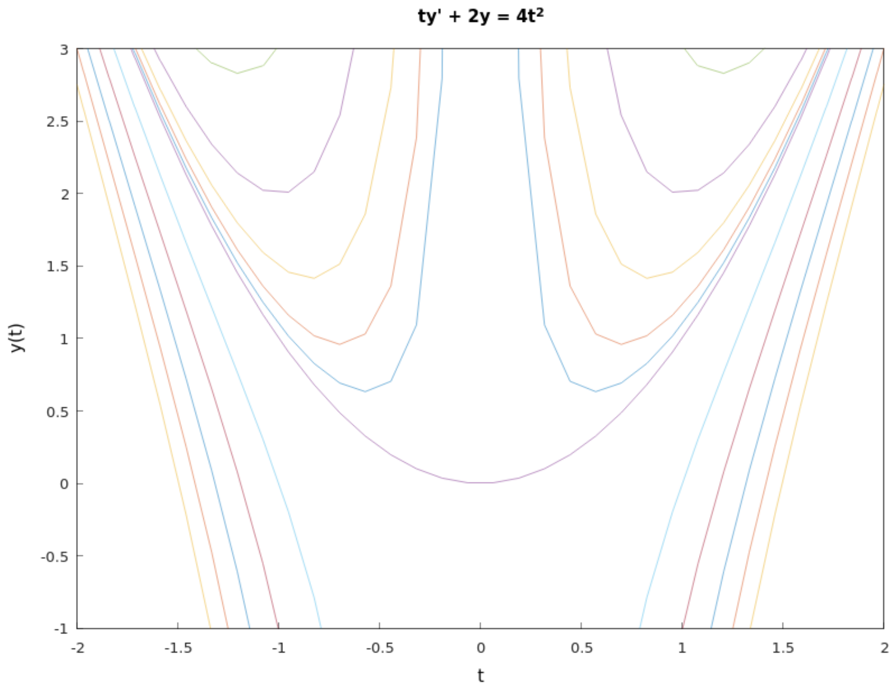


Figure 5.3: *The graphs shows the solutions for $ty' + 2y = 4t^2$.*

The solution becomes unbounded and is asymptotic to the positive y-axis as $t \to 0$ from the right. This is the effect of the infinite discontinuity in the coefficient $p(t)$ at the origin.

The function $y = t^2 + \frac{1}{t^2}$ for $t < 0$ is not part of the solution of this initial value problem. The solution fails to exist for some values of t. This is due to the infinite discontinuity in $p(t)$ at $t = 0$, which restricts the solution to the interval $0 < t < \infty$.

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;

int main() {
        using namespace hamzstplot;

        std::vector<double> t = linspace(-2 * pi, 2 * pi);
        std::vector<double> y1 = transform(t, [](auto t) { return
            pow(t,2) + 0.1/(pow(t,2)); });
        std::vector<double> y2 = transform(t, [](auto t) { return
            pow(t,2) + 0.23/(pow(t,2)); });
        std::vector<double> y3 = transform(t, [](auto t) { return
            pow(t,2) + 0.5/(pow(t,2)); });
        std::vector<double> y4 = transform(t, [](auto t) { return
            pow(t,2) + 1/(pow(t,2)); });
        std::vector<double> y5 = transform(t, [](auto t) { return
```

```cpp
        pow(t,2) + 2/(pow(t,2)); });
    std::vector<double> y6 = transform(t, [](auto t) { return pow(t,2) + -1/(pow(t,2)); });
    std::vector<double> y7 = transform(t, [](auto t) { return pow(t,2) + -2/(pow(t,2)); });
    std::vector<double> y8 = transform(t, [](auto t) { return pow(t,2) + -3/(pow(t,2)); });
    std::vector<double> y9 = transform(t, [](auto t) { return pow(t,2) + -4/(pow(t,2)); });
    std::vector<double> y10 = transform(t, [](auto t) { return pow(t,2) + -5/(pow(t,2)); });
    std::vector<double> y11 = transform(t, [](auto t) { return pow(t,2); });

    plot(t, y1, t, y2, t, y3, t, y4, "-", t, y5,"-", t,y6, "-", t,y7, "-", t,y8, "-", t,y9, "-", t,y10, "-", t, y11, "-");
    axis({-2, 2, -1, 3});
    title("ty' + 2y = 4t^2");
    xlabel("t");
    ylabel("y(t)");
    show();
    return 0;
}
```

Code 20: *Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Solutions Curves for Method of Integrating Factors/main.cpp*

- Separable Equations

[H*] The general first order equation is

$$\frac{dy}{dx} = f(x,y) \tag{5.19}$$

By setting $M(x,y) = -f(x,y)$ and $N(x,y) = 1$ we will have

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0 \tag{5.20}$$

If it happens that M is a function of x only and N is a function of y only, then it becomes

$$M(x) + N(y)\frac{dy}{dx} = 0 \tag{5.21}$$

Such an equation is said to be separable, because if it is written in the differential form

$$M(x)\ dx + N(y)\ dy = 0 \tag{5.22}$$

The differential form (5.22) is also more symmetric and tends to suppress the distinction between independent and dependent variables.

A separable equation can be solved by integrating the functions M and N.

[H*] **Example:**
Show that the equation

$$\frac{dy}{dx} = \frac{x^2}{1-y^2}$$

is separable, and then find an equation for its integral curves.

Solution:

If we write the equation as

$$-x^2 + (1-y^2)\frac{dy}{dx} = 0$$

then it is separable. Recall from calculus that if y is a function of x, then by the chain rule

$$\frac{d}{dx}f(y) = \frac{d}{dy}f(y)\frac{dy}{dx} = f'(y)\frac{dy}{dx}$$

If $f(y) = y - \frac{y^3}{3}$, then

$$\frac{d}{dx}\left(y - \frac{y^3}{3}\right) = (1-y^2)\frac{dy}{dx}$$

Thus,

$$\begin{aligned} -x^2 + (1-y^2)\frac{dy}{dx} &= 0 \\ (1-y^2)dy &= x^2\,dx \\ y - \frac{y^3}{3} &= \frac{x^3}{3} + c \\ -x^3 + 3y - y^3 &= c \end{aligned}$$

Any differentiable function $y = \phi(x)$ that satisfies $-x^3 + 3y - y^3 = c$ is a solution of $\frac{dy}{dx} = \frac{x^2}{1-y^2}$.

An equation of the integral curve passing through a particular point (x_0, y_0) can be found by substituting x_0 and y_0 for x and y, respectively, in $-x^3 + 3y - y^3 = c$ and determining the corresponding value of c.

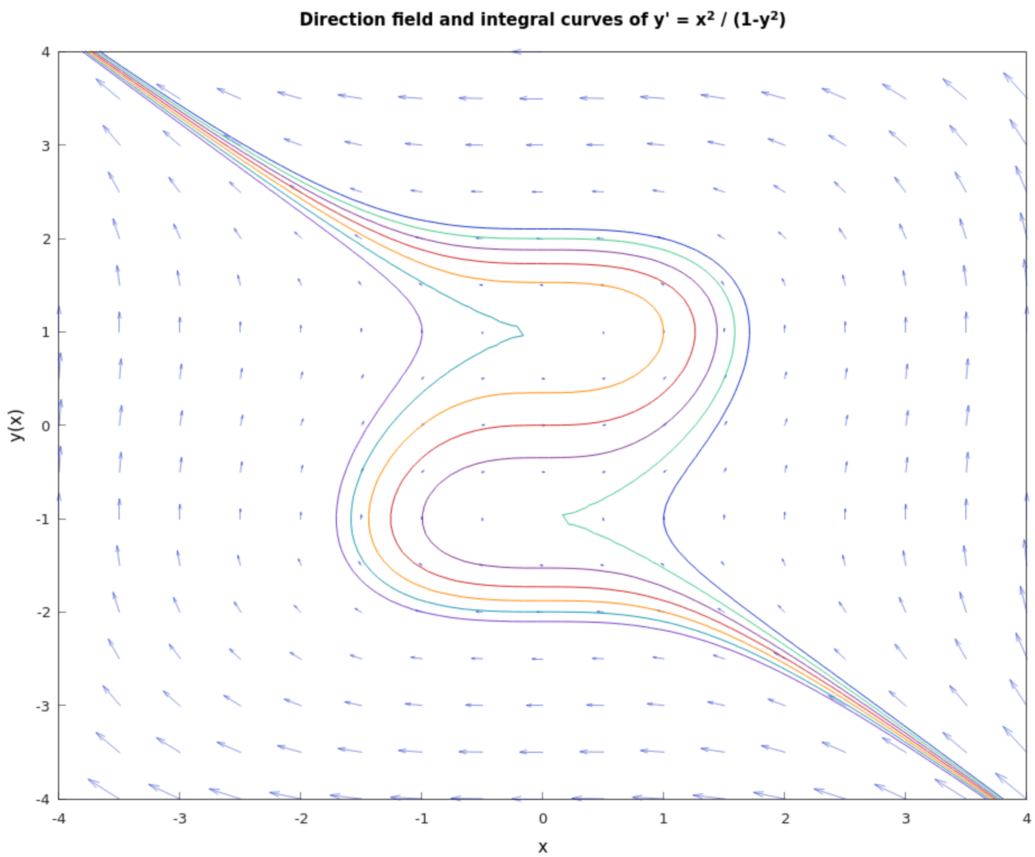


Figure 5.4: *The graphs shows the direction field and integral curves of* $y' = \frac{x^2}{1-y^2}$.

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;

int main() {
        using namespace hamzstplot;

        vector_2d newcolors = {{0.83, 0.14, 0.14},
                {1.00, 0.54, 0.00},
                {0.10, 0.6, 0.70},
                {0.47, 0.25, 0.80},
                {0.50, 0.2, 0.60},
                {0.25, 0.80, 0.54}};
        colororder(newcolors);

        auto [x, y] = meshgrid(iota(-4, 0.5, 5), iota(-4, 0.5, 4))
            ;
        vector_2d u = transform(x, y, [](double x, double y) {
```

```
            return 1-pow(y,2) ; }); // dx
        vector_2d v = transform(x, y, [](double x, double y) {
            return pow(x,2) ; }); // dy

        quiver(x, y, u, v);
        hold(on);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) ; },"-")->line_width(1);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) - 1; },"-")->line_width(1);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) - 2; },"-")->line_width(1);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) - 3; },"-")->line_width(1);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) + 1; },"-")->line_width(1);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) + 2; },"-")->line_width(1);
        fimplicit([](double x, double y) { return -pow(x, 3) + 3*y
             - pow(y, 3) + 3; },"-")->line_width(1);
        hold(off);

        axis({-4, 4, -4, 4});
        title("Integral curves of y' = x^{2} / (1-y^{2})");
        xlabel("x");
        ylabel("y(x)");
        show();
        return 0;
}
```

Code 21: *Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Direction Field and Solutions Curves for Separable Equations/main.cpp*

The coding here is using **fimplicit** since we are inputting $-x^3 + 3y - y^3 - c = 0$ as the function to be drawn, it is the implicit function. We also draw the direction field so we are using **hold(on)** then closing it with **hold(off)**.

- Modeling with First Order Equations

[H*] Differential equations are of interest to nonmathematicians primarily because of the possibility of using them to investigate a wide variety of problems in the physical, biological, and social sciences.

One reason for this is that mathematical models and their solutions lead to equations relating the variables and parameters in the problem. These equations often enable you to make predictions about how the natural process will behave in various circumstances.

Mathematical modeling and experiment or observation are both critically important and have somewhat complementary roles in scientific investigations. Mathematical models are validated by comparison of their predictions with experimental results.

[H*] Three identifiable steps that are always present in the process of mathematical modeling:

1. Construction of the Model.
2. Analysis of the Model.
3. Comparison with Experiment or Observation.

[H*] **Compound Interest**

Suppose that a sum of money is deposited in a bank that pays interest at an annual rate r. The value $S(t)$ of the investment at any time t depends on the frequency with which interest is compounded as well as on the interest rate.

Financial institutions have various policies concerning compounding: some compound monthly, some weekly, some even daily.

If we assume that compounding takes place continuously, then we can set up a simple initial value problem that describes the growth of the investment.

The rate of change of the value of the investment is $\frac{dS}{dt}$, and this quantity is equal to the rate at which interest accrues, which is the interest rate r times the current value of the investment $S(t)$. Thus

$$\frac{dS}{dt} = rS \tag{5.23}$$

is the differential equation that governs the process. Suppose that we also know the value of the investment at some particular time, say,

$$S(0) = S_0 \tag{5.24}$$

Then the solution of the initial value problem at Eq. (5.23) and Eq. (5.24) gives the balance $S(t)$ in the account at any time t. This initial value problem is readily solved, since the differential equation at Eq. (5.23) is both linear and separable. Consequently, by solving Eqs. (5.23) and (5.24), we find that

$$S(t) = S_0 e^{rt} \tag{5.25}$$

Thus a bank account with continuously compounding interest grows exponentially.

Let us suppose that there may be deposits or withdrawals in addition to the accrual of interest, dividends, or capital gains. If we assume that the deposits or withdrawals take place at a constant rate k, then Eq. (5.23) is replaced by

$$\frac{dS}{dt} = rS + k \tag{5.26}$$

where k is positive for deposits and negative for withdrawals.

Eq. (5.26) is linear with the integrating factor e^{-rt}, so its general solution is

$$S(t) = ce^{rt} - \frac{k}{r} \tag{5.27}$$

where c is an arbitrary constant. To satisfy the initial condition Eq. (5.24), we must choose $c = S_0 + \frac{k}{r}$. Thus the solution of the initial value problem is

$$S(t) = S_0 e^{rt} + \left(\frac{k}{r}\right)(e^{rt} - 1) \tag{5.28}$$

The first term in the expression (5.28) is the part of $S(t)$ that is due to the return accumulated on the initial amount S_0, and the second term is the part that is due to the deposit or withdrawal rate k.

The advantage of stating the problem in this general way without specific values for S_0, r, or k lies in the generality of the resulting formula (5.28) for $S(t)$. With this formula we can readily compare the results of different investment programs or different rates of return.

[H*] **Retirement Plan Example:**
Suppose that one person opens an individual retirement account (IRA) at age 25 and makes annual investments of USD 2000 thereafter in a continuous manner. Assuming a rate of return of 8%, what will be the balance in the IRA at age 65?

Solution:

We have

$$\begin{aligned} S_0 &= 0 \\ r &= 0.08 \\ k &= 2000 \end{aligned}$$

and we wish to determine $S(40)$. From Eq. (5.28) we have

$$\begin{aligned} S(40) &= (0)e^{(0.08)(40)} + \left(\frac{2000}{0.08}\right)(e^{(0.08)(40)} - 1) \\ &= (25,000)(e^{3.2} - 1) \\ &= 588,313 \end{aligned}$$

It is interesting to note that the total amount invested is USD 80,000 ($40 \times 2,000$), so the remaining amount of USD 508,313 results from the accumulated return on the investment. The balance after 40 years is also fairly sensitive to the assumed rate.

For the C++ code, we are using Hamzstlab Mathematics for the first time as the introduction, this is to complement Hamzstplot beautifully. Hamzstlab Mathematics is using ImGUI and Implot so it can make the interactive GUI like in this example.

We can of course plot $S(t)$ easily with various rate with Hamzstplot, but we want a slider that can change the plot at an instant, that is why we are using Hamzstlab Mathematics, it can do the job better and faster.

We create a few examples for the first order ordinary differential equations section for Hamzstlab Mathematics. The code to create this is longer, not only one CPP file, but more than one, and it is a lot different than Hamzstplot, a lot of learning, you have to read a lot for the codes, read about ImGUI, Implot, steeper curve of learning, but it will worth your time since it can sustain not only decades but till another 100 years for the durability.

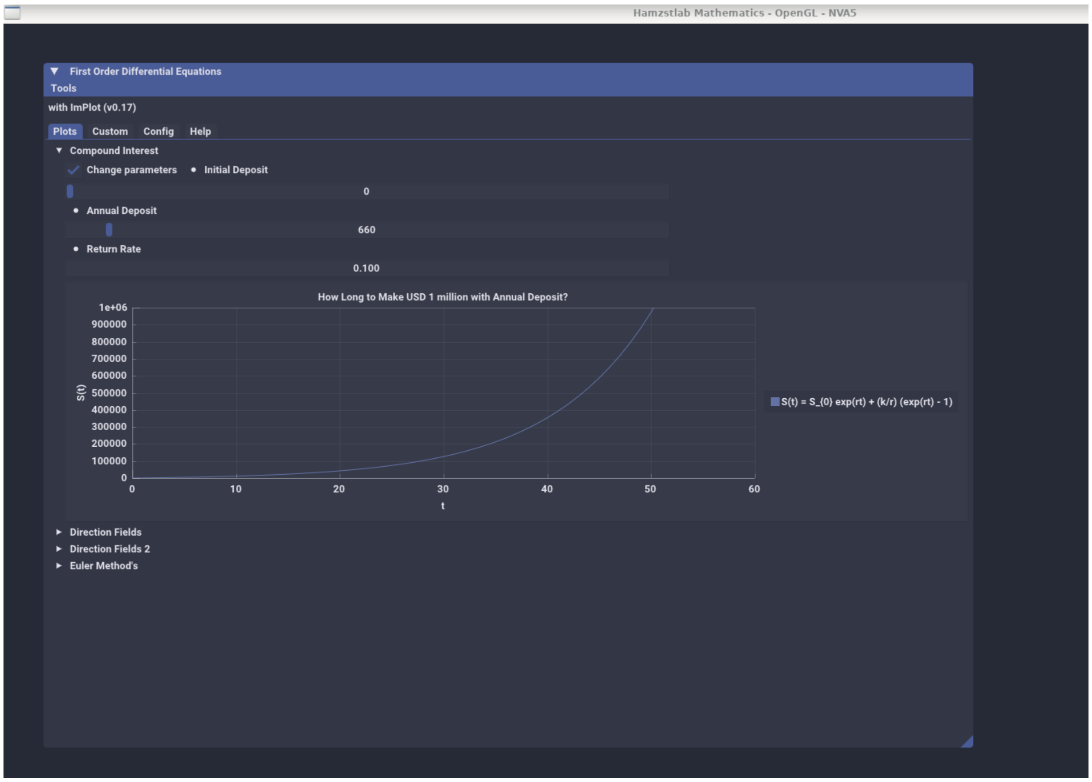


Figure 5.5: *The interactive plot of $S(t)$ with Hamzstlab-Mathematics you can change the parameters and the plot will adjust with the parameters that you just input (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

```
...
...

void Demo_CompoundInterest() {
static float xs1[1001], ys1[1001];
static int S0 = 0;
static float r = 0.08;
static int k = 2000;
for (int i = 0; i < 1001; ++i) {
xs1[i] = i;
ys1[i] = S0 * exp(r*i) + (k/r)*(exp(r*i) - 1) ;
}

static bool range = false;
ImGui::Checkbox("Change parameters", &range);
```

```
if (range) {
ImGui::SameLine();
ImGui::SetNextItemWidth(200);
ImGui::BulletText("Initial Deposit");
ImGui::SliderInt("##Initial Deposit", &S0, 0, 10000);
ImGui::SetNextItemWidth(200);
ImGui::BulletText("Annual Deposit");
ImGui::SliderInt("##Annual Deposit", &k, 1, 10000);
ImGui::SetNextItemWidth(200);
ImGui::BulletText("Return Rate");
ImGui::DragFloat("##Return rate", &r, 0.01f, 0.2f);
ImGui::SetNextItemWidth(200);
}
if (ImPlot::BeginPlot("How Long to Make USD 1 million with Annual
    Deposit?")) {
ImPlot::SetupAxes("t","S(t)");
ImPlot::SetupAxesLimits(0, 60, 0, 1000000);
ImPlot::SetupLegend(ImPlotLocation_East, ImPlotLegendFlags_Outside
    );

ImPlot::PlotLine("S(t) = S_{0} exp(rt) + (k/r) (exp(rt) - 1)",
    xs1, ys1, 1001);
ImPlot::EndPlot();
}
}
...
...
```

Code 22: *Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp*

We only include the snippet of code to plot the $S(t)$, the full code can be seen at Hamzstlab Mathematics from the repository of Hamzstlab Mathematics at the **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**, to run the example with Makefile you can go to this directory **Hamzstlab-Mathematics/examples/First Order Differential Equation**.

[H*] **Mortgage Loan Example:**
A home buyer can afford to spend no more than USD 500 / month on mortgage payments. Suppose that the interest rate is 6%, that interest is compounded continuously, and that payments are also made continuously.

(a) Determine the maximum amount that this buyer can afford to borrow on a 20-year mortgage.

(b) Determine the total interest paid during the term of the mortgage.

Solution:

(a) The amount of money $S(t)$ that the buyer has to pay changes in time due to two factors, the compound interest and its' continuous payments. The rate of growth for compounding is rS, and the rate of decay due to the continuous payments is k.

$$\frac{dS}{dt} = rS - k$$

$$\frac{dS}{dt} - rS = -k$$

This is a linear first-order inhomogeneous ODE, so it can be solved by multiplying both sides by an integrating factor $\mu(t)$

$$\mu(t) = e^{\int^t (-r)\ ds} = e^{-rt}$$

proceed with the multiplication and solve it for $S(t)$

$$\begin{aligned}
e^{-rt}\frac{dS}{dt} - re^{-rt}S &= -ke^{-rt} \\
\frac{d}{dt}(e^{-rt}S) &= -ke^{-rt} \\
e^{-rt}S &= \frac{k}{r}e^{-rt} + C \\
S(t) &= \frac{k}{r} + Ce^{rt}
\end{aligned}$$

Apply the initial condition $S(0) = S_0$ to determine C

$$\begin{aligned}
S(0) &= \frac{k}{r} + Ce^{rt} \\
S_0 &= \frac{k}{r} + Ce^{rt} \\
C &= S_0 - \frac{k}{r}
\end{aligned}$$

Therefore, the amount of money the buyer has to pay after t years is

$$\begin{aligned}
S(t) &= \frac{k}{r} + \left(S_0 - \frac{k}{r}\right)e^{rt} \\
&= \frac{k}{r}(1 - e^{rt}) + S_0e^{rt}
\end{aligned}$$

For a year the home buyer will need to pay $k = 500 \times 12 = USD\ 6,000$ year and $r = 6\% = 0.06$.

$$S(t) = \frac{6000}{0.06}(1 - e^{0.06t}) + S_0e^{0.06t}$$

For a 20-year mortgage, $S(t) = 0$ at $t = 20$.

$$\begin{aligned}
0 &= \frac{6000}{0.06}(1 - e^{0.06t}) + S_0e^{0.06t} \\
S_0 &= 100000 - 100000e^{-1.2} \\
S_0 &\approx USD\ 69,880.58
\end{aligned}$$

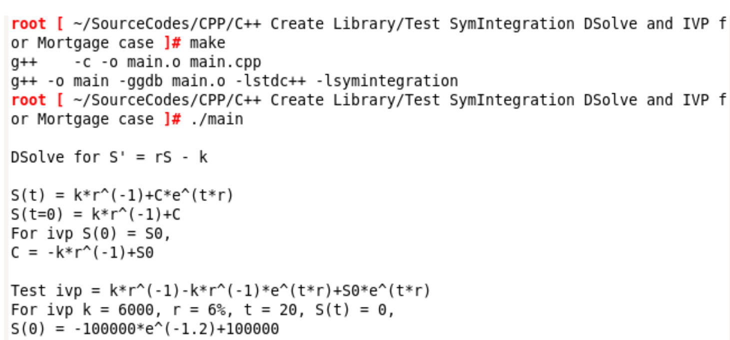
The value of S_0 is how much the buyer can afford to borrow initially.

(b) For the 20-year mortgage, the home buyer pays a total of

$$USD\ 6,000 \times 20 = USD\ 120,000$$

so the total interest paid is

$$USD\ 120,000 - USD\ 69,880.58 = USD\ 50,119.42$$



```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP f
or Mortgage case ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP f
or Mortgage case ]# ./main

DSolve for S' = rS - k

S(t) = k*r^(-1)+C*e^(t*r)
S(t=0) = k*r^(-1)+C
For ivp S(0) = S0,
C = -k*r^(-1)+S0

Test ivp = k*r^(-1)-k*r^(-1)*e^(t*r)+S0*e^(t*r)
For ivp k = 6000, r = 6%, t = 20, S(t) = 0,
S(0) = -100000*e^(-1.2)+100000
```

Figure 5.6: *The function* ***dsolve*** *and* ***ivp*** *in SymIntegration is able to compute the solution $S(t)$ for the ordinary differential equation of $\frac{dS}{dt} = Sr - k$ (**SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Mortgage case/main.cpp**).*

The symbolic computation to obtain the result can be done with **SymIntegration**, a C++ library that we create from branching out of SymbolicC++ v 3.35, we have been working on this project since April 2025.

For the visualization part, we will use Hamzstlab Mathematics to make an interactive graph.

Figure 5.7: *The interactive shaded line plot for this mortgage loan case example, you can change the time of the loan, the amount you can pay monthly and the interest rate, this can also be applied to other industry like life insurance or car insurance. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

```
...
...

void Demo_MortgageLoan() {
	static int t = 20, t1 = 1;
	static float xs1[100], ys1[100], ys2[100], ys3[100];
	static float r = 0.06;
	static int k = 500;
	for (int i = 1; i <= t; ++i) {
		xs1[i] = i;
		ys1[i] = (12*k/r) - (12*k/r)*(exp(-r*i)) ;
		ys2[i] = i*(12*k) - ys1[i];
		ys3[i] = i*(12*k); // Total payment
	}

	static ImVec4 color1 = ImVec4(0.133,0.545,0.133,1); // Forest
	    green
	static ImVec4 color2 = ImVec4(0,0.808,0.82,1); // Dark turquoise
	static ImVec4 color3 = ImVec4(0.855,0.647,0.125,1); // Goldenrod
	static float thickness = 1;
	static int shade_mode = 0;
	static ImPlotShadedFlags flags = 0;
	static bool show_lines = true;
	static bool show_fills = true;
```

```
static bool range = false, details = false;
ImGui::Text("Monthly payment = %4.0d", k);
ImGui::Text("Annual payment = %4.0d ", k*12);
ImGui::Checkbox("Change parameters", &range);
ImGui::Checkbox("Payment details", &details);
ImGui::Checkbox("Lines",&show_lines); ImGui::SameLine();
ImGui::Checkbox("Fills",&show_fills);

if (range) {
       ImGui::SetNextItemWidth(200);
       ImGui::BulletText("Monthly Payment");
       ImGui::SliderInt("##Monthly Payment", &k, 100, 10000);
       ImGui::SetNextItemWidth(200);
       ImGui::BulletText("Length of Mortgage");
       ImGui::SliderInt("##Length of Mortgage", &t, 5, 50);
       ImGui::SetNextItemWidth(200);
       ImGui::BulletText("Return Rate");
       ImGui::DragFloat("##Return rate", &r, 0.01f, 0.2f);
       ImGui::SetNextItemWidth(200);
}
if (details) {
       ImGui::SetNextItemWidth(200);
       ImGui::BulletText("Time / t");
       ImGui::SliderInt("##Time / t", &t1, 1, t);
       ImGui::Text("Total amount borrowed = %.3f ", ys1[t1]);
       ImGui::Text("Interest paid = %.3f ", ys2[t1]);
       ImGui::Text("Total payment = %.3f ", ys3[t1]);
}

if (ImPlot::BeginPlot("Mortgage Loan Accumulation")) {
       ImPlot::SetupAxes("t","S(t)");
       ImPlot::SetupAxesLimits(0, 30, 0, 200000);
       ImPlot::SetupLegend(ImPlotLocation_East,
           ImPlotLegendFlags_Outside);


       if (show_fills) {
              ImPlot::PushStyleVar(ImPlotStyleVar_FillAlpha, 0.23f
                  );
              ImPlot::SetNextFillStyle(color2);
              ImPlot::PlotShaded("Amount borrowed", xs1, ys1, t+1,
                   shade_mode == 0, flags);
              ImPlot::SetNextFillStyle(color3);
              ImPlot::PlotShaded("Interest paid", xs1, ys2, t+1,
                  shade_mode == 0, flags);
              ImPlot::SetNextFillStyle(color1);
              ImPlot::PlotShaded("Total payment", xs1, ys3, t+1,
                  shade_mode == 0, flags);
```

```
                ImPlot::PopStyleVar();
            }
            if (show_lines) {
                ImPlot::SetNextLineStyle(color2, thickness);
                ImPlot::PlotLine("Amount borrowed", xs1, ys1, t+1);
                ImPlot::SetNextLineStyle(color3, thickness);
                ImPlot::PlotLine("Interest paid", xs1, ys2, t+1);
                ImPlot::SetNextLineStyle(color1, thickness);
                ImPlot::PlotLine("Total payment", xs1, ys3, t+1);


            }
            ImPlot::EndPlot();
        }
}

...
```

Code 23: *Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp*

[H*] **Radiocarbon Dating Example:**
An important tool in archeological research is radiocarbon dating, developed by the American chemist Willard F. Libby. This is a means of determining the age of certain wood and plant remains, hence of animal or human bones or artifacts found buried at the same levels.

Radiocarbon dating is based on the fact that some wood or plant remains contain residual amounts of carbon-14, a radioactive isotope of carbon. This isotope is accumulated during the lifetime of the plant and begins to decay at its death. Since the half-life of carbon-14 is long (approximately 5730 years), measurable amounts of carbon-14 is still present, then by appropriate laboratory measurements the proportion of the original amount of carbon-14 that remains can be accurately determined. In other words, if $Q(t)$ is the amount of carbon-14 at time t and Q_0 is the original amount, then the ratio $\frac{Q(t)}{Q_0}$ can be determined, as long as this quantity is not too small. Present measurement techniques permit the use of this method for time periods of 50,000 years or more.

(a) Assuming that Q satisfies the differential equation $Q' = -rQ$, determine the decay constant r for carbon-14.
(b) Find an expression for $Q(t)$ at any time t, if $Q(0) = Q_0$
(c) Suppose that certain remains are discovered in which the current residual amounts of carbon-14 is 20% of the original amount. Determine the age of the remains.

Solution:

(a) The rate that the mass of carbon-14 decreases is assumed to be proportional to how much is present at any given time

$$\frac{dQ}{dt} \propto -Q$$

Change this proportionality to an equation by introducing a constant r on the right side.

$$Q' = -rQ$$

Divide both sides by Q.

$$\frac{Q'}{Q} = -r$$

The left side can be written as the derivative of a logarithm by the chain rule.

$$\begin{aligned}\frac{d}{dt}\ln\ Q &= -r\\ \ln\ Q &= -rt + C\\ Q(t) &= e^{-rt+C}\\ &= ce^{-rt}\end{aligned}$$

(b) Suppose that there is a certain amount of mass Q_0 initially. Then the initial condition is $Q(0) = Q_0$. We will use it to determine c.

$$\begin{aligned}Q(0) &= ce^{-r(0)}\\ c &= Q_0\end{aligned}$$

Consequently, the mass decays exponentially. Substituting back we will have the solution to the initial value problem as

$$Q(t) = Q_0 e^{-rt}$$

(c) From the formula that we obtain at (b) we know that r can be determined with knowledge of the half-life. For carbon-14, specifically, we know that half the initial mass is lost after 5730 years.

$$\frac{Q_0}{2} = Q_0 e^{-r(5730)}$$

Solve this equation for r

$$\begin{aligned}e^{-5730r} &= \frac{1}{2}\\ \ln|e^{-5730r}| &= \ln\left|\frac{1}{2}\right|\\ r &= -\frac{\ln\left|\frac{1}{2}\right|}{5730}\\ &= \frac{\ln|2|}{5730}\\ r &\approx 0.0001210\end{aligned}$$

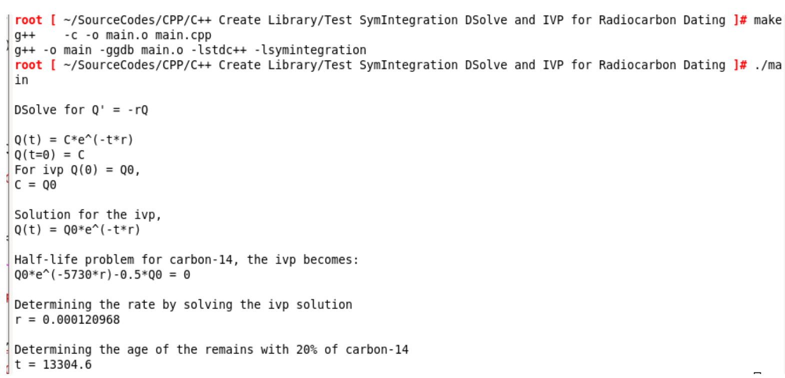
r is the rate of the decreasing mass per year. Therefore, the mass of a sample of carbon-14 is

$$Q(t) = Q_0 e^{-\frac{\ln|2|}{5730}t}$$

where t is in years. If some remains have 20% of the original amount of carbon-14, then $Q(t) = 0.2Q_0$. Solve the resulting equation for t to determine how old the remains are.

$$
\begin{aligned}
0.2Q_0 &= Q_0 e^{-\frac{\ln|2|}{5730}t} \\
e^{-\frac{\ln|2|}{5730}t} &= 0.2 \\
\ln|e^{-\frac{\ln|2|}{5730}t}| &= \ln|0.2| \\
-\frac{\ln|2|}{5730}t &= \ln\frac{1}{5} \\
-\frac{\ln|2|}{5730}t &= -\ln|5| \\
t &= \frac{\ln|5|}{\ln|2|}5730 \\
t &\approx 13,305
\end{aligned}
$$

so the age of the remains are around 13,305 years old.



```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Radiocarbon Dating ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Radiocarbon Dating ]# ./ma
in

DSolve for Q' = -rQ

Q(t) = C*e^(-t*r)
Q(t=0) = C
For ivp Q(0) = Q0,
C = Q0

Solution for the ivp,
Q(t) = Q0*e^(-t*r)

Half-life problem for carbon-14, the ivp becomes:
Q0*e^(-5730*r)-0.5*Q0 = 0

Determining the rate by solving the ivp solution
r = 0.000120968

Determining the age of the remains with 20% of carbon-14
t = 13304.6
```

Figure 5.8: *The computation of the rate and the age of remains with SymIntegration′* ***dsolve*** *and* ***ivp*** *functions. (****SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Radiocarbon Dating/main.cpp****).*

Figure 5.9: *The interactive line plot for this radiocabon dating, you can change the half-time and the percentage amount of the carbon-14 in the remains. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

```
...

double division(double x, double y)
{
       return x/y;
}
void Demo_RadiocarbonDating() {
       static int half_life=5730, t1 = 20;
       static double xt1[50000], qt[50000];
       double r = -(division(1,half_life))*log(0.5);
       for (int i = 1; i <= 50000; ++i) {
              xt1[i] = i;
              qt[i] = 1*exp(-r*i) ;
       }

       //static ImVec4 color1 = ImVec4(0.133,0.545,0.133,1); // Forest
           green
       static ImVec4 color2 = ImVec4(0,0.808,0.82,1); // Dark turquoise
       //static ImVec4 color3 = ImVec4(0.855,0.647,0.125,1); // Goldenrod
       static float thickness = 1;
       static bool range = true;

       ImGui::Checkbox("Change parameters", &range);

       if (range) {
              ImGui::SetNextItemWidth(200);
```

```
            ImGui::BulletText("Half-life of carbon-14 (years)");
            ImGui::SliderInt("##Half-life of carbon-14 (years)", &
                half_life, 100, 10000);
            ImGui::SetNextItemWidth(200);
            ImGui::BulletText("Residual amount of cabon-14 in the
                remains (%)");
            ImGui::SliderInt("##Residual amount of carbon-14 in the
                remains (%) ", &t1, 1, 100);
            ImGui::Text("Age of remains = %.5f ", log(division(t1,100))
                /(-r));
            ImGui::Text("Rate of decay = %.10f ", r);
    }

    if (ImPlot::BeginPlot("Radiocarbon Dating with carbon-14")) {
            ImPlot::SetupAxes("t","Q(t)");
            ImPlot::SetupAxesLimits(0, 30000, 0, 1);
            ImPlot::SetupLegend(ImPlotLocation_East,
                ImPlotLegendFlags_Outside);

            ImPlot::SetNextLineStyle(color2, thickness);
            ImPlot::PlotLine("Q(t)=Q_{0}exp(-rt)", xt1, qt, 50000);

            ImPlot::EndPlot();
    }
}
...
```

Code 24: *Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp*

[H*] **Newton's Law of Cooling Example:**
Newton's law of cooling states that the temperatureof an object changesat a rate proportional to the difference between its temperature and that of its surrounding. Suppose that the temperature of a cup of coffee obeys Newton's law of cooling. If the coffee has a temperature of 200^0F when freshly poured, and 1 minute later has cooled to 190^0in a room at 70^F, determine when the coffee reaches a temperature of 150^0F.

Solution:
Newton's law of cooling can be expressed mathematically as a proportionality. Let $T = T(t)$ be the temperature of the coffee; $\frac{dT}{dt}$ then represents the rate the temperature changes. Also, let T_0 be the temperature of the surroundings.

$$\frac{dT}{dt} \propto -(T - T_0)$$

The minus sign in front of the parentheses accounts for the fact that the temperature of an object will decease if it's hotter than the environment. Introduce a proportionality constant k on the right to change this to an equation we can use

$$\frac{dT}{dt} = -k(T - T_0)$$

Solve this ODE by separating variables.

$$\begin{aligned}
\frac{dT}{T-T_0} &= -k\,dt \\
\ln|T-T_0| &= -kt+c \\
|T-T_0| &= e^{-kt+c} \\
|T-T_0| &= e^{c}e^{-kt} \\
T-T_0 &= \pm Ce^{-kt} \\
T(t) &= T_0+Ce^{-kt}
\end{aligned}$$

Use the initial condition $T(0)=200$ to determine C.

$$\begin{aligned}
T(0) &= T_0+C \\
C &= 200-T_0
\end{aligned}$$

Also, use the fact that the room temperature is $T_0=70$.

$$T(t)=70+130e^{-kt}$$

After 1 minute, the coffee has cooled to 190^0F.

$$\begin{aligned}
T(1) &= 70+130e^{-k} \\
190 &= 70+130e^{-k} \\
e^{-k} &= \frac{12}{13} \\
\ln e^{-k} &= \ln\frac{12}{13} \\
k &= \ln\frac{13}{12} \\
k &\approx 0.0800\frac{1}{\text{minute}}
\end{aligned}$$

So then

$$T(t)=70+130e^{-t\ln\left(\frac{13}{12}\right)}$$

Set $T(t)=150$ and solve the resulting equation for t to find when the coffee reaches

150^0F.

$$
\begin{aligned}
T(t) &= 70 + 130e^{-t\ln\left(\frac{13}{12}\right)} \\
150 &= 70 + 130e^{-t\ln\left(\frac{13}{12}\right)} \\
80 &= 130e^{-t\ln\left(\frac{13}{12}\right)} \\
e^{-t\ln\left(\frac{13}{12}\right)} &= \frac{8}{13} \\
\ln\left(e^{-t\ln\left(\frac{13}{12}\right)}\right) &= \ln\left(\frac{8}{13}\right) \\
-t\ln\left(\frac{13}{12}\right) &= \ln\left(\frac{8}{13}\right) \\
t\ln\left(\frac{13}{12}\right) &= \ln\left(\frac{13}{8}\right) \\
t &= \frac{\ln\left(\frac{13}{8}\right)}{\ln\left(\frac{13}{12}\right)} \\
&\approx 6.07 \text{ minute}
\end{aligned}
$$

Therefore, it takes about 6 minutes and 4 seconds for the coffee to cool to 150^0F.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Newton's Law of Cooling ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Newton's Law of Cooling ]# ./main

DSolve for T' = -k(T - Troom)

T(t) = Troom+C*e^(-t*k)

T(t) = C*e^(-t*k)+70

Solution for the ivp T(0) = 200,
C = -Troom*e^(t*k)+200*e^(t*k)
C = 130

T(t) = 130*e^(-t*k)+70

Detemining rate k:
T(1) = 130*e^(-k)+70
k = 0.0800427

The coffee reaches temperature T(t) = 150 Fahrenheit at
t = 6.06561 minutes
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Newton's Law of Cooling ]#
```

Figure 5.10: *The computation of initial value problem solution for the Newton's law of cooling with SymIntegration' **dsolve** function. (**SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Newton's Law of Cooling/main.cpp**).*

Figure 5.11: *The interactive line plot for Newton's law of cooling. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

[H*] **Pollutants in a Lake Example:**
Consider a lake of constant volume V containing at time t an amount $Q(t)$ of pollutant, evenly distributed throughout the lake with a concentration $c(t)$, where $c(t) = \frac{Q(t)}{V}$. Assume that water containing a concentration k of pollutant enters the lake at a rate r, and that water leaves the lake at the same rate. Suppose that pollutants are also added directly to the lake at a constant rate P.

Note that given assumptions neglect a number of factors that may, in some cases, be important-for example, the water added or lost by precipitation, absorption, and evaporation; the stratifying effect of temperature differences in a deep lake; the tendency of irregularities in the coastline to produce sheltered bays; and the fact that pollutants are not deposited evenly throughout the lake but (usually) at isolated points around its periphery. The results below must be interpreted in the light of the neglect of such factors as these.

(a) If at time $t = 0$ the concentration of pollutant is c_0, find an expression for the concentation $c(t)$ at any time. What is the limiting concentration as $t \to \infty$?
(b) If the addition of pollutants to the lake is terminated ($k = 0$ and $P = 0$ for $t > 0$), determine the time interval T that must elapse before the concentration of pollutants is reduced to 50% of its original value; to 10% of its original value.
(c) Table 5.1 contains data for several of the Great Lakes. Using these data, determine from part (b) the time T necessary to reduce the contamination of each of these lakes to 10% of the original value.

Solution:

(a) The conservation law that governs the mass of pollutant in the lake is as follows.

Lake	V($km^3 \times 10^3$)	r($km^3/year$)
Superior	12.2	65.2
Michigan	4.9	158
Erie	0.46	175
Ontario	1.6	209

Table 5.1: *Volume and Flow Data for the Great Lakes*

rate of mass accumulation = rate of mass in - rate of mass out

the rate of mass in is the sum of rk and P, and the rate of mass out is $rc(t)$.

$$\begin{aligned}\frac{dQ}{dt} &= rk + P - rc(t)\\ &= rk + P - \frac{r}{V}Q(t)\end{aligned}$$

$$\frac{dQ}{dt} + \frac{r}{V}Q = rk + P$$

This is a first-order linear inhomogeneous ODE, so it can be solved by multiplying both sies by an integrating factor $\mu(t)$.

$$\mu(t) = e^{\int \frac{r}{V}\, ds} = e^{\frac{rt}{V}}$$

$$\begin{aligned}e^{\frac{rt}{V}}\frac{dQ}{dt} + \frac{r}{V}e^{\frac{rt}{V}}Q &= rke^{\frac{rt}{V}} + Pe^{\frac{rt}{V}}\\ \frac{d}{dt}(e^{\frac{rt}{V}}Q) &= rke^{\frac{rt}{V}} + Pe^{\frac{rt}{V}}\\ e^{\frac{rt}{V}}Q &= \int^t (rke^{\frac{rs}{V}} + Pe^{\frac{rs}{V}})\, ds + C\\ &= rk\left(\frac{V}{r}\right)e^{\frac{rt}{V}} + P\left(\frac{V}{r}\right)e^{\frac{rt}{V}} + C\\ e^{\frac{rt}{V}}Q &= kVe^{\frac{rt}{V}} + \frac{PV}{r}e^{\frac{rt}{V}} + C\\ Q(t) &= kV + \frac{PV}{r}e^{\frac{rt}{V}} + Ce^{\frac{-rt}{V}}\end{aligned}$$

Divide both sides by V to obtain the concentration.

$$c(t) = \frac{Q(t)}{V} = k + \frac{P}{r} + \frac{C}{V}e^{\frac{-rt}{V}}$$

Apply the initial condition $c(0) = c_0$ to determine C.

$$\begin{aligned}c(0) &= k + \frac{P}{r} + \frac{C}{V}\\ c_0 &= k + \frac{P}{r} + \frac{C}{V}\\ \frac{C}{V} &= c_0 - k - \frac{P}{r}\end{aligned}$$

Therefore, the concentration is

$$c(t) = \frac{Q(t)}{V} = k + \frac{P}{r} + \left(c_0 - k - \frac{P}{r}\right) e^{\frac{-rt}{V}}$$

Because of the decaying exponential function, the limit of $c(t)$ as $t \to \infty$ is

$$\lim_{t\to\infty} c(t) = k + \frac{P}{r}$$

(b) If the pollution stops, then the rate of mass in becomes zero in the conservation law.

rate of mass accumulation = - rate of mass out

The rate of mass out is still $rc(t)$.

$$\begin{aligned}
\frac{dQ}{dt} &= -rc(t) \\
&= -\frac{r}{V}Q(t) \\
\frac{\frac{dQ}{dt}}{Q} &= -\frac{r}{V} \\
\frac{dQ}{Q} &= -\frac{r}{V}\,dt \\
\ln Q &= -\frac{rt}{V} + C_1 \\
Q &= e^{-\frac{rt}{V} + C_1} \\
Q(t) &= C_2 e^{-\frac{rt}{V}}
\end{aligned}$$

Since Q is the mass of pollutant, divide both sides by V to get the concentration.

$$c(t) = \frac{Q(t)}{V} = \frac{A_1}{V} e^{-\frac{rt}{V}}$$

Use the initial condition $c(0) = c_0$ to determine C_2

$$c(0) = \frac{C_2}{V} = c_0$$

Therefore

$$c(t) = c_0 e^{-\frac{rt}{V}}$$

Set $c(t) = 0.5c_0$ and solve for $t = T$ to find how long it will take for the concentration to reach half its initial value.

$$\begin{aligned}
0.5c_0 &= c_0 e^{-\frac{rT}{V}} \\
e^{-\frac{rT}{V}} &= 0.5 \\
\ln e^{-\frac{rT}{V}} &= \ln 0.5 \\
-\frac{rT}{V} &= \ln\frac{1}{2} \\
T &= \frac{V}{r}\ln 2
\end{aligned}$$

On the other hand, set $c(t) = 0.1c_0$ and solve for $t = T$ to find how long it will take for the concentration to reach one-tenth its initial value.

$$\begin{aligned}
0.1c_0 &= c_0 e^{-\frac{rt}{V}} \\
e^{-\frac{rt}{V}} &= 0.1 \\
\ln e^{-\frac{rt}{V}} &= \ln 0.1 \\
-\frac{rT}{V} &= \ln \frac{1}{10} \\
T &= \frac{V}{r} \ln 10
\end{aligned}$$

(c) Plug in the numbers fo V and r into the formula at part (b).
Lake Superior:

$$T = \frac{V}{r} \ln 10 = \frac{12.2 \times 10^3}{65.2} \ln 10 \approx 431 \ \textit{years}$$

Lake Michigan:

$$T = \frac{V}{r} \ln 10 = \frac{4.9 \times 10^3}{158} \ln 10 \approx 71.4 \ \textit{years}$$

Lake Erie:

$$T = \frac{V}{r} \ln 10 = \frac{0.46 \times 10^3}{175} \ln 10 \approx 6.05 \ \textit{years}$$

Lake Ontario:

$$T = \frac{V}{r} \ln 10 = \frac{1.6 \times 10^3}{209} \ln 10 \approx 17.6 \ \textit{years}$$

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Pollutant in Great Lakes ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Pollutant in Great Lakes ]# ./main

DSolve for Q' = rk +  P - rc(t) ,        c(t) = Q(t)/V

Q(t) = k*V+P*V*r^(-1)+C*e^(-V^(-1)*t*r)
c(t) = k+P*r^(-1)+C*e^(-V^(-1)*t*r)*V^(-1)
c(0) = k+P*r^(-1)+C*V^(-1)
c0 - c(0) = c0-k-P*r^(-1)-C*V^(-1)
For ivp c(0) = c0,
C/V = c0-k-P*r^(-1)

Therefore, the concentration is,
c(t) = k+P*r^(-1)+c0*e^(-V^(-1)*t*r)-k*e^(-V^(-1)*t*r)-P*r^(-1)*e^(-V^(-1)*t*r)

The limit of c(t) as t -> ∞
lim_{t -> ∞} c(t) = k+P*r^(-1)+c0*e^(-inf*V^(-1)*r)-k*e^(-inf*V^(-1)*r)-P*r^(-1)*e^(-inf*V^(-1)*r)

If the pollution stops, then
Q(t) = C*e^(-V^(-1)*t*r)

The concentration when the pollution stops is
c(t) = C*e^(-V^(-1)*t*r)*V^(-1)

For ivp c(0) = c0,
c0-C*V^(-1) = 0

Therefore the ivp solution will be,
c(t) = c0*e^(-V^(-1)*t*r)

Set c(t) = 0.5 c0 and solve for t=T,
T = 0.693147*V*r^(-1)

Set c(t) = 0.1 c0 and solve for t=T,
T = 2.30259*V*r^(-1)

Time to reduce the contamination in each lakes to 10 percent of the original value

Lake Superior:  T = 430.852

Lake Michigan:  T = 71.4093

Lake Erie:      T = 6.05251

Lake Ontario:   T = 17.6274
```

Figure 5.12: *The computation of initial value problem solution for the pollutant concentration in a lake, $c(t)$, and the time to clean up a lake to make the pollutant concentration less than 10% with SymIntegration' **dsolve** function. (**SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Pollutant in Great Lakes/main.cpp**).*

Figure 5.13: *The interactive line plot for pollutant in great lakes. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

[H*] **Fluid Dynamic Example:**
A body falling in a relatively dense fluid, oil for example, is acted on by three forces: a resistive force R, a buoyant force B, and its weight w due to gravity. The buoyant force is equal to the weight of the fluid displaced by the object. For a slowly moving spherical body of radius a, the resistive force is given by Stokes's law, $R = 6\pi\mu a|v|$, where v is the velocity of the body, and μ is the coefficient of viscosity of the surrounding fluid.

(a) Find the limiting velocity of a solid sphere of radius a and density ρ falling freely in a medium of density ρ' and coefficient of viscosity μ.

(b) In 1910 R.A. Millikan studied the motion of tiny droplets of oil falling in an electric field. A field of strength E exerts a force Ee on a droplet with charge e. Assume that E has been adjusted so the droplet is held stationary $(v = 0)$ and that w and B are as given above. Find an expression for e. Millikan repeated this experiment many times, and from the data that he gathered he was able to deduce the charge on an electron.

Solution:
According to Newton's second law, the equation of motion for the mass is

$$\sum \mathbb{F} = m$$

This is a vector equation: it consists of three scalar equations-one for each direction in the chosen coordinate system. Since the mass is projected vertically upward, only the equation in the y-direction is relevant.

$$\sum F_y = ma_y$$

(a) We will use the free-body diagram to write the equation of motion. (Take the positive y-axis to point downward in the direction of the motion.)

$$w - R - B = ma_y \tag{5.29}$$

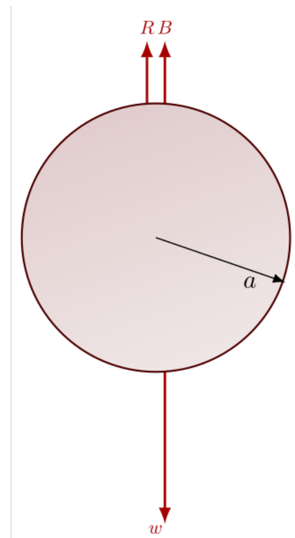


Figure 5.14: *The free-body diagram for a sphere shape body falling in a dense fluid.*

Now, we'll write expressions for each of the variables. w is the weight of the solid ball; multiply its volume by its density by g to get force.

$$w = \frac{4}{3}\pi a^3 \times \rho \times g$$

R is the resistive force given by Stokes's law

$$R = 6\pi\mu a v$$

B is the buoyant force; multiply the volume of the ball by the density of the medium by g to get force.

$$B = \frac{4}{3}\pi a^3 \times \rho' \times g$$

m is the mass of the ball; multiply its volume by its density to get the mass.

$$m = \frac{4}{3}\pi a^3$$

Finally, replace a_y with dv/dt. As a result, equation (5.29) becomes

$$\frac{4}{3}\pi a^3 \rho g - 6\pi\mu a v - \frac{4}{3}\pi a^3 \rho' g = \frac{4}{3}\pi a^3 \rho \frac{dv}{dt}$$

Simplify both sides of the equation.

$$\begin{aligned}
\frac{4}{3}\pi a^3(\rho-\rho')g - 6\pi\mu a v &= \frac{4}{3}\pi a^3\rho\frac{dv}{dt} \\
\left(1-\frac{\rho'}{\rho}\right)g - \frac{9\mu}{2a^2\rho}v &= \frac{dv}{dt} \\
\frac{dv}{dt} + \frac{9\mu}{2a^2\rho}v &= \left(1-\frac{\rho'}{\rho}\right)g
\end{aligned}$$

This is a first-order linear inhomogeneous ODE, so it can be solved by multiplying both sides by an integrating factor $\psi(t)$

$$\psi(t) = e^{\int^t \frac{9\mu}{2a^2\rho}\ ds} = e^{\frac{9\mu t}{2a^2\rho}}$$

Now, to find the general solution

$$\begin{aligned}
e^{\frac{9\mu t}{2a^2\rho}}\frac{dv}{dt} + \frac{9\mu}{2a^2\rho}e^{\frac{9\mu t}{2a^2\rho}}v &= \left(1-\frac{\rho'}{\rho}\right)ge^{\frac{9\mu t}{2a^2\rho}} \\
\frac{d}{dt}\left[e^{\frac{9\mu t}{2a^2\rho}}v\right] &= \left(1-\frac{\rho'}{\rho}\right)ge^{\frac{9\mu t}{2a^2\rho}} \\
\int \frac{d}{dt}\left[e^{\frac{9\mu t}{2a^2\rho}}v\right]\ dt &= \int \left(1-\frac{\rho'}{\rho}\right)ge^{\frac{9\mu t}{2a^2\rho}}\ dt \\
e^{\frac{9\mu t}{2a^2\rho}}v &= \frac{2a^2\rho}{9\mu}\left(1-\frac{\rho'}{\rho}\right)ge^{\frac{9\mu t}{2a^2\rho}} + C_1 \\
v(t) &= \frac{2a^2}{9\mu}(\rho-\rho')g + C_1e^{-\frac{9\mu t}{2a^2\rho}}
\end{aligned}$$

In the limit as $t \to \infty$, the exponential function decays to zero, and only the first term remains.

$$\lim_{t\to\infty} v(t) = \frac{2a^2}{9\mu}(\rho-\rho')g$$

(b) Since the oil droplet is stationary, the sum of the forces must be equal to zero.

$$\begin{aligned}
w - F_e - B &= 0 \\
w &= \frac{4}{3}\pi a^3\rho g \\
F_e &= eE \\
B &= \frac{4}{3}\pi a^3\rho' g
\end{aligned}$$

Therefore, the electron charge is

$$\begin{aligned}
\frac{4}{3}\pi a^3\rho g - eE - \frac{4}{3}\pi a^3\rho' g &= 0 \\
eE &= \frac{4}{3}\pi a^3(\rho-\rho')g \\
e &= \frac{4}{3E}\pi a^3(\rho-\rho')g
\end{aligned}$$

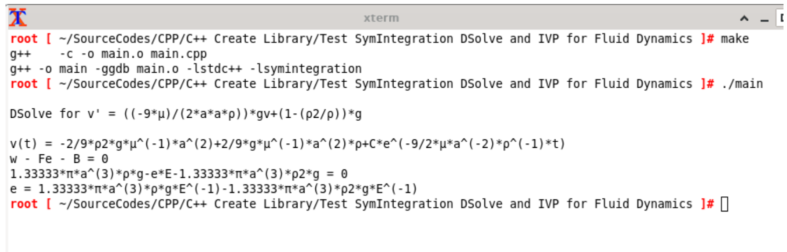


Figure 5.15: *The computation of initial value problem solution for the case of a body falling in a relatively dense fluid with SymIntegration' **dsolve** function. (**SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Fluid Dynamics/main.cpp**).*

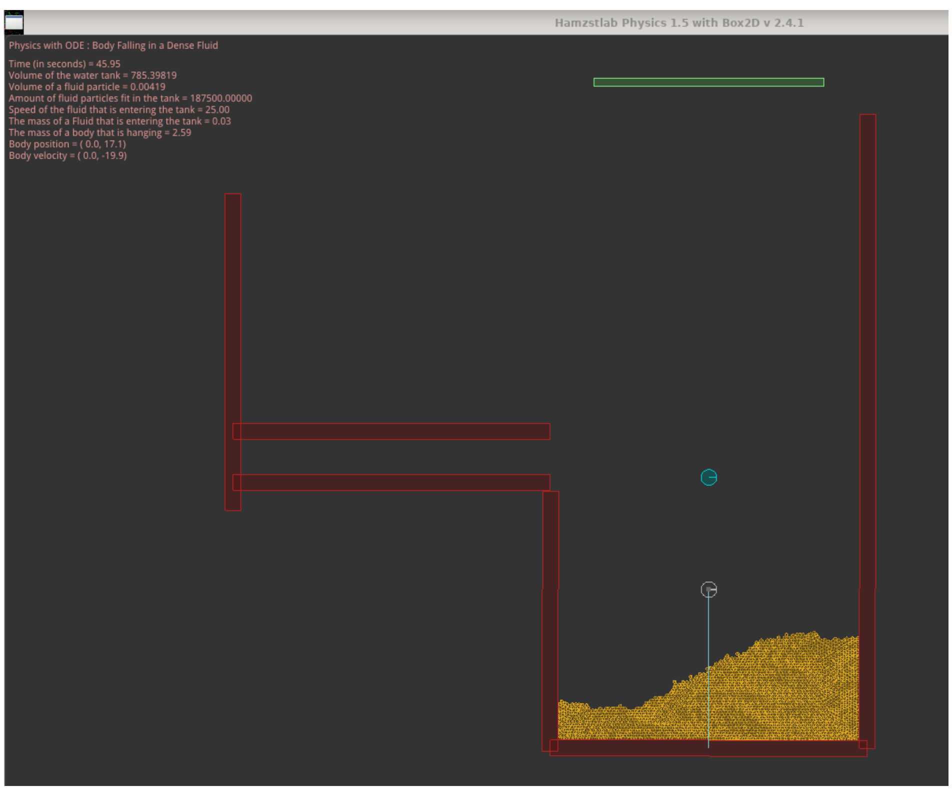


Figure 5.16: *The 2D simulation of body falling in a relatively dense fluid with Hamzstlab Physics 2D that is using Box2D version 2.4.2 as the backend library (**Hamzstlab-Physics2D/tests/ode_bodyfallinginadensefluid.cpp**).*

Figure 5.17: *The 2D simulation of body that has already fallen in a relatively dense fluid (**HamzstlabPhysics2D/tests/ode_bodyfallinginadensefluid.cpp**).*

[H*] **Sky Diver Example:**
A sky diver weighing 180 lb (including equipment) falls vertically downward from an altitude of 5000 ft and opens parachute after 10 s of free fall. Assume that the force of air resistance is $0.75|v|$ when the parachute is closed and $12|v|$ when the parachute is open, where the velocity v is measured in ft/s.

(a) Find the speed of the sky diver when the parachute opens.
(b) Find the distance fallen before the parachute opens.
(c) What is the limiting velocity v_L after the parachute opens?
(d) Determine how long the sky diver is in the air after the parachute opens.
(e) Plot the graph of velocity versus time from the beginning of the fall until the skydiver reaches the ground.

Solution:
Split up the diver's motion into two parts: one where the parachute is closed and the other where the parachute is open. According to Newton's second law, the equation of motion for the diver is

$$\sum \mathbb{F} = m$$

This is a vector equation, so it actually represents three scalar equations-one for each direction in the chosen coordinate system. Since the diver only moves vertically, only the y-equation is relevant.

$$\sum F_y = ma_y$$

We will replace a_y with $\frac{dv}{dt}$.

Minus signs are placed in front of v, thus it becomes $-0.75v$, because the positive y-axis points upward and the diver is moving downward.

(a) Before the parachute opens:

$$
\begin{aligned}
-0.75v - W &= ma_y \\
\frac{dv}{dt} + \frac{0.75}{m}v &= -\frac{W}{m} \\
e^{\frac{0.75t}{m}}\frac{dv}{dt} + \frac{0.75}{m}e^{\frac{0.75t}{m}}v &= -\frac{W}{m}e^{\frac{0.75t}{m}} \\
\frac{d}{dt}\left(e^{\frac{0.75t}{m}}v\right) &= -\frac{W}{m}e^{\frac{0.75t}{m}} \\
e^{\frac{0.75t}{m}}v &= -\frac{W}{0.75}e^{\frac{0.75t}{m}} + C_1 \\
v(t) &= -\frac{W}{0.75} + C_1e^{-\frac{0.75t}{m}}
\end{aligned}
$$

Integrate the velocity to get the position.

$$
y(t) = -\frac{W}{0.75}t - \frac{C_1m}{0.75}e^{-\frac{0.75t}{m}} + C_4
$$

After the parachute opens:

$$
\begin{aligned}
-12v - W &= ma_y \\
\frac{dv}{dt} + \frac{12}{m}v &= -\frac{W}{m} \\
e^{\frac{12t}{m}}\frac{dv}{dt} + \frac{12}{m}e^{\frac{12t}{m}}v &= -\frac{W}{m}e^{\frac{12t}{m}} \\
\frac{d}{dt}\left(e^{\frac{12t}{m}}v\right) &= -\frac{W}{m}e^{\frac{12t}{m}} \\
e^{\frac{12t}{m}}v &= -\frac{W}{12}e^{\frac{12t}{m}} + C_2 \\
v(t) &= -\frac{W}{12} + C_2e^{-\frac{12t}{m}} \\
v(t) &= -\frac{W}{12} + C_3e^{-\frac{12(t-10)}{m}}
\end{aligned}
$$

Integrate the velocities to get the positions.

$$
\begin{aligned}
y(t) &= -\frac{W}{12}t - \frac{C_3m}{12}e^{-\frac{12(t-10)}{m}} + C_5 \\
y(t) &= -\frac{W}{12}(t-10) - \frac{C_3m}{12}e^{-\frac{12(t-10)}{m}} + C_6
\end{aligned}
$$

Our task now is to determine the integration constants, C_1, C_3, C_4, and C_6. Assuming that the diver falls from rest, the initial condition $v(0) = 0$ can be used to find C_1.

Before the parachute opens:

$$\begin{aligned} v(0) &= 0 \\ -\frac{W}{0.75} + C_1 &= 0 \\ C_1 &= \frac{W}{0.75} \end{aligned}$$

So then the velocity will be

$$v(t) = -\frac{W}{0.75} + \frac{W}{0.75}e^{-\frac{0.75t}{m}} = -\frac{W}{0.75}(1 - e^{-\frac{0.75t}{m}})$$

The parachute opens at 10 seconds. The velocity at this time is

$$v(10) = -\frac{W}{0.75}(1 - e^{-\frac{7.5}{m}})$$

and it serves as the initial condition that we can use to find C_3.

After the parachute opens:

$$\begin{aligned} v(10)_{before} &= v(10)_{after} \\ -\frac{W}{0.75}(1 - e^{-\frac{7.5}{m}}) &= -\frac{W}{12} + C_3 e^{-\frac{12(10-10)}{m}} \\ -\frac{W}{0.75}(1 - e^{-\frac{7.5}{m}}) &= -\frac{W}{12} + C_3 \\ C_3 &= \frac{W}{12} - \frac{W}{0.75}(1 - e^{-\frac{7.5}{m}}) \end{aligned}$$

So then the velocity after the parachute opens becomes

$$v(t) = -\frac{W}{12}\left[\frac{W}{12} - \frac{W}{0.75}(1 - e^{-\frac{7.5}{m}})\right]e^{-\frac{12(t-10)}{m}}$$

With these values of C_1 and C_3, the position before the parachute opens becomes

$$y(t) = -\frac{W}{0.75}\left(t + \frac{m}{0.75}e^{-\frac{0.75t}{m}}\right) + C_4$$

and the position after the parachute opens becomes

$$y(t) = -\frac{W}{12}(t - 10) - \left[\frac{W}{12} - \frac{W}{0.75}(1 - e^{-\frac{7.5}{m}})\right]\frac{m}{12}e^{-\frac{12(t-10)}{m}} + C_6$$

Use the fact that the diver starts at 5000 feet to find C_4.

$$\begin{aligned} y(0) &= -\frac{W}{0.75}\left(\frac{m}{0.75}\right) + C_4 \\ 5000 &= -\frac{W}{0.75}\left(\frac{m}{0.75}\right) + C_4 \\ C_4 &= 5000 + \frac{Wm}{0.5625} \end{aligned}$$

So then, before the parachute opens:

$$y(t) = -\frac{W}{0.75}\left(t + \frac{m}{0.75}e^{-\frac{0.75t}{m}}\right) + 5000 + \frac{Wm}{0.5625}$$

The position of the diver at 10 seconds (when the parachute is opened) is

$$y(10) = -\frac{W}{0.75}\left(10 + \frac{m}{0.75}e^{-\frac{7.5}{m}}\right) + 5000 + \frac{Wm}{0.5625}$$

Because the position of the diver is continuous, this serves as the initial condition to find C_6.

After the parachute opens, the position will be:

$$\begin{aligned} y(10)_{before} &= y(10)_{after} \\ -\left[\frac{W}{12} - \frac{W}{0.75}(1 - e^{-\frac{7.5}{m}})\right]\frac{m}{12} + C_6 &= -\frac{W}{0.75}\left(10 + \frac{m}{0.75}e^{-\frac{7.5t}{m}}\right) + 5000 + \frac{Wm}{0.5625} \\ C_6 &= -\frac{W}{0.75}\left(10 + \frac{m}{0.75}e^{-\frac{7.5}{m}}\right) + 5000 + \frac{Wm}{0.5625} \\ &\quad + \left[\frac{W}{12} - \frac{W}{0.75}(1 - e^{-\frac{7.5}{m}})\right]\frac{m}{12} \end{aligned}$$

Now, the numbers w and m will be plugged in. The weight is $W = 180$ lb, and the mass is

$$m = \frac{W}{g} = \frac{180}{9.81 m.s^2 \times 3.28 ft/m} \approx 5.59 \frac{lb \cdot s^2}{ft}$$

As a result, the constants are

$$\begin{aligned} C_1 &= 240 \\ C_3 &\approx -162.2 \\ C_4 &\approx 6790 \\ C_6 &\approx 3846 \end{aligned}$$

and the position and velocity are

$$y(t) \approx \begin{cases} 6790 - 240(7.459e^{-0.13407t} + t), & 0 \le t \le 10 \\ 3846 - 15(t-10) + 75.61e^{-2.145(t-10)}, & 10 \le t \lessapprox 266.4 \end{cases}$$

$$v(t) \approx \begin{cases} -240(1 - e^{-0.13407t} + t), & 0 \le t \le 10 \\ -15 - 162.2e^{-1.2(t-10)}, & 10 \le t \lessapprox 266.4 \end{cases}$$

266.4 seconds is the time it takes for the diver to reach the ground. It was found by using Newton's method :

$$\begin{aligned} y(t)_{after} &= 0 \\ 3846 - 15(t-10) + 75.61e^{-2.145(t-10)} &= 0 \\ t &\approx 266.40608 \end{aligned}$$

we can find the solution of t with Newton's method (a numerical method), since it has a very complicated form to be solved by analytical method.

Now, we already determine the general solution for the position and velocity and obtain all the constants, we can compute the speed of the sky diver when the parachute opens, it is

$$\begin{aligned}|v(10)_{before}| &= |v(10)_{after}| \\ &= |-240(1 - e^{-0.13407(10)} + 10)| \\ |v(10)| &= 177.2 \ \frac{ft}{s}\end{aligned}$$

(b) To determine the distance fallen before the parachute opens, we can use the general solution of the position formula:

$$\begin{aligned}y(10)_{before} &= 6790 - 240(7.459e^{-0.13407(10)} + 10) \\ &= 3921.7 \ ft\end{aligned}$$

thus, since the sky diver starts at 5000 ft, then the sky diver will have travelled a distance of $5000 - 3921.7 = 1078.3$ ft after 10 seconds.

(c) The limiting velocity after the parachute opens can be determined with taking the limit of t to ∞

$$\lim_{t \to \infty} v(t)_{after} = \lim_{t \to \infty} -15 - 162.2e^{-1.2(t-10)} = -15 \ \frac{ft}{s}$$

the minus sign means the sky diver is going downward, and we already make the positive y-axis as upward movement.

(d) The time to reach the ground is 266.4 seconds, and it takes 10 seconds before the parachute opens, thus the time the sky diver is in the air after the parachute opens is

$$266.4 - 10 = 254.4 \ s$$

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Sky Diver ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve and IVP for Sky Diver ]# ./main

DSolve for v' + (0.75/m)v = -W/m

Before parachute opens,
v(t) = -1.33333*W+C1*e^(-0.75*m^(-1)*t)

After parachute opens,
v(t) = -1/12*W+C3*e^(-12*m^(-1)*t+120*m^(-1))

Before parachute opens,
y(t) = -1.33333*W*t-1.33333*C1*e^(-0.75*m^(-1)*t)*m+C4

After parachute opens,
y(t) = -1/12*W*t+5/6*W-1/12*C3*e^(-12*m^(-1)*t+120*m^(-1))*m+C6

C1 = 1.33333*W

Before parachute opens,
v(t) = -1.33333*W+1.33333*W*e^(-0.75*m^(-1)*t)

When the parachute opens,
v(10)_{before} = -1.33333*W+1.33333*W*e^(-7.5*m^(-1))

v(10)_{after} = -1/12*W+C3

C3 = -1.25*W+1.33333*W*e^(-7.5*m^(-1))

So then,
v(t)_{after} = -1/12*W-1.25*W*e^(-12*m^(-1)*t+120*m^(-1))+1.33333*W*e^(112.5*m^(-1)-12*m^(-1)*t)

Thus,
y(t)_{before} = -1.33333*W*t-1.77778*W*e^(-0.75*m^(-1)*t)*m+C4

y(t)_{after} = -1/12*W*t+5/6*W+0.104167*W*e^(-12*m^(-1)*t+120*m^(-1))*m-0.111111*W*e^(112.5*m^(-1)-12*m^(-1)*t)*m+C6

The diver starts at 5000 feet, thus
C4 = 1.77778*W*m+5000

The position of the diver after the parachute opens at 10 seconds,
y(10)_{before} = -13.3333*W-1.77778*W*e^(-7.5*m^(-1))*m+1.77778*W*m+5000

y(10)_{after} = 0.104167*W*m-0.111111*W*e^(-7.5*m^(-1))*m+C6

Thus,
C6 = -13.3333*W-1.66667*W*e^(-7.5*m^(-1))*m+1.67361*W*m+5000
```

Figure 5.18: *The computation of initial value problem solution for the case of sky diver with SymIntegration, part 1. (**SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Sky Diver/main.cpp**).*

```
We have W = 180 lb, g = 9.81 m/s^2, then
m = 5.59409

As a result, the constants are
C1 = 240
C3 = 240*e^(-1.3407)-225 = -162.201
C4 = 6790.11
C6 = -1678.23*e^(-1.3407)+4285.22 = 3846.09

*** Newton' Method ***


f(x) = -15*x+75.614*e^(-2.14512*x+21.4512)+3996.09

f'(x) = -162.201*e^(-2.14512*x+21.4512)-15

     n          p_{n}
     0          15.707963943481
     1          266.39306020632
     2          266.40608181366
     3          266.40608181366
The procedure was successful.
*****************************

The time till the diver reaches the ground:
t = 266.40608181366

The general solution for the position:
For 0 <= t <= 10,
y(t) = -240*t-1790.1096442157*e^(-0.13407*t)+6790.1096442157

For 10 <= t <= 266.4,
y(t) = -15*t+75.613958296091*e^(-2.14512*t+21.4512)+3996.0912272048

The general solution for the velocity:
For 0 <= t <= 10,
v(t) = 240*e^(-0.13407*t)-240

For 10 <= t <= 266.4,
v(t) = -162.20101422011*e^(-2.14512*t+21.4512)-15

The speed of the sky diver at 10 second:
v(10) = -177.20101422011

The position of the sky diver at 10 second:
y(10) = 3921.7051855009
```

Figure 5.19: *The computation of initial value problem solution for the case of sky diver with SymIntegration, part 2. (**SymIntegration/Examples/Differential Equations/Test SymIntegration DSolve and IVP for Sky Diver/main.cpp**).*

(e) The plot of $y(t)$ and $v(t)$ can be done with either Hamzstplot or Hamzstlab Mathematics, we will choose Hamzstlab Mathematics this time for the interactive feature, we code more but we gain more, if we change the parameters such as the weight of the sky diver or the starting height at the runtime, the plot will adjust in an instant. It is very useful in sky diving field and early warning system.

Figure 5.20: *The interactive plots for the general solution of $y(t)$ and $v(t)$ for the case of sky diver with Hamzstlab Mathematics. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

[H*] **Baseball Example:**
Let $v(t)$ and $w(t)$, respectively, be the horizontal and vertical components of the velocity of the batted (or thrown) baseball. the equations of motion are

$$\frac{dv}{dt} = -rv, \quad \frac{dw}{dt} = -g - rw$$

where r is the coefficient of resistance

(a) Determine $v(t)$ and $w(t)$ in terms of initial speed u and initial angle of elevation A.

(b) Find $x(t)$ and $y(t)$ if $x(0) = 0$ and $y(0) = h$.

(c) Plot the trajectory of the ball for $r = \frac{1}{5}\ s^{-1}, u = 125$ ft/s and $h = 3$ ft, and for several values of A. How do the trajectories differ from those with $r = 0$?

(d) Assuming that $r = \frac{1}{5}\ s^{-1}$ and $h = 3$ ft, find the minimum initial velocity u and the optimal angle A for which the ball will clear a wall that is 350 ft distant and 10 ft high.

Solution:

(a) Both the ODE for v and w can be solved by multiplying both sides by an integrating factor.

$$\begin{aligned}
\frac{dv}{dt} &= -rv \\
\frac{dv}{dt} + rv &= 0 \\
e^{rt}\frac{dv}{dt} + re^{rt}v &= 0 \\
\frac{d}{dt}(e^{rt}v) &= 0 \\
e^{rt}v &= C_1 \\
v(t) &= C_1e^{-rt}
\end{aligned}$$

$$\begin{aligned}
\frac{dw}{dt} &= -g - rw \\
\frac{dw}{dt} + rw &= -g \\
e^{rt}\frac{dw}{dt} + re^{rt}w &= -ge^{rt} \\
\frac{d}{dt}(e^{rt}w) &= -ge^{rt} \\
e^{rt}w &= -\frac{g}{r}e^{rt} + C_2 \\
w(t) &= -\frac{g}{r} + C_2e^{-rt}
\end{aligned}$$

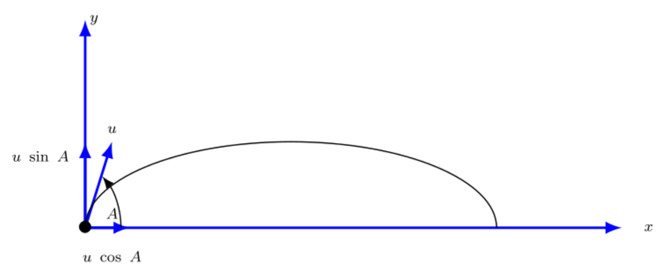


Figure 5.21: *A schematic of the baseball being launched with initial speed u at an angle A, the initial velocity then being decomposed into its components along the x- and y- axes.*

We can then use the initial velocity to determine C_1 and C_2.

$$\begin{aligned}
v(0) &= u\cos\ A \\
C_1e^{-r(0)} &= u\cos\ A \\
C_1 &= u\cos\ A
\end{aligned}$$

$$\begin{aligned}
w(0) &= u\ \sin\ A \\
-\frac{g}{r} + C_2e^{-r(0)} &= u\sin\ A \\
C_2 &= \frac{g}{r} + u\sin\ A
\end{aligned}$$

Therefore,

$$\begin{aligned} v(t) &= ue^{-rt}\cos\ A \\ w(t) &= -\frac{g}{r} + \left(\frac{g}{r} + u\sin\ A\right)e^{-rt} \end{aligned}$$

(b) Replace v with $\frac{dx}{dt}$ and replace w with $\frac{dy}{dt}$.

$$\begin{aligned} \frac{dx}{dt} &= ue^{-rt}\cos\ A \\ \frac{dy}{dt} &= -\frac{g}{r} + \left(\frac{g}{r} + u\sin\ A\right)e^{-rt} \end{aligned}$$

Integrate the velocities to get the positions

$$\begin{aligned} x(t) &= -\frac{u}{r}e^{-rt}\cos\ A + C_3 \\ y(t) &= -\frac{g}{r}t - \frac{1}{r}\left(\frac{g}{r} + u\sin\ A\right)e^{-rt} + C_4 \end{aligned}$$

Apply the initial conditions to determine C_3 and C_4.

$$\begin{aligned} x(0) &= 0 \\ -\frac{u}{r}e^{-r(0)}\cos\ A + C_3 &= 0 \\ C_3 &= \frac{u}{r}\cos\ A \end{aligned}$$

$$\begin{aligned} y(0) &= h \\ -\frac{g}{r}(0) - \frac{1}{r}\left(\frac{g}{r} + u\sin\ A\right)e^{-r(0)} + C_4 &= h \\ C_4 &= h + \frac{1}{r}\left(\frac{g}{r} + u\sin\ A\right) \end{aligned}$$

Therefore,

$$\begin{aligned} x(t) &= -\frac{u}{r}(1 - e^{-rt})\cos\ A \\ y(t) &= h - \frac{g}{r}t + \frac{1}{r}\left(\frac{g}{r} + u\sin\ A\right)(1 - e^{-rt}) \end{aligned}$$

(c) The trajectory plot with air drag:

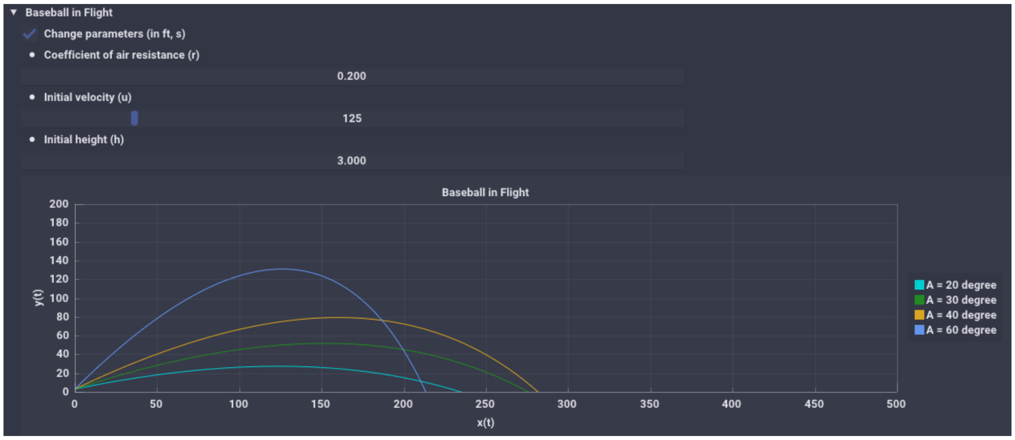


Figure 5.22: *The interactive plots is a parametric plot of $(x(t), y(t))$ with air drag for the case of baseball in flight with Hamzstlab Mathematics. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/-main.cpp**).*

for comparison, without air drag:

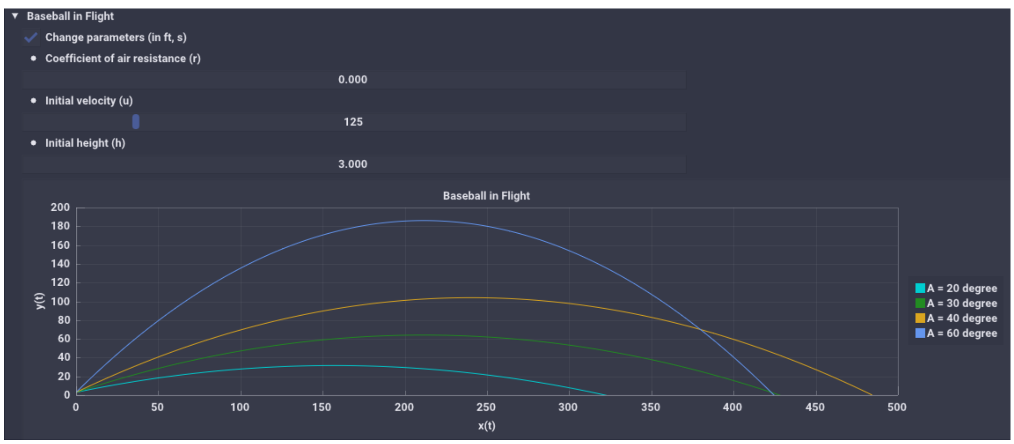


Figure 5.23: *The interactive plots is a parametric plot of $(x(t), y(t))$ without air drag for the case of baseball in flight with Hamzstlab Mathematics. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

(d) Suppose the outfield wall is at a distance L and has height H. From the result in part (b), we set $x(t) = L$ and $y(t) = H$.

$$L = -\frac{u}{r}(1 - e^{-rt}) \cos\ A$$

$$H = h - \frac{g}{r}t + \frac{1}{r}\left(\frac{g}{r} + u \sin\ A\right)(1 - e^{-rt})$$

Now, We will set $x(t) = 350$ ft, $y(t) = 10$ ft, $r = \frac{1}{5}s^{-1}$, $h = 3$ ft, and $g = 32\ ft/s^2$.

$$350 = 5u(1 - e^{-\frac{t}{5}} \cos\ A)$$

$$10 = 3 - 160t + 5(160 + u \sin\ A)(1 - e^{-\frac{t}{5}})$$

Since we want a formula involving u and A, we will eliminate t. Solve the first equation for $e^{-\frac{t}{5}}$ and t.

$$\begin{aligned} 350 &= 5u(1 - e^{-\frac{t}{5}} \cos\ A) \\ e^{-\frac{t}{5}} &= 1 - \frac{350}{5u \cos\ A} \\ t &= -5 \ln \left(1 - \frac{350}{5u \cos\ A} \right) \end{aligned}$$

and then plug these formulas into the second equation to eliminate t.

$$\begin{aligned} 10 &= 3 + 800 \ln \left(1 - \frac{350}{5u \cos\ A} \right) + 5(160 + u \sin\ A) \left(\frac{350}{5u \cos\ A} \right) \\ 7 &= 800 \ln \left(1 - \frac{70}{u \cos\ A} \right) + \frac{56000}{u \cos\ A} + 350 \tan\ A \end{aligned}$$

In order to clear the wall that is 350 ft distant and 10 ft high, we have to find the solution for this function of 2 variables:

$$f(u, A) = 800 \ln \left(1 - \frac{70}{u \cos\ A} \right) + \frac{56000}{u \cos\ A} + 350 \tan\ A - 7$$

We will use downhill simplex method for $-f(u, A)$ instead of $f(u, A)$, because if we minimize $f(u, A)$ then the ball will never clear the wall as it will looks for minimum value of $f(u, A)$.

When we apply the downhill simplex to

$$-f(u, A) = -\left(800 \ln \left(1 - \frac{70}{u \cos\ A} \right) + \frac{56000}{u \cos\ A} + 350 \tan\ A - 7 \right)$$

It is enough until the iteration showing the value of $-f(u, A)$ surpassing zero to the negative value. This is the logical explanation, we want to have this condition:

$$f(u, A) \geq 0$$

downhill simplex will find the value of u and A that will make the value of $f(u, A)$ minimum, that is, if $f(u, A) = 0$ or if $f(u, A) < 0$ then the condition will be fulfilled. So to make the downhill simplex working properly in this context we apply it to $-f(u, A)$ to have this condition:

$$-f(u, A) < 0$$

the more negative the value of $-f(u, A)$ is better, meaning the ball will surpass the wall more than expected, like doubling the home run distance. But we will only need the value of u and A when the downhill simplex compute the value when $-f(u, A)$ barely pass 0 towards negative value. This is the minimum value for u and optimum angle for A (the explanation for the downhill simplex method is available at chapter 11 in this book). We use this method because computing the derivative of $f(u, A)$ is not easy.

After 20 iterations we obtain the vector

$$\begin{bmatrix} u \\ A \end{bmatrix} = \begin{bmatrix} 145.381 \\ 0.6375 \end{bmatrix}$$

as the minimum value for u and A(in radian) to clear the wall. We can of course add the number of iteration or change the initial value when computing with downhill simplex method to obtain the value that is negative and near zero, but this is enough for now. Just to demonstrate the technique to solve this problem with downhill simplex method.

Now, if we input $u = 145.381$ and $A = 0.6375$ radian (equal to 36.526^0) we will obtain

$$f(u, A) = 0.18157$$

which satisfy the requirement to clear the wall of 350 ft distant and 10 ft high.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Eigen for Baseball in Flight ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb  main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Eigen for Baseball in Flight ]# ./main
f(x,y) = -800*ln(-70*x^(-1)*cos(y)^(-1)+1)-56000*x^(-1)*cos(y)^(-1)-350*sin(y)*cos(y)^(-1)+7

Total iteration:          20
Reflection:              3
Expansion:               7
Outside contraction:     0
Inside contraction:      10
Shrink:                  0

x_{0}:
145.431
 0.6375
x_{1}:
145.403
0.64375
x_{2}:
145.381
 0.6375

f(x,y) from x_{0}= -0.329485723352
f(x,y) from x_{1}= -0.182805997698
f(x,y) from x_{2}= -0.574296479311
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Eigen for Baseball in Flight ]# 
```

Figure 5.24: *The computation of the minimum velocity u and optimum angle A to clear a wall of 350 ft distant and 10 ft high for the baseball in flight case with downhill simplex method. (**SymIntegration/Examples/Differential Equations/Test Downhill Simplex with Eigen for Baseball in Flight/main.cpp**).*

[H*] **Brachistochrone Example:**

One of the famous problems in the history of mathematics is the brachistochrone problem; to find the curve along which a particle will slide without friction in the minimum time from one given point P to another Q, the second point being lower than the first but not directly beneath it. This problem was posed by Johann bernoulli in 1696 as a challenge problem to the mathematicians of his day. Correct solutions were found by Johann Bernoulli and his brother Jakob Bernoulli and by Isaac Newton, Gottfried Leibniz, and the Marquis de L'Hospital. The brachistochrone problem is important in the development of mathematics as one of the forerunners of the calculus of variations.

In solving this problem, it is convenient to take the origin as the upper point P and to orient the axes. The lower point Q has coordinates (x_0, y_0). It is then possible to show that the curve of minimum time is given by a function $y = \phi(x)$ that satisfies the differential equation

$$(1 + y'^2)y = k^2 \tag{5.30}$$

where k^2 is a certain positive constant to be determined later.

(a) Solve Eq. (5.30) for y'.Why is it necessary to choose the positive square root?

(b) Introduce the new variable t by the relation

$$y = k^2 \sin^2 t \tag{5.31}$$

Show that the equation found in part (a) then takes the form

$$2k^2 \sin^2 t \; dt = dx \tag{5.32}$$

(c) Letting $\theta = 2t$, show that the solution of Eq. (5.32) for which $x = 0$ when $y = 0$ is given by

$$\begin{aligned} x &= k^2 \frac{\theta - \sin\ \theta}{2} \\ y &= k^2 \frac{1 - \cos\ \theta}{2} \end{aligned} \tag{5.33}$$

Equations (5.33) are parametric equations of the solution of Eq. (5.30) that passes through $(0, 0)$. The graphs of Eqs. (5.33) is called a cycloid.

(d) If we make a proper choice of the constant k, then the cycloid also passes through the point (x_0, y_0) and is the solution of the brachistochrone problem. Find k if $x_0 = 1$ and $y_0 = 2$.

Solution:

(a) We will have

$$\begin{aligned} (1 + y'^2)y &= k^2 \\ 1 + y'^2 &= \frac{k^2}{y} \\ y'^2 &= \frac{k^2}{y} - 1 \\ y' &= \pm\sqrt{\frac{k^2}{y} - 1} \end{aligned}$$

We choose the positive sign for y' (the slope) because Q is lower than P and the positive y-axis points downward. Therefore

$$y' = \sqrt{\frac{k^2}{y} - 1}$$

(b) Let $y = k^2 \sin^2 t$. Differentiate both siddes with respect to t then use chain rule for

$\frac{dy}{dt}$

$$\begin{aligned}
\frac{dy}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\frac{dy}{dx}\frac{dx}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\sqrt{\frac{k^2}{y} - 1}\frac{dx}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\sqrt{\frac{k^2}{k^2 \sin^2 t} - 1}\frac{dx}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\sqrt{\csc^2 t - 1}\frac{dx}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\sqrt{\cot^2 t}\frac{dx}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\cot\ t\frac{dx}{dt} &= 2k^2 \sin\ t\ \cos\ t \\
\frac{dx}{dt} &= 2k^2 \sin^2 t \\
dx &= 2k^2 \sin^2 t\ dt
\end{aligned}$$

(c) Continue from the result of part (b)

$$\begin{aligned}
dx &= 2k^2 \sin^2 t\ dt \\
x &= \int 2k^2 \sin^2 t\ dt \\
&= 2k^2 \int \sin^2 t\ dt \\
&= 2k^2 \int \frac{1}{2}(1 - \cos\ 2t)\ dt \\
&= k^2 \left(t - \frac{1}{2}\sin\ 2t\right) + C_1
\end{aligned}$$

We also have

$$\begin{aligned}
y &= k^2 \sin^2 t \\
&= \frac{k^2}{2}(1 - \cos\ 2t)
\end{aligned}$$

In order to have $x = 0$ when $y = 0$ (that is, $t = 0$), C_1 must be zero:

$$C_1 = 0$$

so

$$\begin{aligned}
x(t) &= k^2 \left(t - \frac{1}{2}\sin\ 2t\right) \\
y(t) &= \frac{k^2}{2}(1 - \cos\ 2t)
\end{aligned}$$

Now replace $\theta = 2t$ to get the desired result

$$x(t) = k^2 \left(\frac{\theta}{2} - \frac{1}{2}\sin\ \theta\right)$$
$$y(t) = \frac{k^2}{2}(1 - \cos\ \theta)$$

Therefore,

$$x(t) = \frac{k^2}{2}\left(\theta - \sin\ \theta\right)$$
$$y(t) = \frac{k^2}{2}(1 - \cos\ \theta)$$

(d) The cycloid has to go through the point $(x = 1, y = 2)$ at some value of θ, say θ_0. Solve the resulting system of equation for k and θ_0.

$$\begin{aligned} 1 &= \frac{k^2}{2}\left(\theta_0 - \sin\ \theta_0\right) \\ 2 &= \frac{k^2}{2}(1 - \cos\ \theta_0) \end{aligned} \tag{5.34}$$

Now to solve the equations above for k and θ_0

$$2x(t) = y(t)$$
$$k^2(\theta_0 - \sin\ \theta_0) = \frac{k^2}{2}(1 - \cos\ \theta_0)$$
$$2\theta_0 - 2\sin\ \theta_0 = 1 - \cos\ \theta_0$$
$$2\theta_0 - 2\sin\ \theta_0 + \cos\ \theta_0 - 1 = 0$$

We can then find the root for the equation above to determine θ_0. There are two ways to find θ_0:

- Find the intersection point between $2\theta_0 - 2\sin\ \theta_0$ and $1 - \cos\ \ theta_0$, we can do this with **solve** function.
- Find the root of $2\theta_0 - 2\sin\ \theta_0 + \cos\ \theta_0 - 1$, this is easier since we can apply numerical method here, e.g. bisection method.

Either way will produce the same result which is

$$\theta_0 \approx 1.40149$$

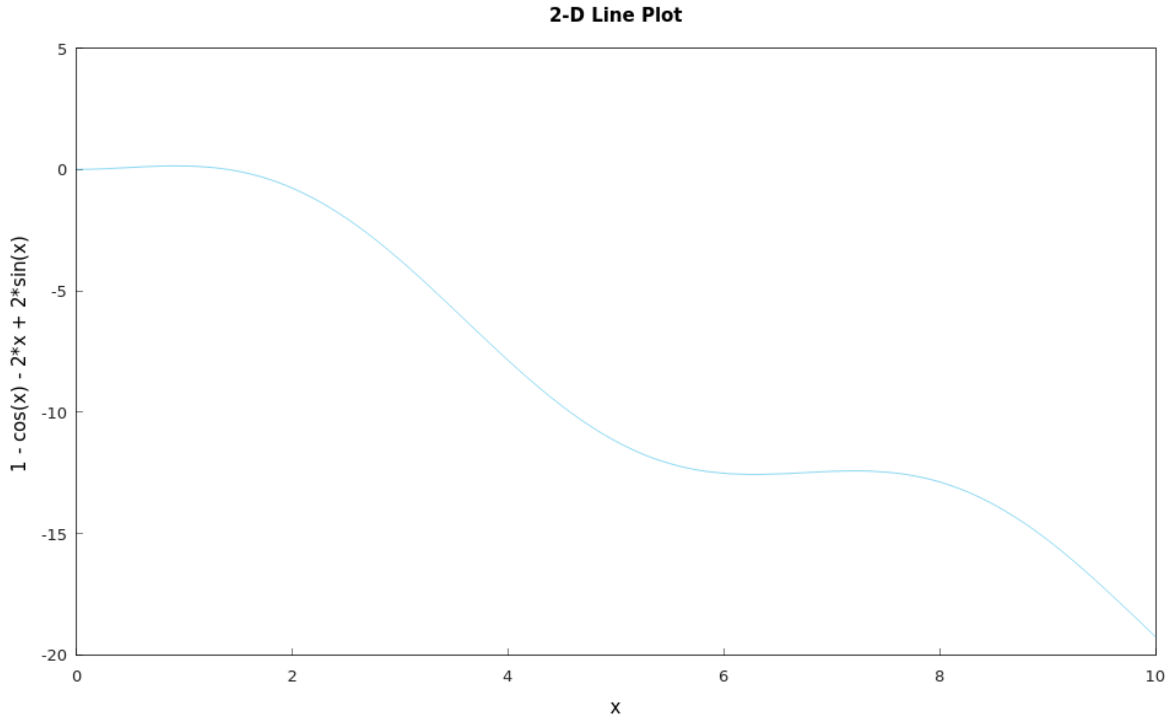


Figure 5.25: *The graph of* $f(\theta_0) = 2\theta_0 - 2\sin\ \theta_0 + \cos\ \theta_0 - 1$ *(**Hamzstplot/Examples with Makefile/Plot 2D Brachistochrone/main.cpp**).*

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve Brachisto
chrone ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration DSolve Brachistochrone ]# ./main
f(t,y) = y'^(2)*y+y-k^(2)

Solve for y'

y' = [ y' == 1/2*(-4*y^(2)+4*y*k^(2))^(1/2)*y^(-1),
  y' == -1/2*(-4*y^(2)+4*y*k^(2))^(1/2)*y^(-1) ]

Take the positive root only.

y' = 1/2*(-4*y^(2)+4*y*k^(2))^(1/2)*y^(-1)

dy/dt = 2*k^(2)*sin(t)*cos(t)

dy/dx = cos(t)*(csc(t))

dx/dt = -k^(2)*cos(2*t)+k^(2)

x(t) = -0.5*k^(2)*sin(2*t)+k^(2)*t
y(t) = -0.5*k^(2)*cos(2*t)+0.5*k^(2)

x(t) = -0.5*k^(2)*sin(θ)+0.5*k^(2)*θ
y(t) = -0.5*k^(2)*cos(θ)+0.5*k^(2)

The cycloid has to go through the point (x=1, y=2) at some value of θ, say θ0.

1 = -0.5*k^(2)*sin(θ)+0.5*k^(2)*θ
2 = -0.5*k^(2)*cos(θ)+0.5*k^(2)

-k^(2)*sin(θ)+k^(2)*θ = -0.5*k^(2)*cos(θ)+0.5*k^(2)
f(θ) = -2*sin(θ)+2*θ+cos(θ)-1
k = 2.19648

Time taken by function: 2143224 microseconds
```

Figure 5.26: *The computation to obtain the solution for Brachistochrone:* $x(t)$ *and* $y(t)$ *and then to compute the (**SymIntegration/Examples/Differential Equations/Test SymIntegration Solve Brachistochrone/-main.cpp**).*

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method with Function ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method with Function ]# ./main
f = -cos(θ0)-2*θ0+2*sin(θ0)+1
iteration               a                       b                       p                       f(p)
     1                  1                       2                       1.5                     -0.0757472

     2                  1                       1.5                     1.25                    0.0826469

     3                  1.25                    1.5                     1.375                   0.0172384

     4                  1.375                   1.5                     1.4375                  -0.0256436

     5                  1.375                   1.4375                  1.40625                 -0.00331926

     6                  1.375                   1.40625                 1.39062                 0.00717788

     7                  1.39062                 1.40625                 1.39844                 0.00198421

     8                  1.39844                 1.40625                 1.40234                 -0.000653763

     9                  1.39844                 1.40234                 1.40039                 0.000668658

    10                  1.40039                 1.40234                 1.40137                 8.30682e-06

    11                  1.40137                 1.40234                 1.40186                 -0.000322513

    12                  1.40137                 1.40186                 1.40161                 -0.00015705

    13                  1.40137                 1.40161                 1.40149                 -7.43579e-05
Procedure completed successfully
solution = 1.40149

Time taken by function: 30471 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method with Function ]# 
```

Figure 5.27: *The computation to obtain θ_0 from function $f(\theta_0) = 2\theta_0 - 2\sin\ \theta_0 + \cos\ \ theta_0 - 1$ with bisection method (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Bisection Method with Function for Brachistochrone/main.cpp**).*

Now, we can plug $\theta_0 = 1.4$ into one of the equation from Eqs. (5.34) to determine k

$$\begin{aligned}
2 &\approx \frac{k^2}{2}(1 - \cos\ 1.4) \\
k &\approx \sqrt{\frac{4}{1 - \cos\ 1.4}} \\
k &\approx 2.193
\end{aligned}$$

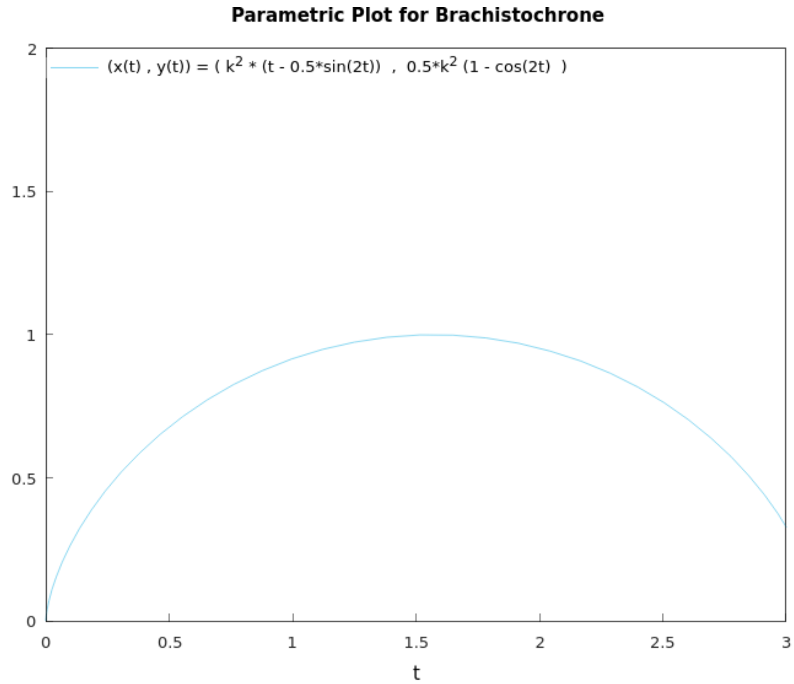


Figure 5.28: *The parametric plot of* $(x(t), y(t)) = \left(k^2\left(t - \frac{1}{2}\sin\ 2t\right), \frac{k^2}{2}(1-\cos\ 2t)\right)$ *(**Hamzstplot/Examples with Makefile/Plot 2D Parametric Plot for Brachistochrone/main.cpp**).*

- Differences Between Linear and Nonlinear Equations

[H*] If you encounter an initial value problem in linear differential equation, you are guaranteed that a solution exists.

Theorem 5.1: Existence and Uniqueness of Solutions for Linear Differential Equation

If the functions p and g are continuous on an open interval $I : \alpha < t < \beta$ containing point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the differential equation

$$y' + p(t)y = g(t) \tag{5.35}$$

for each t in I, and that also satisfies the initial condition

$$y(t_0) = y_0 \tag{5.36}$$

where y_0 is an arbitrary prescribed initial value.

The theorem states that the solution exists throughout any interval I containing the initial point t_0 in which the coefficients p and g are continuous.

The solution can be discontinuous or fail to exist only at points where at least one of p and g is discontinuous.

[H*] For the nonlinear differential equations, we will use more general theorem

Theorem 5.2: Existence and Uniqueness of Solutions for Nonlinear Differential Equations

Let the functions f and $\frac{\partial f}{\partial y}$ be continuous in some rectangle $\alpha < t < \beta, \gamma < y < \delta$ containing the point (t_0, y_0). Then, in some interval $t_0 - h < t < t_0 + h$ contained in $\alpha < t < \beta$, there is a unique solution $y = \phi(t)$ of the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \tag{5.37}$$

[H*] **Example:**
Solve the initial value problem

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}, \quad y(0) = -1$$

The differential equation is nonlinear, so

$$\begin{aligned} f(x,y) &= \frac{3x^2+4x+2}{2(y-1)} \\ \frac{\partial f}{\partial y}(x,y) &= -\frac{3x^2+4x+2}{2(y-1)^2} \end{aligned}$$

Thus each of these functions is continuous everywhere except on the line $y = 1$. Consequently, a rectangle can be drawn about the initial point $(0, -1)$ in which both f and $\frac{\partial f}{\partial y}$ are continuous.

The theorem guarantes that the initial value problem has a unique solution in some interval about $x = 0$. However, even though the rectangle can be stretched infinitely far in both the positive and negative x directions, this does not necessarily mean that the solution exists for all x.

We can use separable equations technique

$$\begin{aligned} \frac{dy}{dx} &= \frac{3x^2+4x+2}{2(y-1)} \\ 2(y-1)dy &= (3x^2+4x+2)dx \\ y^2 - 2y &= x^3 + 2x^2 + 2x + c \end{aligned}$$

Now to solve the initial value problem we substitute $x = 0$ and $y = -1$ so we will obtain $c = 3$, thus

$$\begin{aligned} y^2 - 2y - (x^3 + 2x^2 + 2x + 3) = 0 \\ y &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-(x^3+2x^2+2x+3))}}{2(1)} \\ y &= \frac{2 \pm \sqrt{4 + (4x^3 + 8x^2 + 8x + 12)}}{2} \\ y &= 1 \pm \sqrt{x^3 + 2x^2 + 2x + 4} \end{aligned}$$

To choose the plus or minus sign we need to check back again with the initial condition.

$$y(0) = 1 - \sqrt{(0)^3 + 2(0)^2 + 2(0) + 4}$$
$$y(0) = -1$$

so we choose the minus sign for the solution. The initial value problem of $\frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)}$, $y(0) = -1$ has the solution of

$$y(x) = 1 - \sqrt{x^3 + 2x^2 + 2x + 4} \tag{5.38}$$

We must find the interval in which the quantity under the radical is positive (under the square root), since if it is negative then it will return complex number. The only real zero of this expression is $x = -2$, so the desired interval is $x > -2$, that means the solution exists only for $x > -2$.

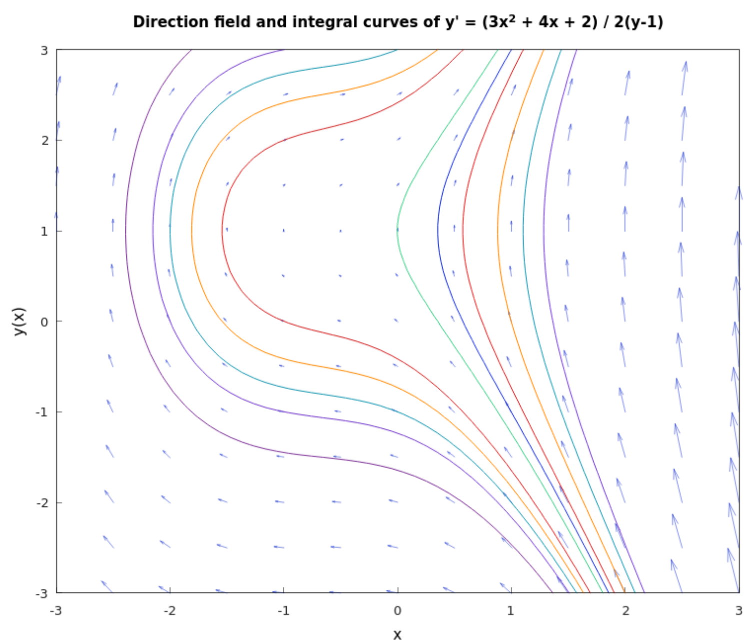


Figure 5.29: *The direction field and integral curves of* $\frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)}$ *(**Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Direction Field and Solutions Curves for Separable Equations Example 2/main.cpp**).*

[H*] **Bernoulli Equations**
Sometimes it is possible to solve a nonlinear equation by making a change of the dependent variable that converts it into a linear equation. The most important such equation has the form

$$y' + p(t)y = q(t)y^n \tag{5.39}$$

it is called a Bernoulli equation after Jakob Bernoulli.

[H*] **Example:**

(a) Solve Bernoulli' equation when $n = 0$ and when $n = 1$.

(b) Show that if $n \neq 0, 1$, then the substitution $v = y^{1-n}$ reduces Bernoulli's equation to a linear equation. This method of solution was found by Leibniz in 1696.

Solution:

(a) For $n = 0$ we will have

$$y' + p(t)y = q(t)$$

it is linear, so the solution is

$$y = \frac{1}{\mu(t)}\left[\int_{t_0}^{t} \mu(s)q(s)\ ds + c\right] \tag{5.40}$$

with

$$\mu(t) = e^{\int p(t)\ dt} \tag{5.41}$$

For $n = 1$ we will have

$$y' + p(t)y = q(t)y$$

it is linear again, so the solution is

$$y = \frac{1}{\mu(t)}[c] \tag{5.42}$$

with

$$\mu(t) = e^{\int p(t)-q(t)\ dt} \tag{5.43}$$

(b) Let $v = y^{1-n}$, then

$$\begin{aligned}\frac{dv}{dt} &= (1-n)y^{-n}\frac{dy}{dt}\\ \frac{dy}{dt} &= \frac{1}{1-n}y^{n}\frac{dv}{dt}\end{aligned}$$

which makes sense when $n \neq 0, 1$. Substituting into the differential equation yields

$$\begin{aligned}\frac{y^n}{1-n}\frac{dv}{dt} + p(t)y &= q(t)y^n\\ v' + (1-n)p(t)y^{1-n} &= (1-n)q(t)\end{aligned} \tag{5.44}$$

Setting $v = y^{1-n}$ then yields a linear differential equation for v. The Eq. (5.39) can be solved with method of integrating factors, it can also be written like this

$$v' + (1-n)p(t)v = (1-n)q(t) \tag{5.45}$$

[H*] **Example:**

The given equation is Bernoulli equation. In each case solve it by using the substitution method found by Leibniz in 1696.

(a) $y' = \epsilon y - \sigma y^3$, $\epsilon > 0$ and $\sigma > 0$. This equation occurs in the study of the stability of fluid flow.

(b) $y' = (\Gamma \cos\ t + T)y - y^3$, where Γ and T are constants. This equation also occurs in the study of the stability of fluid flow.

Solution:

(a) Since $n = 3$, set $v = y^{-2}$. It follows that

$$\begin{aligned} \frac{dv}{dt} &= -2y^{-3}\frac{dy}{dt} \\ \frac{dy}{dt} &= -\frac{y^3}{2}\frac{dv}{dt} \end{aligned}$$

Substituting into the differential equation yields

$$\begin{aligned} y' &= \epsilon y - \sigma y^3 \\ -\frac{y^3}{2}\frac{dv}{dt} &= \epsilon y - \sigma y^3 \\ -\frac{y^3}{2}\frac{dv}{dt} - \epsilon y &= -\sigma y^3 \\ \frac{dv}{dt} + 2\epsilon y^{-2} &= 2\sigma \\ v' + 2\epsilon v &= 2\sigma \end{aligned}$$

This differential equation $v' + 2\epsilon v = 2\sigma$ is linear, and can be written as

$$(v \cdot e^{2\epsilon t})' = 2\sigma e^{2\epsilon t} \tag{5.46}$$

with the integrating factor $\mu(t) = e^{2\epsilon t}$. The solution is given by

$$v(t) = \frac{\sigma}{\epsilon} + ce^{-2\epsilon t} \tag{5.47}$$

Converting back to the original dependent variable, we will have

$$\begin{aligned} y &= \pm v^{-\frac{1}{2}} \\ y(t) &= \pm\frac{1}{\sqrt{\frac{\sigma}{\epsilon} + ce^{-2\epsilon t}}} \end{aligned} \tag{5.48}$$

(b) Since $n = 3$, set $v = y^{-2}$. It follows that

$$\begin{aligned} \frac{dv}{dt} &= -2y^{-3}\frac{dy}{dt} \\ \frac{dy}{dt} &= -\frac{y^3}{2}\frac{dv}{dt} \end{aligned}$$

The differential equation is written as

$$\begin{aligned} y' &= (\Gamma \cos\ t + T)y - y^3 \\ -\frac{y^3}{2}\frac{dv}{dt} &= (\Gamma \cos\ t + T)y - y^3 \\ -\frac{y^3}{2}\frac{dv}{dt} - (\Gamma \cos\ t + T)y &= -y^3 \\ \frac{dv}{dt} + 2(\Gamma \cos\ t + T)y^{-2} &= 2 \\ v' + 2(\Gamma \cos\ t + T)v &= 2 \end{aligned}$$

This ordinary differential equation is linear, with integrating factor

$$\mu(t) = e^{2(\Gamma \sin\ t + Tt)}$$

Thus, we will have

$$(v \cdot e^{2(\Gamma \sin\ t+Tt)})' = 2e^{2(\Gamma \sin\ t+Tt)} \tag{5.49}$$

Then we can find the solution with the method of integrating factor

$$v(t) = \frac{1}{\mu(t)} \left[\int_{t_0}^{t} 2e^{2(\Gamma \sin\ s+Ts)}\ ds + c \right] \tag{5.50}$$

the solution is given by

$$v(t) = \frac{1}{e^{2(\Gamma \sin\ t+Tt)}} \left[\int_{t_0}^{t} 2e^{2(\Gamma \sin\ s+Ts)}\ ds + c \right] \tag{5.51}$$

the integral $\int_{t_0}^{t} 2e^{2(\Gamma \sin\ s+Ts)}\ ds$ is not an easy one to be integrated, even with SymPy in 2025, so we will keeping it that way. We can also write it in this form

$$v(t) = (e^{-2(\Gamma \sin\ t+Tt)}) \int_{t_0}^{t} 2e^{2\Gamma \sin\ s} e^{2Ts}\ ds + c(e^{-2(\Gamma \sin\ t+Tt)}) \tag{5.52}$$

- Autonomous Equations and Population Dynamics

[H*] An important class of first order equations consists of those in which the independent variable does not appear explicitly. Such equations are called autonomous and have the form

$$\frac{dy}{dt} = f(y) \tag{5.53}$$

Equation (5.53) is separable, but the main purpose of this section is to show how geometrical methods can be used to obtain important qualitative information directly from the differential equation, without solving the equation.

[H*] **Exponential Growth**

Let $y = \phi(t)$ be the population of the given species at time t. The simplest hypothesis concerning the variation of population is that the rate of change of y is proportional to the current value of y; that is,

$$\frac{dy}{dt} = ry \tag{5.54}$$

where the constant of proportionality r is called the rate of growth or decline, depending on whether it is positive or negative. Here, we assume that $r > 0$, so the population is growing.

Solving Eq. (5.54) subject to the initial condition

$$y(0) = y_0 \tag{5.55}$$

we obtain

$$y = y_0 e^{rt} \tag{5.56}$$

Thus the mathematical model consisting of the initial value problem (5.54), (5.55) with $r > 0$ predicts that the population will grow exponentially for all time. However, it is clear that such ideal conditions cannot continue indefinitely; eventually, limitations on space, food supply, or other resources will reduce the growth rate and bring an end to uninhibited exponential growth.

[H*] **Logistic Growth**
Consider the differential equation that the growth rate actually depends on the population

$$\frac{dy}{dt} = h(y)y \tag{5.57}$$

We now want to choose $h(y)$ so that $h(y) \equiv r > 0$ when y is small, $h(y)$ decreases as y grows larger, and $h(y) < 0$ when y is sufficiently large. The simplest function that has these properties is

$$h(y) = r - ay$$

where a is also a positive constant. Using this function in Eq. (5.57), we obtain

$$\frac{dy}{dt} = (r - ay)y \tag{5.58}$$

Equation (5.58) is known as the Verhulst equation or the logistic equation. It is often convenient to write the logistic equation in the equivalent form

$$\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right)y \tag{5.59}$$

where $K = r/a$. The constant r is called the intrinsic growth rate, that is, the growth rate in the absence of any limiting factors.

We first seek solutions of Eq. (5.59) of the simplest possible type, that is, constant functions.

For such a solution $\frac{dy}{dt} = 0$ for all t, so any constant solution of Eq. (5.59) must satisfy the algebraic equation

$$r\left(1 - \frac{y}{K}\right)y = 0$$

Thus the constant solutions are $y = \phi_1(t) = 0$ and $y = \phi_2(t) = K$. These solutions are called equilibrium solutions of Eq. (5.59) because they correspond to no change or variation in the value of y as t increases.

In the same way, any equilibrium solutions of the more general Eq. (5.53) can be found by locating the roots of $f(y) = 0$. The zeros of $f(y)$ are also called critical points.

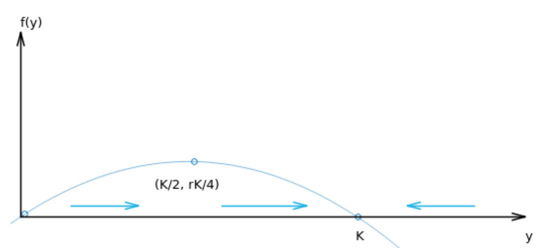


Figure 5.30: *The equilibrium solution $f(y) = 0$ or $r(1 - y/K)y = 0$ to visualize other solutions of Eq. (5.59) (**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Logistic Growth/main.cpp**).*

The graph of $f(y)$ versus y above is a parabola with the intercepts are $(0,0)$ and $(K,0)$, corresponding to the critical points of Eq. (5.59), and the vertex of the parabola is

$(K/2, rK/4)$.

Observe that $\frac{dy}{dt} > 0$ for $0 < y < K$; therefore, y is an increasing function of t when y is in this interval; this is indicated by the rightward-pointing arrows near the y-axis. Similarly, if $y > K$, then $\frac{dy}{dt} < 0$; hence y is decreasing, as indicated by the leftward-pointing arrow.

The y-axis is often called the phase line, and it is usually drawn with vertical line (rotating it 90 degree counterclockwise).

The dots at $y = 0$ and $y = K$ are the critical points or equilibrium solutions. The arrows again indicate that y is increasing whenever $0 < y < K$ and that y is decreasing whenever $y > K$.

Note that, if y is near zero or K, then the slope $f(y)$ is near zero, so the solution curves are relatively flat. They become steeper as the value of y leaves the neighborhood of zero or K.

To sketch the graphs of solutions of Eq. (5.59) in the ty-plane, we start with the equilibrium solutions $y = 0$ and $y = K$; then we draw other curves that are increasing when $0 < y < K$, decreasing when $y > K$, and flatten out as y approaches either the values 0 or K. Thus, the graphs of solutions of Eq. (5.59) must have the general shape regardless of the values of r and K.

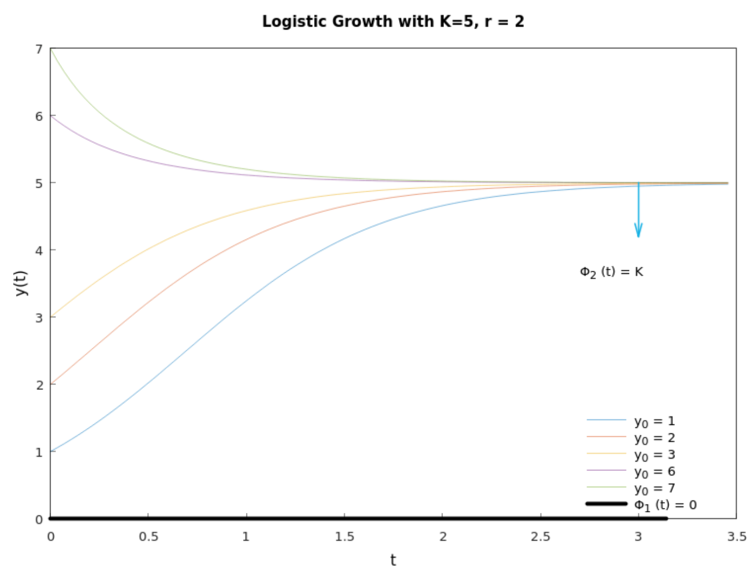


Figure 5.31: *The solution curves for* $y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}}$ *to visualize all possible solutions of Eq. (5.59) with different* y_0 *(**Hamzstplot/Examples with Makefile/Plot 2D Solution Curves for Logistic Growth/-main.cpp**).*

It may seems that other solutions intersect the equilibrium solution at $y(t) = K$ in Figure (5.31), but it is not possible.

Based on the uniqueness part of Theorem 4.2, the fundamental existence and uniqueness theorem, states that only one solution can pass through a given point in the ty-plane. Thus, although other solutions may be asymptotic to the equilibrium solution as $t \to \infty$, they cannot intersect it at any finite time.

We can determine the concavity of the solution curves and the location of inflection points by finding $\frac{d^2y}{dt^2}$. From the differential equation (5.53) we obtain (using the chain rule)

$$\frac{d^2y}{dt^2} = \frac{d}{dt}\frac{dy}{dt} = \frac{d}{dt}f(y) = f'(y)\frac{dy}{dt} = f'(y)f(y) \tag{5.60}$$

The graph of y versus t is concave up when $y'' > 0$, that is, when f and f' have the same sign.

Similarly, it is concave down when $y'' < 0$, which occurs when f and f' have opposite signs. The signs of f and f' can be easily identified from the graph of $f(y)$ versus y. Inflection points may occur when $f'(y) = 0$.

In the case of Eq. (5.59), solutions are concave up for $0 < y < K/2$ where f is positive and increasing, so that both f and f' are positive. Solutions are also concave up for $y > K$ where f is negative and decreasing (both f and f' are negative). For $K/2 < y < K$, solutions are concave down since here f is positive and decreasing, so f is positive but f' is negative. There is an inflection point whenever the graph of y versus t crosses the line $y = K/2$.

Finally, observe that K is the upper bound that is approached, but not exceeded, by growing populations starting below this value. Thus it is natural to refer to K as the saturation level, or the environmental carrying capacity, for the given species.

It reveals that solutions of the nonlinear equation (5.59) are strikingly different from those of the linear equation (5.53), at least for large values of t. Regardless of the value of K, that is, no matter how small the nonlinear term in Eq. (5.59), solutions of that equation approach a finite values as $t \to \infty$, whereas solutions of Eq. (5.53) grow (exponentially) without bound as $t \to \infty$. Thus, even a tiny nonlinear term in the differential equation (5.59) has a decisive effect on the solution for large t.

In many situations it is sufficient to have the qualitative information about a solution $y = \phi(t)$ of Eq. (5.59). This information was obtained entirely from the graph of $f(y)$ versus y, and without solving the differential equation (5.59). However, if we wish to know the value of the population at some particular time, then we must solve Eq. (5.59) subject to the initial condition (5.55). Provided that $y \neq 0$ and $y \neq K$, we can write Eq. (5.59) in the form

$$\begin{aligned} \frac{dy}{\left(1 - \frac{y}{K}\right) y} &= r\,dt \\ \left(\frac{1}{y} - \frac{1/K}{1 - y/K}\right) dy &= r\,dt \\ \ln|y| - \ln\left|1 - \frac{y}{K}\right| &= rt + c \end{aligned} \tag{5.61}$$

where c is an arbitrary constant of integration to be determined from the initial condition

$y(0) = y_0$. We have already noted that if $0 < y_0 < K$, then y remains in this interval for all time. Thus in this case we can remove the absolute value bars in Eq. (5.61), and by taking the exponential of both sides, we find that

$$\frac{y}{1 - (y/K)} = Ce^{rt} \tag{5.62}$$

where $C = e^c$. In order to satisfy the initial condition $y(0) = y_0$, we must choose $C = \frac{y_0}{1-(y_0/K)}$. Using this value of C in Eq. (5.62) and solving for y, we obtain

$$y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}} \tag{5.63}$$

We have derived the solution (5.63) under the assumption that $0 < y_0 < K$. If $y_0 > K$, then the details of dealing with Eq. (5.61) are only slightly different, and Eq, (5.63) is also valid in this case.

Note that Eq. (5.63) also contains the equilibrium solutions $y = \phi_1(t) = 0$ and $y = \phi_2(t) = K$ corresponding to the initial conditions $y_0 = 0$ and $y_0 = K$, respectively.

In particular, if $y_0 = 0$, then Eq. (5.63) requires that $y(t) = 0$ for all t. If $y_0 > 0$, and if we let $t \to \infty$ in Eq. (5.63), then we obtain

$$\lim_{t\to\infty} y(t) = \frac{y_0 K}{y_0} = K$$

Thus, for each $y_0 > 0$, the solution approaches the equilibrium solution $y = \phi_2(t) = K$ asymptotically as $t \to \infty$. Therefore we say that the solution $\phi_2(t) = K$ is an asymptotically stable solution of Eq. (5.59) or that the point $y = K$ is an asymptotically stable equilibrium or critical point.
After a long time, the population is close to the saturation level K regardless of the initial population size, as long as it is positive. Other solutions approach the equilibrium solution more rapidly as r increases.

On the other hand, the situation for the equilibrium solution $y = \phi_1(t) = 0$ is quite different. Even solutions that start very near zero grow as t increases and, as we have seen, approach K as $t \to \infty$. We say that $\phi_1(t) = 0$ is an unstable equilibrium solution or that $y = 0$ is an unstable equilibrium critical point. This means that the only way to guarantee that the solution remains near zero is to make sure its initial value is exactly equal to zero.

If we want to find the time, t, when the population reach certain level $y(t)$, e.g. when the tuna population in Pacific Ocean reach 85% ($y(t) = 0.85K$) of its maximum limit that the environment can carry, then we will use this formula from derivating of Eq. (5.63):

$$t = -\frac{1}{r} \ln \frac{\left(\frac{y_0}{K}\right)\left[1 - \left(\frac{y}{K}\right)\right]}{\left(\frac{y}{K}\right)\left[1 - \left(\frac{y_0}{K}\right)\right]} \tag{5.64}$$

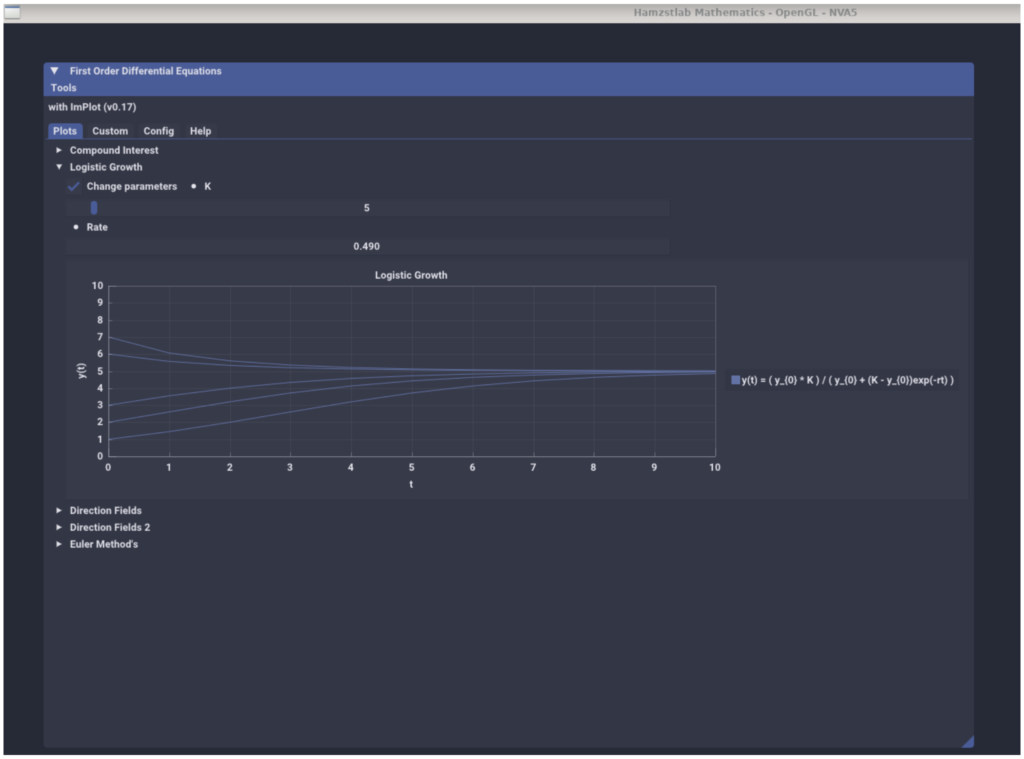


Figure 5.32: *The solution curves for $y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}}$ we set y_0 with fixed values at first and the parameters K and r can be changed at in instant in Hamzstlab Mathematics (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

[H*] **A Critical Threshold**
We now turn to a consideration of the equation

$$\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)y \tag{5.65}$$

where r and T are given positive constants. Observe that this equation differs from the logistic equation (5.59) in the presence of the minus sign on the right side, along with T replacing K.

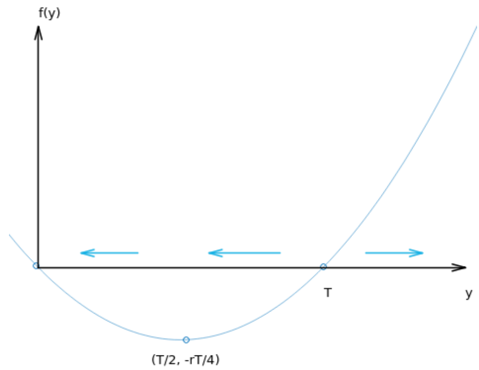


Figure 5.33: *The graph of $f(y)$ for $\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)y$ (**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Critical Threshold/main.cpp**).*

The intercepts on the y-axis are the critical points $y = 0$ and $y = T$, corresponding to the equilibrium solutions $\phi_1(t) = 0$ and $\phi_2(t) = T$.

The dots at $y = 0$ and $y = T$ are the critical points, or equilibrium solutions, and the arrows indicate where solutions are either increasing or decreasing.

If $0 < y < T$, then $\frac{dy}{dt} < 0$, and y decreases as T increases. Thus $\phi_1(t) = 0$ is an asymptotically stable equilibrium solution and $\phi_2(t) = T$ is an unstable one.

Further, $f'(y)$ is negative for $0 < y < T/2$ and positive for $T/2 < y < T$, so the graph of y versus t is concave up and concave down, respectively, in these intervals. Also, $f'(y)$ is positive for $y > T$, so the graph of y versus t is also concave up here.

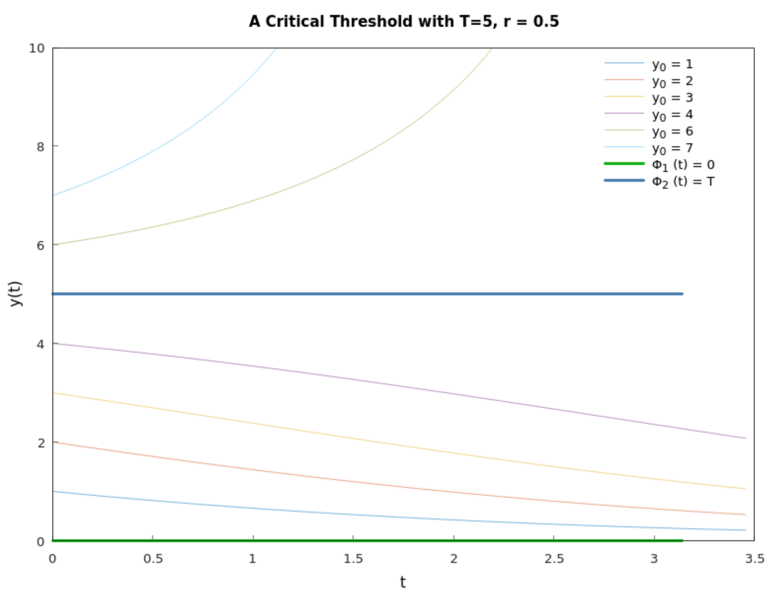


Figure 5.34: *The solution curves for growth with a threshold for $\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)y$ (**Hamzstplot/Examples with Makefile/Plot 2D Solution Curves for Critical Threshold/main.cpp**).*

Figure (5.34) shows the solution curves of Eq. (5.65). After we draw the two thick horizontal line at $y = 0$ and $y = T$, we can then sketch the curves in the strip $0 < y < T$ that are decreasing as t increases and change concavity as they cross the line $y = T/2$.

Next draw some curves above $y = T$ that increase more and more steeply as t and y increase. Make sure that all curves become flatter as y approaches either zero or T.

From this figure it appears that as time increases, y either approaches zero or grows without bound, depending on whether the initial value y_0 is less than or greater than T. Thus T is a threshold level, below which growth does not occur.

By solving the differential equation (5.65) with separating the variables and integrating we will obtain

$$y(t) = \frac{y_0 T}{y_0 + (T - y_0)e^{rt}} \tag{5.66}$$

which is the solution of Eq. (5.65) subject to the initial condition $y(0) = y_0$.

If $0 < y_0 < T$, then it follows from Eq. (5.66) that $y \to 0$ as $t \to \infty$. This agrees with our qualitative geometric analysis. If $y_0 > T$, then the denominator on the right side of Eq. (5.66) is zero for a certain finite value of t. We denote this value by t^* and calculate it from

$$y_0 - (y_0 - T)e^{rt^*} = 0$$

which gives

$$t^* = \frac{1}{r} \ln \frac{y_0}{y_0 - T} \tag{5.67}$$

Thus, if the initial population y_0 is above the threshold T, the threshold model predicts that the graph of y versus t has a vertical asymptote at $t = t^*$; in other words, the population becomes unbounded in a finite time, whose value depends on y_0, T, and r. The existence and location of this asymptote were not apparent from the geometric analysis, so in this case the explicit solution yields additional important qualitative, as well as quantitative, information.

The population of some species exhibit the threshold phenomenon. If too few are present, then the species cannot propagate itself successfully and the population becomes extinct. However, if the population is alrger than the threshold level, then further growth occurs. Of course, the population cannot become unbounded, so eventually Eq. (5.65) must be modified to take this into account.

Critical thresholds also occur in other circumstances. For example, in fluid mechanics, equations of the form (5.59) and (5.65) often govern the evolution of a small disturbance y in a laminar (or smooth) fluid flow.

For instance, if Eq. (5.65) holds and $y < T$, then the disturbance is damped out and laminar flow persists.
However, if $y > T$, then the disturbance grows larger and the laminar flow breaks up into a turbulent one. In this case T is referred to as the critical amplitude. Experimenters speak of keeping the disturbance level in a wind tunnel sufficiently low so taht they can study laminar flow over an airfoil, for example.

[H*] **Logistic Growth with a Threshold**

This model comes after we modified the threshold model (5.65) so that unbounded growth does not occur when y is above the threshold T. The simplest way to do this is to introduce another factor that will have the effect of making $\frac{dy}{dt}$ negative when y is large. Thus we consider

$$\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{K}\right)y \tag{5.68}$$

where $r > 0$ and $0 < T < K$.

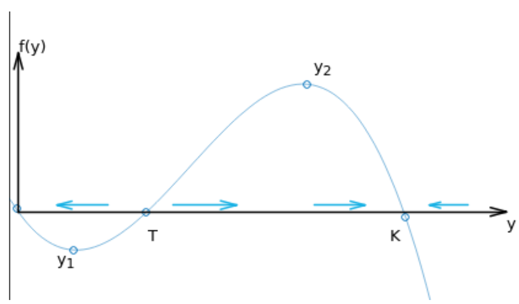


Figure 5.35: *The graph of $f(y)$ versus y for logistic growth with a threshold for $\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{K}\right)y$ (**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Logistic Growth with a Threshold/main.cpp**).*

The equilibrium solutions are found by setting $\frac{dy}{dt} = 0$.
In this problem there are three critical points, $y = 0, y = T$, and $y = K$, corresponding to the equilibrium solutions $\phi_1(t) = 0, \phi_2(t) = T$, and $\phi_3(t) = K$, respectively.

From Figure 5.35 we observe that $\frac{dy}{dt} > 0$ for $T < y < K$, and consequently y is increasing there. Further, $\frac{dy}{dt} < 0$ for $y < T$ and for $y > K$, so y is decreasing in these intervals.

Consequently, the equilibrium solutions $\phi_1(t)$ and $\phi_3(t)$ are asymptotically stable, and the solution $\phi_2(t)$ is unstable.

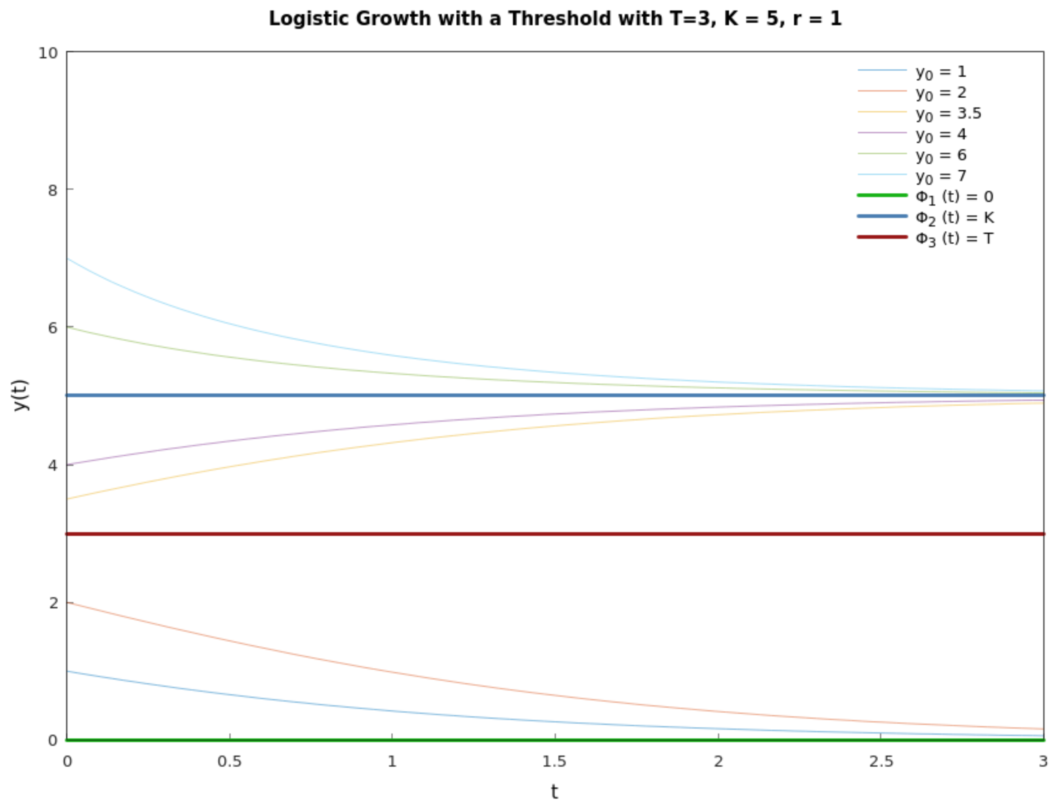


Figure 5.36: *The solution curves for logistic growth with a threshold for* $\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{K}\right)y$ *(**Hamzstplot/Examples with Makefile/Plot 2D Solution Curves for Logistic Growth with a Threshold/main.cpp**).*

You should make sure that you understand the relation between these two Figure (5.35) and Figure (5.36).

We see that if y starts below the threshold T, then y declines to ultimate extinction. On the other hand, if y starts above T, then y eventually approaches the carrying capacity K. The inflection points on the graphs of y versus t in Figure (5.36) correspond to the maximum and minimum points y_1 and y_2, respectively, on the graphy of $f(y)$ versus y in Figure (5.35). These values can be obtained by differentiating the right side of Eq. (5.68) with respect to y, setting the result equal to zero, and solving for y. We obtain

$$y_{1,2} = \frac{\left(K + T \pm \sqrt{K^2 - KT + T^2}\right)}{3} \tag{5.69}$$

where the plus sign yields y_1 and the minus sign yields y_2.

The logistic equation with a threshold population at Eq. (5.68) can be solved using the method of separation of variables. However, it is very difficult to get the solution as an explicit function of t, so we are opting for the implicit function instead.

$$\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{K}\right)y$$
$$\frac{dy}{\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{K}\right)y} = -r\,dt$$
$$\ln|y| - \ln|T - y| - \ln|K - y| = -rt + C$$

Solving for C with the initial value of $y(0) = y_0$, we will obtain

$$C = \ln|y_0| - \ln|T - y_0| - \ln|K - y_0|$$

thus

$$\ln|y(t)| - \ln|T - y(t)| - \ln|K - y(t)| = -rt + \ln|y_0| - \ln|T - y_0| - \ln|K - y_0| \quad (5.70)$$

where

* $y(t)$ represents the population size at time t.
* r is a positive constant representing the intrinsic growth rate.
* T is the threshold population below which the population declines towards extinction.
* K is the carrying capacity, the maximum sustainable population size.
* $0 < T < K$

The good thing is, sometimes you can solve a very complex problem by dividing it into smaller partition and see the pattern at each partition, in C++ code we can use **if.. then** conditional that can be very useful for drawing the solution curves of this type. Without even solving the Eq. (5.68), we can use Eq. (5.63) and Eq. (5.66) under the **if .. then .. if else .. then ..** conditional.

[H*] **Phase Line Analysis**

In order to determine the stability for single variable function, we use phase line analysis.

First, find the equilibrium solutions by finding the root of $f(y) = 0$ from

$$\frac{dy}{dt} = f(y)$$

Then, draw a horizontal line representing the variable (e.g. y). Mark the equilibrium points on the line.

Analyze the sign of the derivative $\frac{dy}{dt}$ on either side of each equilibrium point.

If the derivative changes from positive to negative as y increases, the equilibrium point is a sink (stable).

If it changes from negatives to positive, it's a source (unstable).

If there's no sign change, it's a node (semistable).

To draw the phase line you can just rotate the horizontal line from above counterclockwise 90^0 to become a vertical line.

The benefits of phase line analysis:

1. Phase lines provide a visual understanding of the system's behavior without solving the differential equation.
2. They clearly show which equilibrium points are stable (solutions approach them) and which are unstable (solutions move away).
3. They reveal how solutions behave as time approaches infinity.

[H*] **Harvesting a Renewable Resource**
Suppose that the population of y of a certain species of fish (for example, tuna or halibut) in a given area of the ocean is described by the logistic equation

$$\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right)y \tag{5.71}$$

Although it is desirable to utilize this source of food, it is intuitively clear that if too many fish are caught, then the fish population may be reduced below a useful level and possibly even driven to extinction.

The solution to Eq. (5.71) with initial value problem $y(0) = y_0$ is

$$y(t) = \frac{y_0 K e^{rt}}{(y_0 - K)\left(\frac{y_0 e^{rt}}{y_0 - K} - 1\right)} \tag{5.72}$$

[H*] **Example:**
At a given level of effort, it is reasonable to assume that the rate at which fish are caught depends on the population y: the more fish there are, the easier it is to catch them. Thus we assume that the rate at which fish are caught is given by Ey, where E is a positive constant, with units of 1/time, that measures the total effort made to harvest the given species of fish. To include this effect, the logistic equation is replaced by

$$\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right)y - Ey \tag{5.73}$$

This equation is known as the Schaefer model after the biologist M.B. Schaefer, who applied it to fish populations.

(a) Show that if $E < r$, then there are two equilibrium points $y_1 = 0$ and $y_2 = K\left(1 - \frac{E}{r}\right) > 0$.
(b) Determine the stability for the equilibrium points.
(c) A sustainable yield Y of the fishery is a rate at which fish can be caught indefinitely. It is the product of the effort E and the asymptotically stable population y_2. Find Y as a function of the effort E; the graph of this function is known as the yield-effort curve.
(d) Determine E so as to maximize Y and thereby find the maximum sustainable yield Y_m.

Solution:

(a) To determine the equilibrium points or the critical points we need to locate the roots of $f(y) = 0$

$$\begin{aligned} f(y) &= 0 \\ r\left(1 - \frac{y}{K}\right)y - Ey &= 0 \\ -\frac{ry^2}{K} + (r - E)y &= 0 \end{aligned}$$

the roots are

$$y_{1,2} = \frac{-(r-E) \pm \sqrt{(r-E)^2 - 4\left(-\frac{r}{K}\right)(0)}}{2\left(-\frac{r}{K}\right)}$$
$$= \frac{-(r-E) \pm (r-E)}{-\frac{2r}{K}}$$

$$y_{1,2} = \begin{cases} 0 \\ \frac{-2r+2E}{-2r/K} = \frac{K(E-r)}{-r} = K\left(1 - \frac{E}{r}\right) \end{cases}$$

If $E < r$ then $K\left(1 - \frac{E}{r}\right) > 0$ and is a possible equilibrium population.

(b) To determine which equilibrium solution that is stable and unstable we will check with inputting y as several values in $\frac{dy}{dt} = f(y)$

$$f(y) = -\frac{ry^2}{K} + (r-E)y$$
$$= \left(y - K\left(1 - \frac{E}{r}\right)\right) y$$

If $0 < y < K\left(1 - \frac{E}{r}\right)$ then $\frac{dy}{dt} < 0$, so y is decreasing on these intervals.
If $y > K\left(1 - \frac{E}{r}\right)$ then $\frac{dy}{dt} > 0$, so y is increasing on these intervals.

The equilibrium at $y = 0$ is a sink (asymptotically stable) and the equilibrium at $K\left(1 - \frac{E}{r}\right)$ is a source (unstable).

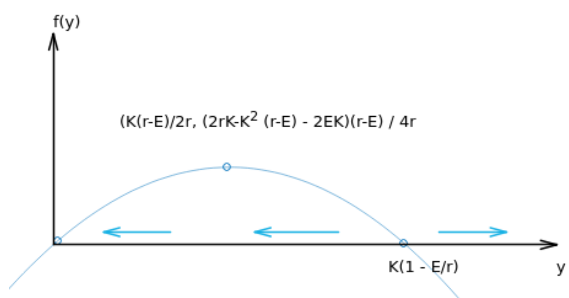


Figure 5.37: *The equilibrium points and the stability analysis for* $\frac{dy}{dt} = f(y) = \left(y - K\left(1 - \frac{E}{r}\right)\right) y$ *(**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Schaefer Model/main.cpp**).*

To determine the vertex of the parabola you will have to compute

$$\frac{df(y)}{dy} = 0$$
$$\frac{d\left(-\frac{ry^2}{K} + (r-E)y\right)}{dt} = 0$$
$$y = \frac{K(r-E)}{2r}$$

then input y back into $f(y)$ again to obtain the vertex fully

$$\begin{aligned} f(y) &= -\frac{r\left(\frac{K(r-E)}{2r}\right)^2}{K} + (r-E)\left(\frac{K(r-E)}{2r}\right) \\ &= \frac{(2rK - K^2(r-E) - 2EK)(r-E)}{4r} \end{aligned}$$

thus the vertex is $\left(\frac{K(r-E)}{2r}, \frac{(2rK-K^2(r-E)-2EK)(r-E)}{4r}\right)$.

(c) The sustainable yield is the effort times the asymptotically stable population.

$$Y(E) = EK\left(1 - \frac{E}{r}\right)$$

To maximize the sustainable yield, take the derivative of it, set it equal to zero, and solve the resulting equation for E.

$$\begin{aligned} Y'(E) &= 0 \\ K\left(1-\frac{E}{r}\right) + EK\left(-\frac{1}{r}\right) &= 0 \\ K\left(1-\frac{E}{r}\right) &= \frac{EK}{r} \\ K(r-E) &= EK \\ rK - EK &= EK \\ 2EK &= rK \\ E &= \frac{r}{2} \end{aligned}$$

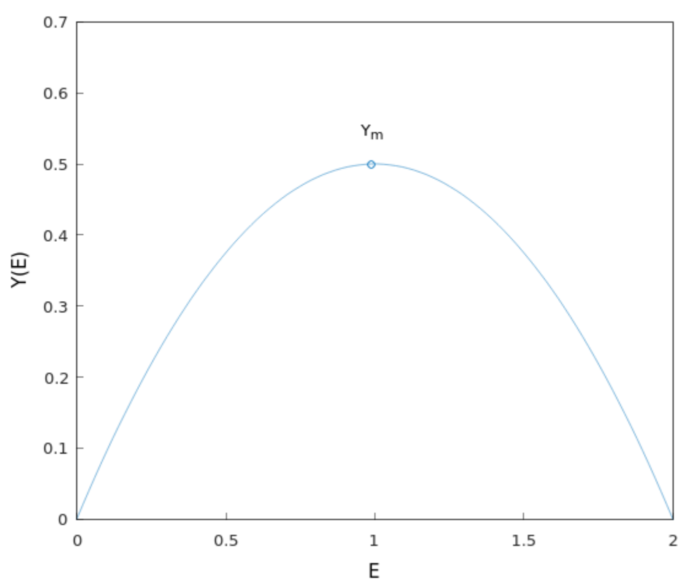


Figure 5.38: *The yield-effort curve for $Y(E) = EK\left(1-\frac{E}{r}\right)$ with $r = 2$ and $K = 1$ (**Hamzstplot/Examples with Makefile/Plot 2D Yield-Effort Curve for Schaefer Model/main.cpp**).*

(d) The maximum sustainable yield is therefore

$$
\begin{aligned}
Y_m &= Y\left(E = \frac{r}{2}\right) \\
&= \left(\frac{r}{2}\right) K \left(1 - \frac{1}{2}\right) \\
&= \frac{rK}{4}
\end{aligned}
$$

[H*] **Example:**

We assume that fish are caught at a constant rate h independent of the size of the fish population. Then y satisfies

$$\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right) y - h \tag{5.74}$$

The assumption of a constant catch rate h may be reasonable when y is large but becomes less so when y is small.

(a) If $h < rK/4$, show that Eq. (5.74) has two equilibrium points y_1 and y_2 with $y_1 < y_2$; determine these points.

(b) Show that y_1 is unstable and y_2 is asymptotically stable.

(c) From a plot of $f(y)$ versus y, show that if the initial population $y_0 > y_1$, then $y \to y_2$ as $t \to \infty$, but that if $y_0 < y_1$, then y decreases as t increases. Note that $y = 0$ is not an equilibrium point, so if $y_0 < y_1$, then extinction will be reached in a finite time.

(d) If $h > rK/4$, show that y decreases to zero as t increases regardless of the value of y_0.

(e) If $h = rK/4$, show that there is a single equilibrium point $y = K/2$ and that this point is semistable. Thus the maximum sustainable yield is $h_m = rK/4$, corresponding to the equilibrium value $y = K/2$. Observe that h_m has the same value as Y_m in the previous example. The fishery is considered to be overexploited if y is reduced to a level below $K/2$.

Solution:

(a) To determine the equilibrium points we will find the roots of $f(y) = 0$

$$
\begin{aligned}
f(y) &= 0 \\
r\left(1 - \frac{y}{K}\right) y - h &= 0 \\
-\frac{r}{K} y^2 + ry - h &= 0
\end{aligned}
$$

with quadratic roots we will obtain

$$
\begin{aligned}
y_{1,2} &= \frac{-r \pm \sqrt{r^2 - 4\left(-\frac{r}{K}\right)(-h)}}{2\left(-\frac{r}{K}\right)} \\
&= \frac{-r \pm \sqrt{r^2 - \left(\frac{4rh}{K}\right)}}{-\frac{2r}{K}} \\
&= \frac{r \pm \sqrt{r^2 - \left(\frac{4rh}{K}\right)}}{\frac{2r}{K}}
\end{aligned}
$$

$$y_{1,2} = \begin{cases} \frac{r+\sqrt{r^2-\left(\frac{4rh}{K}\right)}}{\frac{2r}{K}} = \frac{K+\sqrt{K^2-\frac{4Kh}{r}}}{2} \\ \frac{r-\sqrt{r^2-\left(\frac{4rh}{K}\right)}}{\frac{2r}{K}} = \frac{K-\sqrt{K^2-\frac{4Kh}{r}}}{2} \end{cases}$$

These are the equilibrium solutions as long as

$$r^2 - \frac{4rh}{K} > 0$$
$$h < \frac{rK}{4}$$

(b) We will try to plot $f(y) = r\left(1 - \frac{y}{K}\right)y - h$ versus y with $r = 1, K = 1$, and $h = 1/8$.

With the chosen parameters, we will have

$$f(y) = (1-y)y - \frac{1}{8}$$

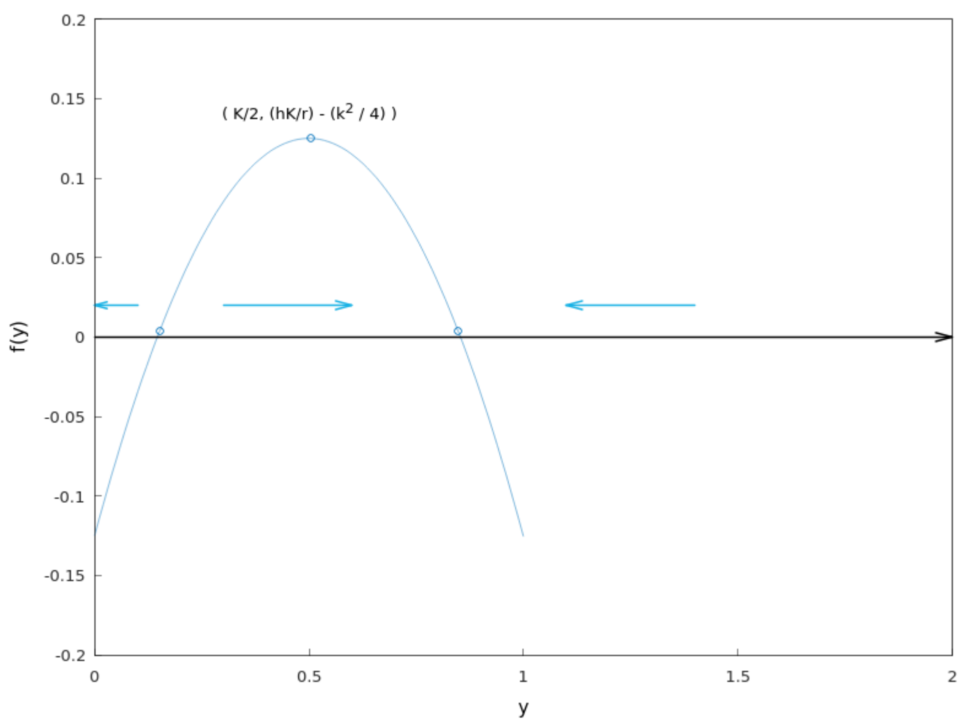


Figure 5.39: *The plot for $f(y) = (1-y)y - \frac{1}{8}$ (**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Schaefer Model Variation/main.cpp**).*

To save more time, we do not need to compute the exact numerical equilibrium points of y_1 and y_2 here, we only need to know $f(y) = (1-y)y - \frac{1}{8}$ and then plot $f(y)$ versus y, from the plot we will be able to determine the stability by approximation. For example, it looks very clearly that $y = 0$ is less than $y = y_1$ so we can compute $f(0)$ to see $\frac{dy}{dt}$ behavior in that interval, if $y < y_1$ then $\frac{dy}{dt} < 0$ and the arrow is pointing to the left on the horizontal line, or pointing downward in the phase line.

We see that the equilibrium point y_1 is a source (unstable) and the equilibrium point at $y = y_2$ is a sink (asymptotically stable).

(c) Suppose that the initial condition is

$$y(0) = y_0$$

As indicated by the arrows from Figure (5.39), if $y_0 > y_1$, then the population will tend to $y(t) = y_2$ as $t \to \infty$.

If $y_0 = y_1$, then the population will remain there.

If $y_0 < y_1$, then the population will tend to zero in a finite time because there are no equilibrium points to the left of $y = y_1$.

(d) Suppose that $h > \frac{rK}{4}$, for example, using $r = 1, K = 1$, and $h = 1$. Then the entire graph of $f(y) = r\left(1 - \frac{y}{K}\right) y - h$ versus y lies below the y-axis, which means all the arrows point to the left.

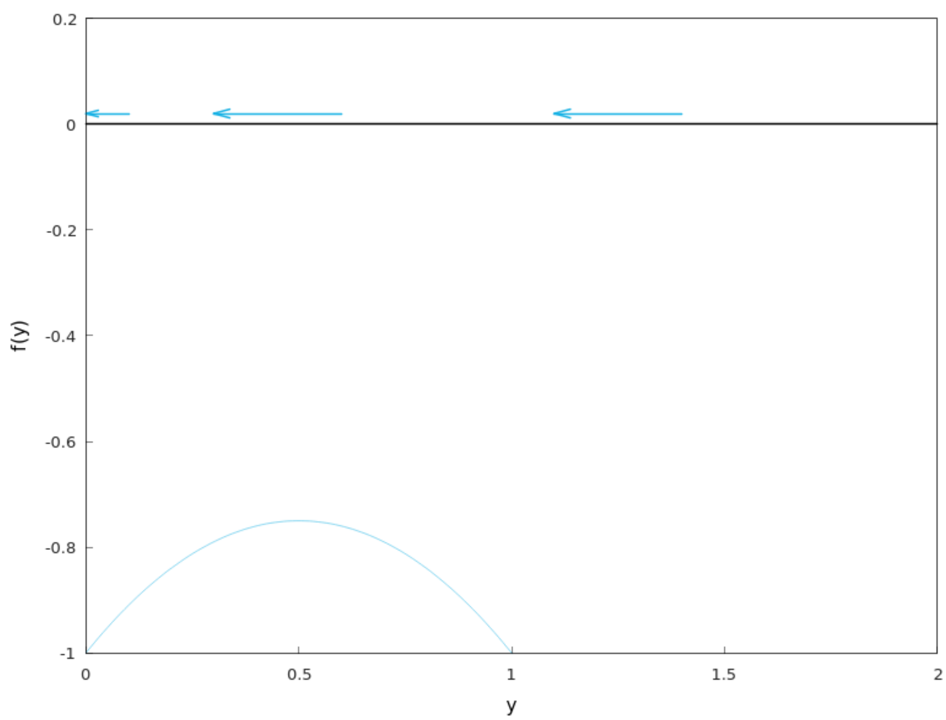


Figure 5.40: *The plot for $f(y) = (1 - y)y - 1$ (**Hamzstplot/Examples with Makefile/Plot 2D No Equilibrium Solution for Schaefer Model Variation/main.cpp**).*

This happens because $h > \frac{rK}{4}$ and this condition does not fit the requirement to have equilibrium solution from part (a).

The population tends to zero regardless of what y_0 is.

(e) Suppose that $h = \frac{rK}{4}$, then there will only be one equilibrium point

$$y = \frac{K \pm \sqrt{K^2 - \frac{4Kh}{r}}}{2} = \frac{K \pm \sqrt{0}}{2} = \frac{K}{2}$$

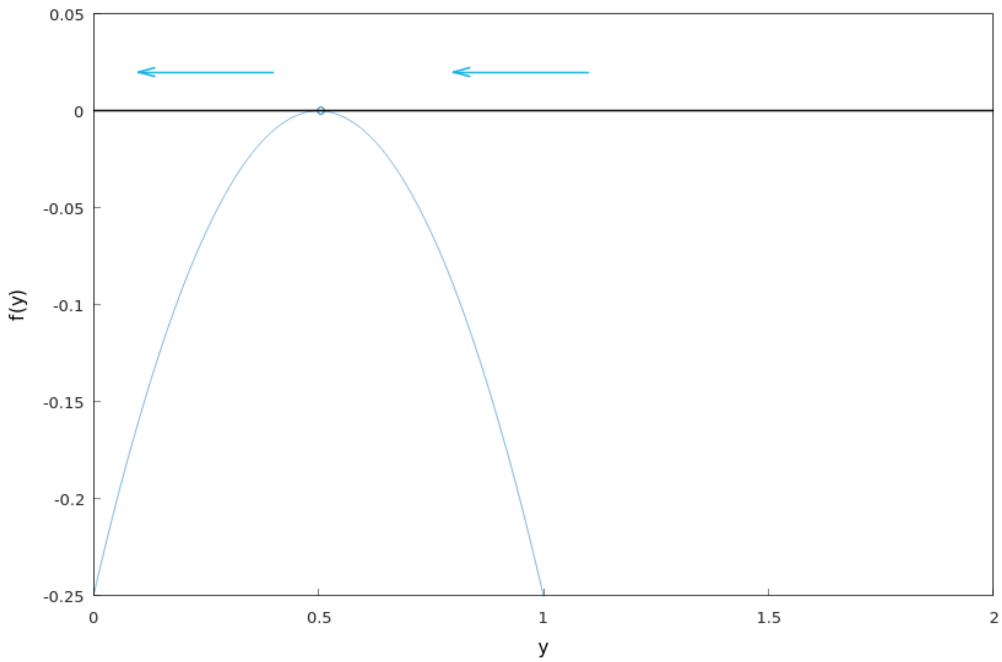


Figure 5.41: *The plot for $f(y) = (1 - y)y - \frac{1}{4}$ (**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution Only 1 for Schaefer Model Variation/main.cpp**).*

Even though the graph lies below the y-axis, the difference between part (d) is that there's now an equilibrium point at $y = \frac{K}{2}$.

If $y_0 > \frac{K}{2}$, then the population will tend to $y(t) = \frac{K}{2}$ as $t \to \infty$.

If $y_0 = \frac{K}{2}$, then the population will remain there.

If $y_0 < \frac{K}{2}$, then the population will tend to zero in a finite time because there are no equilibria to the left of $y = \frac{K}{2}$.

Thus, the equilibrium point here at $y = \frac{K}{2}$ is semistable (stable from one side and unstable from the other side).

[H*] **Epidemics**
The use of mathematical methods to study the spread of contagious diseases goes back at least to some work by Daniel Bernoulli in 1760 on smallpox. In more recent years many mathematical models have been proposed and studied for many different diseases. Similar models have also been used to describe the spread of rumors and of

consumer products.

[H*] **Example:**
Suppose that a given population can be divided into two parts: those who have a given disease and can infect others, and those who do not have it but are susceptible. Let x be the proportion of susceptible individuals and y the proportion of infectious individuals; then $x + y = 1$. Assume that the disease spreads by contact between sick and well members of the population and that the rate of spread $\frac{dy}{dt}$ is proportional to the number of such contacts. Further, assume that members of both groups move about freely among each other, so the number of contacts is proportional to the product of x and y. Since $x = 1 - y$, we obtain the initial value problem

$$\frac{dy}{dt} = \alpha y(1 - y), \quad y(0) = y_0 \tag{5.75}$$

where α is positive proportionality factor, and y_0 is the initial proportion of infectious individuals.

(a) Find the equilibrium points for the differential equation (5.75) and determine whether each is asymptotically stable, semistable, or unstable.

(b) Solve the initial value problem (5.75) and verify that the conclusions you reached in part (a) are correct. Show that $y(t) \to 1$ as $t \to \infty$, which means that ultimately the disease spreads through the entire population.

Solution:

(a) The equilibrium points are found by setting

$$\frac{dy}{dt} = 0$$

and solving the resulting equation for y.

$$\alpha y(1 - y)$$

The equilibrium points are at $y = \phi_1(t) = 0$ and $y = \phi_2(t) = 1$. With phase line analysis we will obtain that the equilibrium solution $y = \phi_1(t) = 0$ is unstable and the equilibrium solution $y = \phi_2(t) = 1$ is asympotically stable.

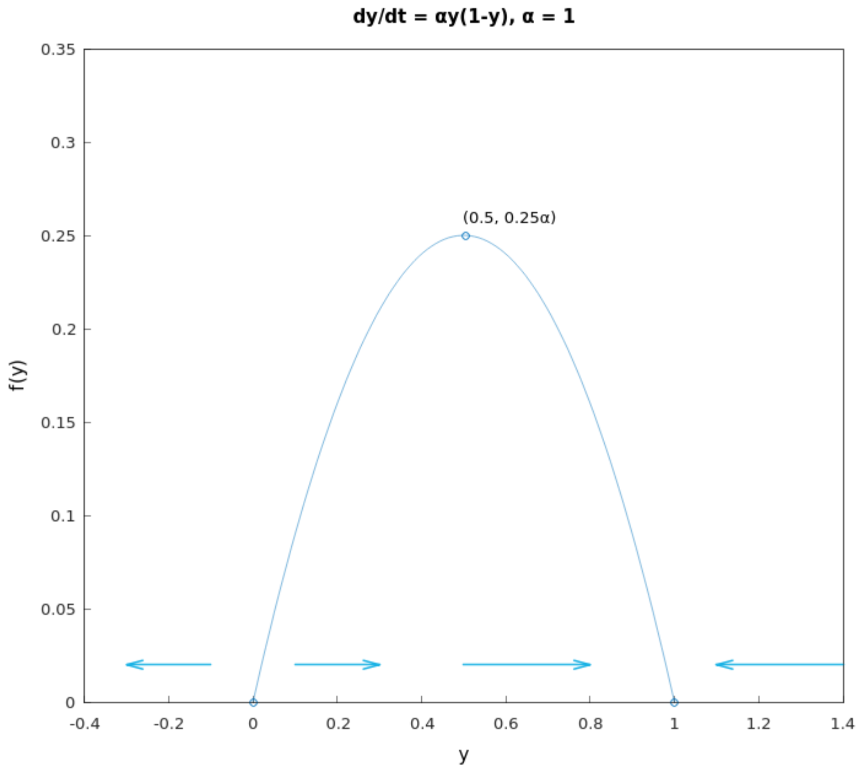


Figure 5.42: *The plot for $f(y) = \alpha y(1-y)$ with $\alpha = 1$ (**Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Smallpox Epidemics Model/main.cpp**).*

What this means is that even if only one individual is infected in a population of eight billion, the entire population will eventually be infected.

(b) The ODE is separable, with

$$[y(1-y)]^{-1} dy = \alpha\ dt$$

Integrating both sides and invoking the initial condition, the solution is

$$\begin{aligned}
\frac{dy}{\alpha y(1-y)} &= dt \\
\frac{1}{\alpha y} + \frac{1/\alpha}{1-y}\ dy &= dt \\
\frac{1}{\alpha} \ln|y| + \frac{1}{\alpha y}(-\ln|y-1|) &= t + c \\
\frac{\ln|y| - \ln|y-1|}{\alpha} &= t + c \\
\ln\left|\frac{y}{y-1}\right| &= \alpha t + C
\end{aligned}$$

To determine C we will apply the initial condition with $y(0) = y_0$, thus

$$C = \ln\left|\frac{y_0}{y_0 - 1}\right|$$

As a result, the previous equation becomes

$$\ln\left|\frac{y}{y-1}\right| = \alpha t + \ln\left|\frac{y_0}{y_0-1}\right|$$

Since y and y_0 are between 0 and 1, the absolute value signs can be removec by writing the denominators as $1-y$ and $1-y_0$.

$$\begin{aligned}
\ln\frac{y}{1-y} &= \alpha t + \ln\frac{y_0}{1-y_0} \\
\frac{y}{1-y} &= e^{\alpha t + \ln\frac{y_0}{1-y_0}} \\
&= e^{\alpha t} e^{\ln\frac{y_0}{1-y_0}} \\
&= e^{\alpha t}\left(\frac{y_0}{1-y_0}\right) \\
\frac{y}{1-y}(1-y) &= (1-y)\left(e^{\alpha t}\left(\frac{y_0}{1-y_0}\right)\right) \\
y &= e^{\alpha t}\left(\frac{y_0}{1-y_0}\right) - e^{\alpha t}\left(\frac{y_0}{1-y_0}\right) y \\
y\left[1 + e^{\alpha t}\left(\frac{y_0}{1-y_0}\right)\right] &= e^{\alpha t}\left(\frac{y_0}{1-y_0}\right) \\
y &= \frac{e^{\alpha t}\left(\frac{y_0}{1-y_0}\right)}{1 + e^{\alpha t}\left(\frac{y_0}{1-y_0}\right)} \\
&= \frac{e^{\alpha t} y_0}{(1-y0) + e^{\alpha t} y_0} \\
y(t) &= \frac{y_0}{(1-y_0)e^{-\alpha t} + y_0}
\end{aligned}$$

It is evident that (independent of y_0)

$$\lim_{t\to-\infty} y(t) = 0$$

and

$$\lim_{t\to\infty} y(t) = 1$$

If there are no infected individuals initially ($y_0 = 0$), then $y(t) = 0$. However, if $0 < y_0 < 1$, then the exponential term decays to zero as $t \to \infty$. If at least one person is infected, then ultimately the disease spreads through the entire population.

[H*] **Bifurcation Points**

For an equation of the form

$$\frac{dy}{dt} = f(a, y) \tag{5.76}$$

where a is a real parameter, the critical points (equilibrium solutions) usually depend on the value of a. As a steadily increases or decreases, it often happens that at a certain value of a, called a bifurcation point, critical points come together, or separate, and equilibrium solutions may either be lost or gained.

Bifurcation points are of great interest in many applications, because near them the nature of the solution of the underlying differential equation is undergoing an abrupt change. For example, in fluid mechanics a smooth (laminar) flow may break up and become turbulent. Or an axially loaded column may suddenly buckle and exhibit a large lateral displacement. Or, as the amount of one of the chemicals in a certain mixture is increased, a spiral wave patterns of varying color may suddenly emerge in an originally quiescent fluid.

[H*] **Example:**
Consider the equation

$$\frac{dy}{dt} = a - y^2 \tag{5.77}$$

(a) Find all of the critical points for Eq. (5.77). Observe that there are no critical points if $a < 0$, one critical point if $a = 0$, and two critical points if $a > 0$

(b) Draw the phase line in each case and determine whether each critical point is asymptotically stable, semistable, or unstable.

(c) In each case sketch several solutions of Eq. (5.77) in the ty-plane.

(d) If we plot the location of the critical points as a function of a in the ay-plane, we obtain the bifurcation diagram for Eq. (5.77). The bifurcation at $a = 0$ is called a saddle-node bifurcation.

Solution:

(a) The critical points are found by setting $\frac{dy}{dt} = 0$ and solving the resulting equation for y.

$$\begin{aligned} a - y^2 &= 0 \\ y^2 &= a \\ y &= \pm\sqrt{a} \end{aligned}$$

If a is negative, then there are no critical points; if a is zero, then there is one critical point; and if a is positive, then there are two critical points.

(b) The phase line for the case of $a < 0$:

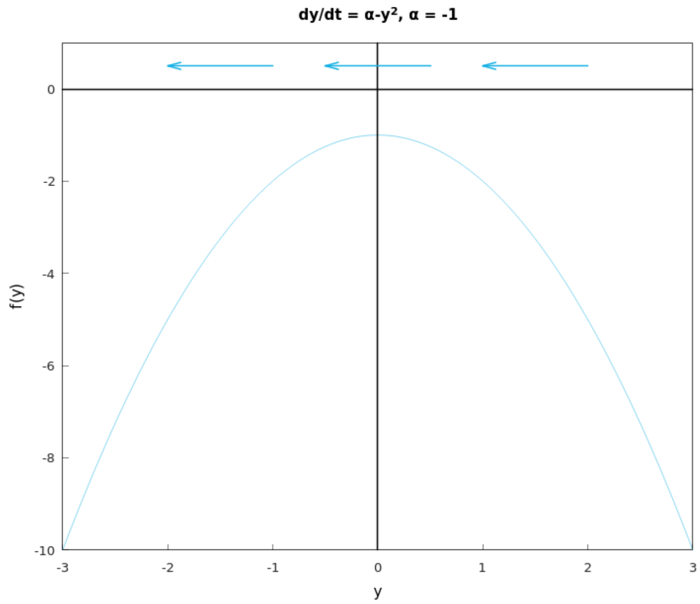


Figure 5.43: *The plot for $f(y) = \alpha - y^2$ with $\alpha = -1$, the phase line is drawn with the blue arrows (**Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 25 No Equilibrium Solution/-main.cpp**).*

There is no critical point here.

The phase line for the case of $a = 0$:

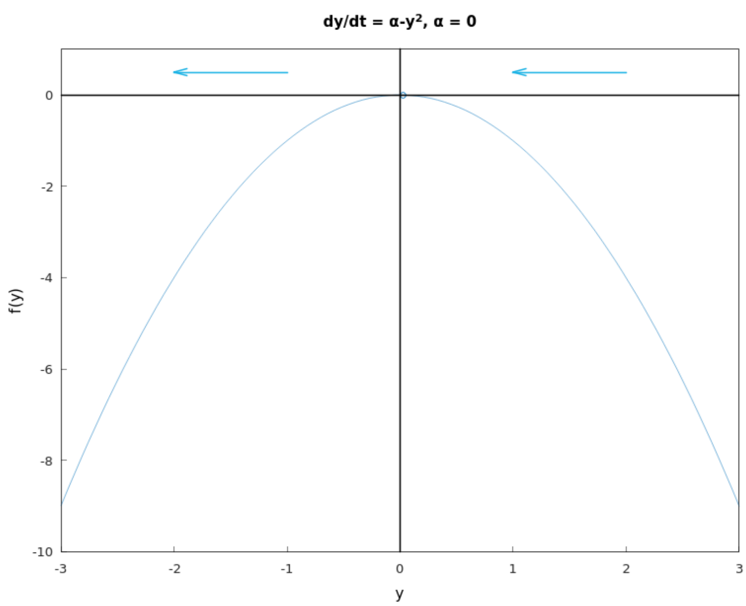


Figure 5.44: *The plot for $f(y) = \alpha - y^2$ with $\alpha = 0$, the phase line is drawn with the blue arrows (**Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 25 Only 1 Equilibrium Solution/main.cpp**).*

There is one critical point here $y = \phi_1(t) = 0$, and it is semistable.

The phase line for the case of $a > 0$:

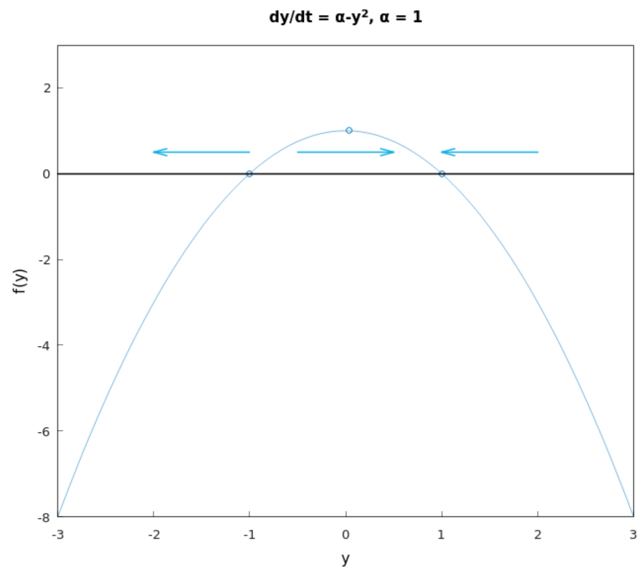


Figure 5.45: *The plot for $f(y) = \alpha - y^2$ with $\alpha = 1$, the phase line is drawn with the blue arrows (**Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 25 2 Equilibrium Solutions/main.cpp**).*

There are two critical points here $y = \phi_1(t) = \sqrt{a} = 1$ that is asymptotically stable and $y = \phi_2(t) = -\sqrt{a} = -1$ that is unstable.

(c) The ty-plane is the direction field for $\frac{dy}{dt}$

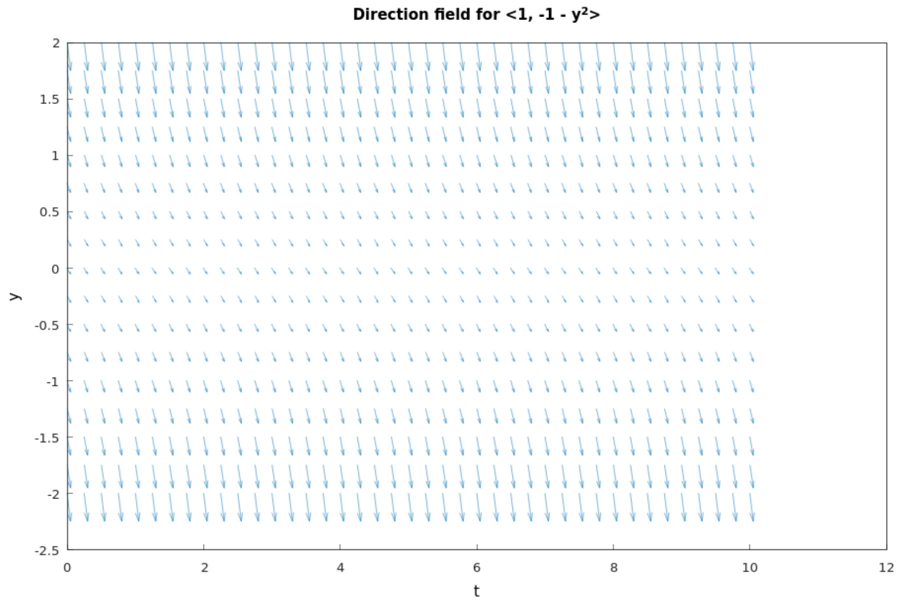


Figure 5.46: *The direction field for* $\frac{dy}{dt} = -1 - y^2$. *Solution curves in the ty-plane corresponding to* $\alpha = -1$ *are tangent to the vectors in the direction field* $< 1, -1 - y^2 >$ *at every point (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25a/main.cpp**).*

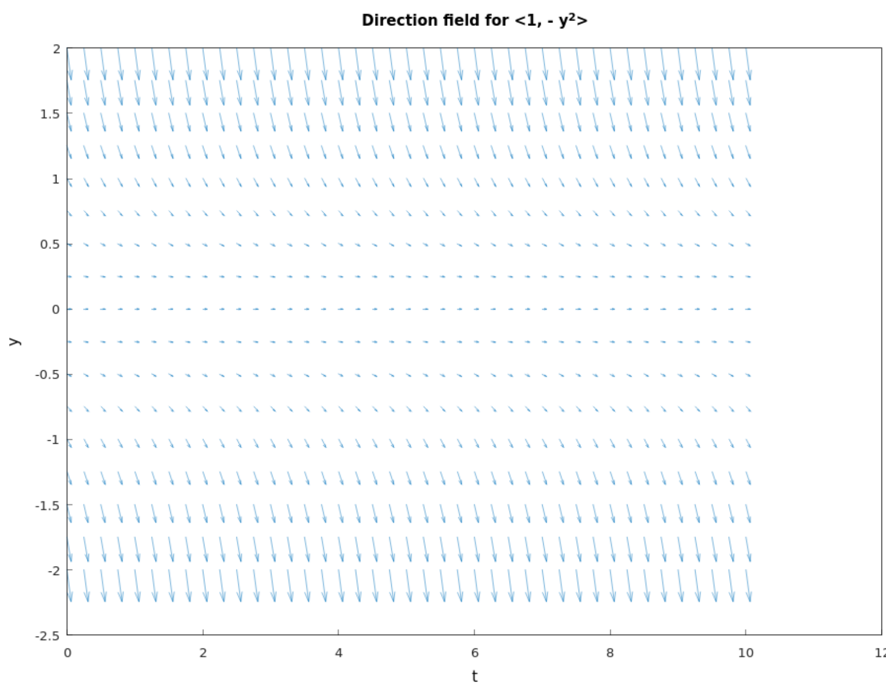


Figure 5.47: *The direction field for* $\frac{dy}{dt} = -y^2$. *Solution curves in the ty-plane corresponding to* $\alpha = 0$ *are tangent to the vectors in the direction field* $< 1, -y^2 >$ *at every point (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25b/main.cpp**).*

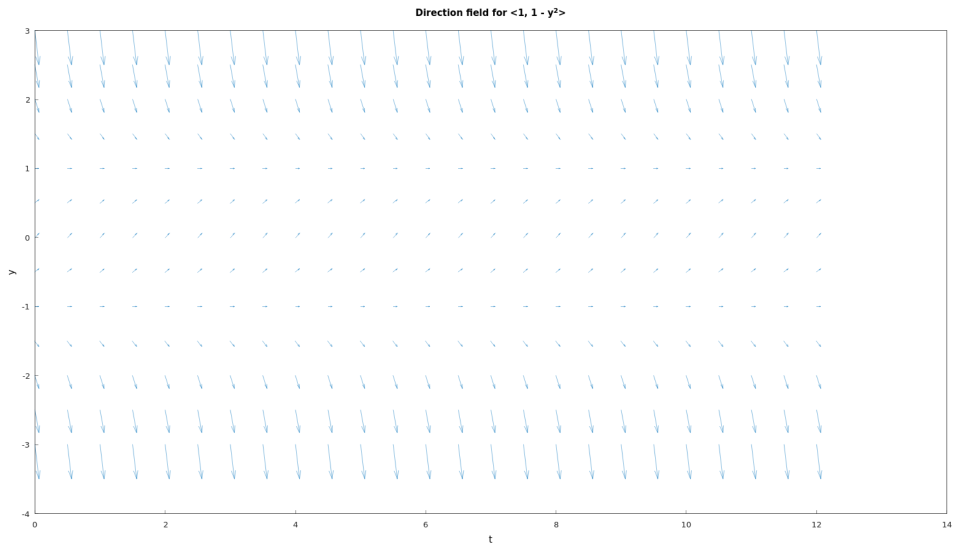


Figure 5.48: *The direction field for* $\frac{dy}{dt} = 1 - y^2$*. Solution curves in the ty-plane corresponding to* $\alpha = 1$ *are tangent to the vectors in the direction field* $< 1, 1 - y^2 >$ *at every point (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp**).*

(d) We are going to plot the location of the critical points as a function of a, it is

$$y(a) = \pm\sqrt{a}$$

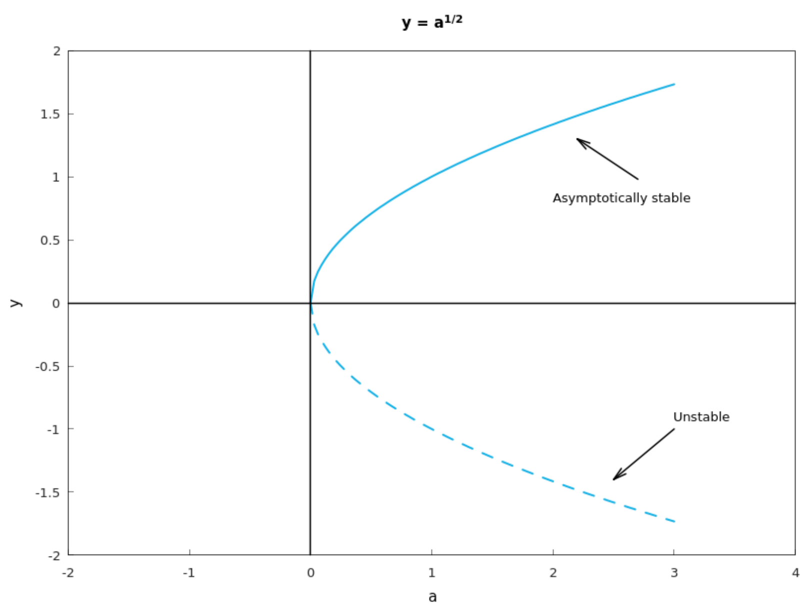


Figure 5.49: *The bifurcation diagram for* $\frac{dy}{dt} = a - y^2$*, the bifurcation point at* $a = 0$ *is called a saddle-node bifurcation where the solution then differ into two kind of solution: stable and unstable (**Hamzstplot/Examples with Makefile/Plot 2D Bifurcation Diagram Problem 25/main.cpp**).*

[H*] **Example:**
Consider the equation

$$\frac{dy}{dt} = ay - y^3 = y(a - y^2) \tag{5.78}$$

(a) Consider the cases $a < 0, a = 0$, and $a > 0$. In each case find all of the critical points for Eq. (5.78), draw the phase line, and determine whether each critical point is asympotically stable, semistable, or unstable.

(b) In each case sketch several solutions of Eq. (5.78) in the ty-plane.

(c) Draw the bifurcation diagram of Eq. (5.78), that is, plot the location of the critical points versus a. For Eq. (5.78) the bifurcation point at $a = 0$ is called a pitchfork bifurcation.

Solution:

(a) The critical points are found by setting $\frac{dy}{dt} = 0$ and solving the resulting equation for y.

$$y(a - y^2) = 0$$
$$y = \{0, \pm\sqrt{2}\}$$

If a is negative, then there is only one critical point at $y = 0$; if a is zero, then there is still only one critical point at $y = 0$; and if a is positive, then there are three critical points.

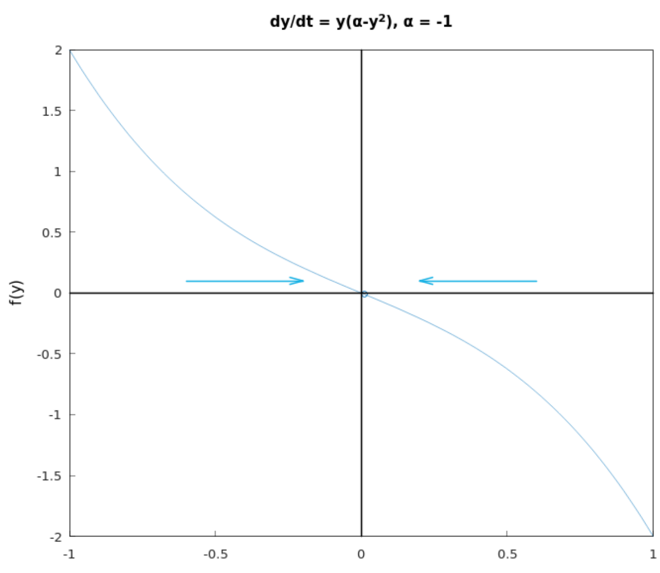


Figure 5.50: *The plot for $f(y) = y(\alpha - y^2)$ with $\alpha = -1$, the phase line is drawn with the blue arrows (**Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 26 Only 1 Equilibrium Solution/main.cpp**).*

There is one critical point at $y = 0$, and it is a sink (stable).

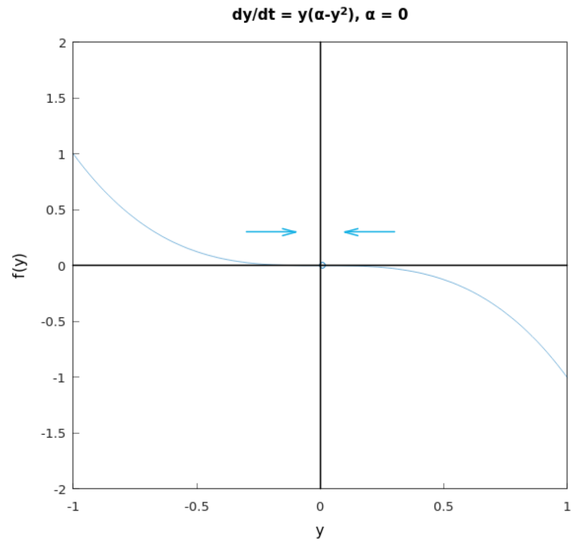


Figure 5.51: *The plot for $f(y) = y(\alpha - y^2)$ with $\alpha = 0$, the phase line is drawn with the blue arrows (**Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 26 Only 1 Equilibrium Solution/main.cpp**).*

There is one critical point at $y = 0$, and it is a sink (stable).

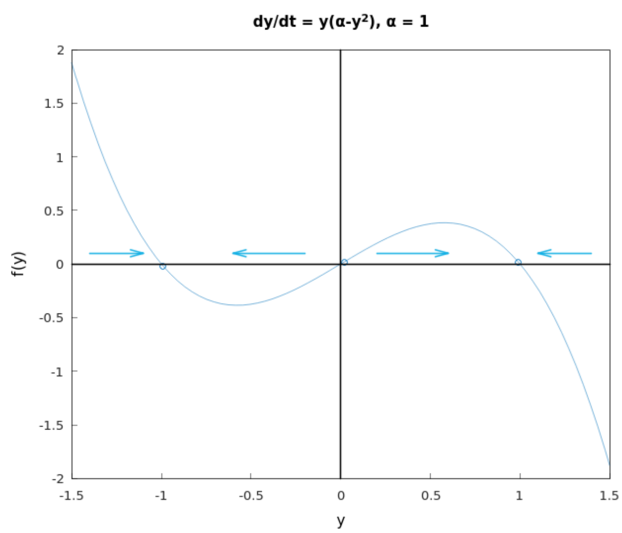


Figure 5.52: *The plot for $f(y) = y(\alpha - y^2)$ with $\alpha = 1$, the phase line is drawn with the blue arrows (**Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 26 2 Equilibrium Solutions/main.cpp**).*

There are now three critical points at $y = -\sqrt{a}, y = 0$, and $y = \sqrt{a}$, and they are stable, unstable, and stable, respectively.

(b) We will draw the solutions in the ty-plane / the direction field.

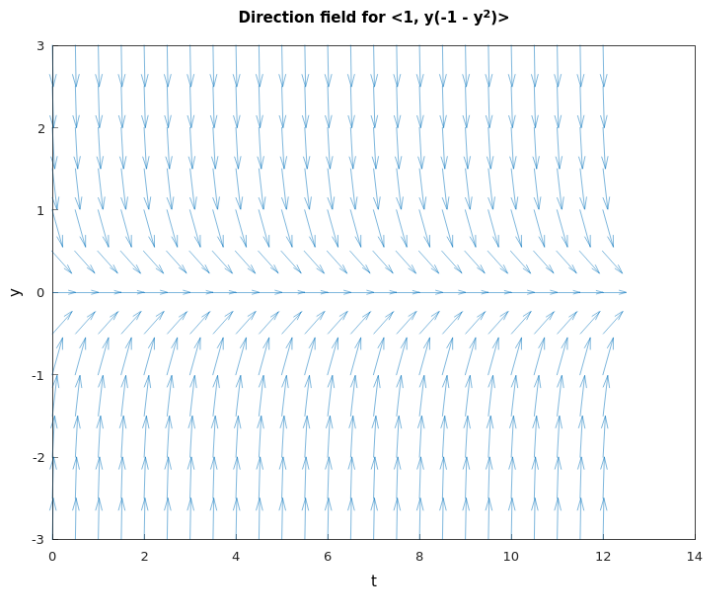


Figure 5.53: *The direction field for $\frac{dy}{dt} = y(-1 - y^2)$. Solution curves in the ty-plane corresponding to $\alpha = 1$ are tangent to the vectors in the direction field $< 1, y(-1 - y^2) >$ at every point (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp**).*

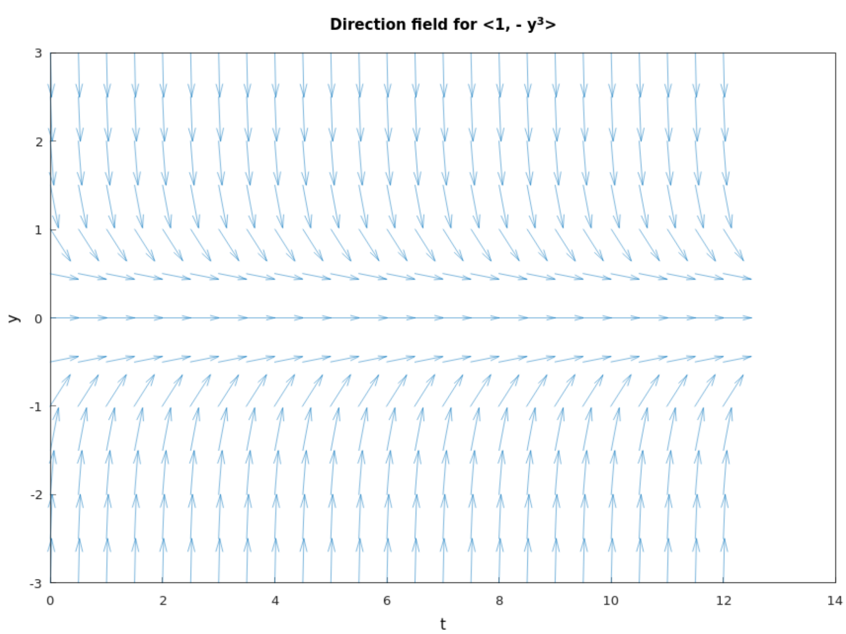


Figure 5.54: *The direction field for $\frac{dy}{dt} = -y^3$. Solution curves in the ty-plane corresponding to $\alpha = 0$ are tangent to the vectors in the direction field $< 1, -y^3 >$ at every point (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp**).*

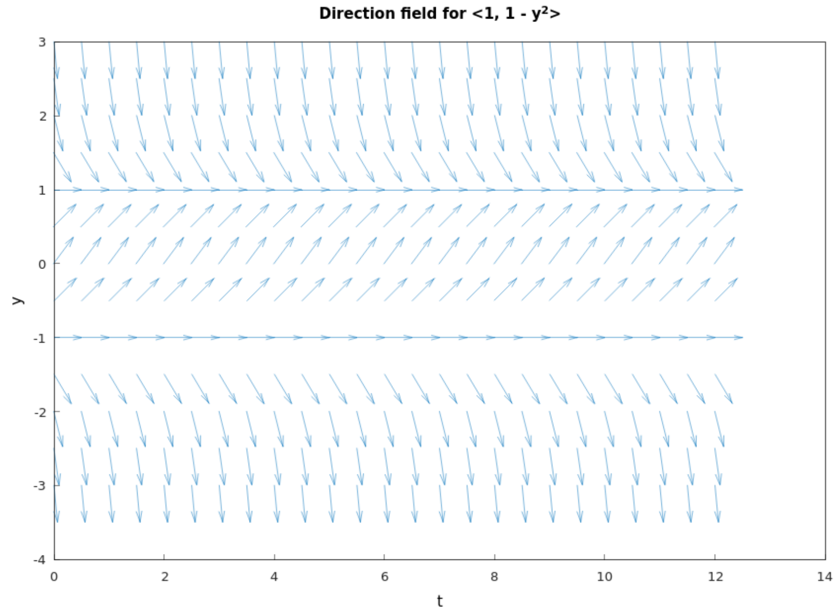


Figure 5.55: *The direction field for $\frac{dy}{dt} = y(1 - y^2)$. Solution curves in the ty-plane corresponding to $\alpha = 1$ are tangent to the vectors in the direction field $< 1, y(1 - y^2) >$ at every point (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp**).*

(c) This is the bifurcation diagram, looks like a pitchfork.

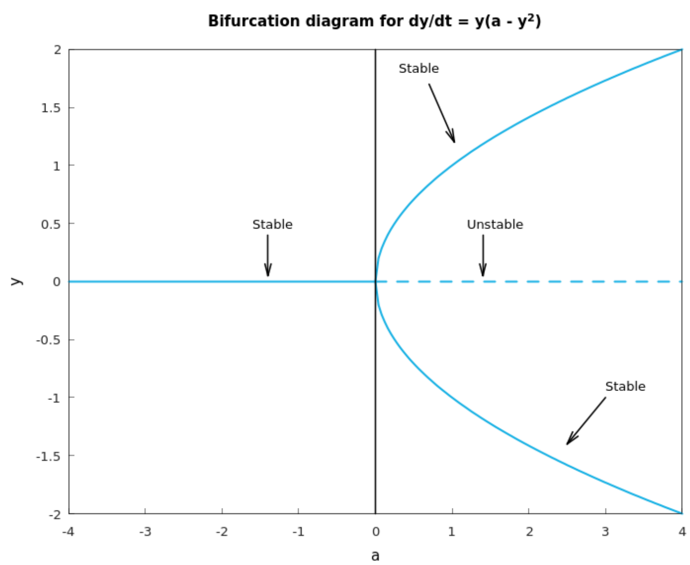


Figure 5.56: *The bifurcation diagram for $\frac{dy}{dt} = y(a - y^2)$, the bifurcation point at $a = 0$ is called a saddle-node bifurcation where the solution then differ into two kind of solution: stable and unstable (**Hamzstplot/Examples with Makefile/Plot 2D Bifurcation Diagram Problem 26/main.cpp**).*

[H*] **Example:**

Consider the equation

$$\frac{dy}{dt} = ay - y^2 = y(a - y) \tag{5.79}$$

(a) Consider the cases $a < 0, a = 0$, and $a > 0$. In each case find all of the critical points for Eq. (5.79), draw the phase line, and determine whether each critical point is asympotically stable, semistable, or unstable.

(b) In each case sketch several solutions of Eq. (5.79) in the ty-plane.

(c) Draw the bifurcation siagram of Eq. (5.79). Observe that for Eq. (5.79) there are the same number of critical points for $a < 0$ and $a > 0$ but that their stability. has changed. For $a < 0$ the equilibrium solution $y = 0$ is asympotically stable and $y = a$ is unstable, while for $a > 0$ the situation is reversed. Thus there has been an exchange of stability as a passes through the bifurcation point $a = 0$. This type of bifurcation is called a transcritical bifurcation.

Solution:

(a) .

[H*] **Chemical Reactions Example:**
A second order chemical reaction involves the interaction (collision) of one molecule of a substance P with one molecule of a substance Q to produce one molecule of a new substance X; this is denoted by

$$P + Q \to X$$

Suppose that p and q, where $p \neq q$, are the initial concentrations of P and Q, respectively, and let $x(t)$ be the concentration of X at time t. Then $p - x(t)$ and $q - x(t)$ are the concentrations of P and Q at time t, and the rate at which the reaction occurs is given by the equation

$$\frac{dx}{dt} = \alpha(p - x)(q - x) \tag{5.80}$$

where α is a positive constant.

(a) If $x(0) = 0$, determine the limiting value of $x(t)$ as $t \to \infty$ without solving the differential equation. Then solve the initial value problem and find $x(t)$ for any t.

(b) If the substances P and Q are the same, then $p = q$ and Eq. (5.80) is replaced by

$$\frac{dx}{dt} = \alpha(p - x)^2 \tag{5.81}$$

If $x(0) = 0$, determine the limiting value of $x(t)$ as $t \to \infty$ without solving the differential equation. Then solve the initial value problem and determine $x(t)$ for any t.

Solution:

(a) The chemical reaction for this problem is

$$P + Q \to X$$

The (stoichiometric) coefficients of P and Q are both 1, so the molecules of P and Q combine in a one-to-one fashion. A sample direction field

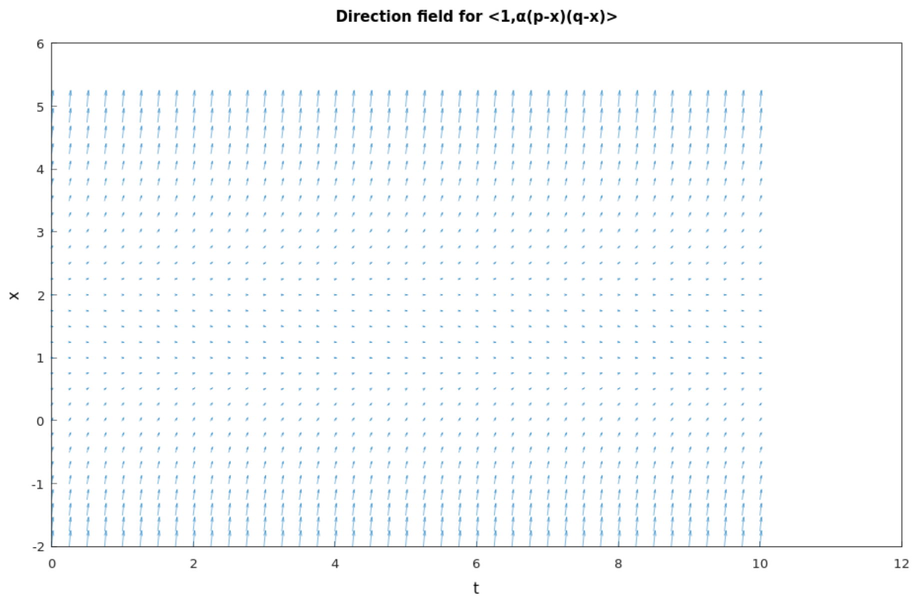


Figure 5.57: *The direction field for* $< 1, \alpha(p-x)(q-x) >$ *with* $\alpha = 1$, $p = 1$, *and* $q = 2$ (***Hamzstplot/Examples with Makefile/Plot Direction Fields for Chemical Reactions Problem 28a/main.cpp***).

We see that the solution $x(t)$ tends to 1 as $t \to \infty$ for the initial condition $x(0) = 0$. We conclude then that $x(t)$ tends to p or q, whichever of the two is smaller, as $t \to \infty$.

$$\lim_{t\to\infty} x(t) = \begin{cases} p, & \text{if } p < q \\ q, & \text{if } q < p \end{cases}$$

Solve the ODE now by separating variables

$$\begin{aligned}
\frac{dx}{dt} &= \alpha(p-x)(q-x) \\
\frac{dx}{(p-x)(q-x)} &= \alpha\ dt \\
\int \frac{dx}{(p-x)(q-x)} &= \alpha t + C \\
\int \left(\frac{1/(q-p)}{p-x} + \frac{1/(p-q)}{q-x} \right) dx &= \alpha t + C \\
\frac{1}{q-p} \int \frac{dx}{p-x} + \frac{1}{p-q} \int \frac{dx}{q-x} &= \alpha t + C \\
\frac{1}{p-q} \int \frac{dx}{x-p} + \frac{1}{q-p} \int \frac{dx}{x-q} &= \alpha t + C \\
\frac{1}{p-q} \ln|x-p| + \frac{1}{q-p} \ln|x-q| &= \alpha t + C \\
\ln|x-p|^{\frac{1}{p-q}} + \ln|x-q|^{\frac{1}{q-p}} &= \alpha t + C \\
-\ln|x-p|^{\frac{1}{q-p}} + \ln|x-q|^{\frac{1}{q-p}} &= \alpha t + C \\
\ln\left|\frac{x-q}{x-p}\right|^{\frac{1}{q-p}} &= \alpha t + C
\end{aligned}$$

Apply the initial condition $x(0) = 0$ now to determine C

$$\ln\left|\frac{-q}{-p}\right|^{\frac{1}{q-p}} = C$$

$$C = \ln\left|\frac{q}{p}\right|^{\frac{1}{q-p}}$$

The previous equation is then

$$\begin{aligned}
\ln\left|\frac{x-q}{x-p}\right|^{\frac{1}{q-p}} &= \alpha t + \ln\left|\frac{q}{p}\right|^{\frac{1}{q-p}} \\
\frac{1}{q-p} \ln\left|\frac{x-q}{x-p}\right| &= \alpha t + \frac{1}{q-p} \ln\left|\frac{q}{p}\right| \\
\ln\left|\frac{x-q}{x-p}\right| &= \alpha t(q-p) + \ln\left|\frac{q}{p}\right| \\
\left|\frac{x-q}{x-p}\right| &= e^{\alpha t(q-p)} \left(\frac{q}{p}\right) \\
\frac{x-q}{x-p} &= \pm \frac{q}{p} e^{\alpha t(q-p)} \\
x - q &= \frac{q}{p} e^{\alpha t(q-p)} x - q e^{\alpha t(q-p)} \\
x\left[1 - \frac{q}{p} e^{\alpha t(q-p)}\right] &= q - q e^{\alpha t(q-p)} \\
x(t) &= \frac{q(1 - e^{\alpha t(q-p)})}{1 - \frac{q}{p} e^{\alpha t(q-p)}}
\end{aligned}$$

from the $\pm$ sign, we choose the plus sign so that the initial condition remains satisfied. Thus, we simplify again one time and the general solution can be written as

$$x(t) = \frac{pq(1 - e^{\alpha t(q-p)})}{p - qe^{\alpha t(q-p)}} \tag{5.82}$$

(b) The direction field below shows that the solution curves, which lie tangent to the direction field vectors at every point, show that the solution $x(t)$ tends to 1 as $t \to \infty$ with the initial value of $x(0) = 0$ and $p = 1, \alpha = 1$. We conclude that

$$\lim_{t\to\infty} x(t) = p$$

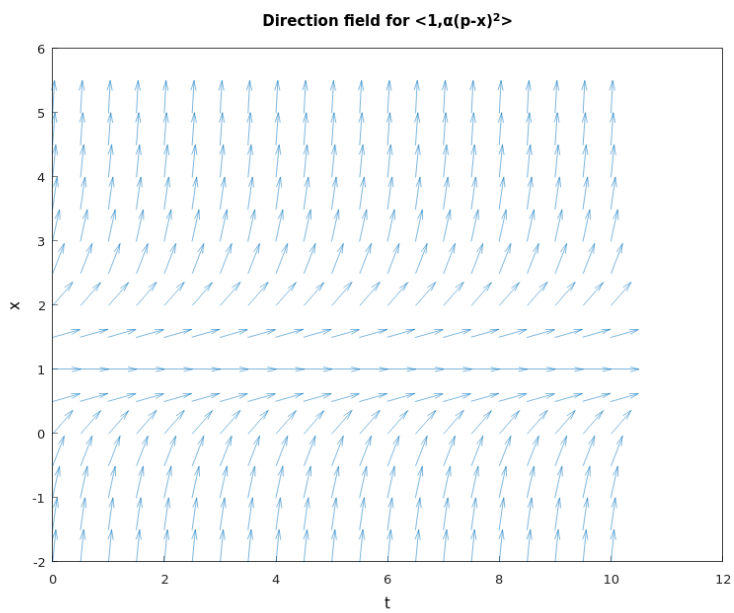


Figure 5.58: *The direction field for $< 1, \alpha(p-x)^2 >$ with $\alpha = 1$ and $p = 1$ (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Chemical Reactions Problem 28b/main.cpp**).*

Solve the ODE now by separating variables

$$\begin{aligned}
\frac{dx}{dt} &= \alpha(p-x)^2 \\
\frac{dx}{(p-x)^2} &= \alpha\, dt \\
\int \frac{dx}{(p-x)^2} &= \alpha t + C_1 \\
\int \frac{dx}{(x-p)^2} &= \alpha t + C_1 \\
-\frac{1}{x-p} &= \alpha t + C_1
\end{aligned}$$

Apply the initial condition $x(0) = 0$ now to determine C_1.

$$-\frac{1}{-p} = C_1$$
$$C_1 = \frac{1}{p}$$

As a result, the previous equation becomes

$$\begin{aligned}
-\frac{1}{x-p} &= \alpha t + \frac{1}{p} \\
-(x-p) &= \frac{1}{\alpha t + \frac{1}{p}} \\
x - p &= -\frac{1}{\alpha t + \frac{1}{p}} \\
x(t) &= p - \frac{1}{\alpha t + \frac{1}{p}} \\
&= p - \frac{p}{\alpha p t + 1} \\
&= \frac{\alpha p^2 t + p - p}{\alpha p t + 1} \\
x(t) &= \frac{\alpha p^2 t}{\alpha p t + 1}
\end{aligned}$$

- Exact Equations and Integrating Factors

[H*] Exact equations are a class of equations for which there is also a well-defined method of solution.

[H*] **Example:**
Solve the differential equation

$$2x + y^2 + 2xyy' = 0 \tag{5.83}$$

The equation is neither linear nor separable, so the methods suitable for those types of equations are not applicable here. However, observe that the function $\psi(x, y) = x^2 + xy^2$ has the property that

$$\begin{aligned}
2x + y^2 &= \frac{\partial \psi}{\partial x} \\
2xy &= \frac{\partial \psi}{\partial y}
\end{aligned} \tag{5.84}$$

Therefore the differential equation can be written as

$$\frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y}\frac{dy}{dx} = 0 \tag{5.85}$$

Assuming that y is a function of x and calling upon the chain rule, we can write Eq. (5.85) in the equivalent form

$$\frac{\partial \psi}{\partial x} = \frac{d}{dx}(x^2 + xy^2) = 0 \tag{5.86}$$

Therefore

$$\psi(x, y) = x^2 + xy^2 = c \tag{5.87}$$

where c is an arbitrary constant, is an equation that defines solutions of Eq. (5.83) implicitly.

The key step here was the recognition that there is a function ψ that satisfies Eqs. (5.84).
[H*] More generally, let the differential equation

$$M(x, y) + N(x, y)y' = 0 \tag{5.88}$$

be given. Suppose that we can identify a function ψ such that

$$\begin{aligned} \frac{\partial \psi}{\partial x}(x, y) &= M(x, y) \\ \frac{\partial \psi}{\partial y}(x, y) &= N(x, y) \end{aligned} \tag{5.89}$$

and such that $\psi(x, y) = c$ defines $y = \phi(x)$ implicitly as a differentiable function of x. Then

$$M(x, y) + N(x, y)y' = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y}\frac{dy}{dx} = \frac{d}{dx}\psi[x, \phi(x)]$$

and the differential equation (5.88) becomes

$$\frac{d}{dx}\psi[x, \phi(x)] = 0 \tag{5.90}$$

In this case Eq. (5.88) is said to be an exact differential equation. Solutions of Eq. (5.88), or the equivalent Eq. (5.90), are given implicitly by

$$\psi(x, y) = c \tag{5.91}$$

where c is an arbitrary constant.

Theorem 5.3: Exact Differential Equation in a Region R

Let the functions M, N, M_y, and N_x, where subscripts denote the partial derivatives, be continous in the rectangular region $R : \alpha < x < \beta,\ \gamma < y < \delta$. Then Eq. (5.88)

$$M(x, y) + N(x, y)y' = 0$$

is an exact differential equation in R if and only if

$$M_y(x, y) = N_x(x, y) \tag{5.92}$$

at each point of R. That is, there exists a function ψ satisfying Eqs. (5.89),

$$\begin{aligned} \psi_x(x, y) &= M(x, y) \\ \psi_y(x, y) &= N(x, y) \end{aligned}$$

if and only if M and N satisfy Eq. (5.92).

- Numerical Approximations: Euler's Method

[H*] Numerical methods for ordinary differential equations are methods used to find numerical approximations to the solutions of ordinary differential equations. Their use is also known as "numerical integration."

Many differential equations cannot be solved exactly. For practical purposes, however - such as in engineering - a numeric approximation to the solution is often sufficient. The algorithms studied here can be used to compute such an approximation. An alternative method is to use techniques from calculus to obtain a series expansion of the solution.

Ordinary differential equations occur in many scientific disciplines, including physics, chemistry, biology, and economics. In addition, some methods in numerical partial differential equations convert the partial differential equation into an ordinary differential equation, which must then be solved.

[H*] **Euler Method**

From any point on a curve, you can find an approximation of a nearby point on the curve by moving a short distance along a line tangent to the curve.

Suppose we have a first-order differential equation with initial value problem of the form

$$y'(t) = \frac{dy}{dt} = f(t, y(t)), \quad y(t_0) = y_0 \tag{5.93}$$

Starting with the differential equation Eq. (5.93), we replace the derivative y' by the finite difference approximation

$$y'(t) \approx \frac{y(t+h) - y(t)}{h} \tag{5.94}$$

which when re-arranged yields the following formula

$$y(t+h) \approx y(t) + hy'(t)$$

and using Eq. (5.93) gives

$$y(t+h) \approx y(t) + hf(t, y(t)) \tag{5.95}$$

This formula is usually applied in the following way.
We choose a step size h, and we construct the sequence t_0, t_1

$$\begin{aligned} &t_0 \\ &t_1 = t_0 + h \\ &t_2 = t_0 + 2h \\ &\vdots \end{aligned}$$

Then we denote y_n, as the numerical estimate of the exact solution $y(t_n)$.
We compute these estimates by the following recursive scheme

$$y_{n+1} = y_n + hf(t_n, y_n) \tag{5.96}$$

This is the Euler method (or forward Euler method). The method is named after Leonhard Euler who described it in 1768.

The Euler method is an example of an explicit method. This means that the new value y_{n+1} is defined in terms of things that are already known, like y_n.

[H*] Consider this initial value problem

$$\frac{dy}{dt} = f(t, y(t)), \quad y(t_0) = y_0 \tag{5.97}$$

First, if f and $\frac{\partial f}{\partial y}$ are continuous, then the initial value problem (5.97) has a unique solution $y = \phi(t)$ in some interval surrounding the initial point $t = t_0$. Second, it is usually not possible to find the solution ϕ by symbolic manipulations of the differential equation.

We have learned differential equations that are linear, separable, or exact, or that can be transformed into one of these types. Nevertheless, it remains true that solutions of the vast majority of first order initial value problems cannot be found by analytical means.

To see how the Euler method works, let us consider how we might use tangent lines to approximate the solution $y = \phi(t)$ of Eqs. (5.97) near $t = t_0$. We know that the solution passes through the initial point (t_0, y_0), and from the differential equation, we also know that its slope at this point is $f(t_0, y_0)$. Thus, we can write down an equation for the line tangent to the solution curve at (t_0, y_0), namely

$$y = y_0 + f(t_0, y_0)(t - t_0) \tag{5.98}$$

The tangent line is a good approximation to the actual solution curve on an interval short enough so that the slope of the solution oes not change appreciably from its value at the initial point.
Thus, if t_1 is close enough to t_0, we can approximate $\phi(t_1)$ by the value y_1 determined by substituting $t = t_1$ into the tangent line approximation at $t = t_0$; thus

$$y_1 = y_0 = f(t_0, y_0)(t_1 - t_0) \tag{5.99}$$

To proceed further, we can try to repeat the process. Unfortunately, we do not know the value $\phi(t_1)$ of the solution at t_1. The best we can do is to use the approximate value y_1 instead. Thus we construct the line through (t_1, y_1) with the slope $f(t_1, y_1)$,

$$y = y_1 + f(t_1, y_1)(t - t_1) \tag{5.100}$$

To approximate the value of $\phi(t)$ at a nearby point t_2, we use Eq. (5.100) instead, obtaining

$$y_2 = y_1 + f(t_1, y_1)(t_2 - t_1) \tag{5.101}$$

Continuing in this manner, we use the value of y calculated at each step to determine the slope of the approximation for the next step. The general expression for the tangent line starting at (t_n, y_n) is

$$y = y_n + f(t_n, y_n)(t - t_n) \tag{5.102}$$

hence the approximation value y_{n+1} at t_{n+1} in terms of t_n, t_{n+1}, and y_n is

$$y_{n+1} = y_n + f(t_n, y_n)(t_{n+1} - t_n), \quad n = 0, 1, 2, \ldots \tag{5.103}$$

If we introduce the notation $f_n = f(t_n, y_n)$, then we can rewrite Eq. (5.103) as

$$y_{n+1} = y_n + f_n(t_{n+1} - t_n), \quad n = 0, 1, 2, \ldots \tag{5.104}$$

Finally, if we assume that there is a uniform step size h between points $t_0, t_1, t_2, \ldots,$ then $t_{n+1} = t_n + h$ for each n, and we obtain Euler's formula in the form

$$y_{n+1} = y_n + f_n h, \quad n = 0, 1, 2, \ldots \tag{5.105}$$

To use Euler's method, you simply evaluate Eq. (5.104) or Eq. (5.105) repeatedly, depending on whether or not the step size is constant, using the result of each step to execute the next step. In this way you will generate a sequence of values $y_1, y_2, y_3, \ldots$ that approximate the values of the solution $\phi(t)$ at the points $t_1, t_2, t_3, \ldots$.

If, instead of a sequence of points, you need a function to approximate the solution $\phi(t)$, then you can use the piecewise linear function constructed fom the collection of tangent line segments. That is, let y be given in $[t_0, t_1]$ by Eq. (5.102) with $n = 0$, in $[t_1, t_2]$ by Eq. (5.102) with $n = 1$, and so on.

[H*] **Backward Euler Method**

If, instead of Eq. (5.94), we use the approximation

$$y'(t) \approx \frac{y(t) - y(t-h)}{h} \tag{5.106}$$

we get the backward Euler method:

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}) \tag{5.107}$$

The backward Euler method is an implicit method, meaning that we have to solve an equation to find y_{n+1}. One often uses fixed-point iteration or the Newton-Raphson method to achieve this.

It costs more time to solve this equation than explicit methods; this cost must be taken into consideration when one selects the method to use.

The advantage of implicit methods are that they usually more stable for solving a stiff equation, meanin that a larger step size h can be used.

The limitation to the Euler method is that it can be prone to significant errors, especially with larger step sizes or highly nonlinear functions. Furthermore, the Euler method is often not accurate enough. In more precise terms, it only has order one. The first-order accuracy means that the error is proportional to the step size h. This caused mathematicians to look for higher-order methods.

One possibility is to use not only the previously computed value y_n to determine y_{n+1}, but to make the solution depend on more past values. This yields a so-called multistep method. Another possibility is to use more points in the interval $[t_n, t_{n+1}]$. This leads to the family of Runge-Kutta methods. One of their fourth-order methods is especially popular.

[H*] Numerical analysis is not only the design of numerical methods, but also their analysis. Three central concepts in this analysis are:

1. Convergence: whether the method approximates the solution.
   A numerical method is said to be convergent if the numerical solution approaches the exact solution as the step size h goes to 0. More precisely, we require that

for every ordinary differential equations with type of Eq. (5.93) with a Lipschitz function f and every $t^* > 0$,

$$\lim_{h \to 0^+} \max_{n=0,1,\ldots,[t^*/h]} ||y_{n,h} - y(t_n)|| = 0$$

2. Order: how well it approximates the solution.
   Suppose the numerical method is

   $$y_{n+k} = \psi(t_{n+k}; y_n, y_{n+1}, \ldots, y_{n+k-1}; h)$$

   The local (truncation) error of the method is the error committed by one step of the method. That is, it is the difference between the result given by the method, assuming that no error was made in earlier steps, and the exact solution:

   $$\delta^h_{n+k} = \psi(t_{n+k}; y_n, y_{n+1}, \ldots, y_{n+k-1}; h) - y(t_{n+k})$$

   The method is said to be consistent if

   $$\lim_{h \to 0} \frac{\delta^h_{n+k}}{h} = 0$$

   The method has order p if

   $$\delta^h_{n+k} = O(h^{p+1})$$

   as $h \to 0$.
   Hence a method is consistent if it has an order greater than 0. The forward Euler method and the backward Euler method both have order 1, so they are consistent. Most methods being used in practice attain higher order. Consistency is a necessary condition for convergence, but not sufficient; for a method to be convergent, it must be both consistent and zero-stable.

   A related concept is the global(truncation) error, the error sustained in all the steps one needs to reach a fixed time t. Explicitly, the global error at time t is $y_N - y(t)$ where $N = \frac{t-t_0}{h}$. The global error of a pth order one-step method is $O(h^p)$; in particular, such a method is convergent. This statement is not necessarily true for multi-step methods.
3. Stability: whether errors are damped out.
   For some differential equations, application of standard methods-such as the Euler method, explicit Runge-Kutta methods, multistep methods (e.g., Adams-Bashforth methods)-exhibit instability in the solutions, though other methods may produce stable solutions.

   This "difficult behavior" in the equation (which may not necessarily be complex itself) is described as stiffness, and is often caused by the presence of different time scales in the underlying problem.

   For example, a collision in a mechanical system like in an impact oscillator typically occurs at much smaller time scale than the time for the motion of objects; this discrepancy makes for very "sharp turns" in the curves of the state parameters.

Stiff problems are ubiquitous in chemical kinetics, control theory, solid mechanics, weather forecasting, biology, plasma physics, and electronics. One way to overcome stiffness is to extend the notion of differential equation to that of differential inclusion, which allows for and models non-smoothness.

[H*] **Example:**
Consider the initial value problem

$$\frac{dy}{dt} = 3 - 2t - \frac{1}{2}y, \quad y(0) = 1 \tag{5.108}$$

Use Euler's method with step size $h = 0.2$ to find approximate values of the solution of Eqs (5.99) at $t = 0.2, 0.4, 0.6, 0.8$ and 1. Compare them with the corresponding values of the actual solution of the initial value problem.

Solution:
Note that the differential equation in the given initial value problem is linear and can be solved using the integrating factor $e^{t/2}$.

The resulting solution of the initial value problem (5.108) is

$$y = \phi(t) = 14 - 4t - 13e^{-t/2} \tag{5.109}$$

To approximate this solution by means of Euler's method, note that in this case

$$f(t, y) = 3 - 2t - \frac{y}{2}$$

Using the initial values $t_0 = 0$ and $y_0 = 1$, we find that

$$f_0 = f(t_0, y_0) = f(0, 1) = 3 - 0 - 0.5 = 2.5$$

and then, the tangent line approximation near $t = 0$ is

$$y = 1 + 2.5(t - 0) = 1 + 2.5t \tag{5.110}$$

Setting $t = 0.2$ in Eq. (5.110), we find the approximate value y_1 of the solution at $t = 0.2$, namely,

$$y_1 = 1 + (2.5)(0.2) = 1.5$$

At the next step we have

$$\begin{aligned}
f_1 &= f(0.2, 1.5) = 3 - 2(0.2) - 0.5(1.5) = 3 - 0.4 - 0.75 = 1.85 \\
y &= 1.5 + 1.85(t - 0.2) = 1.13 + 1.85t \\
y_2 &= 1.13 + 1.85(0.4) = 1.87
\end{aligned}$$

In SymIntegration, we can use this function:
eulermethod(f, y, t, y0, t0, t1, h)
with f as the function $\frac{dy}{dt} = f(t, y)$, $y0$ is the value $y(t_0)$, $t0$ is the initial point t_0, and $t1$ is the point t_1 where we want to approximate the solution at.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Solve Initial Value Problem with Euler Method ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Solve Initial Value Problem with Euler Method ]# ./main
For dy/dt = 3 - 2*t -0.5*y
y(t) = -4*t+C*e^(-0.5*t)+14
ivp for y(0)=1
y(t) = -4*t-13*e^(-0.5*t)+14
Euler' Method

f(x) = -2*t-0.5*y+3

     t          Euler approximation y_{i}                   Tangent line
     0          1                                   2.5*t+1
   0.2          1.5                                 1.85*t+1.13
   0.4          1.87                                1.265*t+1.364
   0.6          2.123                               0.7385*t+1.6799
   0.8          2.2707                              0.26465*t+2.05898
     1          2.32363                             -0.161815*t+2.48545

solution =
2.32363

Time taken by function: 253124 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Solve Initial Value Problem with Euler Method ]#
```

Figure 5.59: *The computation of initial value problem solution with Euler's method. The first column contains the t-values separated by the step size $h = 0.2$. The second column shows the corresponding y-values computed from Euler's formula (5.105). In the third column are the tangent line approximations found from Eq. (5.102) (**SymIntegration/Examples/Differential Equations/Test SymIntegration Solve Initial Value Problem with Euler Method/main.cpp**).*

One way to achieve more accurate results is to use a smaller step size, with a corresponding increase in the number of computational steps.

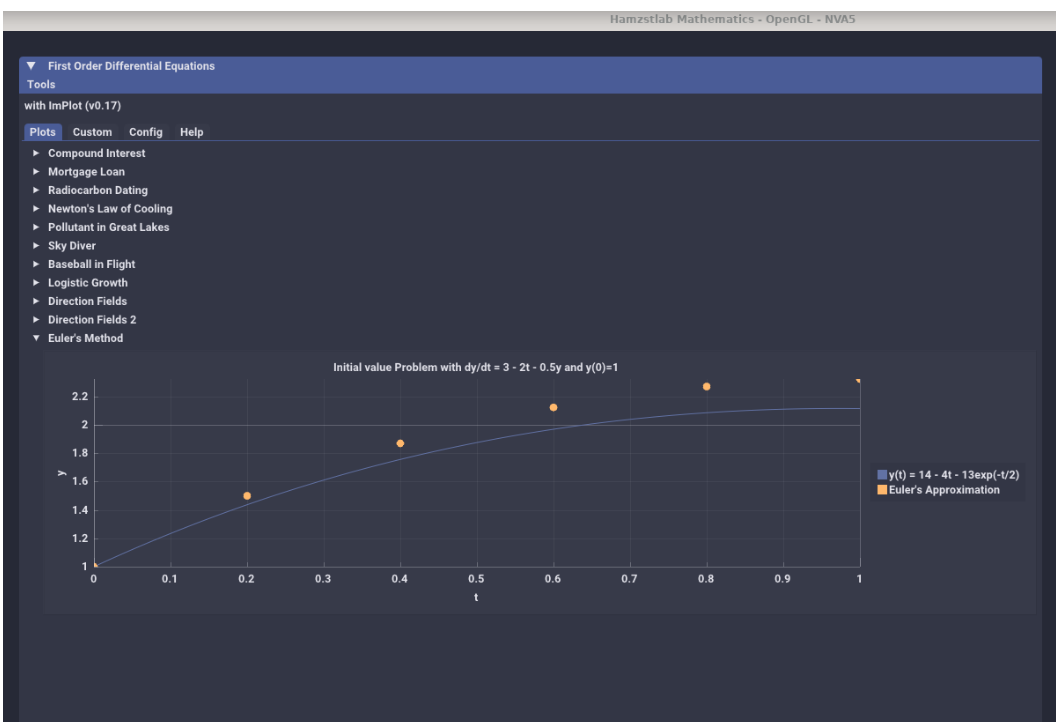


Figure 5.60: *The solution curve for $\frac{dy}{dt} = 3 - 2t - \frac{1}{2}y, \quad y(0) = 1$ and sequence of points from the Euler's method approximation (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

[H*] **Converging and Diverging Solutions**
Draw a direction field for the given differential equation and state whether you think that the solutions are converging or diverging.

(a) $y' = 5 - 3\sqrt{y}$
(b) $y' = y(3 - ty)$
(c) $y' = -ty + 0.1y^3$

Solution:

(a) The direction field is a two-dimensional vector field that shows what the direction of the solution is at every point in a region. Every solution to the differential equation is a curve drawn such that the direction field vectors are tangent to it at every point.

$$< dt, dy >= \langle 1, \frac{dy}{dt} \rangle dt =< 1, 5 - 3\sqrt{y} > dt$$

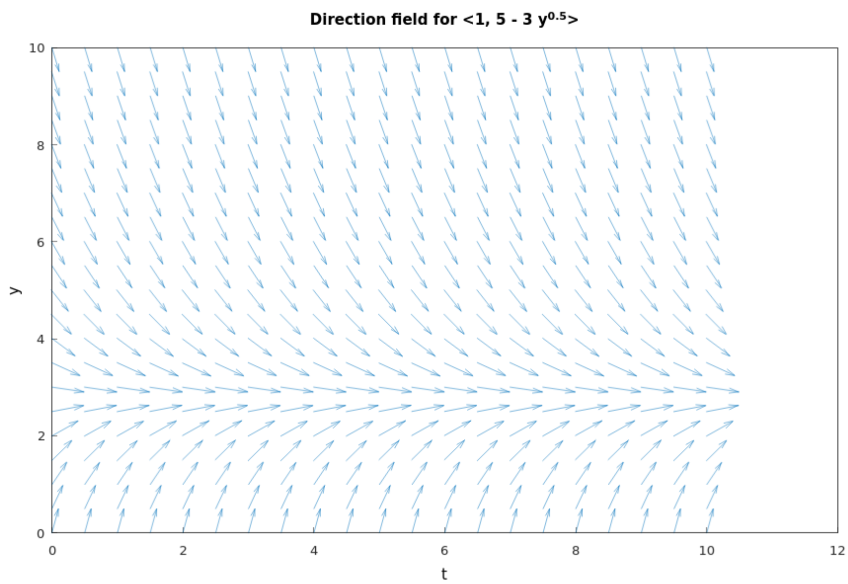


Figure 5.61: *The direction field for $< 1, 5 - 3\sqrt{y} >$, the vectors are possible solutions to the differential equation, depending what the initial condition is (**Hamzstplot/Examples with Makefile/Plot Direction Fields for Euler's Method Problem 7/main.cpp**).*

All solutions appear to converge as $t \to \infty$.

Besides using Hamzstplot, we want to demonstrate how to compute direction field with function in SymIntegration and with manual C++ codes, the resulting computation are 4 columns of data (**direction_field.dat**):

$$x \quad y \quad dx \quad dy$$

the plotting then will be done with gnuplot. To comprehend more how a direction field is made, we need to learn this thoroughly, x and y is the point in two-dimensional space, while dx is the change toward the x axis and dy is the change toward the y axis, both dx and dy influence the arrow head for the vector at that

point.
So if you have $dx > 0$ the arrow will be going to the right, if you have $dx < 0$ the arrow will be going to the left, if you have $dy > 0$ the arrow will be going upward, if you have $dy < 0$ the arrow will be going downward.
For example, if you have $dx = 0.047, dy = -0.047$, the arrow will be going toward southeast.

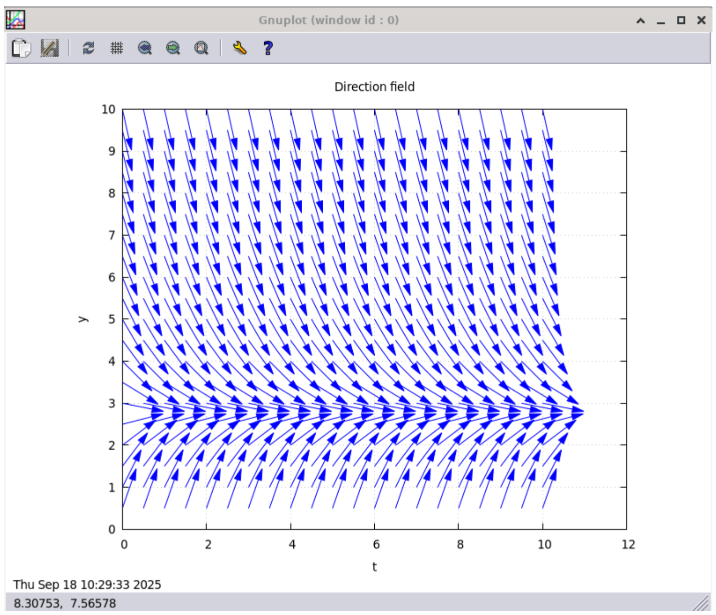


Figure 5.62: *The computation is done with SymIntegration' directionfield() function and then plot the result with gnuplot for $y' = 5 - 3\sqrt{y}$ (**SymIntegration/Examples/Differential Equations/Test SymIntegration Compute Direction Fields/main.cpp**).*

In SymIntegration, we use this function:
directionfield(f, t, y, x_min, x_max, y_min, y_max, step_size, k)

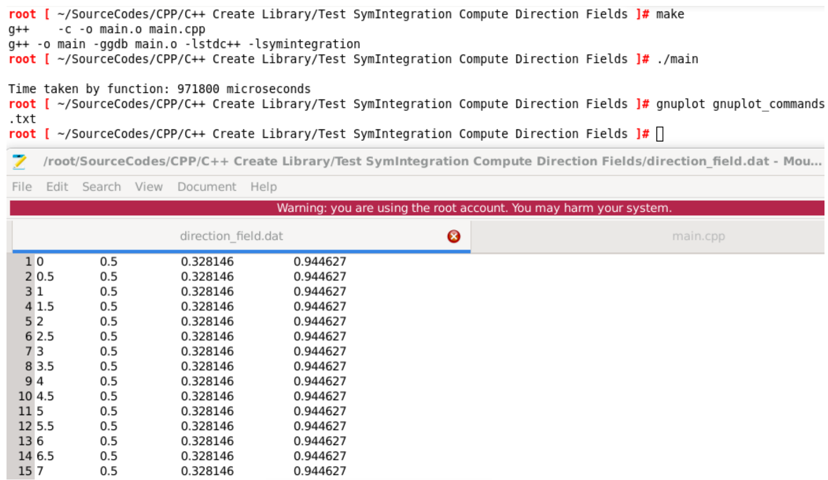


Figure 5.63: *The computation with SymIntegration' directionfield function will produce* ***direction_field.dat*** *and then plotting the result with gnuplot for $y' = 5 - 3\sqrt{y}$ (**SymIntegration/Examples/Differential Equations/Test SymIntegration Compute Direction Fields/main.cpp**).*

You can plot the direction field with gnuplot after the data has been produced / computed at the current working directory that contains **direction_field.dat** with command:
gnuplot gnuplot_commands.txt

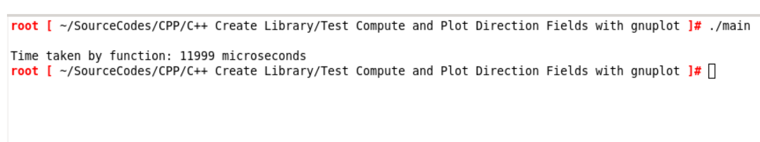


Figure 5.64: *The computation for the direction field of $y' = 5 - 3\sqrt{y}$ with manual C++ code, it will produce* ***direction_field.dat*** *as well with same results but it is faster than SymIntegration since it does not use Symbolic, only double computation, you can also plot this result with gnuplot (**SymIntegration/Examples/Differential Equations/Test Compute and Plot Direction Fields with gnuplot/main.cpp**).*

(b) The direction field is

$$< dt, dy >= \langle 1, \frac{dy}{dt} \rangle dt =< 1, y(3 - ty) > dt$$

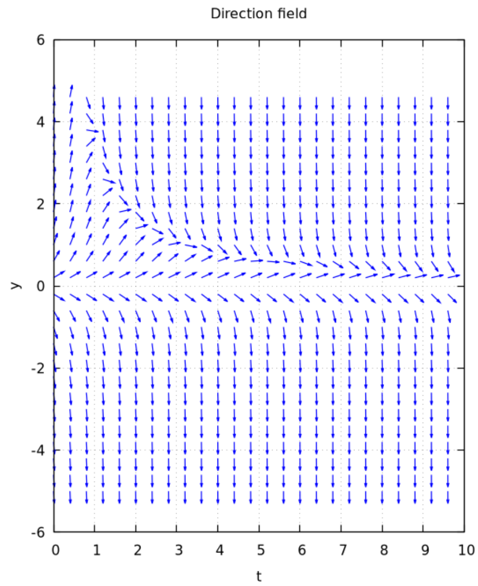


Figure 5.65: *The direction field for $y' = y(3 - ty)$ (**SymIntegration/Examples/Differential Equations/Test SymIntegration Compute Direction Fields Part b/main.cpp**).*

Solutions with initial conditions above $y = 0$ appear to converge as $t \to \infty$, but solutions with initial conditions below $y = 0$ appear to diverge as $t \to \infty$.

(c) The direction field is

$$< dt, dy >= \langle 1, \frac{dy}{dt} \rangle dt =< 1, -ty + 0.1y^3 > dt$$

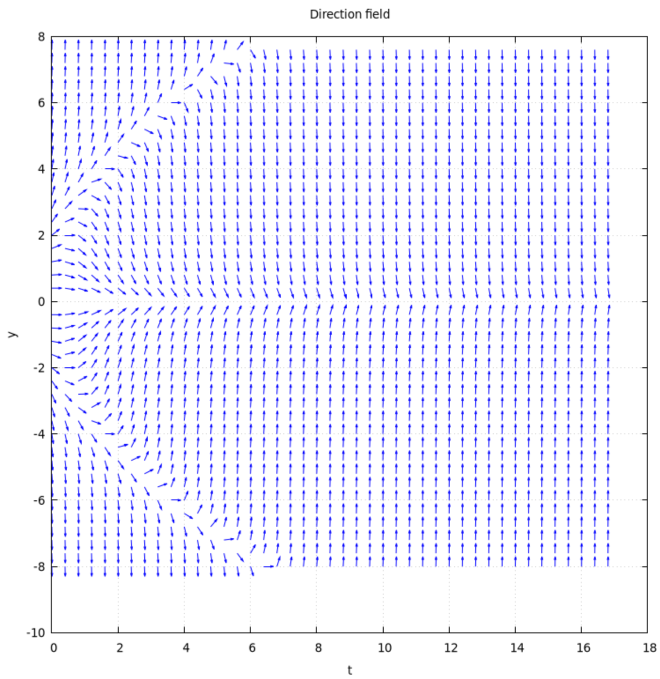


Figure 5.66: *The direction field for $y' = -ty + 0.1y^3$ (**SymIntegration/Examples/Differential Equations/Test SymIntegration Compute Direction Fields Part c/main.cpp**).*

Solutions with initial conditions below $y \approx 2.4$ and above $y \approx -2.4$ appear to converge as $t \to \infty$, but solutions with initial conditions above $y \approx 2.4$ and below $y \approx -2.4$ appear to diverge as $t \to \infty$.

[H*] **Convergence of Euler's Method**

It can be shown that, under suitable conditions on f, the numerical approximations generated by the Euler method for the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0$$

converges to the exact solution as the step size h decreases. this is illustrated by the following example. Consider the initial value problem

$$y' = 1 - t + y, \quad y(t_0) = y_0$$

(a) Show that the exact solution is $y = \phi(t) = (y_0 - t_0)e^{t-t_0} + t$.

(b) Using the Euler formula, show that

$$y_k = (1 + h)y_{k-1} + h - ht_{k-1}, \quad k = 1, 2, \ldots$$

(c) Noting that $y_1 = (1 + h)(y_0 - t_0) + t_1$, show by induction that

$$y_n = (1 + h)^n(y_0 - t_0) + t_n \tag{5.111}$$

for each positive integer n.

(d) Consider a fixed point $t > t_0$ and for a given n choose $h = \frac{(t-t_0)}{n}$. Then $t_n = t$ for every n. Note also that $h \to 0$ as $n \to \infty$. By substituting h in Eq. (5.111) and letting $n \to \infty$, show that $y_n \to \phi(t)$ as $n \to \infty$.

Solution:

(a) The ODE is linear, with general solution $y(t) = t + ce^t$.
Invoking the specified initial condition $y(t_0) = y_0$, we have

$$y_0 = t_0 + ce^{t_0}$$

Hence

$$c = (y_0 - t_0)e^{-t_0}$$

Thus the solution is given by

$$\phi(t) = (y_0 - t_0)e^{t-t_0} + t$$

(b) The Euler formula is

$$y_{n+1} = (1+h)y_n + h - ht_n$$

Now set

$$k = n + 1$$

we will have

$$y_k = (1+h)y_{k-1} + h - ht_{k-1}, \quad k = 1, 2, \dots$$

(c) We have

$$y_1 = (1+h)y_0 + h - ht_0 = (1+h)y_0 + (t_1 - t_0) - ht_0$$

Rearranging the terms,

$$y_1 = (1+h)(y_0 - t_0) + t_1$$

Next, we will compute for y_2

$$\begin{aligned}
y_2 &= (1+h)y_1 + h - ht_1 \\
&= (1+h)\left[(1+h)(y_0 - t_0) + t_1\right] + h - ht_1 \\
&= (1+h)^2(y_0 - t_0) + (1+h)t_1 + h - ht_1 \\
&= (1+h)^2(y_0 - t_0) + t_1 + h \\
&= (1+h)^2(y_0 - t_0) + t_2
\end{aligned}$$

We can also continue on y_3 to detect the pattern, but it is enough till y_2 at this time for the induction beginning step, afterward is the proving step for any number k, now suppose that

$$y_k = (1+h)^k(y_0 - t_0) + t_k$$

for some $k \geq 1$. Then

$$y_{k+1} = (1+h)y_k + h - ht_k$$

Substituting for y_k, we find that

$$\begin{aligned}
y_{k+1} &= (1+h)^{k+1}(y_0 - t_0) + (1+h)t_k + h - ht_k \\
&= (1+h)^{k+1}(y_0 - t_0) + t_k + h \\
&= (1+h)^{k+1}(y_0 - t_0) + t_{k+1}
\end{aligned}$$

Noting that $t_{k+1} = t_k + h$, the result is verified.

(d) Substituting $h = \frac{t-t_0}{n}$ with $t_n = t$,

$$y_n = \left(1 + \frac{t-t_0}{n}\right)(y_0 - t_0) + t$$

taking the limit of both sides as $n \to \infty$, and using the fact that

$$\lim_{n\to\infty}\left(1 + \frac{a}{n}\right)^n = e^a$$

pointwise convergence is proved.

- The Existence and Uniqueness Theorem

[H*] We have discussed a theorem before, which states that under certain conditions on $f(t, y)$, the initial value problem

$$y' = f(t, y(t)), \quad y(t_0) = y_0 \tag{5.112}$$

has a unique solution in some interval containing the point t_0.

In some cases the existence of a solution of the initial value problem (5.112) can be estalished directly by actually solving the problem and exhibiting a formula for the solution. However, in general, this approach is not feasible because there is not method of solving the differential equation that applies in all cases. Therefore, for general case, it is necessary to adopt an indirect approach that demonstrates the existence of a soltion of Eqs. (5.112) but usually does not provide a practical means of finding it.

The heart of this method is the construction of a sequence of functions that converges to a limit function satisfying the initial value problem, although the members of the sequence individually do not. As a rule, it is impossible to compute explicitly more than a few members of the sequence; therefore the limit function can be determined only in rare cases.

First of all, we note that it is sufficient to consider the problem in which the initial point (t_0, y_0) is the origin; that is, we consider the problem

$$y' = f(t, y), \quad y(0) = y_0 \tag{5.113}$$

Theorem 5.4: The Existence and Uniqueness Theorem

If f and $\frac{\partial f}{\partial y}$ are continuous in a rectangle $R : |t| \leq a,\ |y| \leq b$, then there is some interval $|t| \leq h \leq a$ in which there exists a unique solution $y = \phi(t)$ of the initial value problem (5.113).

It is necessary to transform the initial value problem (5.113) into a more convenient form. If we suppose temporarily that there is a differentiable function $y = \phi(t)$ that satisfies the initial value problem, then $f[t, \phi(t)]$ is a continuous function of t only. Hence we can integrate $y' = f(t, y)$ from the initial point $t = 0$ to an arbitrary value of t, obtaining

$$\phi(t) = \int_0^t f[s, \phi(s)]\, ds \tag{5.114}$$

where we have made use of the initial condition $\phi(0) = 0$. We also denote the dummy variable of integration by s.
Since Eq. (5.114) contains an integral of the unknown function ϕ, it is called an integral equation. This integral equation is not a formula for the solution of the initial value problem, but it does provide another relation satisfied by any solution of Eqs. (5.113). Conversely, suppose that there is a continuous function $y = \phi(t)$ that satisfies the integral equation (5.114); then this function also satisfies the initial value problem (5.113).

To show this, we first substitute zero for t in Eq. (5.114), which shows that the initial condition is satisfied. Further, since the integrand in Eq. (5.114) is continuous, it follows from the fundamental theorem of Calculus that ϕ is differentiable, and that $\phi'(t) = f[t, \phi(t)]$. Therefore the initial value problem and the integral equation are equivalent in the sense that any solution of one is also a solution of the other. It is more convenient to show that there is a unique solution of the integral equation in a certain interval $|t| \leq h$. The same conclusion will then hold also for the initial value problem.

One method of showing that the integral equation (5.114) has a unique solution is known as the method of successive approximations or Picard's iteration method. In using this method, we start by choosing an initial function ϕ_0, either arbitrarily or to approximate in some way the solution of the initial value problem. the simplest choice is

$$\phi_0(t) = 0 \tag{5.115}$$

then ϕ_0 at least satisfies the initial condition in Eqs. (5.113), although presumably not the differential equation. The next approximation ϕ_1 is obtained by substituting $\phi_0(s)$ for $\phi(s)$ in the right side of Eq. (5.114) and calling the result of this operation $\phi_1(t)$. Thus

$$\phi_1(t) = \int_0^t f[s, \phi_0(s)]\, ds \tag{5.116}$$

Similarly, ϕ_2 is obtained from ϕ_1:

$$\phi_2(t) = \int_0^t f[s, \phi_1(s)]\, ds \tag{5.117}$$

and, in general,

$$\phi_{n+1}(t) = \int_0^t f[s, \phi_n(s)]\, ds \tag{5.118}$$

In this manner we generate the sequence of functions $\{\phi_n\} = \phi_0, \phi_1, \dots, \phi_n, \dots$. Each member of the sequence satisfies the initial condition, but in general none satisfies the differential equation.

However, if at some stage, say, for $n = k$, we find that $\phi_{k+1}(t) = \phi_k(t)$, then it follows that ϕ_k is a solution of the integral equation (5.114). Hence ϕ_k is also a solution of the initial value problem (5.113), and the sequence is terminated at this point. In general, this does not occur, and it is necessary to consider the entire infinite sequence.

To establish Theorem 5.4, we must answer four principal questions:

1. Do all members of the sequence $\{\phi_n\}$ exist, or may the process break down at some stage?

2. Does the sequence converge?
3. What are the properties of the limit function? In particular, does it satisfy the integral equation (5.114) and hence the initial value problem (5.113)?
4. Is this the only solution, or may there be others?

[H*] **Example:**
Solve the initial value problem

$$y' = 2t(1+y), \quad y(0) = 0 \tag{5.119}$$

by the method of successive approximations.

Solution:
Note first that if $y = \phi(t)$, then the corresponding integral equation is

$$\phi(t) = \int_0^t 2s[1+\phi(s)]\, ds \tag{5.120}$$

If the initial approximation is $\phi_0(t) = 0$, it follows that

$$\phi_1(t) = \int_0^t 2s[1+\phi_0(s)]\, ds = \int_0^t 2s\, ds = t^2 \tag{5.121}$$

Similarly,

$$\phi_2(t) = \int_0^t 2s[1+\phi_1(s)]\, ds = \int_0^t 2s[1+s^2]\, ds = t^2 + \frac{t^4}{2} \tag{5.122}$$

and

$$\phi_3(t) = \int_0^t 2s[1+\phi_2(s)]\, ds = \int_0^t 2s\left[1+s^2+\frac{s^4}{2}\right]\, ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6} \tag{5.123}$$

Equations (5.121), (5.122), and (5.123) suggest that

$$\phi_n(t) = t^2 + \frac{t^4}{2!} + \frac{t^6}{3!} + \cdots + \frac{t^{2n}}{n!} \tag{5.124}$$

for each $n \geq 1$, and this result can be established by mathematical induction, as follows. Equation (5.124) is certainly true for $n = 1$. We must show that if it is true for $n = k$, then it also holds for $n = k+1$. We have

$$\begin{aligned}
\phi_{k+1}(t) &= \int_0^t 2s[1+\phi_k(s)]\, ds \\
&= \int_0^t \left(1+s^2+\frac{s^4}{2!}+\cdots+\frac{s^{2k}}{k!}\right) \\
&= t^2 + \frac{t^4}{2!} + \frac{t^6}{3!} + \cdots + \frac{t^{2k+2}}{(k+1)!}
\end{aligned} \tag{5.125}$$

and the inductive proof is complete.

It follows from Eq. (5.124) that $\phi_n(t)$ is the nth partial sum of the infinite series

$$\sum_{k=1}^{\infty} \frac{t^{2k}}{k!} \tag{5.126}$$

hence $\lim_{n\to\infty} \phi_n(t)$ exists if and only if the series (5.125) converges. Applying the ratio test, we see that, for each t,

$$\left| \frac{t^{2k+2}}{(k+1)!} \frac{k!}{t^{2k}} \right| = \frac{t^2}{k+1} \to 0 \quad \text{as } k \to \infty \tag{5.127}$$

Thus the series (5.126) converges for all t, and its sum $\phi(t)$ is the limit of the sequence $\{\phi_n(t)\}$. Further, since the series (5.126) is a Taylor series, it can be differentiated or integrated term by term as long as t remains within the interval of convergence, which in this case is the entire t-axis.

Therefore, we can verify by direct computation that

$$\phi(t) = \sum_{k=1}^{\infty} \frac{t^{2k}}{k!}$$

is a solution of the integral equation (5.120). Alternatively, by substituting $\phi(t)$ for y in Eqs. (5.119), we can verify that this function satisfies the initial value problem. In this example, it is also possible, from the series (5.126), to identify ϕ in terms of elementary functions, namely, $\phi(t) = e^{t^2} - 1$. However, this is not necessary for the discussion of existence and uniqueness.

Explicit knowledge of $\phi(t)$ does make it possible to visualize the convergence of the sequence of iterates more clearly by plotting $\phi(t) - \phi_k(t)$ for various values of k.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Picard's Iteration for First Order Differential Equation ]# ./main

For dy/dt =2*t*y+2*t

Φ_{0}(t) =2*t
Φ_{1}(t) =t^(2)
Φ_{2}(t) =1/2*t^(4)+t^(2)
Φ_{3}(t) =1/6*t^(6)+1/2*t^(4)+t^(2)
Φ_{4}(t) =1/24*t^(8)+1/6*t^(6)+1/2*t^(4)+t^(2)

Sequence 1: t^(2)
Sequence 2: 1/2*t^(4)
Sequence 3: 1/6*t^(6)
Sequence 4: 1/24*t^(8)

Sequence 4 / sequence 3: 1/4*t^(2)
Sequence 3 / sequence 2: 1/3*t^(2)
Sequence 2 / sequence 1: 1/2*t^(2)

Φ_{n}(t) = t^(2*n)*n!^(-1)

Time taken by function: 739842 microseconds
```

Figure 5.67: *The computation for Picard's iteration method and to determine the pattern by mathematical induction for $\phi_n(t)$ (**SymIntegration/Examples/Differential Equations/Test SymIntegration Picard's Iteration for First Order Differential Equation/main.cpp**).*

If you take a look at the source code **src/differentialequations.cpp** under the function **picarditeration_mathematicalinduction**, we are creating an **if** conditional to determine the pattern thus we can produce the correct $\phi_n(t)$, it can be done by seeing the first 3 sequences of the series.

Finally, to deal with the question of uniqueness, let us suppose that the initial value problem has tow solutions ϕ and ψ. Since ϕ and ψ both satisfy the integral equation (5.120), we have by subtraction that

$$\phi(t) - \psi(t) = \int_0^t 2s[\phi(s) - \psi(s)]\, ds$$

Taking absolute values of both sides, we have, if $t > 0$,

$$|\phi(t) - \psi(t)| = \Big| \int_0^t 2s[\phi(s) - \psi(s)]\, ds \Big| \leq \int_0^t 2s|\phi(s) - \psi(s)|\, ds$$

If we restrict t to lie in the interval $0 \leq t \leq A/2$, where A is arbitrary, then $2t \leq A$, and

$$|\phi(t) - \psi(t)| \leq A \int_0^t |\phi(s) - \psi(s)|\, ds \tag{5.128}$$

It is now convenient ot introduce the function U defined by

$$U(t) = \int_0^t |\phi(s) - \psi(s)|\, ds \tag{5.129}$$

Then it follows at once that

$$U(0) = 0 \tag{5.130}$$

$$U(t) \geq 0, \quad \text{for } t \geq 0 \tag{5.131}$$

Further, U is differentiable, and $U'(t) = |\phi(t) - \psi(t)|$. Hence, by Eq. (5.128),

$$U'(t) - AU(t) \leq 0 \tag{5.132}$$

Multiplying Eq. (5.132) by the positive quantity e^{-At} gives

$$[e^{-At}U(t)]' \leq 0 \tag{5.133}$$

Then, upon integrating Eq. (5.133) from zero to t and using Eq. (5.130), we obtain

$$e^{-At}U(t) \leq 0 \quad \text{for } t \geq 0$$

Hence, $U(t) \leq 0$ for $t \geq 0$, and in conjunction with Eq.(5.131), this requires that $U(t) = 0$ for each $t \geq 0$. Thus $U'(t) \equiv 0$, and therefore

$$\psi(t) \equiv \phi(t)$$

which contradicts the original hypothesis. Consequently, there cannot be two different solutions of the initial value problem for $t \geq 0$. A slight modification of this argument leads to the same conclusion for $t \leq 0$.

- First Order Difference Equations

[H*] Although a continuous model leading to a differential equation is reasonable and attractive for many problems, there are some cases in which a discrete model may be more natural. Sometimes population growth may be described more accurately by a discrete than by a continuous model. This is true, for example, of species whose generations do not overlap and that propagate at regular intervals, such as at particular times of the calendar year. Then the population y_{n+1} of the species in that year $n+1$ is some function of n and the population y_n in the preceding year; that is

$$y_{n+1} = f(n, y_n), \quad n = 0, 1, 2, \ldots \tag{5.134}$$

Equation (5.134) is called a first order difference equation. It is first order because the value of y_{n+1} depends on the value of y_n but not on earlier values y_{n-1}, y_{n-2}, and so forth. As for differential equations, the difference equation (5.134) is linear if f is a linear function of y_n; otherwise, it is nonlinear. A solution of the difference equation (5.134) is a sequence of numbers $y_0, y_1, y_2, \ldots$ that satisfy the equation for each n. In addition to the difference equation itself, there may also be an initial condition

$$y_0 = \alpha \tag{5.135}$$

that prescribes the value of the first term of the solution sequence.
We now assume temporarily that the function f in Eq. (5.134) depends only on y_n, but not on n. Int his case

$$y_{n+1} = f(y_n), \quad n = 0, 1, 2, \ldots \tag{5.136}$$

If y_0 is given, then successive terms of the solution can be found from Eq. (5.136). Thus

$$\begin{aligned} y_1 &= f(y_0) \\ y_2 &= f(y_1) = f[f(y_0)] \end{aligned}$$

The quantity $f[f(y_0)]$ is called the second iterate of the difference equation and is sometimes denoted by $f^2(y_0)$. Similarly, the third iterates y_3 is given by

$$y_3 = f(y_2) = f[f[f(y_0)]] = f^3(y_0)$$

and so on. In general, the nth iterate y_n is

$$y_n = f(y_{n-1}) = f^n(y_0)$$

This procedure is referred to as iterating the difference equation. It is often of primary interet to determine the behavior of y_n as $n \to \infty$.
Solutions for which y_n has the same value for all n are called equilibrium solutions. They are frequently of special importance, just as in the study of differential equations. If equilibrium solutions exist, one can find them by setting y_{n+1} equal to y_n in Eq. (5.136) and solcing the resulting equation

$$y_n = f(y_n) \tag{5.137}$$

for y_n.

[H*] **Linear Equations**

Suppose that the population of a certain species in a given region in year $n+1$, denoted by y_{n+1}, is a positive multiple ρ_n of the population y_n in year n; that is,

$$y_{n+1} = \rho_n y_n, \quad n = 0, 1, 2 \ldots \tag{5.138}$$

Note that the reproduction rate ρ_n may differ from year to year. The difference equation (5.138)is linear and can easily be solved by iteration. We obtain

$$\begin{aligned} y_1 &= \rho_0 y_0 \\ y_2 &= \rho_1 y_1 = \rho_1 \rho_0 y_0 \end{aligned}$$

and, in general,

$$y_n = \rho_{n-1} \ldots \rho_0 y_0, \quad n = 1, 2, \ldots \tag{5.139}$$

Thus, if the initial population y_0 is given, then the population of each succeeding generation is determined by Eq.(5.139). Although for a population problem ρ_n is intrinsically positive, the solution (5.139) is also valid if ρ_n is negative for some or all values of n. Note, however, that if ρ_n is zero for some n, then y_{n+1} and all succeeding values of y are zero; in other words, the species has become extinct.
If the reproduction rate $\rho_n$has the same value ρ for each n, then the difference equation (5.138) becomes

$$y_{n+1} = \rho y_n \tag{5.140}$$

and its solution is

$$y_n = \rho y_0 \tag{5.141}$$

Equation (5.140) also has an equilibrium solution, namely, $y_n = 0$ for all n, corresponding to the initial value $y_0 = 0$. The limiting behavior of y_n is easy to determine from Eq. (5.141). In fact,

$$\lim_{n\to\infty} y_n = \begin{cases} 0, & \text{if } |\rho| < 1; \\ y_0, & \text{if } \rho = 1; \\ \text{does not exist}, & \text{otherwise.} \end{cases} \tag{5.142}$$

In other words, the equilibrium solution $y_n = 0$ is asymptotically stable for $|\rho| < 1$ and unstable if $|\rho| > 1$.

Now we will modify the population model represented by Eq. (5.138) to include the effect of immigration or emigration. If b_n is the net increase in population in year n due to immigration, then the population in year $n+1$ is the sum of those due to natural reproduction and those due to immigration. Thus

$$y_{n+1} = \rho y_n + b_n, \quad n = 0, 1, 2, \ldots \tag{5.143}$$

where we are now assuming that the reproduction rate ρ is constant. We can solve Eq. (5.143) by iteration in the same manner as before. We have

$$\begin{aligned} y_1 &= \rho y_0 + b_0 \\ y_2 &= \rho(\rho y_0 + b_0) + b_1 = \rho^2 y_0 + \rho b_0 + b_1 \\ y_3 &= \rho(\rho^2 y_0 + \rho b_0 + b_1) + b_2 = \rho^3 y_0 + \rho^2 b_0 + \rho b_1 + b_2 \end{aligned}$$

and so forth. In general, we obtain

$$y_n = \rho^n y_0 + \sum_{j=0}^{n-1} \rho^{n-1-j} b_j \tag{5.144}$$

Note that the first term on the right side of Eq. (5.144) represents the descendants of the original population, while the other terms represent the population in year n resulting from immigration in all preceding years.

[H*] **Nonlinear Equations**

Nonlinear difference equations are much more complicated and have much more varied solutions thatn linear equations. We will restrict our attention to a single equation, the logistic difference equation

$$y_{n+1} = \rho y_n \left(1 - \frac{y_n}{k}\right) \tag{5.145}$$

which is analogous to the logistic differential equation

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right) \tag{5.146}$$

Note that if the derivative $\frac{dy}{dt}$ in Eq. (5.146) is replaced by the difference $\frac{(y_{n+1}-y_n)}{h}$, then Eq. (5.146) reduces to Eq. (5.145) with $\rho = 1 + hr$ and $k = \frac{(1+hr)K}{hr}$. To simplify Eq. (5.145) a little more, we can scale the variable y_n by introducing the new variable $u_n = \frac{y_n}{k}$. Then Eq. (5.145) becomes

$$u_{n+1} = \rho u_n(1 - u_n) \tag{5.147}$$

where ρ is a positive parameter.
We begin our investigation of Eq. (5.147) by seeking the equilibrium, or constant, solutions. These can be found by setting u_{n+1} equal to u_n in Eq. (5.147), which corrresponds to setting $\frac{dy}{dt}$ equal to zero in Eq. (5.146). The resulting equation is

$$u_n = \rho u_n - \rho u_n^2 \tag{5.148}$$

so it follows that the equilibrium solutions of Eq. (5.147) are

$$u_n = 0, \quad u_n = \frac{\rho - 1}{\rho} \tag{5.149}$$

The next question is whether the equilibrium solutions are asymptotically stable or unstable. That is, for an initial condition near one of the equilibrium solutions, does the resulting solution sequence approach or depart from the equilibrium solution? One way to examine this question is by approximating Eq. (5.147) by a linear equation in the neighborhood of an equilibrium solution. For example, near the equilibrium solution $u_n = 0$, the quantity u_n^2 is small compared to u_n itself, so we assume that we can neglect the quadratic term in Eq. (5.147) in comparison with the linear terms. This leaves us with the linear difference equation

$$u_{n+1} = \rho u_n \tag{5.150}$$

which is presumably a good approximation to Eq. (5.147) for u_n sufficiently near zero. However, Eq. (5.150) is the same as Eq. (5.140), and we have already concluded, in Eq. (5.142), that $u_n \to 0$ as $n \to \infty$ if and only if $|\rho| < 1$, or (since ρ must be positive) for $0 < \rho < 1$. Thus the equilibrium solution $u_n = 0$ is asymptotically stable for the linear approximation (5.150) for this set of ρ values, so we conclude that it is also asymptotically stable for the full nonlinear equation (5.147). This conclusion is correct, although our argument is not complete. What is lacking is a theorem stating that the solutions of the nonlinear equation (5.147) resemble those of the linear equation (5.150) near the equilibrium solution $u_n = 0$.

Now consider the other equilibrium solution $u_n = \frac{\rho-1}{\rho}$. To study solutions in the neighborhood of this point we write

$$u_n = \frac{\rho - 1}{\rho} + v_n \tag{5.151}$$

where we assume that v_n is small. By substituting from Eq. (5.151) in Eq. (5.147) and simplifying the resulting equation, we eventually obtain

$$v_{n+1} = (2-\rho)v_n - \rho v_n^2 \tag{5.152}$$

Since v_n is small, we again neglect the quadratic term in comparison with the linear terms and thereby obtain the linear equation

$$v_{n+1} = (2-\rho)v_n \tag{5.153}$$

Referring to Eq. (5.142) once more, we find that $v_n \to 0$ as $n \to \infty$ for $|2-\rho| < 1$, that is, for $1 < \rho < 3$. Therefore we conclude that, for this range of values of ρ, the equilibrium solution $u_n = \frac{\rho-1}{\rho}$ is asymptotically stable.

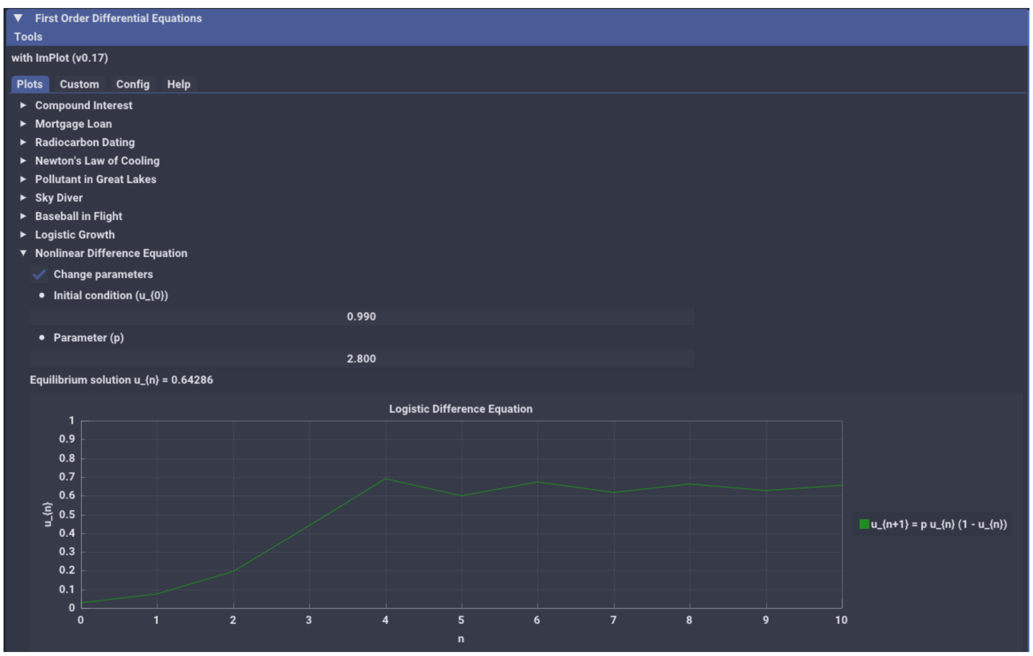


Figure 5.68: *The interactive plot for solutions of nonlinear difference equation for the logistic difference equation at Eq. (5.147) with varying ρ and initial conditions u_0 with Hamzstlab Mathematics. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

If you plot the solution of Eq. (5.147) with a particular initial conditions (e.g. $u_0 = 0.2$) you can observe that the solution converges to zero for $\rho = 0.8$ and to the nonzero equilibriumsolution for $\rho = 1.5$ and $\rho = 2.8$. The convergence is monotone for $\rho = 0.8$ and $\rho = 1.5$, and is oscillatory for $\rho = 2.8$.

Another way of displaying the solution of a difference equation is to use the graphs of the parabola $y = \rho x(1-x)$ and the straight line $y = x$. The equilibrium solutions correspond to the points of intersection of these two curves.

The difference equation (5.147) has two equilibrium solutions

$$u_n = 0$$
$$u_n = \frac{\rho - 1}{\rho}$$

the former is asymptotically stabe for $0 \leq \rho < 1$, and the latter is asymptotically stable for $1 < \rho < 3$. When $\rho = 1$, the two equilibrium solutions coincide at $u = 0$; this solution can be shown to be asymptotically stable. The interval in which each one is asymptotically stable are indicated by the solid portions of the curves. There is an exchange of stability form one equilibrium solution to the other at $\rho = 1$.

For $\rho > 3$ neither of the equilibtium solutions is stable, and the solutions of Eq. (5.147) exhibit increasing compleity as ρ increases. For ρ somewhat greater than 3, the sequence u_n rapidly approaches a steady oscillation of period 2; that is u_n oscillates back and forth between two distinct values. For n greater than about 20, the solution alternates between the values 0.5130 and 0.7995.

At about $\rho = 3.449$, each state in the oscillation of period 2 separates into two distinct states, and the solution becomes periodic with period 4. As ρ increases further, a periodic solutions of period $8, 16, \ldots$ appear. The appearance of a new solution at a certain parameter value is called a bifurcation.

The ρ-values at which the successive period doublings occur approach a limit that is approximately 3.57. For $\rho > 3.57$ the solutions possess some regularity, but no discernible detailed pattern for most values of ρ. For example, a solution for $\rho = 3.65$ oscilaltes between approximately 0.3 and 0.9, but its fine structure is unpredictable. The term chaotic is used to describe this situation. One of the features of chaotic solutions is extreme sensitivity to the initial conditions.

It is only comparatively recently that chaotic solutions of difference and differential equations have become widely known. Equation (5.146) was one of the first instances of mathematical chaos to be found and studied in detail, by Robert May in 1974. On the basis of his analysis of this equation as a model of the population of certain insect species, May suggested that if the growth rate ρ is too large, then it will be impossible to make effective long-range predictions about these insect populations. The occurrence of chaotic solutions in simple problems has stimulated an enormous amount of research in recent years, but many questions remain unanswered. It is increasingly clear, however, that chaotic solutions are much more common than was suspected at first and that they may be a part of the investigation of a wide range of phenomena.

V. Second Order Linear Equations

- Homogeneous Equations with Constant Coefficients

[H*] Linear equations of second order are of crucial importance in the study of differential equations for two main reasons. The first is that linear equations have a rich theoretical structure that underlies a number of systematic methods of solution. Further, a substantial portion of this structure and of these methods is understandable at a fairly elementary mathematical level. Another reason to study second order linear equations is that they are vital to any serios investigation of the classical areas of mathematical physics. One cannot go very far in the development of fluid mechanics, heat conduction, wave motion, or electromagnetic phenomena without finding it necessary to solve second order linear differential equations.

[H*] A second order ordinary differential equation has the form

$$\frac{d^2y}{dt^2} = f\left(t, y, \frac{dy}{dt}\right) \tag{5.154}$$

where f is some given function. We usually denote the independent variable by t since time is often the independent variable in physical problems, but sometimes we will use x instead. We will use y, or occasionally some other letter, to designate the dependent variable. Equation (5.154) is said to be linear if the function f has the form

$$f\left(t, y, \frac{dy}{dt}\right) = g(t) - p(t)\frac{dy}{dt} - q(t)y \tag{5.155}$$

that is, if f is linear in y and $\frac{dy}{dt}$. In Eq. (5.155) $g, p,$ and q are specified functions of the independent variable t but do not depend on y. In this case we usually rewrite Eq. (5.154) as

$$y'' + p(t)y' + q(t)y = g(t) \tag{5.156}$$

where the primes denote differentiation with respect to t. Instead of Eq. (5.156), we often see the equation

$$P(t)y'' + Q(t)y' + R(t)y = G(t) \tag{5.157}$$

Of course, if $P(t) \neq 0$, we can divide Eq. (5.157) by $P(t)$ and thereby obtain Eq. (5.156) with

$$p(t) = \frac{Q(t)}{P(t)}, \quad q(t) = \frac{R(t)}{P(t)}, \quad g(t) = \frac{G(t)}{P(t)} \tag{5.158}$$

In discussing Eq. (5.156) and in trying to solve it, we will restrict ourselves to intervals in which $p, q,$ and g are continuous functions.

If Eq. (5.154) is not of the form (5.156) or (5.157), then it is called nonlinear. Analytical investigations of nonlinear equations are relatively difficult. Numerical or geometrical approaches are often more appropriate.

We should not be misled by the fact that first order linear equations are easily solved by means of formulas. In general, Eq. (5.156) cannot be solved explicitly in terms of known elementary functions, or even in terms of indicated integrations. To find solutions, it is commonly necessary to resort to infinite processes of one kind or another, usually infinite series.

[H*] An initial value problem consists of a differential equation such as Eq. (5.154), (5.156), or (5.157) together with a pair of initial conditions

$$\begin{aligned} y(t_0) &= y_0 \\ y'(t_0) &= y'_0 \end{aligned} \tag{5.159}$$

where y_0 and y'_0 are given numbers prescribing values for y and y' at the initial point t_0. Observe that the initial conditions for a second order equation identify not only a particular point (t_0, y_0) through which the graph of the solution must pass, but also the slope y'_0 of the graph at that point. It is reasonable to expect that two initial conditions are needed for a second order equation because, roughly speaking, two integrations are required to find a solution and each integration introduces an arbitrary constant. Presumably, two initial conditions will suffice to determine values for these two constants.

[H*] A second order linear equation is said to be homogeneous if the term $g(t)$ in Eq. (5.156), or the term $G(t)$ in Eq. (5.157), is zero for all t. Otherwise, the equation is called nonhomogeneous. As a result, the term $g(t)$, or $G(t)$, is sometimes called the nonhomogeneous term. We begin our discussion with homogeneous equations, which we will write in the form

$$P(t)y'' + Q(t)y' + R(t)y = 0 \tag{5.160}$$

Once the homogeneous equation has been solved, it is always possible to solve the corresponding nonhomogeneous equation (5.157), or at least to express the solution in terms of an integral. Thus the problem of solving the homogeneous equation is the more fundamental one.

[H*] Consider the general second order linear differential equation

$$ay'' + by' + cy = 0 \tag{5.161}$$

whose coefficients a, b, and c are arbitrary (real) constants. We can solve Eq. (5.161) for any values of its coefficients and also satisfy any given set of initial conditions for y and y'. We start by seeking exponential solutions of the form $y = e^{rt}$, where r is a parameter to be determined.Then it follows that

$$\begin{aligned} y' &= re^{rt} \\ y'' &= r^2 e^{rt} \end{aligned}$$

By substituting these expressions for y, y', and y'' in Eq. (5.161), we obtain

$$(ar^2 + br + c)e^{rt} = 0$$

or, since $e^{rt} \neq 0$,

$$ar^2 + br + c = 0 \tag{5.162}$$

Equation (5.162) is called the characteristic equation for the differential equation (5.161). It significance lies in the fact that if r is a root of the polynomial equation (5.162), then $y = e^{rt}$ is a solution of the differential equation (5.161). Since Eq. (5.162) is a quadratic equation with real coefficients, it has two roots, which may be real and different, real but repeated, or complex conjugates.

Assuming that the roots of the characteristic equation (5.162) are real and different, let them be denoted by r_1 and r_2, where $r_1 \neq r_2$. Then $y_1(t) = e^{r_1 t}$ and $y_2(t) = e^{r_2 t}$ are two solutions of Eq. (5.161). It now follows that

$$y = c_1 y_1(t) + c_2 y_2(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t} \tag{5.163}$$

is also a solution of Eq. (5.161). To verify that this is so, we can differentiate the expression in Eq. (5.163); hence

$$y' = c_1 r_1 e^{r_1 t} + c_2 r_2 e^{r_2 t} \tag{5.164}$$

and

$$y'' = c_1 r_1^2 e^{r_1 t} + c_2 r_2^2 e^{r_2 t} \tag{5.165}$$

Substituting these expressions for y, y' and y'' in Eq. (5.161) and rearranging terms, we obtain

$$ay'' + by' + cy = c_1(ar_1^2 + br_1 + c)e^{r_1 t} + c_2(ar_2^2 + br_2 + c)e^{r_2 t} \tag{5.166}$$

The quantity in each of the parentheses on the right side of Eq. (5.166) is zero because r_1 and r_2 are roots of Eq. (5.162); therefore, y as given by Eq. (5.163) is indeed a solution of Eq. (5.161), as we wished to verify.

Now suppose that we want to find the particular member of the family solutions (5.163) that satisfies the initial conditions (5.159)

$$\begin{aligned} y(t_0) &= y_0 \\ y'(t_0) &= y'_0 \end{aligned}$$

By substituting $t = t_0$ and $y = y_0$ in Eq. (5.163), we obtain

$$c_1 e^{r_1 t_0} + c_2 e^{r_2 t_0} = y_0 \tag{5.167}$$

Similarly, setting $t = t_0$ and $y' = y'_0$ in Eq. (5.164) gives

$$c_1 r_1 e^{r_1 t_0} + c_2 r_2 e^{r_2 t_0} = y'_0 \tag{5.168}$$

On solving Eqs. (5.167) and (5.168) simultaneously for c_1 and c_2, we find that

$$\begin{aligned} c_1 &= \frac{y'_0 - y_0 r_2}{r_1 - r_2} e^{-r_1 t_0} \\ c_1 &= \frac{y_0 r_1 - y'_0}{r_1 - r_2} e^{-r_2 t_0} \end{aligned} \tag{5.169}$$

recall that $r_1 - r_2 \neq 0$ so that the expressions in Eq. (5.169) always make sense. Thus, no matter what initial conditions are assigned-that is, regardless of the valueas of t_0, y_0, and y'_0 in Eqs. (5.159)-it is always possible to determine c_1 and c_2 so that the initial conditions are satisfied. Moreover, there is only one possible choice of c_1 and c_2 for each set of initial conditons. With the values of c_1 and c_2 given by Eq. (5.169), the expression (5.163) is the solution of the initial value problem

$$\begin{aligned} ay'' + by' + cy &= 0 \\ y(t_0) &= y_0 \\ y'(t_0) &= y'_0 \end{aligned} \tag{5.170}$$

[H*] **Example:**
Find the solution, the maximum point and maximum value of the initial value problem

$$\begin{aligned} y'' + 5y' + 6y &= 0 \\ y(0) &= 2 \\ y'(0) &= 3 \end{aligned} \tag{5.171}$$

Solution:
We assume that $y = e^{rt}$, and it then follows that r must be a root of the characteristic equation

$$r^2 + 5r + 6 = (r+2)(r+3) = 0$$

Thus the possible values of r are $r_1 = -2$ and $r_2 = -3$; the general solution of Eq. (5.171) is

$$y = c_1 e^{-2t} + c_2 e^{-3t} \tag{5.172}$$

To satisfy the first initial condition, we set $t = 0$ and $y = 2$ in Eq. (5.171); thus c_1 and c_2 must satisfy

$$c_1 + c_2 = 2 \tag{5.173}$$

To use the second initial condition, we must first differentiate Eq. (5.171). This gives

$$y' = -2c_1 e^{-2t} - 3c_2 e^{-3t}$$

Then, setting $t = 0$ and $y' = 3$, we obtain

$$-2c_1 - 3c_2 = 3 \tag{5.174}$$

By solving the two equations above (5.173) and (5.174), we find that

$$\begin{aligned} c_1 &= 9 \\ c_2 &= -7 \end{aligned}$$

Using these values in the expression (5.172), we obtain the solution

$$y = 9e^{-2t} - 7e^{-3t} \tag{5.175}$$

of the initial value problem (5.171).

The solution (5.175) of the initial value problem (5.171) initially increases (because its initial slope is positive) but eventually approaches zero (because both terms involve negative exponential functions). Therefore the solution must have a maximum point, and the graph of y confirms this.

The coordinates of the maximum point can be estimated from the graph, but to find them more precisely, we seek the point where the solution has a horizontal tangent line. By differentiating the solution (5.175), $y = 9e^{-2t} - 7e^{-3t}$, with respect to t, we obtain

$$y' = -18e^{-2t} + 21e^{-3t} \tag{5.176}$$

Setting y' equal to zero and multiplying by e^{3t}, we find that the critical value t_m satisfies $e^t = \frac{7}{6}$; hence

$$t_m = \ln\left(\frac{7}{6}\right) \approx 0.15415 \tag{5.177}$$

The corresponding maximum value $y_m$is given by

$$y_m = 9e^{-2t_m} - te^{-3t_m} = \frac{108}{49} \approx 2.20408 \tag{5.178}$$

In this example, the initial slope is 3, but the solution of the given differential equation behaves in a similar way for any other positive initial slope.

```
root [ ~/SourceCodes/CPP/C++ Create Library/SI Test Clustered/Differential Equations/Test SymIntegration IVPSolve Second Order Linear Equations ]# ./main

IVP Solve for y'' + 5y'+ 6y = 0 with y(0) = 2, y'(0) = 3


The general solution is:
c1*e^(-2*t)+c2*e^(-3*t)

The solution for the initial value problem is:
9*e^(-2*t)-7*e^(-3*t)

y' = -18*e^(-2*t)+21*e^(-3*t)

Critical value t_{m} = 0.154151

Maximum value y_{m} = 2.20408

IVP Solve for y'' - y = 0 with y(0) = 2, y'(0) = -1


The general solution is:
c1*e^t+c2*e^(-t)

The solution for the initial value problem is:
0.5*e^t+1.5*e^(-t)

y' = 0.5*e^t-1.5*e^(-t)

Critical value t_{m} = 0.549306

Maximum value y_{m} = 1.73205

IVP Solve for 4y'' - 8y' +  3y = 0 with y(0) = 2, y'(0) = 1/2


The general solution is:
c1*e^(1.5*t)+c2*e^(0.5*t)

The solution for the initial value problem is:
-0.5*e^(1.5*t)+2.5*e^(0.5*t)

y' = -0.75*e^(1.5*t)+1.25*e^(0.5*t)

Critical value t_{m} = 0.510826

Maximum value y_{m} = 2.15166
```

Figure 5.69: *The computation is done with SymIntegration′* ***ivpsecondorderlinear()*** *function (**SymIntegration/Examples/Differential Equations/Test SymIntegration IVPSolve Second Order Linear Equations/-main.cpp**).*

- Solutions of Linear Homogeneous Equations; the Wronskian

[H*] We will introduce a differential operator notation. Let p and q be continuous functions on an open interval I, that is, for $\alpha < t < \beta$. The cases $\alpha = -\infty$, or $\beta = \infty$, or both, are included. Then, for any function ϕthat is twice differentiable on I we define the differential operator L by the equation

$$L[\phi] = \phi'' + p\phi' + q\phi \tag{5.179}$$

Note that $L[\phi]$ is a function on I. The value of $L[\phi]$ at a point t is

$$L[\phi](t) = \phi''(t) + p(t)\phi'(t) + q(t)\phi(t)$$

For example, if $p(t) = t^2, q(t) = 1 + t$, and $\phi(t) = \sin\ 3t$, then

$$\begin{aligned} L[\phi](t) &= (\sin\ 3t)'' + t^2(\sin\ 3t)' + (1+t)\sin\ 3t \\ &= -9\sin\ 3t + 3t^2\cos\ 3t + (1+t)\sin\ 3t \end{aligned}$$

The operator L is often written as $L = D^2 + pD + q$, where D is the derivative operator.

Since it is customary to use the symbol y to denote $\phi(t)$, we will usually write the second order linear homogeneous equation in the form

$$L[y] = y'' + p(t)y' + q(t)y = 0 \tag{5.180}$$

With Eq. (5.180) we associate a set of initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y'_0 \tag{5.181}$$

where t_0 is any point in the interval I, and y_0 and y'_0 are given real numbers.

Theorem 5.5: Existence and Uniqueness Theorem

Consider the initial value problem

$$\begin{aligned} y'' + p(t)y' + q(t)y &= g(t) \\ y(t_0) &= y_0 \\ y'(t_0) &= y'_0 \end{aligned} \tag{5.182}$$

where p, q, and g are continuous on an open interval I that contains the point t_0. Then there is exactly one solution $y = \phi(t)$ of this problem, and the solution exists throughout the interval I.

[H*] **Example:**
Find the longest interval in which the solution of the initial value problem

$$\begin{aligned} (t^2 - 3t)y'' + ty' - (t+3)y &= 0 \\ y(1) &= 2 \\ y'(1) &= 1 \end{aligned}$$

is certain to exist.

Solution:
If the given differential equation is written in the form of Eq. (5.182), then

$$\begin{aligned} p(t) &= \frac{1}{t-3} \\ q(t) &= -\frac{t+3}{t(t-3)} \\ g(t) &= 0 \end{aligned}$$

The only points of discontinuity of the coefficients are $t = 0$ and $t = 3$. Therefore, the longest open interval, containing the initial point $t = 1$, in which all the coefficients are continuous is $0 < t < 3$. Thus, this is the longest interval in which the existence and uniqueness theorem guarantees that the solution exists.

Theorem 5.6: Principle of Superposition

If y_1 and y_2 are two solutions of the differential equation (5.180),

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

then the linear combination $c_1y_1 + c_2y_2$ is also a solution for any values of the constants c_1 and c_2.

A special case of Principle of Superposition theorem occurs if either c_1 or c_2 is zero. Then we conclude that any constant multiple of a solution of Eq. (5.180) is also a solution. To prove the theorem, we need only substitute

$$y = c_1y_1(t) + c_2y_2(t) \tag{5.183}$$

for y in Eq. (5.180). By calculating the indicated derivatives and rearranging terms, we obtain

$$\begin{aligned}
L[c_1y_1 + c_2y_2] &= [c_1y_1 + c_2y_2]'' + p[c_1y_1 + c_2y_2]' + q[c_1y_1 + c_2y_2] \\
&= c_1y_1'' + c_2y_2'' + c_1py_1' + c_2py_2' + c_1qy_1 + c_2qy_2 \\
&= c_1[y_1'' + py_1' + qy_1] + c_2[y_2'' + py_2' + qy_2] \\
&= c_1L[y_1] + c_2L[y_2]
\end{aligned}$$

Since $L[y_1] = 0$ and $L[y_2] = 0$, it follows that $L[c_1y_1 + c_2y_2] = 0$ also. Therefore, regardless of the values of c_1 and c_2, y as given by Eq. (5.183) does satisfy the differential equation (5.180), and the proof for the theorem is complete.

The theorem states that, beginning with only two solutions of Eq. (5.180), we can construct an infinite family of solutions by means of Eq. (5.183). The next question is whether all solutions of Eq. (5.180) are included in Eq. (5.183) or whether there may be other solutions of a different form. We begin to address this question by examining whether the constants c_1 and c_2 in Eq. (5.183) can be chosen so as to satisfy the initial conditions (5.181). These initial conditions require c_1 and c_2 to satisfy the equations

$$\begin{aligned}
c_1y_1(t_0) + c_2y_2(t_0) &= y_0 \\
c_1y_1'(t_0) + c_2y_2'(t_0) &= y_0'
\end{aligned} \tag{5.184}$$

The determinant of coefficients of the system (5.185) is

$$W = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix} = y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \tag{5.185}$$

If $W \neq 0$, then Eqs (5.184) have a unique solution (c_1, c_2) regardless of the values of y_0 and y_0'. This solution is given by

$$\begin{aligned}
c_1 &= \frac{y_0y_2'(t_0) - y_0'y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)} \\
c_2 &= \frac{-y_0y_1'(t_0) + y_0'y_1(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}
\end{aligned} \tag{5.186}$$

or, in terms of determinants,

$$c_1 = \frac{\begin{vmatrix} y_0 & y_2(t_0) \\ y_0' & y_2'(t_0) \end{vmatrix}}{\begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix}}, \quad c_2 = \frac{\begin{vmatrix} y_1(t_0) & y_0 \\ y_1'(t_0) & y_0' \end{vmatrix}}{\begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix}} \tag{5.187}$$

with these values for c_1 and c_2, the expression (5.183) satisfies the initial conditions (5.181) as well as the differential equation (5.180).

On the other hand, if $W = 0$, then Eqs. (5.184) have no solution unless y_0 and y_0' satisfy a certain additional conditions; in this case there are infinitely many solutions.

The determinant W is called the Wronskian determinant, or simply the Wronskian, of the solutions y_1 and y_2. Sometimes we use the more extended notation $W(y_1, y_2)(t_0)$ to stand for the expression on the right side of Eq. (5.185), thereby emphasizing that the Wronskian depends on the functions y_1 and y_2, and that it is evaluated at the point t_0.

Theorem 5.7: Wronskian for Second Order Linear Differential Equations

Suppose that y_1 and y_2 are two solutions of Eq. (5.180)

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

and that the initial conditions (5.181)

$$y(t_0) = y_0, \quad y'(t_0) = y_0'$$

are assigned. Then it is always possible to choose the constants c_1, c_2 so that

$$y = c_1 y_1(t) + c_2 y_2(t)$$

satisfies the differential equation (5.180) and the initial conditions (5.181) if and only if the Wronskian

$$W = y_1 y_2' - y_1' y_2$$

is not zero at t_0.

Theorem 5.8: Fundamental Set of Solutions

Consider the differential equation (5.180)

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

whose coefficients p and q are continuous on some open interval I. Choose some point t_0 in I. Let y_1 be the solution of Eq. (5.180) that also satisfies the initial conditions

$$y(t_0) = 1, \quad y'(t_0) = 0$$

and let y_2 be the solution of Eq. (5.180) that satisfies the initial conditions

$$y(t_0) = 0, \quad y'(t_0) = 1$$

Then y_1 and y_2 form a fundamental set of solutions of Eq. (5.180). If you compute the Wronskian it will be $W(y_1, y_2)(t_0) = 1$, since the Wronskian is not zero at the point t_0, the functions y_1 and y_2 do form a fundamental set of solutions.

Theorem 5.9: Abel's Theorem

If y_1 and y_2 are solutions of the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

where p and q are continuous on an open interval I, then the Wronskian $W(y_1, y_2)(t)$ is given by

$$W(y_1, y_2)(t) = ce^{-\int p(t)\, dt} \tag{5.188}$$

where c is a certain constant that depends on y_1 and y_2, but not on t. Further, $W(y_1, y_2)(t)$ either is zero for all t in I (if $c = 0$) or else is never zero in I (if $c \neq 0$).

[H*] **Exact Equations**
The equation

$$P(x)y'' + Q(x)y' + R(x)y = 0$$

is said to be exact if it can be written in the form

$$[P(x)y']' + [f(x)y]' = 0$$

where $f(x)$ is to be determined in terms of $P(x), Q(x)$, and $R(x)$. The latter equation can be integrated once immediately, resulting in a first order linear equation for y that can be solved. By equating the coefficients of the preceeding equations and then eliminating $f(x)$, show that a necessary condition for exactness is

$$P''(x) - Q'(x) + R(x) = 0$$

It can be shown that this is also a sufficient condition.

Solution:
Expand the left side of the second ODE

$$P'(x)y' + P(x)y'' + f'(x)y + f(x)y' = 0$$
$$P(x)y'' + \left[P'(x) + f(x)\right] y' + f'(x)y = 0$$

Equate the coefficients of this and the first ODE

$$P'(x) + f(x) = Q(x)$$
$$f'(x) = R(x)$$

Differentiate both sides of the first equation with respect to x.

$$P''(x) + f'(x) = Q'(x)$$

Substitute $R(x)$ for $f'(x)$

$$P''(x) + R(x) = Q'(x)$$

Therefore,

$$P''(x) - Q'(x) + R(x) = 0$$

[H*] **The Adjoint Equation**
If a second order linear homogeneous equation is not exact, it can be made exact by multiplying by an appropriate integrating factor $\mu(x)$. Thus we require that $\mu(x)$ be such that

$$\mu(x)P(x)y'' + \mu(x)Q(x)y' + \mu(x)R(x)y = 0$$

can be written in the form

$$\left[\mu(x)P(x)y'\right]' + [f(x)y]' = 0$$

By equating coefficients in these two equations and eliminating $f(x)$, show that the function μ must satisfy

$$P\mu'' + (2P' - Q)\mu' + (P'' - Q' + R)\mu = 0$$

This equation is known as the adjoint of the original equation and is impotant in the advanced theory of differential equations. In general, the problem of solving the adjoint differential equation is as difficult as that of solving the original equation, so only occasionally is it possible to find an integrating factor for a second order equation.

Solution:

Suppose we have the second-order ODE,

$$P(x)y'' + Q(x)y' + R(x)y = 0$$

To make it exact, multiply both sides by an integrating factor $\mu = \mu(x)$

$$\mu(x)P(x)y'' + \mu(x)Q(x)y' + \mu(x)R(x)y = 0 \tag{5.189}$$

Now that it's exact, it can be written in the form,

$$\left[\mu(x)P(x)y'\right]' + [f(x)y]' = 0$$

Expand the left side

$$\mu'(x)P(x)y' + \mu(x)P'(x)y' + \mu(x)P(x)y'' + f'(x)y + f(x)y' = 0$$

Factor it now

$$\mu(x)P(x)y'' + \left[\mu'(x)P(x) + \mu(x)P'(x) + f(x)\right] y' + f'(x)y = 0$$

Equate the coefficients with those of equation (5.189)

$$\begin{aligned} \mu'(x)P(x) + \mu(x)P'(x) + f(x) &= \mu(x)Q(x) \\ f'(x) &= \mu(x)R(x) \end{aligned}$$

Differentiate both sides of the first equation with respect to x.

$$\mu''(x)P(x) + \mu'(x)P'(x) + \mu(x)P''(x) + f'(x) = \mu'(x)Q(x) + \mu(x)Q'(x)$$

Substitute $\mu(x)R(x)$ for $f'(x)$

$$\mu''(x)P(x) + \mu'(x)P'(x) + \mu(x)P''(x) + \mu(x)R(x) = \mu'(x)Q(x) + \mu(x)Q'(x)$$

Therefore, the adjoint is

$$P\mu'' + (2P' - Q)\mu' + (P'' - Q' + R)\mu = 0$$

[H*] **Example:**
Find the adjoint of the given differential equation
(a) $x^2y'' + xy' + (x^2 - \nu^2)y = 0$ (Bessel's equation)
(b) $(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0$ (Legendre's equation)
(c) $y'' - xy = 0$ (Airy's equation)
Solution:
(a) To make the ODE exact, multiply both sides by an integrating factor $\mu = \mu(x)$

$$x^2\mu(x)y'' + x\mu(x)y' + (x^2 - \nu^2)\mu(x)y = 0 \tag{5.190}$$

Now that it's exact, it can be written in the form

$$\left[x^2\mu(x)y'\right]' + [f(x)y]' = 0$$

Expand the left side

$$2x\mu(x)y' + x^2\mu'(x)y' + x^2\mu(x)y'' + f'(x)y + f(x)y' = 0$$

Factor it now

$$x^2\mu(x)y'' + \left[x^2\mu'(x) + 2x\mu(x) + f(x)\right] y' + f'(x)y = 0$$

Equate the coefficients with those of equation (5.190)

$$\begin{aligned} x^2\mu'(x) + 2x\mu(x) + f(x) &= x\mu(x) \\ f'(x) &= (x^2 - \nu^2)\mu(x) \end{aligned}$$

Differentiate both sides of the first equation with respect to x

$$2x\mu'(x) + x^2\mu''(x) + 2\mu(x) + 2x\mu'(x) + f'(x) = \mu(x) + x\mu'(x)$$

Substitute $(x^2 - \nu^2)\mu(x)$ for $f'(x)$

$$2x\mu'(x) + x^2\mu''(x) + 2\mu(x) + 2x\mu'(x) + (x^2 - \nu^2)\mu(x) = \mu(x) + x\mu'(x)$$

Bring all terms to the left side and combine like terms

$$x^2\mu''(x) + 3x\mu'(x) + (x^2 - \nu^2 + 1)\mu(x) = 0$$

(b) To make the ODE exact, multiply both sides by an integrating factor $\mu = \mu(x)$

$$(1 - x^2)\mu(x)y'' - 2x\mu(x)y' + \alpha(\alpha+1)\mu(x)y = 0 \tag{5.191}$$

Now that it's exact, it can be written in the form

$$\left[(1 - x^2)\mu(x)y'\right]' + [f(x)y]' = 0$$

Expand the left side

$$-2x\mu(x)y' + (1 - x^2)\mu'(x)y' + (1 - x^2)\mu(x)y'' + f'(x)y + f(x)y' = 0$$

Factor it now.

$$(1 - x^2)\mu(x)y'' + \left[(1 - x^2)\mu'(x) - 2x\mu(x) + f(x)\right]y' + f'(x)y = 0$$

Equate the coefficients with those of equation (5.191)

$$\begin{aligned} (1 - x^2)\mu'(x) - 2x\mu(x) + f(x) &= -2x\mu(x) \\ f'(x) &= \alpha(\alpha+1)\mu(x) \end{aligned}$$

Add $2x\mu(x)$ to both sides of the first equation

$$(1 - x^2)\mu'(x) + f(x) = 0$$

Differentiate both sides with respect to x.

$$-2x\mu'(x) + (1 - x^2)\mu''(x) + f'(x) = 0$$

Substitute $\alpha(\alpha+1)\mu(x)$ for $f'(x)$

$$-2x\mu'(x) + (1 - x^2)\mu''(x) + \alpha(\alpha+1)\mu(x) = 0$$

Therefore,

$$(1 - x^2)\mu''(x) - 2x\mu'(x) + \alpha(\alpha+1)\mu(x) = 0$$

(c) To make the ODE exact, multiply both sides by an integrating factor $\mu = \mu(x)$

$$\mu(x)y'' - x\mu(x)y = 0 \tag{5.192}$$

Now that it's exact, it can be written in the form

$$\left[\mu(x)y'\right]' + [f(x)y]' = 0$$

Expand the left side

$$\mu'(x)y' + \mu(x)y'' + f'(x)y + f(x)y' = 0$$

Factor it now

$$\mu(x)y'' + \left[\mu'(x) + f(x)\right] y' + f'(x)y = 0$$

Equate the coefficients with those of equation (5.192)

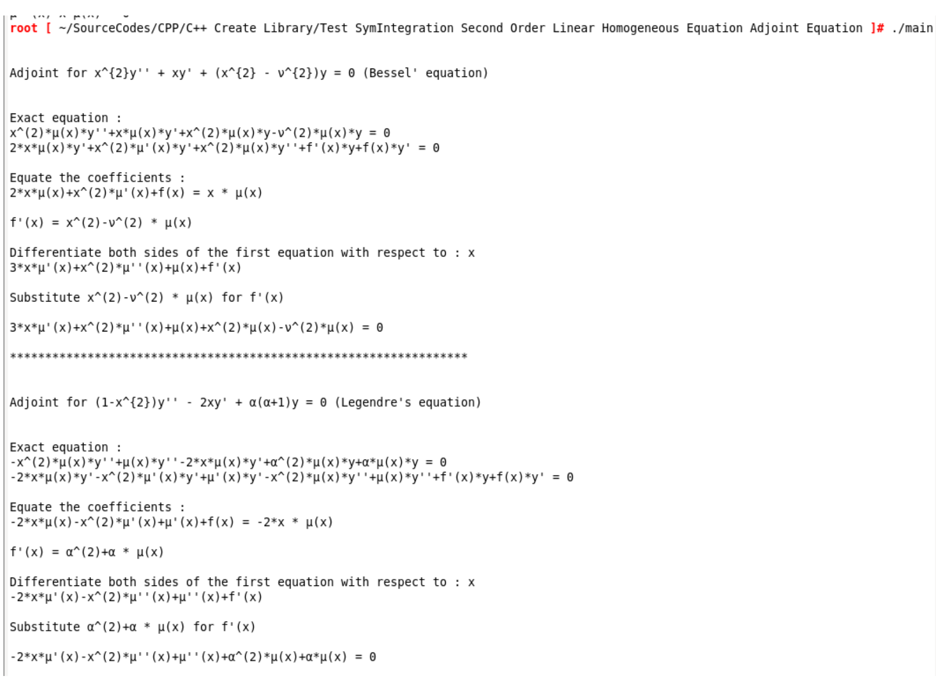
$$\begin{aligned} \mu'(x) + f(x) &= 0 \\ f'(x) &= -x\mu(x) \end{aligned}$$

Differentiate both sides of the first equation with respect to x

$$\mu''(x) + f'(x) = 0$$

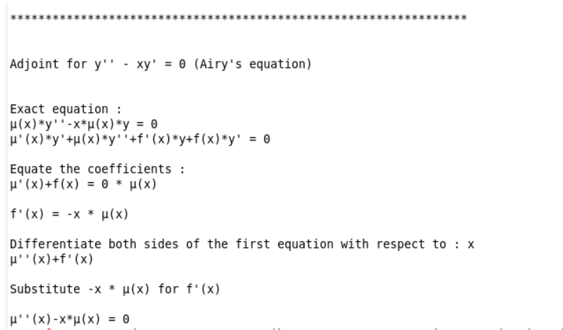
Substitute $-x\mu(x)$ for $f'(x)$

$$\mu''(x) - x\mu(x) = 0$$



```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Second Order Linear Homogeneous Equation Adjoint Equation ]# ./main


Adjoint for x^{2}y'' + xy' + (x^{2} - ν^{2})y = 0 (Bessel' equation)


Exact equation :
x^(2)*µ(x)*y''+x*µ(x)*y'+x^(2)*µ(x)*y-ν^(2)*µ(x)*y = 0
2*x*µ(x)*y'+x^(2)*µ'(x)*y'+x^(2)*µ(x)*y''+f'(x)*y+f(x)*y' = 0

Equate the coefficients :
2*x*µ(x)+x^(2)*µ'(x)+f(x) = x * µ(x)

f'(x) = x^(2)-ν^(2) * µ(x)

Differentiate both sides of the first equation with respect to : x
3*x*µ'(x)+x^(2)*µ''(x)+µ(x)+f'(x)

Substitute x^(2)-ν^(2) * µ(x) for f'(x)

3*x*µ'(x)+x^(2)*µ''(x)+µ(x)+x^(2)*µ(x)-ν^(2)*µ(x) = 0

*****************************************************************


Adjoint for (1-x^{2})y'' - 2xy' + α(α+1)y = 0 (Legendre's equation)


Exact equation :
-x^(2)*µ(x)*y''+µ(x)*y''-2*x*µ(x)*y'+α^(2)*µ(x)*y+α*µ(x)*y = 0
-2*x*µ(x)*y'-x^(2)*µ'(x)*y'+µ'(x)*y'-x^(2)*µ(x)*y''+µ(x)*y''+f'(x)*y+f(x)*y' = 0

Equate the coefficients :
-2*x*µ(x)-x^(2)*µ'(x)+µ'(x)+f(x) = -2*x * µ(x)

f'(x) = α^(2)+α * µ(x)

Differentiate both sides of the first equation with respect to : x
-2*x*µ'(x)-x^(2)*µ''(x)+µ''(x)+f'(x)

Substitute α^(2)+α * µ(x) for f'(x)

-2*x*µ'(x)-x^(2)*µ''(x)+µ''(x)+α^(2)*µ(x)+α*µ(x) = 0
```

Figure 5.70: *The computation of adjoint of the given second order linear differential equation (**SymIntegration/Examples/Differential Equations/Test SymIntegration Second Order Linear Homogeneous Equation Adjoint Equation/main.cpp**).*



```
*****************************************************************

Adjoint for y'' - xy' = 0 (Airy's equation)

Exact equation :
μ(x)*y''-x*μ(x)*y = 0
μ'(x)*y'+μ(x)*y''+f'(x)*y+f(x)*y' = 0

Equate the coefficients :
μ'(x)+f(x) = 0 * μ(x)

f'(x) = -x * μ(x)

Differentiate both sides of the first equation with respect to : x
μ''(x)+f'(x)

Substitute -x * μ(x) for f'(x)

μ''(x)-x*μ(x) = 0
```

Figure 5.71: *The computation of adjoint of the given second order linear differential equation (**SymIntegration/Examples/Differential Equations/Test SymIntegration Second Order Linear Homogeneous Equation Adjoint Equation/main.cpp**).*

[H*] **Euler' Equations**

An equation of the form

$$t^2\frac{d^2y}{dt^2} + \alpha\frac{dy}{dt} + \beta y = 0, \quad t > 0 \tag{5.193}$$

where α and β are real constants, is called an Euler equation.

(a) Let $x = \ln\ t$ and calculate $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ in terms of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(b) Use the results of part (a) to transform Eq. (5.193) into

$$\frac{d^2y}{dx^2} + (\alpha - 1)\frac{dy}{dx} + \beta y = 0 \tag{5.194}$$

Observe that Eq. (5.194) has constant coefficients. If $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions of Eq. (5.194), then $y_1(\ln\ t)$ and $y_2(\ln\ t)$ form a fundamental set of solutions of Eq. (5.193).

Solution:

(a) Make the substitution $x = \ln\ t$ in the ODE. Then

$$e^x = t \to e^{2x} = t^2$$

and the ODE becomes

$$e^{2x}\frac{d^2y}{dt^2} + \alpha e^x\frac{dy}{dt} + \beta y = 0$$

The aim now is to find what the derivatives are in terms of this new variable by using the chain rule.

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} = \frac{dy}{dx}\left(\frac{1}{t}\right) = \frac{dy}{dx}\left(\frac{1}{e^x}\right) = e^{-x}\frac{dy}{dx}$$

$$\begin{aligned}
\frac{d^2y}{dt^2} &= \frac{d}{dt}\left(\frac{dy}{dt}\right) \\
&= \frac{dx}{dt}\frac{d}{dx}\left(e^{-x}\frac{dy}{dx}\right) \\
&= \frac{1}{t}\left(-e^{-x}\frac{dy}{dx} + e^{-x}\frac{d^2y}{dx^2}\right) \\
&= \frac{1}{e^x}\left(-e^{-x}\frac{dy}{dx} + e^{-x}\frac{d^2y}{dx^2}\right)
\end{aligned}$$

(b) Substitute these expressions into the ODE

$$\begin{aligned}
e^{2x}\frac{1}{e^x}\left(-e^{-x}\frac{dy}{dx} + e^{-x}\frac{d^2y}{dx^2}\right) + \alpha e^x\left(e^{-x}\frac{dy}{dx}\right) + \beta y = 0 \\
e^x\left(-e^{-x}\frac{dy}{dx} + e^{-x}\frac{d^2y}{dx^2}\right) + \alpha\frac{dy}{dx} + \beta y = 0 \\
-\frac{dy}{dx} + \frac{d^2y}{dx^2} + \alpha\frac{dy}{dx} + \beta y = 0
\end{aligned}$$

Therefore,

$$\frac{d^2y}{dx^2} + (\alpha - 1)\frac{dy}{dx} + \beta y = 0$$

- Complex Roots of the Characteristic Equation

[H*] Suppose we have the homogeneous second order linear differential equation

$$ay'' + by' + cy = 0 \tag{5.195}$$

We have learned that if we seek solutions of the form $y = e^{rt}$, then r must be a root of the characteristic equation

$$ar^2 + br + c = 0 \tag{5.196}$$

If the roots r_1 and r_2 are real and different, which occurs whenever the discriminant $b^2 - 4ac$ is positive, then the general solution of Eq. (5.195) is

$$y = c_1e^{r_1t} + c_2e^{r_2t} \tag{5.197}$$

Suppose now that $b^2 - 4ac$ is negative. Then the roots of Eq. (5.196) are conjugate complex numbers; we denote them by

$$\begin{aligned}
r_1 &= \lambda + i\mu \\
r_2 &= \lambda - i\mu
\end{aligned} \tag{5.198}$$

where λ and μ are real. The corresponding expressions for y are

$$\begin{aligned}
y_1(t) &= e^{(\lambda+i\mu)t} \\
y_2(t) &= e^{(\lambda-i\mu)t}
\end{aligned} \tag{5.199}$$

[H*] **Euler's Formula**

To assign a meaning to the expressions in Eqs. (5.199), we need to give a definition of the complex exponential function. Recall from Calculus that the Taylor series for e^t about $t = 0$ is

$$e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!}, \quad -\infty < t < \infty \tag{5.200}$$

If we now assume that we can substitute it for t in Eq. (5.200), then we have

$$\begin{aligned} e^{it} &= \sum_{n=0}^{\infty} \frac{(it)^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n)!} + i \sum_{n=0}^{\infty} \frac{(-1)^{n-1} t^{2n-1}}{(2n-1)!} \end{aligned} \tag{5.201}$$

where we have separated the sum into its real and imaginary parts, making use of the fact that $i^2 = -1, i^3 = -i, i^4 = 1$, and so forth. The first series in Eq. (5.201) is precisely the Taylor series for $\cos\ t$ about $t = 0$, and the second is the Taylor series for $\sin\ t$ about $t = 0$. Thus we have

$$e^{it} = \cos\ t + i\ \sin\ t \tag{5.202}$$

Equation (5.202) is known as Euler's formula and is an extremely important mathematical relationship.

There are some variations of Euler's formula that are also worth noting

$$e^{-it} = \cos\ t - i\ \sin\ t \tag{5.203}$$

Further, if t is replaced by μt, then we obtain a generalized version of Euler's formula, namely

$$e^{i\mu t} = \cos\ \mu t + i\ \sin\ \mu t \tag{5.204}$$

Next, we want to extend the definition of the exponential function to arbitrary complex exponents of the form $(\lambda + i\mu)t$. Since we want the usual properties of the exponential function to hold for complex exponents, we certainly want $e^{(\lambda+i\mu)t}$ to satisfy

$$e^{(\lambda+i\mu)t} = e^{\lambda t} e^{i\mu t} \tag{5.205}$$

Then, substituting for $e^{i\mu t}$ from Eq. (5.204), we obtain

$$\begin{aligned} e^{(\lambda+i\mu)t} &= e^{\lambda t}(\cos\ \mu t + i\ \sin\ \mu t) \\ &= e^{\lambda t} \cos\ \mu t + i e^{\lambda t}\ \sin\ \mu t \end{aligned} \tag{5.206}$$

We now take Eq. (5.206) as the definition of $e^{(\lambda+i\mu)t}$. The value of the exponential function with a complex exponent is a complex number whose real and imaginary parts are given by the terms on the right side of Eq. (5.206). Observe that the real and imaginary parts of $e^{(\lambda+i\mu)t}$ are expressed entirely in terms of elementary real-valued functions.

[H*] **Complex Roots; The General Case**

The functions $y_1(t)$ and $y_2(t)$, given by Eqs. (5.199) and with the meaning expressed by Eq. (5.206), are solutions of Eq. (5.195) when the roots of the characteristic equation (5.196) are complex numbers $\lambda \pm i\mu$. Unfortunately, the solutions y_1 and y_2 are complex-valued functions, whereas in general we would prefer to have real-valued solutions, if possible, because the differential equation itself has real coefficients. To find a fundamental set of real-valued solutions let us form the sum and then the difference of y_1 and y_2. We have

$$\begin{aligned} y_1(t) + y_2(t) &= e^{\lambda t}(\cos\ \mu t + i\ \sin\ \mu t) + e^{\lambda t}(\cos\ \mu t - i\ \sin\ \mu t) \\ &= 2e^{\lambda t}\cos\ \mu t \end{aligned}$$

and

$$\begin{aligned} y_1(t) - y_2(t) &= e^{\lambda t}(\cos\ \mu t + i\ \sin\ \mu t) - e^{\lambda t}(\cos\ \mu t - i\ \sin\ \mu t) \\ &= 2ie^{\lambda t}\sin\ \mu t \end{aligned}$$

Hence, neglecting the constant multipliers 2 and $2i$, respectively, we have obtained a pair of real-valued solutions

$$\begin{aligned} u(t) &= e^{\lambda t}\cos\ \mu t \\ v(t) &= e^{\lambda t}\sin\ \mu t \end{aligned} \tag{5.207}$$

Observe that u and v are simply the real and imaginary parts, respectively, of y_1.
By direct computation you can show that the Wronskian of u and v is

$$W(u, v)(t) = \mu e^{2\lambda t} \tag{5.208}$$

Thus, as long as $\mu \neq 0$, the Wronskian W is not zero, so u and v form a fundamental set of solutions. If $\mu = 0$ then the roots are real and repeated. Consequently, if the roots of the characteristic equation are complex numbers $\lambda \pm i\mu$, with $\mu \neq 0$, then the general solution of Eq. (5.195) is

$$y = c_1 e^{\lambda t}\cos\ \mu t + c_2 e^{\lambda t}\sin\ \mu t \tag{5.209}$$

[H*] **Example:**
Find the solution of the initial value problem

$$16y'' - 8y' + 145y = 0, \quad y(0) = -2, \quad y'(0) = 1 \tag{5.210}$$

Solution:

The characteristic equation is $16r^2 - 8r + 145 = 0$ and its roots are $r = \frac{1}{4} \pm 3i$. Thus the general solution of the differential equation is

$$y = c_1 e^{t/4}\cos\ 3t + c_2 e^{t/4}\sin\ 3t \tag{5.211}$$

To apply the first initial condition, we set $t = 0$ in Eq. (5.211); this gives

$$y(0) = c_1 = -2$$

For the second initial condition we must differentiate Eq. (5.211) and then set $t = 0$. In this way we find that

$$y'(0) = \frac{1}{4}c_1 + 3c_2 = 1$$

from which $c_2 = \frac{1}{2}$. Using these values of c_1 and c_2 in Eq. (5.211), we obtain

$$y = -2e^{t/4}\cos\ 3t + \frac{1}{2}e^{t/4}\sin\ 3t \tag{5.212}$$

as the solution of the initial value problem (5.210).

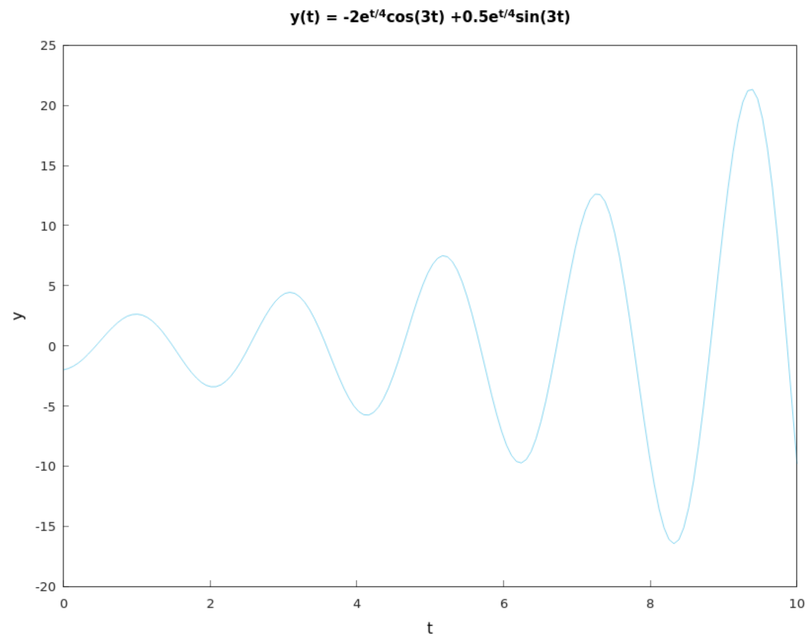


Figure 5.72: *The graph of the solution* $y = -2e^{t/4}\cos\ 3t + \frac{1}{2}e^{t/4}\sin\ 3t$ *of the initial value problem* $16y'' - 8y' + 145y = 0, y(0) = -2, y'(0) = 1$ *(**Hamzstplot/Examples with Makefile/Plot 2D Growing Oscillation/main.cpp**).*

In this case we observe that the solution is a growing oscillation. The trigonometric factors in Eq. (5.212) determine the oscillatory part of the solution, while the exponential factor (with a positive exponent) causes the magnitude of the oscillation to increase with time.

Suppose we want to play around with interactive graph, we can use Hamzstlab-Mathematics and change the parameters of $a, b, c, y0, dy0$ to see the solution graph directly for different initial value problem. The second order linear differential equation will be of the form

$$ay'' + by' + cy = 0$$

with initial value

$$y(0) = y0, \quad y'(0) = dy0$$

Figure 5.73: *The interactive plot for solution of initial value problem for second order differential equation Eq. (5.210), we can change the parameters and the graph will adjust in an instant. (**Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp**).*

- Repeated Roots; Reduction of Order

[H*] We showed how to solve the equation

$$ay'' + by' + cy = 0 \tag{5.213}$$

when the roots of the characteristic equation

$$ar^2 + br + c = 0 \tag{5.214}$$

either are real and different or are complex conjugates. Now we consider the third possibility, namely, that the two roots r_1 and r_2 are equal. This case is transitional between the other two and occurs when the discriminant $b^2 - 4ac$ is zero. Then it follows from the quadratic formula that

$$r_1 = r_2 = -\frac{b}{2a} \tag{5.215}$$

The difficulty is immediately apparent; both roots yield the same solution

$$y_1(t) = e^{-\frac{bt}{2a}} \tag{5.216}$$

of the differential equation (5.213), and it is not obvious how to find a second solution.

To find a second solution, we assume that

$$y = v(t)y_1(t) = v(t)e^{-\frac{bt}{2a}} \tag{5.217}$$

and substitute for y in Eq. (5.213) to determine $v(t)$. We have

$$y' = v'(t)e^{-\frac{bt}{2a}} - \frac{b}{2a}v(t)e^{-\frac{bt}{2a}} \tag{5.218}$$

and

$$y'' = v''(t)e^{-\frac{bt}{2a}} - \frac{b}{a}v'(t)e^{-\frac{bt}{2a}} + \frac{b^2}{4a^2}v(t)e^{-\frac{bt}{2a}} \tag{5.219}$$

Then, by substituting in Eq. (5.213), we obtain

$$\left[a\left(v''(t) - \frac{b}{a}v'(t) + \frac{b^2}{4a^2}v(t)\right) + b\left(v'(t) - \frac{b}{2a}v(t)\right) + cv(t)\right]e^{-\frac{bt}{2a}} = 0 \tag{5.220}$$

Canceling the factor $e^{-\frac{bt}{2a}}$, which is nonzero, and rearranging the remaining terms, we find that

$$av''(t) + (-b+b)v'(t) + \left(\frac{b^2}{4a} - \frac{b^2}{2a} + c\right)v(t) = 0 \tag{5.221}$$

The term involving $v'(t)$ is obviously zero. Further, the coefficient of $v(t)$ is $c - \left(\frac{b^2}{4a}\right)$, which is also zero because $b^2 - 4ac = 0$ (repeated roots). Thus, Eq. (5.221) reduces to

$$v''(t) = 0$$

therefore

$$v(t) = c_1 + c_2 t$$

Hence, from Eq. (5.217), we have

$$y = c_1 e^{-\frac{bt}{2a}} + c_2 t e^{-\frac{bt}{2a}} \tag{5.222}$$

Thus y is a linear combination of the two solutions

$$y_1(t) = e^{-\frac{bt}{2a}}, \quad y_2(t) = te^{-\frac{bt}{2a}} \tag{5.223}$$

The Wronskian of these two solutions is

$$W(y_1, y_2)(t) = \begin{vmatrix} e^{-\frac{bt}{2a}} & te^{-\frac{bt}{2a}} \\ -\frac{b}{2a}e^{-\frac{bt}{2a}} & \left(1 - \frac{bt}{2a}\right)e^{-\frac{bt}{2a}} \end{vmatrix} = e^{-\frac{bt}{a}} \tag{5.224}$$

Since $W(y_1, y_2)(t)$ is never zero, the solutions y_1 and y_2 given by Eq. (5.223) are a fundamental set of solutions. Further Eq. (5.222) is the general solution of Eq. (5.213) when the roots of the characteristic equation are equal.

[H*]

VI. Higher Order Linear Equations

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

VII. Series Solutions of Second Order Linear Equations

- Power Series

[H*] The first
[H*]

- Series Solutions Near an Ordinary Point

[H*] The first
[H*]

VIII. Laplace Transform

- Definition

[H*] The first
[H*]

- Solution of Initial Value Problems

[H*] The first
[H*]

IX. Systems of First Order Linear Equations

- Introduction

[H*] The first
[H*]

- Linear Algebraic Equations

[H*] The first
[H*]

X. Numerical Methods

- The Runge-Kutta Method

[H*] The first
[H*]

- Multistep Methods

[H*] The first
[H*]

XI. Nonlinear Differential Equations and Stability

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

XII. Boundary Value Problems and Sturm-Liouville Theory

- The Occurrence of Two-Point Boundary Value Problems

[H*] The first
[H*]

- Nonhomogeneous Boundary Value Problems

[H*] The first
[H*]

Chapter 6

Partial Differential Equations

*"PDE makes my Mathematics Learning more colorful..." - **Glanz***

*"Learn from basic Ordinary Differential Equation, don't jump to PDE yet (but still help Glanz learning PDE on 2023)" - **Sentinel***

There is always a funny story behind every chapter in Lasthrim Projection, it is only 3 days before Partial Differential Equation exam, and after surgery on Glanz' left leg, miss 2 weeks of classes, she thinks a chapter dedicated for PDE(Partial Differential Equation) is better than include just a snippet in the **Differential Equation** chapter, actually Freya is the one ordered her to add this chapter. We shall see why this chapter exists, the book bought by Glanz has her drawing of Stacy (Valhalla Projection' subordinate in Coppito, L'Aquila and Todai / University of Tokyo).

There are a lot of books talking about Partial Differential Equations, a lot of people wrote in their blogs and websites, and create the video to explain it. We read and watch a lot then try to summarize it after we able to comprehend. Hopefully, after certain time if we read it again, the memory will be refreshed quickly.

I. Preliminaries

[H*] In many important physical problems there are two or more independent variables governing the process, thus the corresponding mathematical models involve partial differential equations. For solving partial differential equations we use a method separation of variables, with its essential feature to replace the partial differential equation by a set of ordinary differential equations, which then will be solved by the given initial or boundary conditions.

The desired solution of the partial differential equations is expressed as a sum, or an infinite series, formed from solutions of the ordinary differential equations.

We will call Partial Differential Equation with PDE, interchangeably.

[H*] The solution of partial differential equations, both analytical and numerical, is the single, most important topic in applied analysis. Virtually all areas of science and engineering phrase problems in the language of PDEs, and every practicing professional is expected to be adept at their solution.

The basic ideas and concepts related to solutions of PDEs are much the same as those used to solve ordinary differential equations (ODEs). Because PDEs are concerned with differential equations in two or more independent variables, it is more common to encounter computational problems.

Typically, for PDEs the memory and computation time requirements are increased by an order of magnitude, round-off error is more likely a concer, patholofies in the solution are more frequent, and overall, the calculation is likely to be more complicated, bigger, and more expensive.

II. Preliminaries Concepts

i. The Standard Form

[H*] The standard form of a second-order, linear, partial differential equation in two independent variables is

$$A\frac{\partial^2 u}{\partial x^2} + 2B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} = F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right) \tag{6.1}$$

where $u(x, y)$ is the dependent variable, and $F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$ is an arbitrary function of x and y, and is linear in u and its first derivatives.

		A	B	C	F
$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$	One-dimensional wave equation	1	0	- $\frac{1}{c^2}$	0
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	Two-dimensional Laplace's equation	1	0	1	0
$D\frac{\partial^2 T}{\partial x^2} = -\frac{\partial T}{\partial t}$	One-dimensional diffusion equation	D	0	0	$-\frac{\partial T}{\partial t}$

Table 6.1: *Homogeneous, second-order partial differential equations.*

ii. Boundary Conditions

[H*] As with ordinary differential equations, the method of solution of a PDE is dictated by the types of boundary conditions specified in the problem. For ODEs, the choices are either initial value problems or boundary value problems. In either case, boudnary values are required to guarantee that the equation has a unique and stable solution.

In initial value problems, sufficient information regarding the function and its derivatives are given at a single point. Then using finite difference expressions of the form

$$y_{i+1} = y_i + \frac{dy}{dx}\Big|_i \triangle x + \frac{1}{2}\frac{d^2 y}{dx^2}\Big|_i (\triangle x)^2 + \ldots \tag{6.2}$$

the solution is constructed in the neighborhood of the point, in a stepping manner. (The initial conditions specify the function and its first derivatives, and the differential equation is used to determine higher-order derivatives in the series). Thus, the solution is constructed emanating from the initial point, and the natural method of solution is a stepping procedure.

iii. Classification of PDEs

[H*] To specify the appropriate boundary conditions sufficient to guarantee a unique, stable solution to a particular PDE, the equation is first classified as being one of three types: elliptic, parabolic, or hyperbolic, by evaluating the quantity $(B^2 - AC)$, where A, B, and C are the coefficients in the standard form of the PDE. The classification is the

$$B^2 - AC < 0 \quad \text{(Elliptic)}$$
$$B^2 - AC = 0 \quad \text{(Parabolic)}$$
$$B^2 - AC > 0 \quad \text{(Hyperbolic)}$$

III. Initial Value Type Methods

[H*]

IV. Derivation of Partial Differential Equations

[H*] Partial differential equations arise in many physical problems. Three known partial differential equations are Heat equation, wave equation, and Laplace's equation.
[H*] The key defining property of a partial differential equation (PDE) [25] is that

- There is more than one independent variable $x, y, \ldots$
- There is a dependent variable that is an unknown function of these variables $u(x, y, \ldots)$
- $\partial u/\partial x$ denoted as the partial derivative of the dependent variable toward the independent variable x.

[H*] When a solution function has two or more independent variables, say $u(x, y)$, then this function can make a partial differential equation by specifying a relation between its' derivatives [14], such as

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Another way of writing partial differential equation is to see it as an identity that relates the independent variables, the dependent variable u, and the derivatives of u. It can be written as

$$F(x, y, u, u_x, u_y) = 0 \tag{6.3}$$

with $u = u(x, y)$, a function in x and y terms as the independent variables.
The general second-order PDE in two independent variables is

$$F(x, y, u, u_x, u_y, u_{xx}, u_{xy}, u_{yy}) = 0 \tag{6.4}$$

A solution of a PDE is a function $u(x, y, \ldots)$ that satisfies the equation identically.
[H*] For PDE, the distinction between the independent variables and the dependent variable (the unknown / the solution) is always maintained.

[H*] An example of a general linear second-order partial differential equation is like this:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} = 0$$

The general nonlinear second-order partial differential equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} + u^2 = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} + \sin(u) = 0$$

The differences are:

1. The linear partial differential equation will contain the unknown / the solution with power of 1, the nonlinear will contain the unknown / the solution with power other than 1 (could be 2 or argument of $\exp, \log, \sin$).

[H*] A linear partial differential equation is homogeneous if the unknown or the solution that satisfies the partial differential equation is 0, for example:

$$\frac{\partial^2 u}{\partial t^2} + c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

and the solution of the partial differential equation above is $u(x,t) = 0$, thus the partial differential equation is homogeneous.

A homogeneous linear differential equation has an important property:

$$\alpha u + \beta v \tag{6.5}$$

are solutions for $\alpha, \beta \in \mathcal{R}$, whenever u and v are solutions of the equation.

A linear partial differential equation is non-homogeneous if the unknown or the solution that satisfies the partial differential equation is not 0, for example:

$$\frac{\partial u}{\partial t} = 2u + \sin(4\pi t)$$

is equivalent to

$$\frac{\partial u}{\partial t} - 2u = \sin(4\pi t)$$

When all terms involving the unknown / the solution on the left side of equation and the right side is not 0 then it is non-homogeneous

[H*] If u and v is any function and $\mathcal{L}$ is an operator, then $\mathcal{L}v$ is a new function. Thus the definition for linearity is

$$\mathcal{L}(u+v) = \mathcal{L}v + \mathcal{L}v\mathcal{L}(cu) = c(u) \tag{6.6}$$

Hence

$$\mathcal{L}(u) = 0 \tag{6.7}$$

is called a homogeneous linear equation. Then

$$\mathcal{L}(u) = g \tag{6.8}$$

where $g \neq 0$ is a given function of the independent variables, is called a non-homogeneous linear equation, for any functions u, v and any constant c.

[H*] Recall this Taylor series expansion of a function $f(z)$ about the point $z = a$, the expansion is given by

$$f(z) = f(a) + f'(a) + f'(a)(z-a) + f''(a)\frac{(z-a)^2}{2!} + f'''(a)\frac{(z-a)^3}{3!}$$

where $f'(a)$ means the derivative of $f(z)$ with respect to z evaluated at the point $z = a$, $f''(a)$ is the second derivative of $f(z)$ with respect to z evaluated at the point $z = a$, etc. If, in the above equation we let

$$z = a + h \quad \text{and} \quad a = x$$

we obtain

$$f(x+h) = f(x) + f'(x)h$$

Thus the value of $f(x)$ at the point $x + h$ (near to x) is given by the value of the function at the point x plus the value of its first derivative evaluated at x, multiplied by the distance $[(x+h) - x = h]$ between the two points.

[H*] The partial derivative of a function $u(x, y)$ with respect to x at the point x, is denoted by

$$\frac{\partial u}{\partial x}$$

[H*] The general solution of the partial differential equation involves as many arbitrary functions as the order of the equation (the order of the highest partial differential coefficient in the equation), for example:

$$\frac{\partial u(x,t)}{\partial t} = -D\frac{\partial^2 u(x,t)}{\partial x^2}$$

has order of 2, we can also call it second-order partial differential equation, this form of PDE is known as the Heat Equation.

[H*] If $w(x, y, z, t)$ is a function of time and the three rectangular coordinates of a point in space, then the following are partial differential equations with order of 2:

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = 0$$

$$a^2\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) = \frac{\partial w}{\partial t}$$

$$a^2\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) = \frac{\partial^2 w}{\partial t^2}$$

The first equation is called Laplace's equation, the second is the heat equation in 3-dimension, and the third is the wave equation in 3-dimension.

[H*] The conditions of partial differential equations are classified into two categories [1]:

1. **Initial Value Problem (IVP)**
   A partial differential equation with initial conditions will have a dependent variable and a sufficient number of its derivatives are prescribed at the initial point of domain.
2. **Boundary Value Problem (BVP)**
   A partial differential equations with its dependent variable and a sufficient number o its derivatives are prescribed at the boundary of domain.

For example consider this heat equation or a metal rod:

$$\frac{\partial u(x,t)}{\partial t} = -D\frac{\partial^2 u(x,t)}{\partial x^2}, \quad 0 \leq x \leq 20,\ t > 0$$

with $u(x,t)$ denotes the temperature of the rod at length-x at time t. The length of the rod is 20 cm.
A lot of people sometimes write u to replace $u(x,t)$, it could cause confusion to those newly learning differential equation, we prefer to write the whole as $u(x,t)$ to denote all the dependent variables that is used by the function u. Just like when we first learning calculus we write $f(x)$ not only writing f.

Now for the initial condition, it will be written like this:

$$u(x,0) = \begin{cases} x, & 0 \leq x \leq 10 \\ 20 - x, & 10 \leq x \leq 20 \end{cases}$$

and the boundary conditions:

$$u(0,t) = 0$$
$$u(L,t) = 0$$

both ends of the metal rod will be cooled at 0^0 perhaps attaching both ends at ice cube. From this we can learn how the temperature will behave on the rod after certain period of time.

[H*] Consider this PDE:

$$\frac{\partial^2 u(x,t)}{\partial t^2} - \frac{\partial^2 u(x,t)}{\partial x^2} = 0, \quad 0 \leq x \leq L,\ t > 0$$

There are four broad categories of boundary conditions:

1. **Dirichlet boundary conditions**
   On the boundary, the values of the dependent variable are specified.
   $$u(0,t) = 0, \quad u(L,t) = 0$$
2. **Neumann boundary conditions**
   On the boundary, the normal derivative of the dependent variable is specified.
   $$\frac{\partial u(0,t)}{\partial x} = 0, \quad \frac{\partial u(L,t)}{\partial x} = 0$$
3. **Cauchy boundary conditions**
   On the boundary, both the values of the dependent variable and its normal derivative are specified.
   $$u(0,t) = 0, \quad \frac{\partial u(L,t)}{\partial x} = 0$$
4. **Robin boundary conditions**
   On the boundary, a linear combination of the dependent variable and its normal derivatives are specified.
   $$u(0,t) = 1, \quad \frac{\partial u(L,t)}{\partial x} + u(L,t) = 0$$

[H*] A second-order partial differential equation of the form

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu < 0$$

is elliptic, while

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$$

is parabolic, and

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu > 0$$

is hyperbolic.

[H*] To have a unique, stable solution, each class of PDEs requires a different class of boundary conditions

1. Dirichlet or Neumann boundary conditions on a closed boundary surrounding the region of interest are the requirements to elliptic equations. Other boundary conditions are either insufficient to determine a unique solution, overly restrictive or lead to instabilities.
2. Cauchy boundary conditions on an open surface are the requirements to elliptic equations. Other boundary conditions are either too restrictive for a solution to exist, or insufficient to determine a unique solution.
3. Parabolic equations require Dirichlet or Neumann boundary conditions on an open surface. Other boundary conditions are too restrictive.

- Two-Point Boundary Value Problems

[H*] Consider this differential equation of second order

$$y'' + p(t)y' + q(t)y = g(t) \tag{6.9}$$

with the initial conditions

$$y(t_0) = y_0 \quad y'(t_0) = y'_0 \tag{6.10}$$

[H*] Consider this independent variables: $x, y, z, \ldots$
and now an unknown function of these variables can be written as below

$$u(x, y, \ldots) \tag{6.11}$$

We denote the derivative with subscript, for example a derivative of u with respect to the independent variable x is

$$\frac{\partial u}{\partial x} = u_x \tag{6.12}$$

for y:

$$\frac{\partial u}{\partial y} = u_y \tag{6.13}$$

[H*] **Partial Differential Equation**
A PDE is an identity that relates the independent variables $(x, y, \ldots)$ with the dependent variable u, and the partial derivatives of u. It can be written as

$$F(x, y, u(x, y), u_x(x, y), u_y(x, y)) = F(x, y, u, u_x, u_y) = 0 \tag{6.14}$$

This is the most general PDE in two independent variables x and y, of first order.

[H*] **Order**
The order of an equation is the highest derivative that appears.

For example:

* $u_t + 2u_x$ is a first order partial differential equation
* $u_{xx} - u_t$ is a second order partial differential equation

[H*] **Solution**
A solution to a PDE is a function $u(x, y, \dots)$ that satisfies the equation identically.

In PDEs, the distinction between the independent variables and the dependent variable is always maintained.

V. Fourier Series

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

VI. Even and Odd Functions

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

VII. The Wave Equation

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

VIII. The Heat Equation

- General Theory of nth Order Linear Equations

[H*] This heat equation is one of the most famous partial differential equation that we will try to simulate, the applications are tremendous, from Black-Scholes equation derived from this, create a long lasting ice cream, to be able to create better computer components will need the comprehension of this equation as well.

[H*]

- The Method of Variation of Parameters

[H*] The first

[H*]

IX. The Laplace Equation

- General Theory of nth Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

X. The Navier-Stokes Equation

- Introduction

[H*] The Navier-Stokes equations are partial differential equations which describe the motion of viscous fluid substances, named after French engineer and physicist Claude-Louis Navier and Anglo-Irish physicist and mathematician Georrge Gabriel Stokes. They were developed over several decades of progressively building the theories, from 1822 (Navier) to 1842-1850 (Stokes)

[H*] The Navier-Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. They are sometimes accompanied by an equation of state relating pressure, temperature and density. They are arise from applying Isaac Newton's second law to fluid motion, together with the assumption that the stress in the fluid is the sum of a diffusing viscous term (proportional to the gradient of velocity) and a pressure term-hence describing viscous flow.

The difference between them and the closely related Euler equations is that Navier-Stokes equations take viscosity into account while the Euler equations model only inviscid flow. As a result, the Navier-Stokes are a parabolic equation and therefore have better analytic properties, at the expense of having less mathematical structure (e.g. they are never completely integrable).

[H*] The Navier-Stokes equations are useful because they describe the physics of many phenomena of scientific and engineering interest. They may be used to model the weather, ocean currents, water flow in a pipe and air flow around a wing. The Navier-Stokes equations, in their full and simplified forms, help with the design of aircraft and cars, the study of blood flow, the design of power stations, the analysis of pollution, and many other things. Coupled with Maxwell's equations, they can be used to model and study magnetohydrodynamics.

Despite their wide range of practical uses, it has not yet been proven whether smooth solutions always exist in three dimensions, i.e., whether they are infinitely differentiable (or even just bouned) at all points in the domain.

- Flow Velocity

[H*] The solution of the equations is a flow velocity. It is a vector field-to every point in a fluid, at any moment in a time interval, it gives a vector whose direction and magnitude are those of the velocity of the fluid at that point in space and at that moment in time.

[H*]

- Cross-Sectional Area Computation

[H*] The cross-sectional area is the area of a two-dimensional shape that results from slicing a three-dimensional object perpendicular to a given axis. Most of the time in three dimensions we will need to determine the cross-sectional area for engineering problem (electrical conductivity, strength of materials), in Navier-Stokes it will be used to compute the cross-sectional area of a fluid that is flowing through a pipe or a hose.

Example: Loaf of bread, the surface of a slice of bread is the cross-sectional area.

[H*] To take a cross-sectional area, first you decide the plane that will be the slice / the area that the water will go through, in three dimension we have 3 planes: xy-plane, yz-plane, and xz-plane.

Then cut at the axis that is not the basis of the plane / building the plane. For example, xy-plane is built by x axis and y axis, so the axis that is not the basis is z.

Now we will use a cylinder in three dimension, for better illustration.

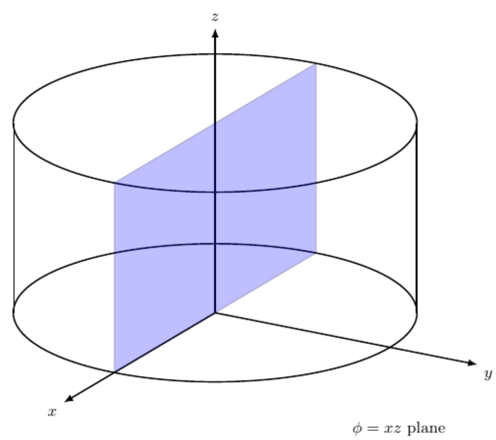


Figure 6.1: *The cross-sectional area of the cylinder in this figure is the blue plane ϕ = xz-plane, it is built by x and z axis, and we cut it at $y = 0$.*

The cross-sectional area of Figure (6.1) is

$$A = 2r * h$$

with r as the radius of the circle and h is the cylinder height. It is only a theory to choose this cross-sectional area, if we make a fluid flow through a cylinder along the y-axis, then the cylinder will have to turn into a box, to comply with the cross-sectional area. As a rule of thumb, if we have a cylinder in three dimension, the water will always flow through the cross-sectional area of a circle, which in the Figure (6.1) it is the area made by xy-plane, then cut at $z = c$,with c as constant.

Chapter 7

Probability and Statistics

"LAM VAM RAM YAM HAM OM AH." - ***7 Chakras Chanting***

In 1747 the government of Sweden conducted its first population census to find out how many citizens were living in the country. Today, governments all over the world use regular censuses to collect data about their citizensso that they have sufficient information to plan the sevices needed now and in the future.

Once information has been collected, it needs to be displayed in a meaningful way and then analysed if it is to be used constructively. Most of the time statistics can be easilly communicated with diagrams, graphs or tables (after transforming and processing tons of data into a knowledge).

In 1854 when Russia was at war with Turkey, England and France went to Turkey's aid and sent soldiers to fight in the Crimea. Florence Nightingale went to Scutari, one of the worst hospitals, there she kept records of all the patients, documenting the reasons for their death or recovery. She realised that pages of figures and tables are difficult to interpret and to explain. So she drew one of the first circular histograms, or 'rose diagrams,' to illustrate the causes of soldiers' death during the war.

The impact was immediate: pages of figures were condensed into a single, clear graphic image. It is fair to imagine that Florence Nightingale's diagram was an important part of her campaign to improcve the conditions im military hospitals [6].

I. Processing and Summarizing Data with Statistics

[H*] Statistical methods are used to analyze data from an industrial process (manufacturing, energy sources, drug testing). Statistical methods are involved with gathering of information or **scientific data**, which then will be processed to extract the knowledge from the data and improve the **quality** of the process.

[H*] In practice, statisticians and data scientists often collect data in various ways, including experimental studies, observational studies, longitudinal studies, survey sampling, and data scraping. Then to gain insight from the data, one may consider different data configurations such as [15]

1. **Single sample**

A case where all observations are considered to represent items from a homogeneous population. The configuration of the data takes the form: $x_1, x_2, \dots, x_n$.

2. **Single sample over time (time-series)**
The configuration of the data takes the form $x_{t_1}, x_{t_2}, \dots, x_{t_n}$ with time points $t_1 < t_2 < \dots < t_n$.

3. **Two samples**
Similar to the single sample case, only now there are two populations. The configuration of the data takes the form: $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$.

4. **Generalizations from two samples to k samples**
Each to potentially different sample size $n_1, n_2, \dots, k$.

5. **Observations in pairs (2-tuples)**
In this case, although similar to the two sample case, each observation is a tuple of points, (x, y). Hence, the configuration of the data is $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.

6. **Generalizations from pairs to vectors of observations**
$(x_{11}, \dots, x_{1p}), \dots (x_{n1}, \dots, x_{np})$.

7. **Other configurations**
Including relationship data (graphs of connection), images, and many more.

Definition 7.1: Histograms

In both the single sample and time-series case, considering frequencies of occurrences is generally an insightful way to visualize the data. The most standard mechanism for plotting frequencies is the histogram. Mathematically, a histogram can be defined as follows. First denote the support of the observations via $[l, m]$, where l is the minimal observation and m is the maximal observation. Then the interval $[l, m]$ is partitioned into a finite set of bins $B_1, B_2, \dots, B_L$, and the frequency in each bin is recorded via

$$f_j = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_i \in B_j, \quad \text{for } j = 1, \dots, L\} \tag{7.1}$$

Here $\mathbf{1}\{\}$ is 1 if $x_i \in B_j$ or 0 if $x_i \notin B_j$. We have that $\sum f_j = 1$ and hence $f_1, \dots, f_L$ is a discrete probability distribution. Histogram is then just a visual representation of this discrete probability distribution (the frequencies).

To represent the widths for each bin, we use a histogram function $h(x)$ which is a scaled plot of the frequencies $f_1, \dots, f_L$. The function $f(x)$ is defined for any $x \in [l, m]$. It is constructed by staying constant on all calues $x \in B_j$, at a height of $\frac{f_j}{|B_j|}$, where $|B_j|$ is the width of the bin j.

i. Descriptive Statistics Computation

Suppose we have a univariate data of a Mathematics examination result for 25 students, compute its' mean, median, mode, standard deviation, and variance.

Solution:

To compute the mean we will use this formula:

$$\bar{x} = \frac{\sum x}{n} \tag{7.2}$$

Mode is the value that occur frequently, and Median is the middle value after the data is sorted, if the size of the data is even, then the median is the average of the two middle data.

Standard deviation can be computed with this formula:

$$S_x = \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2} \tag{7.3}$$

Standard deviation is the square root of variance.

In SymIntegration there is a function that can be used to compute the standard descriptive statistics, it is:
double descriptivestatistics(vector<double>)
the input is a vector with double as the data type, since it can also represents integer.

We have saved the grade data in a textfile named **vectorx.txt**, we can just load / call it from the **main.cpp** file.

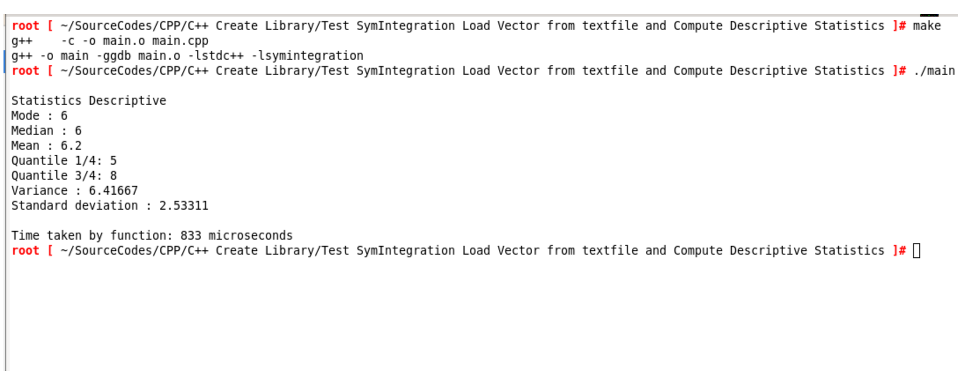


Figure 7.1: *The computation of descriptive statistics with SymIntegration took 833 microseconds (**SymIntegration/Examples/Statistics/Test SymIntegration Load Vector from textfile and Compute Descriptive Statistics/main.cpp**).*

II. Variability

[H*] Life would be simple for scientists if the observed product were always the same and were always on target, there would be no need for for statistical methods. Statisticians make use of fundamental laws of probability and statistical inference to draw conclusions about scientific systems.

Information is gathered in the form of **samples**, or collections of **observations**. Samples are collected from **populations**, which are collections of all individuals or individual items of a particular type.

Research scientists gain much from scientific data. Data provide understanding of scientific phenomena.

III. Correlation

[H*] In Darrell Huff's book **How to Lie with Statistics**, he refers to a study that claimed that non-smoking college students gained better grades than students who smoked. The study concluded that 'smoking leads to poor college grades.' Huff questioned the findings of the study, arguing that instead of smoking being the cause of poor grades, perhaps 'low grades depressed students and caused them to smoke,' or maybe 'students with lower grades were more sociable and therefore more likely to smoke.' Measuring the relationship between two sets of data is a useful tool to establish how one variable changes with another, but it is limited; it does not tell you why this relationship exists. the data in the study clearly showed a relationship between smoking and poor grades, but it does not explain why this relationship exists.

Data that consists of measurements of two variables taken from each individual in a sample is called bivariate data; the relationship between the two variables is referred to as the correlation.

For bivariate data we try to classify one of two variables as 'independent' and the other as 'dependent.' The independent variable is the one that can be controlled by the person conducting the experiment or study; it is hypothesised to 'cause' some kind of effect to the dependent variable. the dependent variable is the variable that is just observed without being controlled, and is supposed to show the 'effect.'

[H*] The line of best fit on a scatter diagram is drawn to give the best representation of the correlation between the two variables. The line of best fit is the one that has an approximately even spread of the data points either side of it.

[H*] **Pearson's product moment correlation coefficient**

Pearson product moment correlation coefficient (PMCC) is a number, usually denoted by r, which can take any value between -1 and $+1$. Its sign indicates the type of correlation, and its magnitude (size) indicates the strength of correlation.

1. $r = +1$ means that there is perfect positive correlation
2. $r = 0$ means that there is no correlation
3. $r = -1$ means that there is perfect negative correlation

If $-0.5 < r < 0.5$, it is difficult to draw a line of best fit that has any meaning, because the data is just too scattered.

The formula to compute the PMCC is

$$r = \frac{S_{xy}}{S_x S_y} \tag{7.4}$$

where S_{xy} is the covariance of x and y, S_x is the standard deviation of the x values, and S_y is the standard deviation of the y values.

The formula that act as the foundation blocks are:

$$\bar{x} = \frac{\sum x}{n} \tag{7.5}$$

$$\bar{y} = \frac{\sum y}{n} \tag{7.6}$$

$$S_x = \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2} \tag{7.7}$$

$$S_y = \sqrt{\frac{\sum y^2}{n} - (\bar{y})^2} \tag{7.8}$$

$$S_{xy} = \frac{\sum xy}{n} - (\bar{x})(\bar{y}) \tag{7.9}$$

with n as the number of observations / data points.

[H*] The formula for the regression line is

$$y - \bar{y} = \frac{S_{xy}}{(S_x)^2}(x - \bar{x}) \tag{7.10}$$

i. Regression Line Plot and Pearson's Correlation Computation

Suppose we have a bivariate data as follow Compute the Pearson's correlation (PMCC), then plot

Distance, x (km)	4	8	5	10	6
Time, y (minutes)	15	35	12	40	24

Table 7.1: *Distance travelled to school and travel time*

the data on a scatter diagram and draw the line of best fit.

Solution:

We will have

$$\begin{aligned}
\bar{x} &= \frac{\sum x}{n} = 6.6 \\
\bar{y} &= \frac{\sum y}{n} = 25.2 \\
S_x &= \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2} = 2.154 \\
S_y &= \sqrt{\frac{\sum y^2}{n} - (\bar{y})^2} = 10.907 \\
S_{xy} &= \frac{\sum xy}{n} - (\bar{x})(\bar{y}) = 22.48
\end{aligned}$$

Hence

$$r = \frac{S_{xy}}{S_x S_y} = 0.957$$

The value of r is close to $+1$, showing that there is strong positive correlation between the distance travelled to school and the time taken.

Now we will compute the regression line formula

$$\begin{aligned}
y - \bar{y} &= \frac{S_{xy}}{(S_x)^2}(x - \bar{x}) \\
y - 25.2 &= \frac{22.48}{(2.154)^2}(x - 6.6) \\
y &= 4.85x - 6.78
\end{aligned}$$

To plot the bivariate data on a scatter diagram and the regression line we can use Hamzstlab Mathematics, and the backend computation can be done with SymIntegration.

In SymIntegration the function to compute Pearson's correlation is:
rpearson(M,n)
with M as the matrix of size $n \times 2$, the first column of the matrix represents the independent variable, x, while the second column of the matrix represents the dependent variable, y.

The function to compute the regression line is:
regressionline(M,n)
with M as the matrix of size $n \times 2$, the result will be a symbolic function, $y(x)$.

We also use **Armadillo** to load the matrix in textfile (**matrixA.txt**) then save them into SymbolicMatrix that will be the input for **rpearson()** and **regressionline()** functions.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Pearson's Product Moment Correlation and Regression Line with
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -larmadillo -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Pearson's Product Moment Correlation and Regression Line with
Matrix A:
    4.0000   15.0000
    8.0000   35.0000
    5.0000   12.0000
   10.0000   40.0000
    6.0000   24.0000
W:
[ 4 15]
[ 8 35]
[ 5 12]
[10 40]
[ 6 24]

x bar: 6.6
y bar: 25.2
sum x: 33
sum y: 126
sum xy: 944
sum x^2: 241
sum y^2: 3770

Pearson's correlation coefficient, r :
0.956835

Regression line, y = 4.84483*x-6.77586

Time taken by function: 11442 microseconds
```

Figure 7.2: *The computation of Pearson's correlation (PMCC), r, and the regression line equation with SymIntegration and Armadillo took 11442 microseconds. (**SymIntegration/Examples/Statistics/Test SymIntegration Pearson's Product Moment Correlation and Regression Line with Armadillo/main.cpp**).*

Figure 7.3: *The scatter plots and the regression line $y = 4.85x - 6.78$ (**Hamzstlab-Mathematics/examples/Statistics/main.cpp**).*

As a comparison, we use another code without **Armadillo** to load the two textfiles for vector x and vector y, the computation is faster 37.63% than with **Armadillo**.

```
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
./main ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Load Vector from textfile and Compute Regression Line ]#
W:
[ 4 15]
[ 8 35]
[ 5 12]
[10 40]
[ 6 24]

x bar: 6.6
y bar: 25.2
sum x: 33
sum y: 126
sum xy: 944
sum x^2: 241
sum y^2: 3770

Pearson's correlation coefficient, r :
0.956835

Regression line, y = 4.84483*x-6.77586

Time taken by function: 7136 microseconds
```

Figure 7.4: *The computation of Pearson's correlation (PMCC), r, and the regression line equation with SymIntegration took 7136 microseconds, it is 37.63% faster. (**SymIntegration/Examples/Statistics/Test SymIntegration Load Vector from textfile and Compute Regression Line/main.cpp**).*

IV. Statistical Inference

[H*] The action of statistical inference involves using mathematical techniques to make conclusions about unknown population parameters based on collected data. The field of statistical inference employs a variety of stochastic models to analyze and put forward efficient methods for carrying out such analyses.

Analysis and methods if statistical inference can be categorized as either frequentist (also known as classical) or Bayesian. The former is based on the assumption that population parameters of some underlying distribution, or probability law, exist and are fixed, but are yet unknown. The process of statistical inference then deals with making conclusions about these parameters based on sampled data. In the latter Bayesian case, it is only assumed that there is a prior distribution of the parameters. In this case, the key process deals with analyzing a posterior distribution (of the parameters)-an outcome of the inference process.

In general, a statistical inference process involves data, a model, and analysis. The data is assumed to be comprised of random samples from the model. The goal of the analysis is then to make informed statements about population parameters of the model based on the data. Such statements typically take one of the following forms:

1. **Point Estimation**
   Determination of a single value (or vector of values) representing a best estimate of the parameter / parameters. In this case, the notion of "best" can be defined in different ways.

2. **Confidence Intervals**
   Determination of a range of values where the parameter lies. Under the model and the statistical process used, it is guaranteed that the parameter lies within this range with a pre-specified probability.

3. **Hypothesis Tests**
   The process of determining if the parameter lies in a given region, in the complement of that region, or fails to take on a specific value. Such tests often represent a scientific hypothesis in a very natural way

[H*] **A Random Sample**
When carrying out (frequentist) statistical inference, we assume there is some underlying distribution $F(x, \theta)$ from which we are sampling, where θ is the scalar or vector-valued unknown parameter we wish to know. Importantly, we assume that each observation is statistically independent and identically distributed as the rest. That is, from a probabilistic perspective, the observations are taken as independent and identically distributed (i.i.d.) random variables. In mathematical statistics language, this is called a random sample. We denote the random variables of the observations by $X_1, \ldots, X_n$ and their respective values $x_1, \ldots, x_n$.

Typically, we compute statistics from the random sample. For example, two common standard statistics include the sample mean and sample variance. However, we can model these statistics as random variables

$$\begin{aligned}
\overline{X} &= \frac{1}{n}\sum_{i=1}^{n} X_i \\
S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2
\end{aligned} \tag{7.11}$$

Note that for S^2, the denominator is $n-1$ to make S^2 an unbiased estimator.

In general, the phrase statistic implies a quantity calculated based on the sample. When working with data, the sample mean and sample variance are nothing but numbers computed from our sample observations. However, in the statistical inference paradigm, we associate random variables to these values, since they themselves are functions of the random sample.

V. Hypothesis Testing

[H*]

Chapter 8

Optimization and Operations Research

"My only place is by your side. If you want to carry out your wish, I will follow you. That is my Nature." -
Sarah (Suikoden III)

Optimization is the act of obtaining the best result under given circumstances. in design, construction, and maintenance of any engineering system, engineers have to take many technological and managerial decisions at several stages. The ultimate goal of all such decisions is to either minimize the effort required or maximize the desired benefit. There is no single method available for solving all optimization problems efficiently. Hence a number of optimization methods have been deveoped for solving different types of optimization problems [20].

The optimum seeking methods are also known as mathematical programming techniques and are generally studied as a part of operations research. Operations research is a branch of mathematics which is concerned with the application of scientific methods and techniques to decision making problems and with establishing the best or optimal solutions.

Methods of operations research:

- Mathematical programming techniques
    1. Calculus methods
    2. Calculus of variations
    3. Nonlinear programming
    4. Geometric programming
    5. Quadratic programming
    6. Linear programming
    7. Dynamic programming
    8. Integer programming
    9. Stochastic programming
    10. Separable programming
    11. Multiobjective programming

12. Network methods: CPM and PERT
13. Game theory

- Stochastic process techniques
  1. Statistical decision theory
  2. Markov processes
  3. Queueing theory
  4. Renewal theory
  5. Simulation methods
  6. Reliability theory
- Statistical methods
  1. Regression analysis
  2. Cluster analysis, pattern recognition
  3. Design of experiments
  4. Discriminate analysis (factor analysis)

Optimization, also known as mathematical programming, collection of mathematical principles and methods used for solving quantitative problems in many disciplines, including physics, biology, engineering, economics, and business. The subject grew from a realization that quantitative problems in manifestly different disciplines have important mathematical elements in common. Because of this commonality, many problems can be formulated and solved by using the unified set of ideas and methods that make up the field of optimization.

Optimization problems typically have three fundamental elements. The first is a single numerical quantity, or objective function, that is to be maximized or minimized. The objective may be the expected return on a stock portfolio, a company's production costs or profits, the time of arrival of a vehicle at a specified destination, or the vote share of a political candidate. The second element is a collection of variables, which are quantities whose values can be manipulated in order to optimize the objective. Examples include the quantities of stock to be bought or sold, the amounts of various resources to be allocated to different production activities, the route to be followed by a vehicle through a traffic network, or the policies to be advocated by a candidate. The third element of an optimization problem is a set of constraints, which are restrictions on the values that the variables can take. For instance, a manufacturing process cannot require more resources than are available, nor can it employ less than zero resources. Within this broad framework, optimization problems can have different mathematical properties. Problems in which the variables are continuous quantities (as in the resource allocation example) require a different approach from problems in which the variables are discrete or combinatorial quantities (as in the selection of a vehicle route from among a predefined set of possibilities).

Optimization, in its broadest sense, can be applied to solve any engineering problem. Some typical applications are given below:

1. Design of aircraft and aerospace structures for minimim weight.
2. Finding the optimal trajectories of space vehicles.
3. Design of civil engineering structures like frames, foundations, bridges, towers, chimneys and dams for minimum cost.

4. Minimum weight design of structures for earthquake, wind and other types of random loading.
5. Design of water resources systems for maximum benefit.
6. Optimal plastic design of structures.
7. Optimum design of linkages, cams, gears, machine tools and other mechanical components.
8. Selection of machining conditions in metal cutting processes for minimum production cost.
9. Design of material handling equipment like conveyors, trucks and cranes for minimum cost.
10. Design of pumps, turbines, and heat transfer equipment for maximum efficiency.
11. Optimum design for electrical machinery like motors, generators and transformers.
12. Optimum design of electrical networks.
13. Shortest route taken by a salesman visiting different cities during one tour.
14. Optimal production planning, controlling and scheduling.
15. Analysis of statistical data and building empirical models from experimental results to obtain the most accurate representation of the physical phenomenon.
16. Optimum design of chemical processing equipment and plants.
17. Design of optimum pipe line networks and process industries.
18. Selection of site for an industry.
19. Planning of maintenance and replacement of equipment to reduce operating costs.
20. Inventory control.
21. Allocation of resources of services among several activities to maximize the benefit.
22. Controlling the waiting and idle times and queueing in production lines to reduce the costs.
23. Planning the best strategy to obtain maximum profit in the presence of a competitor.
24. Optimum design of control systems.

An optimization of a mathematical programming problem can be stated as follows.

Find

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ which minimizes } f(\boldsymbol{x}) \tag{8.1}$$

subject to constraints

$$g_j(\boldsymbol{x}) \leq 0, \quad j = 1, 2, \ldots, m$$

and

$$l_j = 0, \quad j = 1, 2, \ldots, p$$

where $\boldsymbol{x}$ is an n-dimensional vector called the design vector, $f(\boldsymbol{x})$ is called the objective function and $g_j(\boldsymbol{x})$ and $l_j(\boldsymbol{x})$ are, respectively, the inequality and the equality constraints.

I. Linear Programming

[H*] Linear programming is concerned with the optimization (minimization or maximization) of a linear function while satisfying a set of linear equality and/or inequality constraints or restrictions. The linear programming problem was first conceived by George B. Dantzig around 1947 while he was working as a Mathematical Advisor to the United States Air Force Comptroller on developing a mechanized planning tool for a time-staged deployment, training, and logistical supply program.

In 1949, George B. Dantzig published the "simplex method" for solving linear programs. Since that time a number of individuals have contributed to the field of linear programming in many different ways, including theoretical development, computational aspects, and exploration of new applications of the subject [2].

Linear programming is an optimization problem, a problem that is well rooted as a principle underlying the analysis of many complex decision or allocation problems. For example, if we have limited resources and we want to achieve a certain goal, we need to learn to optimize our resources with either linear programming or nonlinear programming depends on our problems at hand. Like how we decide to buy groceries depending on our needs / diet (the goal), and the money we have (the constraints).

One of the early industrial applications of linear programming have been made in the petroleum refineries. In general, an oil refinery has a choice of buying crude oil from several different sources with differing compositions and at differing prices. It can manufacture different products like aviation fuel, diesel fuel and gasolinein varying quantities. The constraints may be due to the restrictions on quantity of the crude oil available from a particular source, the capacity of the refinery to produce a particular product, etc. A mix of the purchased crude oil and the manufactured products is sought that gives the maximum profit.

In food processing industry, linear programming has been used to determine the optimal shipping plan for the distribution of a particular product from the different manufacturing plants to the various warehouses. In the iron and steel industry, the linear programming, the linear programming was used to decide the types of products to be made in their rolling mills to maximize the profit. Metal working industries use linear programming for shop loading and for determining the choice between producing and buying a part. Paper mills use it to decrease the amount of trim loses. The optimal routing of messages in a communication network and the routing

[H*] Any general linear programming problem may be manipulated into this form.

Consider the following linear programming problem.

Minimize

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

Subject to

$$\begin{array}{ccccc} a_{11}x_1 & +a_{12}x_2 & +\dots & +a_{1n}x_n & \geq b_1 \\ a_{21}x_1 & +a_{22}x_2 & +\dots & +a_{2n}x_n & \geq b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1}x_1 & +a_{m2}x_2 & +\dots & +a_{mn}x_n & \geq b_m \\ x_1, & x_2, & \dots, & x_n & \geq 0 \end{array}$$

(The form above is not a standard form) Here $c_1x_1 + c_2x_2 + \cdots + c_nx_n$ is the objective function (or criterion function) to be minimized and will be denoted by z. The coefficients $c_1, c_2, \dots, c_n$ are the (known) cost coefficients and $x_1, x_2, \dots, x_n$ are the decision variables to be determined.

The inequality $\sum_{j=1}^{n} a_{ij}x_j \geq b_i$ denotes the ith constraint (or restriction or functional, structural, or technological constraint). The coefficients a_{ij} for $i = 1, 2, \dots, m,\ j = 1, 2, \dots, n$ are called the technological coefficients. These technological coefficients form the constraint matrix A.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_n \end{bmatrix}$$

The column vector whose ith component is b_i, which is referred to as the right-hand-side vector, represents the minimal requirements to be satisfied. The constraints $x_1, x_2, \dots, x_n \geq 0$ are the nonnegativity constraints. A set of variables $x_1, x_2, \dots, x_n$ satisfying all the constraints is called a feasible point or a feasible vector. The set of all such points constitutes the feasible region or the feasible space.

We stated that the linear programming problem above is not in standard form, the characteristics of the linear programming stated in standard form are:

1. The objective function is of the minimization type.
2. All the constraints are of the equality type.
3. All the decision variables are nonnegative.

Any linear programming problem can be put in the standard form, for example the maximization of a function $f(x_1, x_2, \dots, x_n)$ is equivalent to the minimization of the negative of the same function. If the constraints equations are not of the equality type, then we can add a dummy / slack variable to make it become an equality type.

[H*] The standard form of linear programming in matrix form:
Minimize

$$\boldsymbol{c}^t \boldsymbol{X} \tag{8.2}$$

subject to the constraints

$$A\boldsymbol{X} = \boldsymbol{b} \tag{8.3}$$

and

$$\boldsymbol{X} \geq \boldsymbol{0} \tag{8.4}$$

where

$$\boldsymbol{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \quad \boldsymbol{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

It can be seen that there are m equations and n decision variables in a linear programming problem.

We may assume that $m < n$, for if $m > n$ there would be $m - n$ redundant equations which could be eliminated (it is the basic knowledge of linear algebra, read more here [22]). The case where $m = n$ is of no interest, for then there is either a unique solution $\boldsymbol{X}$ which satisfies Eqs. (8.3) and (8.4) (in which case there can be no optimization) or no solution, in which case the constraints are inconsistent.

the case $m < n$ corresponds to an underdetermined set of linear equations which, if they have one solution, have an infinite number of solutions. The problem of linear programming is to find one of these solutions satisfying Eqs.(8.3) and (8.4) and yielding the minimum of f.

[H*] A linear programming problem may have

1. A unique and finite optimum solution.
2. An infinite number of optimal solutions.
3. An unbounded solution.
4. No solution.
5. A unique feasible point.

Assuming that the linear programming is properly formulated, the following general geometrical characteristics can be noted from the graphical solution.

1. The feasible region is a convex polygon.
2. The optimum value occurs at an extreme point or vertex of the feasible region.

[H*] To represent an optimization problem as a linear program, several assumptions that are implicit in the linear programming formulation are needed, they are:

1. **Proportionality**
   Given a variable x_j, its contribution to cost is $c_j x_j$ and its contribution to the ith constraint is $a_{ij} x_j$. This means that if x_j is doubled, so is its contribution to cost and to each of the constraints.
2. **Additivity**
   This assumption guarantees that the total cost is the sum of the individual costs, and that the total contribution to the ith restriction is the sum of the individual contributions of the individual activities. In other words, there are no substitution or interaction effects among the activities.
3. **Divisibility**
   This assumption ensures that the decision variables can be divided into any fractional levels so that noninteger values for the decision variables are permitted.
4. **Deterministic**
   The coefficients c_j, a_{ij}, and b_i are all known deterministically. Any probabilistic or stochastic element inherent in demand, costs, prices, resource availabilities, usages, and so on are all assumed to be approximated by these coefficients through some deterministic equivalent.

It is important to recognize that if a linear programming problem is being used to model a given situation, then the aforementioned assumptions are implied to hold, at least over some anticipated operating range for the activities.

Despite the seemingly restrictive assumptions, linear programs are among the most widely used models today. They represent several systems quite satisfactorily, and they are capable of providing a large amount of information besides simply a solution. Moreover, they are also often used to solve certain types of nonlinear optimization problems via (successive) linear approximations and constitute an important tool in solution methods for linear discrete optimization problems having integer restricted variables.

[H*] **Problem Manipulation**

Recall that a linear program is a problem of minimizing or maximizing a linear function in the presence of linear inequality and/or equality constraints. By simple manipulation the problem can be transformed from one form to another equivalent form.

1. **Inequalities and Equations**
   An inequality can be easily transformed into an equation. To illustrate, consider the constraint given by
   $$\sum_{j=1}^{n} a_{ij}x_j \geq b_i$$
   this constraint can be put in an equation form by subtracting the nonnegative surplus or slack variable x_{n+i} (sometimes denoted by s_i)leading to
   $$\sum_{j=1}^{n} a_{ij}x_j - x_{n+i} = b_i$$
   and
   $$x_{n+i} \geq 0$$
   Similarly, the constraint
   $$\sum_{j=1}^{n} a_{ij}x_j \leq b_i$$
   is equivalent to
   $$\sum_{j=1}^{n} a_{ij}x_j + x_{n+i} = b_i$$
   and
   $$x_{n+i} \geq 0$$

[H*] **Definitions and Theorems**

Some of the more poweful methods for solving linear programming problems take advantage of these characteristics.

1. Point in n-dimensional space. A point $\boldsymbol{X}$ in an n-dimensional space is characterized by an ordered set of n values of coordinates $(x_1, x_2, \ldots, x_n)$. The coordinates of $\boldsymbol{X}$ are also called the components of $\boldsymbol{X}$.
2. Line segment in n-dimensions (L). If the coordinates of two points A and B are given by $x_j^{(1)}$ and $x_j^{(2)}$ ($j = 1, 2, \ldots, n$), the line segment (L) joining these points is the collection of points $\boldsymbol{X}(\lambda)$ whose coordinates are given by
   $$x_j = \lambda x_j^{(1)} + (1-\lambda)x_j^{(2)}, \quad j = 1, 2, \ldots, n$$

where $0 \le \lambda \le 1$.

Thus

$$L = \{\boldsymbol{X} | \boldsymbol{X} = \lambda \boldsymbol{X}^{(1)} + (1-\lambda)\boldsymbol{X}^{(2)}\} \tag{8.5}$$

In one dimension, for example, it is easy to see that the definition is in accordance with our experience

$$x(\lambda) - x^{(1)} = \lambda(x^{(2)} - x^{(1)}), \quad 0 \le \lambda \le 1 \tag{8.6}$$

hence

$$x(\lambda) = (1-\lambda)x^{(1)} + \lambda x^{(2)} \tag{8.7}$$

3. In n-dimensional space, the set of points coordinates satisfy a linear equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = \boldsymbol{a}^T\boldsymbol{x} = b \tag{8.8}$$

is called a hyperplane.

Thus, the hyperplane H is given by

$$H(A, b) = \{\boldsymbol{x} | A^T\boldsymbol{x} = b\} \tag{8.9}$$

A hyperplane has $(n-1)$ dimensions in n-dimensional space. For example, in a 3-dimensional space, it is a plane and in 2-dimensional space, it is a line. The set of points whose coordinates satisfy a linear inequality like $a_1x_1 + \cdots + a_nx_n \le b$ is called a closed half space, closed due to the inclusion of equality sign in the above inequality.

A hyperplane partitions E^n into two closed half-spaces so that

$$H^+ = \{\boldsymbol{x} | A^T\boldsymbol{x} \ge b\} \tag{8.10}$$

and

$$H^- = \{\boldsymbol{x} | A^T\boldsymbol{x} \le b\} \tag{8.11}$$

4. Convex set is a collection of points such that if $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$ are any two points in the collection, the line segment joining them is also in the collection. If S denotes the convex set, it can be defined mathematically as follows:

$$\text{if } \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} \in S, \text{ then } \boldsymbol{x} \in S$$

where

$$\boldsymbol{x} = \alpha\boldsymbol{x}^{(2)} + (1-\alpha)\boldsymbol{x}^{(1)}, \quad 0 \le \alpha \le 1$$

As an assumption, the set containing only one point is convex.

A convex polygon consists of a set of points having the property that the line segment joining any two points in the set is entirely in the convex set. In problems having more than two decision variables, the feasible region is called a convex polyhedron.

5. Convex polyhedron is a set of points common to one or more half-spaces. In particular, a convex polygon is the intersection of one or more half planes.

6. Vertex is a point in the convex set which does not lie on a line segment joining two toher points of the set.
   Thus, for example, every point on the circumference of a circle and each corner point of a polygon can be called a vertex or extreme point.

7. Feasible solution is any solution in linear programming problem that satisfies the constraints

$$A\boldsymbol{x} = \boldsymbol{b} \tag{8.12}$$

   and

$$\boldsymbol{x} \geq \boldsymbol{0} \tag{8.13}$$

8. Basic solution is a solution in which $(n - m)$ variables are set equal to zero.
   The basic solution can be obtained by setting $(n - m)$ variables to zero and solving the constraint Eqs. () simultaneously.

9. Basis is the collection of variables that are not set equal to zero to obtain the basic solution.

10. Basic feasible solution is a basic solution which satisfies the nonnegativity conditions of Eqs. ().

11. Nondegenerate basic feasible solution is a basic feasible solution which has got exactly m positive x_i.

12. Optimal solution is a feasible solution which optimizes the objective function.

13. Optimal basic solution is a basic feasible solution for which the objective function is optimal.

Theorem 8.1: Intersection of Convex Sets

The intersection of any number of convex sets is also convex.

Theorem 8.2: Feasible Region in Linear Programming

The feasible region of a linear programming problem is convex.

Theorem 8.3: Local Minimum in Linear Programming

Any local minimum solution is global for a linear programming problem.

Theorem 8.4: Basic Feasible Solution in Linear Programming

Every basic feasible solution is an extreme point of the convex set of feasible solutions.

Theorem 8.5: Closed and Bounded Convex Polyhedron

Let S be a closed, bounded convex polyhedron with $\boldsymbol{x}_{i^e}$, $i = 1$ to p as the set of its extreme points. Then any vector $\boldsymbol{x} \in S$ can be written as

$$\begin{aligned} \boldsymbol{x} &= \sum_{i=1}^{p} \lambda_i \boldsymbol{x}_{i^e} \\ \lambda_i &\geq 0 \\ \sum_{i=1}^{p} \lambda_i &= 1 \end{aligned} \tag{8.14}$$

Theorem 8.6: Closed Convex Polyhedron

Let S be a closed convex polyhedron. Then the minimum of a linear function over S is attained at an extreme point of S.

$$\begin{aligned} \boldsymbol{x} &= \sum_{i=1}^{p} \lambda_i \boldsymbol{x}_{i^e} \\ \lambda_i &\geq 0 \\ \sum_{i=1}^{p} \lambda_i &= 1 \end{aligned} \tag{8.15}$$

For the proof of each theorem above, you can read it here [20].

[H*] **Farmer' Problem**

A farmer has a choice of planting potato, corn, rice or chili on her 200 acre farm (1 acre $\approx 4046.86\ m^2 \approx \frac{2}{5}\ ha$). The anticipated labour, water and fertilizer requirements, and yields per acre are given in the following table The farmer can also give a land for lease in which

Crop	Labour costs(USD)	Water($cu.m$)	Fertilizer(kg)	Yields(kg)	Selling price(USD/kg)
Potato	300	10000	100	1500	2.5
Corn	200	7000	120	3000	1
Rice	250	6000	160	2500	1.5
Chili	360	8000	200	2000	2

Table 8.1: *The anticipated labour, water, fertilizer requirements, yields per acre, and the expected market price for each commodity for this farm.*

case she gets USD 800 per acre. The cost of water is USD 0.02 per $cu.m$, and the cost of fertilizer is USD 2 per kg. The farmer can get a maximum loan of USD 200,000 from the land mortgage bank at an interest of 8%. She can repay the loan after six months. Further, the irrigation canal cannot supply more than 700,000 $cu.m$ of water. Formulate the problem of finding the planting schedule for maximizing the anticipated returns of the farmer.

Solution:

We will define the design variables:

$$\begin{aligned}
x_1 &= \text{acres of potato}\\
x_2 &= \text{acres of corn}\\
x_3 &= \text{acres of rice}\\
x_4 &= \text{acres of chili}\\
x_5 &= \text{acres leased}
\end{aligned}$$

Then the costs per acre:

$$\begin{aligned}
\text{Potato} &: 300 + 10000(0.02) + 100(2) = 700\\
\text{Corn} &: 200 + 7000(0.02) + 120(2) = 580\\
\text{Rice} &: 250 + 6000(0.02) + 160(2) = 690\\
\text{Chili} &: 360 + 8000(0.02) + 200(2) = 920
\end{aligned}$$

Total farming cost:

$$700x_1 + 580x_2 + 690x_3 + 920x_4$$

Using the interest of 8%, the total cost is:

$$1.08(700x_1 + 580x_2 + 690x_3 + 920x_4)$$

Revenue per acre (yields multiplied by selling price)

$$\begin{aligned}
\text{Potato} &: 3750\\
\text{Corn} &: 3000\\
\text{Rice} &: 3750\\
\text{Chili} &: 4000
\end{aligned}$$

The net return will be

$$(3750 - 1.08 * 700)x_1 + (3000 - 1.08 * 580)x_2 + (3750 - 1.08 * 690)x_3 + (4000 - 1.08 * 920)x_4 + 800x_5$$

so the objective function is:
maximize

$$f = 2994x_1 + 2373.6x_2 + 3004.8x_3 + 3006.4x_4 + 800x_5$$

with constraints:

$$\begin{aligned}
x_1 + x_2 + x_3 + x_4 + x_5 &= 200\\
700x_1 + 580x_2 + 690x_3 + 920x_4 &\leq 200000\\
10000x_1 + 7000x_2 + 6000x_3 + 8000x_4 &\leq 700000\\
x_i &\geq 0, \quad i = 1, 2, 3, 4, 5
\end{aligned}$$

i. Simplex Method Computation

Find the amount of each type of food to be purchased in order to meet exactly the daily requirements of a person at minimum cost. Assume that a person, on the average, requires 3000 calories and 100 grams of protein.

	Bread	Meat	Potatoes	Cabbage	Milk
Calories	2500	3000	600	100	600
Protein($grams$)	80	150	20	10	40
Cost (USD/kg)	3	10	1	2	3

Table 8.2: *The valorie values and the protein contents of five types of foods along with their costs.*

Solution:

Let x_1, x_2, x_3, x_4, and x_5 denote the amount of bread, meat, potatoes, cabbage, and milk to be purchased in kg per day. Then the objective function to be minimized is given by

$$f = 3x_1 + 10x_2 + x_3 + 2x_4 + 3x_5 \tag{8.16}$$

The constraints due to the calorie and protein requirements are respectively given by

$$\begin{aligned} 2500x_1 + 3000x_2 + 600x_3 + 100x_4 + 600x_5 = 3000 \\ 80x_1 + 150x_2 + 20x_3 + 10x_4 + 40x_5 = 100 \end{aligned} \tag{8.17}$$

The nonnegativity requirements of the decision variables are given by

$$x_i \geq 0, \quad i = 1, 2, 3, 4, 5 \tag{8.18}$$

Since $m = 2$ (the number of equality constraints) and $n = 5$ (the number of decision variables), a basic solution can be obtained by setting any of the $(n - m)$ variables equal to zero and solving the constraint equations. The number of basic solutions is

$$\frac{5!}{3!2!} = 10$$

Out of these ten basic solutions, the solution which has all the variables nonnegative and makes the objective function given by Eq. (8.16) assume a minimum value will be the optimum.

The simplex computation makes use of Gaussian elimination to compute the values of the basic variables and the objective function for all basic solutions.

This is the initial simplex tableau:

$$S = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & f & C \\ 2500 & 3000 & 600 & 100 & 600 & 0 & 3000 \\ 80 & 150 & 20 & 10 & 40 & 0 & 100 \\ -3 & -10 & -1 & -2 & -3 & 1 & 0 \end{bmatrix} \tag{8.19}$$

Thus we make this matrix A that represents the constraints equations only:

$$A = \begin{bmatrix} 2500 & 3000 & 600 & 100 & 600 & 0 & 3000 \\ 80 & 150 & 20 & 10 & 40 & 0 & 100 \end{bmatrix} \tag{8.20}$$

The matrix A is the one that will be the input for the C++ codes, the objective function will be inputted as a vector, to create the matrix A we make two options, the first one is loading the matrix from textfile (**matrix.txt**) and the second one is to input the coefficients manually in the C++ source code (**main.cpp**).

To load a **std::vector** from a text file in C++, we need to include **<fstream>** for file operations, **<vector>** for using vector, **<string>** for reading lines, and **<iostream>** for input/output.

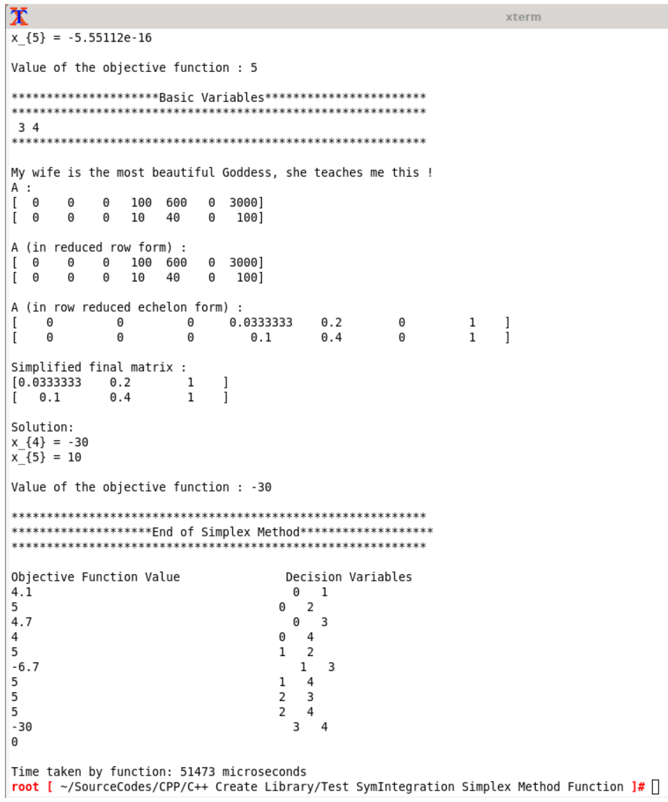
The loaded vector will then be saved as Symbolic matrix **B_mat** that will be used as the input to do the simplex computation.

To do the simplex computation we use this function in SymIntegration:
Symbolic simplexmethod(const SymbolicMatrix &A, vector<double> f, int C, int R, int n)

with A as the SymbolicMatrix that represent the constraint equations that we have defined in Eq. (8.20), f as the vector of objective function, C as the number of column for matrix A, R as the number of row for matrix A, and n is the number of decision variables.

```
xterm
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Simplex Method Function ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Simplex Method Function ]# ./main
Simplex Method :
A :
[2500 3000  600  100  600   0  3000]
[ 80   150  20   10   40    0   100]


********************Basic Variables**********************
*********************************************************
 0 1
*********************************************************

My wife is the most beautiful Goddess, she teaches me this !
A :
[2500 3000   0    0    0    0  3000]
[ 80   150   0    0    0    0   100]

A (in reduced row form) :
[2500 3000   0    0    0    0  3000]
[  0   54    0    0    0    0    4 ]

A (in row reduced echelon form) :
[    1        1.2        0         0         0         0        1.2    ]
[    0         1         0         0         0         0     0.0740741]

Simplified final matrix :
[    1        1.2       1.2    ]
[    0         1     0.0740741]

Solution:
x_{1} = 1.11111
x_{2} = 0.0740741

Value of the objective function : 4.07407

********************Basic Variables**********************
*********************************************************
 0 2
*********************************************************

My wife is the most beautiful Goddess, she teaches me this !
A :
[2500   0   600   0    0    0  3000]
[ 80    0   20    0    0    0   100]

A (in reduced row form) :
[2500   0   600   0    0    0  3000]
[  0    0   0.8   0    0    0    4 ]
```

Figure 8.1: *The function in SymIntegration to compute the linear programming problem with simplex method (**SymIntegration/Examples/Operations Research/Test SymIntegration Simplex Method Function/main.cpp**).*



```
x_{5} = -5.55112e-16

Value of the objective function : 5

********************Basic Variables**********************
*********************************************************
 3 4
*********************************************************

My wife is the most beautiful Goddess, she teaches me this !
A :
[  0    0    0   100  600   0  3000]
[  0    0    0   10   40    0   100]

A (in reduced row form) :
[  0    0    0   100  600   0  3000]
[  0    0    0   10   40    0   100]

A (in row reduced echelon form) :
[    0         0         0     0.0333333    0.2        0         1    ]
[    0         0         0        0.1       0.4        0         1    ]

Simplified final matrix :
[0.0333333    0.2        1    ]
[   0.1       0.4        1    ]

Solution:
x_{4} = -30
x_{5} = 10

Value of the objective function : -30

*********************************************************
********************End of Simplex Method******************
*********************************************************

Objective Function Value               Decision Variables
4.1                                      0   1
5                                      0   2
4.7                                      0   3
4                                      0   4
5                                      1   2
-6.7                                      1   3
5                                      1   4
5                                      2   3
5                                      2   4
-30                                      3   4
0

Time taken by function: 51473 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Simplex Method Function ]#
```

Figure 8.2: *The function in SymIntegration to compute the linear programming problem with simplex method takes 51473 microseconds (**SymIntegration/Examples/Operations Research/Test SymIntegration Simplex Method Function/main.cpp**).*

Due to the C++ coding the meaning of decision variable 0 is for x_1, 1 for x_2 and so on, from the result we know that the feasible and optimum solution occurs when we use basic variables 0 and 4, or, x_1 and x_5, referring to bread and milk. With the values for the basic variables:

$$\begin{aligned} x_1 &= 1.15385 \\ x_5 &= 0.192308 \\ f &= 4.03846 \end{aligned}$$

II. Nonlinear Programming

[H*]

Chapter 9

Introduction to Theory of Interest and Financial Mathematics

"In my world everything what would be it wouldn't be, and what wouldn't be it would." (Alice in Wonderland 1951) - ***Alice***

Mathematical finance, also known as quantitative finance and financial mathematics, is a field of applied mathematics, concerned with mathematical modeling of financial markets.

In general, there exist two separate branches of finance that require advanced quantitative techniques: derivatives pricing on the one hand, and risk and portfolio management on the other. Mathematical finance overlaps heavily with the fields of computational finance and financial engineering. The latter focuses on applications and modeling, often by help of stochastic asset models, while the former focuses, in addition to analysis, on building tools of implementation for the models. Also related is quantitative investing, which relies on statistical and numerical models (and lately machine learning) as opposed to traditional fundamental analysis when managing portfolios.

French mathematician Louis Bachelier's doctoral thesis, defended in 1900, is considered the first scholarly work on mathematical finance. But mathematical finance emerged as a discipline in the 1970s, following the work of Fischer Black, Myron Scholes and Robert Merton on option pricing theory. Mathematical investing originated from the research of mathematician Edward Thorp who used statistical methods to first invent card counting in blackjack and then applied its principles to modern systematic investing.

I. Bond Pricing

[H*] A bond is a debt instrument requiring the issuer (also called the debtor or borrower) to repat to the lender/investor the amount borowed plus interest over a specified period of time.

A typical ("plain vanilla") bond issued in the United States specifies:

1. A fixed date when the amount borrowed (the principal) is due.
2. The contractual amount of interest, which typically is paid every six months.

The date on which the principal is required to be repaid is called the maturity date. Assuming that the issuer does not default or redeem the issue prior to the maturity date, an investor holding this bond until the maturity date is assured of a known cash flow pattern [10].

[H*] For a variety of reasons, since 1980s there was a wide range of bond structures being developed. In the residential mortgage market particularly, new types of mortgage designs were introduced. The practice of pooling of individual mortgages to form mortgage pass-through securities grew dramatically. Using the basic instruments in the mortgage market (mortgages and mortgage pass-through securities), issuers created derivative instruments such as collateralized mortgage obligations and stripped mortgage-backed securities that met specific investment needs of a broadening range of institutional investors.

[H*] The U.S. bond market is the largest bond market in the world. The market is divided into six sectors:

1. **U.S. Treasury sector**
   This sector includes securities issued by the U.S. government. These securities include Treasury bills, notes, and bonds. The U.S. Treasury is the largest issuer of securities in the world. This sector plays a key role in the valuation of securities and the determination of interest rates throughout the world.
2. **The agency sector**
   This sector includes securities issued by federally related institutions and government sponsored enterprises. The securities issued are not backed by any collateral and are referred to as agency debenture securities.
3. **The municipal sector**
   This sector is where the state and local governments and their authorities raise funds. The two major sectors within the municipal sector are general obligation sector and the revenue sector. Bonds issued in this sector typically are exempt from federal income taxes.
4. **The corporate sector**
   It includes securities issued by U.S. corporations and non U.S. corporations issued in the United States. The latter securities are referred to as Yankee bonds. The corporate sector is divided into the investment grade and noninvestment grade sectors.
5. **Asset-backed securities sector**
   Int his sector, a corporate issuer pools loans or receivables and uses the pool of assets of collateral for the issuance of a security.
6. **The mortgage sector**
   It is the sector where securities are backed by mortgage loans. These are loans obtained by borrowers in order to purchase residential property or an entity to purchase commercial propety.

[H*] These are some important features of bonds:

1. **Type of issuer**
   A key feature of a bond is the nature of the issuer. Each issuer has different abilities to satisfy their contractual obligation to lenders.
2. **Term to maturity**
   The term to maturity of a bond is the number of years over which the issuer has promised to meet the conditions of the obligation.
   There are three reasons why the term to maturity of a bond is important. The most obvious is that it indicates the time period over which the holder of the bond can expect

to receive the coupon payments and the number of years before the principal will be paid in full. The second reason is that the yield on a bond depends on it. Finally, the price of a bond will fluctuate over its life as yields in the market change, the volatility of a bond's price is dependent on its maturity.

3. **Principal and coupon rate**
   The principal value of a bond is the amount that the issuer agrees to repay the bondholder at the maturity date. This amount is also referred to as the redemption value, maturity value, par value, or face value.
   The coupon rate, also called the nominal rate, is the interest that the issuer agrees to pay each year. The annual amount of the interest payment made to owners during the term of the bond is called the coupon. The coupon rate multiplied by the principal of the bond provides the dollar amount of the coupon.
4. **Amortization feature**
   The principal repayment of a bond issue can call for either the total principal to be repaid at maturity or the principal repaid over the life of the bond. In the latter case, there is a schedule of principal repayments. This schedule is called an amortization schedule. Loans that have this feature are automobile loans and home mortgage loans.
5. **Embedded options**
   It is common for a bond issue to include a provision in the indenture that gives either the bondholder and/or the issuer an option to take some action against the other party. The most common type of option embedded in a bond is a call feature. This provision grants the issuer the right to retire the debt, fully or partially, before the scheduled maturity date. Inclusion of a call feature benefits bond issuers by allowing them to replace an old bond issue with a lower-interest cost issue if interest rates in the market decline. A call provision effectively allows the issuer to alter the maturity of a bond. A call provision is detrimental to the bondholder's interests.

   An issue may also include a provision that allows the bondholder to change the maturity of a bond. An issue with a put provision included in the indenture grants the bondholder the right to sell the issue back to the issuer at par value on designated dates. Here the advantage to the investor is that if interest rates rise after the issue date, thereby reducing a bond's price, the investor can force the issuer to redeem the bond at par value.

   A convertible bond is an issue giving the bondholder the right to exchange the bond for a specified number of shares of common stock. Such a feature allows the bondholder to take advantage of favorable movements in the price of the issuer's common stock. An exchangeable bond allows the bondholder to exchange the issue for a specified number of common stock shares of a corporation different from the issuer of the bond.

[H*] Risks associated with investing in bonds:

1. **Interest-rate risk**
   The price of a typical bond will change in the opposite direction from a change in interest rates: As interest rates rise, the price of a bond will fall; as interest rates fall, the price of a bond will rise. If an investor has to sell a bond prior to the maturity date, an increase in interest rates will mean the realization of a capital loss (selling the bond below the purchase price). This risk is referred to as interest-rate risk or market risk. This risk is by far the major risk faced by an investor in the bond market.

2. **Reinvestment risk**
   Calculation of the yield of a bond assumes that the cash flows received are reinvested. The additional income from such reinvestment, sometimes called interest-on-interest, depends on the prevailing interest-rate levels at the time of reinvestment, as well as on the reinvestment strategy. Variability in the reinvestment rate of a given strategy because of changes in market interest rates is called reinvestment risk. Reinvestment risk is greater for longer holding periods, as well as for bonds with large, early, cash flows, such as high-coupon bonds.

3. **Call risk**
   Many bonds include a provision that allows the issuer to retire or "call" all or part of the issue before the maturity date. The issuer usually retains this right in order to have flexibility to refinance the bond in the future if the market interest rate drops below the coupon rate.
   From the investor's perspective, there are three disadvantages to call provisions. First, the cash flow pattern of a callable bond is not known with certainty. Second, because the issuer will call the bonds when interest rates have dropped, the investor is exposed to reinvestment risk. Finally, the capital appreciation potential of a bond will be reduced, because the price of a callable bond may not rise much above the price at which the issuer will call the bond.

4. **Default risk**
   Also referred as credit risk, refers to the risk that the issuer of a bond may default (i.e., will be unable to make timely principal and interest patments on the issue). Default risk is gauged by quality ratings assigned by four nationally recognized rating companies: Moody's Investors Service, Standard & Poor's Corporation, Duff & Phelps Credit Rating Company, and Fitch IBCA, as well as the credit research staffs of securities firms.
   Because of this risk, bonds with default risk trade in the market at a price that is lower than comparable U.S. Treasury securities, which are considered free of default risk.

5. **Inflation risk**
   This purchasing-power risk arises because of the variation in the value of cash flows from a security due to inflation, as measured in terms of purchasing power. For example, if investors purchase a bond on which they can realize a coupon rate of 6% but the rate of inflation is 10%, the purchasing power of the cash flow actually has declined.

6. **Exchange-rate risk**
   A non-dollar-denominated bond (i.e., a bond whose payments occur in a foreign currency) has unknown U.S. dollar cash flows. The dollar cash flows are dependent on the exchange rateat the time the payments are received. For example, suppose that an investor purchases a bond whose payments are in Japanese Yen / JPY. If the JPY depreciates relative to the USD, fewer dollars will be received. The risk of this occurring is referred to as exchange-rate or currency risk. Of course, should the JPY appreciate relative to the USD, the investor will benefit by receiving more dollars.

7. **Liquidity risk**
   Liquidity or marketability risk depends on the case with which an issue can be sold at or near its value. The primary measure of liquidity is the size of the spread between the bid and the ask price quoted by a dealer. The wider the dealer spread, the more the liquidity risk. For an investor who plans to hold the bond until the maturity date, liquidity risk is less important.

8. **Volatility risk**
   The risk that a change in volatility will affect the price of a bond adversely. The price of a bond with certain types of embedded options depends on the level of interest rates and factors that influence the value of the embedded option. One of these factors is the expected volatiliy of interest rates. Specifically, the value of an option rises when expected interest-rate volatility increases. In the case of a bond that is callable or a mortgage-backed security, in which the investor has granted the borrower an option, the price of the security falls, because the investor has given away a more valuable option.

[H*] The notion that money has a time value is one of the basic concepts in the analysis of any financial instrument. Money has time value because of the opportunity to invest it at some interest rate.

Future Value
To determine the future value of any sum of money invested today, we can use this equation

$$P_n = P_0(1+r)^n \tag{9.1}$$

where n is the number of periods, P_n is the future value, P_0 is the original principal, and r is the interest rate per period.

Present Value
This time we will show how to determine the amount of money that must be invested today in order to realize a specific future value. This amount is called the present value. The formula is

$$PV = P_n \left[\frac{1}{(1+r)^n}\right] \tag{9.2}$$

the term in brackets is the present value of USD 1.

In most applications in portfolio management a financial instrument will offer a series of future values. To determine the present value of a series of future values, the present value of each future value must first be computed. Then these present values are added together to obtain the present value of the entire series of future values. The formula can be expressed as follows:

$$PV = \sum_{t=1}^{n} \frac{P_t}{(1+r)^t} \tag{9.3}$$

Present Value of an Ordinary Annuity
When the same dollar amount of money is received each period or paid each year, the series is referred to as an annuity. When the first payment is received one period from now, the annuity is called an ordinary annuity. When the first payment is immediate, the annuity is called an annuity due. For bond, we shall deal with ordinary annuities.

To compute the present value of an ordinary annuity, the present value of each future value can be computed and then summed. Alternatively, a formula fo the present value of an ordinary annuity can be used

$$PV = A\left[\frac{1-\frac{1}{(1+r)^n}}{r}\right] \tag{9.4}$$

[H*] **Pricing a Bond**

The price of any financial instrument is equal to the present value of the expected cash flows from the financial instrument. Therefore determining the price requires

1. An estimate of the expected cash flows
2. An estimate of the appropriate required yield

The expected cash flows for some financial instruments are simple to compute; for others, the task is more difficult. The required yield reflects the yield for financial instruments with comparable risk, or alternative (or substitute) investments.

The first step in determining the price of a bond is to determine its cash flows. The cash flows for a bond that the issuer cannot retire prior to its stated maturity date (i.e., a noncallable bond) consist of

1. Periodic coupon interest payments to the maturity date
2. The par (or maturity) value at maturity

The required yield is determined by investigating the yields offered on comparable bonds in the market. By comparable, we mean noncallable bonds of the same credit quality and the same maturity. The required yield typically is expressed as an annual interest rate. When the cash flows occur semiannually, the market convention is to use one-half the annual interest rate as the periodic interest rate with which to discount the cash flows.

The price of a bond is the present value of the cash flows, it is determined by adding these two present values:

1. The present value of the semiannual coupon payments
2. The present value of the par or maturity value at the maturity date

In general, the price of a bond can be computed using the following formula:

$$P = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C}{(1+r)^n} + \frac{M}{(1+r)^n} \tag{9.5}$$

or

$$P = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{M}{(1+r)^n} = C\left[\frac{1-\frac{1}{(1+r)^n}}{r}\right] + \frac{M}{(1+r)^n} \tag{9.6}$$

where
P = the bond price
n = the number of periods
C = the coupon paid every period
r = the interest rate per period
M = maturity value
t = time period when the payment is to be received

[H*] As yields in the marketplace change, the only variable that can change to compensate an investor for the new required yield in the market is the price of the bond. When the coupon rate is equal to the required yield, the price of the bond will be equal to its par value.

When yields in the marketplace rise above the coupon rate at a given point in time, the price of the bond adjusts so that an investor cintemplating the purchase of the bond can realize some additional interest. If it did not, investors would not buy the issue because it offers a below market yield; the resulting lack of demand would cause the price to fall and thus the yield on the bond to increase. This is how a bond's price fall below its par value.

The capital appreciation realized by holding the bond to maturity represents a form of interest to a new investor to compensate for a coupon rate that is lower than the required yield. When a bond sells below its par value, it is said to be selling at a discount. When the required yield is greater than the coupon rate, the price of the bond is always lower than the par value.

When the required yield in the market is below the coupon rate, the bond must sell above its par value. This is because investors who have the opportunity to purchase the bond at par would be getting a coupon rate in excess of what the market requires. As a result, investors would bid up the price of the bond because its yield is so attractive.the price would eventually be bid up to a level where the bond offers the required yield in the market. A bond whose price is above its par value is said to be selling at a premium.

Discount Bond: coupon rate < required yield $\iff$ price < par
Par Bond: coupon rate = required yield $\iff$ price = par
Premium Bond: coupon rate > required yield $\iff$ price > par

i. Bond Pricing Computation

Compute the bond price for a 20-year coupon bond paid semiannually with a 10% coupon rate and a par or maturity value of USD 1,000, for the required yield of 11% coupon rate.

Solution:

The cash flows for this bond are as follows:

1. 40 semiannual coupon payments of USD 50.

$$C = \frac{100}{2} = 50$$

2. A USD 1,000 cash flow 40 six-months periods from now.

$$M = 1000$$

The semianuual or periodic interest rate (or periodic require yield) is $\frac{11\%}{2} = 5.5\%$.

The present value of the 40 semiannual coupon payments of USD 50 discounted 5.5% is

$$\begin{aligned}
PV_{coupon} &= C\left[\frac{1-\frac{1}{(1+r)^n}}{r}\right] \\
&= 50\left[\frac{1-\frac{1}{(1.055)^{40}}}{0.055}\right] \\
&= 50\left[\frac{1-\frac{1}{(8.51332)}}{0.055}\right] \\
&= 50\left[\frac{1-0.117463}{0.055}\right] \\
&= 50[16.04613] \\
&= 802.31
\end{aligned}$$

The present value of the par or maturity value of USD 1,000 received 40 six-months periods from now, discounted at 5.5% is

$$PV_M = \frac{M}{(1+r)^n} = \frac{1000}{(1.055)^{40}} = 117.46$$

the price of the bond is then equal to the sum of the two present values above

$$P = PV_{coupon} + PV_M = 919.77$$

The bond price with 11% required yield is USD 919.77.

The computation can be done with SymIntegration, it has the function **bondpricing(C,M,r,n)** to compute the bond price with formula from Eq. (9.6).

We also include a table of price-yield relationship for the 20-year bond with 10% coupon bond paid semiannually.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Present Value and Bond Pricing ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Present Value and Bond Pricing ]# ./main

A 20-year 10% coupon bond with yield 11% and semiannual coupon payments
Present Value of Coupon payments = 802.306
Present value of par (maturity value) = 117.463
Bond Price = 919.769

Price-Yield Relationship for a 20-year 10% coupon bond

Yield           Price
0.045           1720.32
0.05            1627.57
0.055           1541.76
0.06            1462.3
0.065           1388.65
0.07            1320.33
0.075           1256.89
0.08            1197.93
0.085           1143.08
0.09            1092.01
0.095           1044.41
0.1             1000
0.105           958.531
0.11            919.769
0.115           883.503
0.12            849.537
0.125           817.696
0.13            787.817
0.135           759.752
0.14            733.366
0.145           708.533
0.15            685.14
0.155           663.081
0.16            642.262
0.165           622.592

Time taken by function: 976 microseconds
```

Figure 9.1: *The computation for the price of a 20-year coupon bond paid semiannually with a 10% coupon rate and a par or maturity value of USD 1,000, for the required yield of 11% coupon rate (**SymIntegration/Examples/Financial Mathematics/Test SymIntegration Present Value and Bond Pricing/main.cpp**).*

II. Mortgage Loans

[H*] A mortgage is a loan secured by the colalteral of specified real estate property, which obliges the borrower to make a predetermined series of payments. The mortgage gives the lender (the mortgage) the right of foreclosure on the loan if the borrower (the mortgager) defaults. That is, if the borrower fails to make the contracted payments, the lender can seize the property eo ensure that the debt is paid off.

[H*] A potential homeowner who wants to borrow funds to purchase a home will apply for a loan from a mortgage originator. Upon completion of the application form, which provides financial information about the applicant, and payment of an application fee, the mortgage originator will perform a credit evaluation of the applicant. The two primary factors in determining whether the funds will be lect are the payment-to-income (PTI) ratio and the loan-to-value (LTV) ratio.

The PTI, the ratio of monthly payments (both mortgage and real estate tax payments) to monthly income, is a measure of the ability of the applicant to make monthly payments. The lower this ratio, the greater the likelihood that the applicant will be able to meet the required payments. The difference between the purchase price of the property and the amount borrowed is the borrower's down payment.

The LTV is the ratio of the amount of the loan to the market (or appraised) value of the property. The lower this ratio, the more protection the lender has if the applicant defaults and the property must be repossessed and sold.

[H*] Every mortgage loan must be serviced. Servicing of a mortgage loan involves collecting monthly payments and forwarding proceeds to ownersof the loan, sending payment notices to mortgagors; reminding mortgagors when payments are overdue, maintaining records of principal balances; administering an escrow balance for real estate taxes and insurance purposes, intiating foreclosure proceedings if necessary, and furnishing tax information to mortgagors when applicable.

Mortgage servicers include bank-related entities, thrift-related entities, and mortgage bankers.

[H*] To compute the mortgage payment every period, we use the amortization method. The amortization method is the method where the borrower repays the lender by means of installemnt payments at periodic intervals.

The formula to compute mortgage payment is derived from the present value formula:

$$MP = MB_0 \left[\frac{i(1+i)^n}{(i+1)^n - 1} \right] \tag{9.7}$$

where:
MP = mortgage payment every period
n = number of periods (months, years)
MB_0 = original mortgage balance
i = periodic interest rate

The term in brackets is called the payment factor or annuity factor.

We will represent the present value by v with formula

$$v = \frac{1}{1+i} \tag{9.8}$$

The formula to compute the interest that has to be paid at period-jth is:

$$interest_j = (1 - v^{n-j+1})MP \tag{9.9}$$

with $j = 1, 2, \ldots, n$.
The formula to compute the principal that has to be paid at period-jth is:

$$principal_j = (v^{n-j+1})MP \tag{9.10}$$

with $j = 1, 2, \ldots, n$.

i. Mortgage Loan Computation

Compute the amortization table for a 30-year (360-month) of USD 100,000 monthly mortgage (fixed-rate mortgage) with an 8.125% mortgage rate.

Solution:

Each monthly mortgage payment for a level-payment mortgage is due on the first of each month and consists of

1. Interest of one-twelfth of the fixed annual interest rate times the amount of the outstanding mortgage balance at the beginning of the previous month.
2. A repayment of a portion of the outstanding mortgage balance (principal).

The difference between the monthly mortgage payment adn the portion of the payment that represents interest equals the amount that is applied to reduce the outstanding mortgage balance. The monthly mortgage payment is designed so that after the last scheduled monthly payment of the loan is made, the amount of the outstanding mortgage balance is zero (the mortgage is fully repaid).

At the beginning of month 1, the mortgage balance is USD 100,000, the amount of the original loan. The mortgage payment for month 1 includes interest on the USD 100,000 borrowed for the month.

The monthly payment is

$$MP = MB_0 \left[\frac{i(1+i)^n}{(i+1)^n - 1} \right] = 100000 \left[\frac{0.0067708(1.0067708)^{360}}{(1.0067708)^{360} - 1} \right] = 742.50$$

Because the interest is 8.125%, the monthly interest rate is $\frac{0.08125}{12} = 0.0067708$. Interest for month 1 is therefore

$$interest_1 = (1 - v^{n-1+1})MP = \left(1 - \left(\frac{1}{1.0067708} \right)^{360} \right) 742.50 = 677.08$$

the difference between the monthly mortgage payment of USD 742.50 and the interest of USD 677.08 is the portion of the monthly mortgage payment that represent repayment of principal.

$$principal_1 = (v^{n-1+1})MP = \left(\left(\frac{1}{1.0067708} \right)^{360} \right) 742.50 = 65.41$$

We can also use $principal_1 = MP - interest_1$ formula. This USD 65.41 in month 1 reduces the mortgage balance.

In SymIntegration we have a function that will show the amortization table till the last period, it is:

amortization(L,r,n)

with L as the total loan / mortgage, r as the interest rate per period, and n is the number of periods.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Amortization ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Amortization ]# ./main

Amortization

Period              Payment             Interest            Principal       Outstanding Loan Balance
1                   742.497             677.083             65.4138             99934.6
2                   742.497              676.64             65.8568             99868.7
3                   742.497             676.195             66.3027             99802.4
4                   742.497             675.746             66.7516             99735.7
5                   742.497             675.294             67.2035             99668.5
6                   742.497             674.839             67.6586             99600.8
7                   742.497             674.381             68.1167             99532.7
8                   742.497             673.919             68.5779             99464.1
9                   742.497             673.455             69.0422             99395.1
10                   742.497             672.987             69.5097             99325.6
11                   742.497             672.517             69.9803             99255.6
12                   742.497             672.043             70.4541             99185.1
13                   742.497             671.566             70.9312             99114.2
14                   742.497             671.086             71.4114             99042.8
15                   742.497             670.602              71.895             98970.9
16                   742.497             670.115             72.3817             98898.5
17                   742.497             669.625             72.8718             98825.6
18                   742.497             669.132             73.3652             98752.3
19                   742.497             668.635              73.862             98678.4
20                   742.497             668.135             74.3621             98604.1
21                   742.497             667.632             74.8656             98529.2
22                   742.497             667.125             75.3725             98453.8
23                   742.497             666.614             75.8828             98377.9
24                   742.497             666.101             76.3966             98301.5
25                   742.497             665.583             76.9139             98224.6
26                   742.497             665.063             77.4346             98147.2
27                   742.497             664.538             77.9589             98069.2
28                   742.497              664.01             78.4868             97990.7
29                   742.497             663.479             79.0182             97911.7
30                   742.497             662.944             79.5532             97832.2
```

Figure 9.2: *The amortization table for a 30-year (360-month) of USD 100,000 monthly mortgage (fixed-rate mortgage) with an 8.125% mortgage rate(**SymIntegration/Examples/Financial Mathematics/Test SymIntegration Amortization for Mortgage/main.cpp**).*

```
328                   742.497             148.227             594.271             21297.7
329                   742.497             144.203             598.294             20699.4
330                   742.497             140.152             602.345               20097
331                   742.497             136.074             606.424             19490.6
332                   742.497             131.968              610.53             18880.1
333                   742.497             127.834             614.663             18265.4
334                   742.497             123.672             618.825             17646.6
335                   742.497             119.482             623.015             17023.6
336                   742.497             115.264             627.233             16396.3
337                   742.497             111.017              631.48             15764.9
338                   742.497             106.741             635.756             15129.1
339                   742.497             102.437             640.061               14489
340                   742.497             98.1029             644.394             13844.6
341                   742.497             93.7398             648.757             13195.9
342                   742.497             89.3472              653.15             12542.7
343                   742.497             84.9248             657.572             11885.2
344                   742.497             80.4725             662.025             11223.1
345                   742.497               75.99             666.507             10556.6
346                   742.497             71.4772              671.02             9885.62
347                   742.497             66.9339             675.563             9210.05
348                   742.497             62.3597             680.137             8529.92
349                   742.497             57.7546             684.743             7845.17
350                   742.497             53.1184             689.379              7155.8
351                   742.497             48.4507             694.046             6461.75
352                   742.497             43.7514             698.746                5763
353                   742.497             39.0203             703.477             5059.53
354                   742.497             34.2572              708.24             4351.29
355                   742.497             29.4618             713.035             3638.25
356                   742.497              24.634             717.863             2920.39
357                   742.497             19.7735             722.724             2197.66
358                   742.497               14.88             727.617             1470.05
359                   742.497             9.95345             732.544             737.504
360                   742.497             4.99351             737.504                   0

Total payment:
267299

Time taken by function: 17046 microseconds
```

Figure 9.3: *The amortization table for a 30-year (360-month) of USD 100,000 monthly mortgage (fixed-rate mortgage) with an 8.125% mortgage rate(**SymIntegration/Examples/Financial Mathematics/Test SymIntegration Amortization for Mortgage/main.cpp**).*

III. Financial Derivative: Forwards, Futures, Swaps, and Options

[H*] A financial derivative is an instrument that is related tos ome other asset and whose value is derived from that asset. There are many types of derivatives. Forward/futures and option contracts are two of the most widely used financial derivatives.

[H*] A forward or futures contract on an underlying asset is an agreement between two parties to sell and buy that asset at a specified future date for a specified price.

An option on an asset is a contract that allows, but does not require, the holder of the option to buy the asset (call option) or sell the asset (put option) on or before a specified date at a specified price.

[H*] A financial position is any combination of investments, including long and short positions in any assets, or in risk-free investing or borrowing. When describing the outcome at time T of a financial position created at time 0 we make a distinction between the payoff at time T of the financial position and the profit from time 0 to time 4t4 on the financial position.

The payoff is simply the value of the asset at time T, say S_T. If the cost of the asset at time 0 was S_0, then the accumulated cost to time T of creating the long position at time 0 is $S_0(1+i)^T$, where i, is the (annual effective) risk-free rate of interest. The profit at time T, is

$$S_T - S_0(1+i)^T$$

Definition 9.1: Forward Contract

A forward contract is an agreement to buy or sell a certain asset at a specified future date called the delivery date, for a specific price called the delivery price. Both parties of the forward contract are bound by the contract terms at the time the contract is made, with the contract being settled on the delivery date.

[H*] In general, the forward contract is constructed at time 0 so that no money changes hands at time 0, and the only exchange takes place at time T.

We will denote by S_0 the value of the asset at time 0, and S_T denote its value at time T. The value at time 0 (current price) is also referred to as the spot price. The spot price is the price quoted for imemdiate settlement, payment and delivery of a security or commodity. For a forward contract arranged at time 0 for delivery to take place at time T, we will denote the delivery price by $F_{0,T}$. The payoff on the long position at the time of delivery is

$$S_T - F_{0,T}$$

and the payoff on the short position at the time of delivery is

$$F_{0,T} - S_T$$

Since no money is initially invested for a forward contract, the profit on the delivery date is the same as the payoff on the delivery date. As with any buy/sell transaction, there are two positions that are taken on a forward contract. The party that will take delivery of the asset and pay the delivery price at time T has a long position on the forwad contract, and the party that will deliver the asset and be paid at time 4t4 has a short position on the contract.

Chapter 10

Introduction to Stochastic Processes

"One side will make you taller, the other side will make you shorter" (Alice in Wonderland 1951) - ***The Caterpillar***

In probability theory and related fields, a stochastic or random process is a mathematical object usually defined as a sequence of random variables, where the index of the sequence has the interpretation of time. Stochastic processes are widely used as mathematical models of systems and phenomena that appear to vary in a random manner. Examples include the growth of a bacterial population, an electrical current fluctuating due to thermal noise, or the movement of a gas molecule. Stochastic processes have applications in many disciplines such as biology, chemistry, ecology, neuroscience, physics, image processing, signal processing, control theory, information theory, computer science, and telecommunications. Furthermore, seemingly random changes in financial markets have motivated the extensive use of stochastic processes in finance.

Applications and the study of phenomena have in turn inspired the proposal of new stochastic processes. Examples of such stochastic processes include the Wiener process or Brownian motion process, used by Louis Bachelier to study price changes on the Paris Bourse, and the Poisson process, used by A. K. Erlang to study the number of phone calls occurring in a certain period of time. These two stochastic processes are considered the most important and central in the theory of stochastic processes, and were discovered repeatedly and independently, both before and after Bachelier and Erlang, in different settings and countries.

Chapter 11

Numerical Analysis

Je n'avais pas peur, je suis né pour faire ça - Jeanne d'Arc

Numerical analysis is the study of algorithms that use numerical approximation (as opposed to symbolic manipulations) for the problems of mathematical analysis (as distinguished from discrete mathematics). It is the study of numerical methods that attempt at finding approximate solutions of problems rather than the exact ones. Numerical analysis finds application in all fields of engineering and the physical sciences, and in the 21st century also the life and social sciences, medicine, business and even the arts.

Modern numerical analysis begins by categorizing equations as belonging to different types and then devises methods of solution for generic problems of each type.

I. Solutions of Equations in One Variable

- Root Finding Methods
  In this section, we consider one of the most basic problems of numerical approximation, the root-finding problem. This process involves finding a root, or solution of an equation of the form

$$f(x) = 0$$

for a given function f.

Definition 11.1: Bisection Method

The Bisection Method is based on the Intermediate Value Theorem. Suppose f is a continuous function defined on the interval $[a, b]$ with $f(a)$ and $f(b)$ of opposite sign. By the Intermediate Value Theorem, there exists a number p in (a, b) with

$$f(p) = 0$$

We assume for simplicity that the root in the interval $[a, b]$ is unique. This method calls for a repeated halving of subintervals of $[a, b]$.

[H*] The bisection method is slow to converge, N may become quite large before $|p - p_N|$ is sufficiently small, and a good intermediate approximation can be inadvertently discarded.

[H*] The bisectionmethod has the important property that it always converges to a solution.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method with Function ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method with Function ]# ./main
f = -cos(θ0)-2*θ0+2*sin(θ0)+1
iteration           a               b               p               f(p)
     1              1               2               1.5             -0.0757472

     2              1               1.5             1.25            0.0826469

     3              1.25            1.5             1.375           0.0172384

     4              1.375           1.5             1.4375          -0.0256436

     5              1.375           1.4375          1.40625         -0.00331926

     6              1.375           1.40625         1.39062         0.00717788

     7              1.39062         1.40625         1.39844         0.00198421

     8              1.39844         1.40625         1.40234         -0.000653763

     9              1.39844         1.40234         1.40039         0.000668658

    10              1.40039         1.40234         1.40137         8.30682e-06

    11              1.40137         1.40234         1.40186         -0.000322513

    12              1.40137         1.40186         1.40161         -0.00015705

    13              1.40137         1.40161         1.40149         -7.43579e-05
Procedure completed successfully
solution = 1.40149

Time taken by function: 30471 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method with Function ]# 
```

Figure 11.1: *The computation for bisection method for $f(\theta_0) = 2\theta_0 - 2\sin\ \theta_0 + \cos\quad theta_0 - 1$ with function (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Bisection Method with Function for Brachistochrone/main.cpp**).*

The computation is using SymIntegration, with this function: **bisectionmethod(f, x, a, b,N)** that can be called to compute the the solution of a function with one variable, this method doesn't need to compute the derivative so it can be used to almost all functions.

As a comparison we try to measure the time if we apply the bisection method directly in the main source code, turns out it is faster than using the function.

```
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method ]# ./main
f = -cos(θ0)-2*θ0+2*sin(θ0)+1

a = 1
b = 2
p = 1.5
f(p) = -0.0757472
f(b) = -0.765258
f(a) = 0.14264
f(a)*f(p) =-0.0108046
iteration        a              b              p              f(p)
     1           1              2              1.5            -0.0757472

     2           1              1.5            1.25           0.0826469

     3           1.25           1.5            1.375          0.0172384

     4           1.375          1.5            1.4375         -0.0256436

     5           1.375          1.4375         1.40625        -0.00331926

     6           1.375          1.40625        1.39062        0.00717788

     7           1.39062        1.40625        1.39844        0.00198421

     8           1.39844        1.40625        1.40234        -0.000653763

     9           1.39844        1.40234        1.40039        0.000668658

    10           1.40039        1.40234        1.40137        8.30682e-06

    11           1.40137        1.40234        1.40186        -0.000322513

    12           1.40137        1.40186        1.40161        -0.00015705

    13           1.40137        1.40161        1.40149        -7.43579e-05
Procedure completed successfully

Time taken by function: 28906 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Bisection Method ]# 
```

Figure 11.2: *The computation for bisection method for $f(\theta_0) = 2\theta_0 - 2\sin\ \theta_0 + \cos\ \ theta_0 - 1$ without function (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Bisection Method Manual for Brachistochrone/main.cpp**).*

Definition 11.2: Newton-Raphson Method

The Newton-Raphson Method is one of the most powerful and well known numerical methods for solving a root-finding problem.
Suppose that $f \in \mathbb{C}^2[a,b]$. Let $\bar{x} \in [a,b]$ be an approximate to p such that $f'(\bar{x}) \neq 0$ and $|p-\bar{x}|$ is "small." Consider the first Taylor polynomial for $f(x)$ expanded about $(\bar{x})$,

$$f(x) = f(\bar{x}) + (x-\bar{x})f'(\bar{x}) + \frac{(x-\bar{x})^2}{2} f''(\xi(x))$$

where $\xi(x)$ lies between x and $\bar{x}$. Since $f(p)=0$, this equation with $x=p$ gives

$$0 = f(\bar{x}) + (p-\bar{x})f'(\bar{x}) + \frac{(p-\bar{x})^2}{2} f''(\xi(p))$$

Newton's method is derived by assuming that since $|p-\bar{x}|$ is small, the term involving $(p-\bar{x})^2$ is much smaller, so

$$0 \approx f(\bar{x}) + (p-\bar{x})f'(\bar{x})$$

Solving for p gives

$$p \approx \bar{x} - \frac{f(\bar{x})}{f'(\bar{x})}$$

Thus, the Newton-Raphson method, which starts with an initial approximation p_0 and generate the sequence $\{p_n\}_{n=0}^{\infty}$, is

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad \text{for } n \geq 1 \tag{11.1}$$

Starting with the initial approximation p_0, the approximation p_1 is the x-intercept of the tangent line to the graph of f at $(p_0, f(p_0))$. The approximation p_2 is the x-intercept of the tangent line to the graph of f at $(p_1, f(p_1))$ and so on.

[H*] Newton-Raphson method is a functional iteration technique of the form

$$p_n = g(p_{n-1})$$

for which

$$g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad \text{for } n \geq 1 \tag{11.2}$$

this is the functional iteration technique that was used to give the rapid convergence.

[H*] It is clear that Newton-Raphson method cannot be continued if $f'(p_{n-1}) = 0$ for some n. This method is the most effective when f' is bounded away from zero near p.

Figure 11.3: *The plot for root finding with bisection method (above) for $f(x) = x^3 + 4x^2 - 10$ and the Newton-Raphson method (below) for $f(x) = \cos(x) - x$. (**Hamzstlab-Mathematics/examples/Root Finding/-main.cpp**).*

The computation is using SymIntegration, with this function: **newtonmethod(**f**,**x**,**x_0**,**N**)** that can be called to compute the the solution of a function with one variable, given that the derivative is continuous and exist.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Newton's Method with function ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Newton's Method with function ]# ./main
Newton method :
f(x) = cos(x)-x
f'(x) = -sin(x)-1

     n          p_{n}
     0          0.78539816339742
     1          0.73953613351524
     2          0.73908517810601
     3          0.73908513321516
The procedure was successful.
solution =
0.73908513321516
Time taken by function: 5645 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Newton's Method with function ]# 
```

Figure 11.4: *The computation for root finding Newton-Raphson method for $f(x) = \cos(x) - x$. with function (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Newton's Method with function/main.cpp**).*

As a comparison we try to implement Newton-Raphson manually, in the main source code, it turns out calling the function resulting in faster speed.

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Newton's Method Manual ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Newton's Method Manual ]# ./main
f(x) = cos(x)-x
f(x)[x==pi/4] = -0.0782914

f'(x) = -sin(x)-1
f'(x)[x==pi/4] = -1.70711

     n          p_{n}                         p_{n} in Degree
     0          0.78539816339742              44.999999999999
     1          0.73953613351524              42.372299247846
     2          0.73908517810601              42.346461406149
     3          0.73908513321516              42.346458834093
The procedure was successful.
Time taken by function: 6779 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Newton's Method Manual ]# 
```

Figure 11.5: *The manual computation for root finding Newton-Raphson method for $f(x) = \cos(x) - x$. (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Newton's Method Manual/main.cpp**).*

II. Interpolation and Polynomial Approximation

- Interpolation and the Lagrange Polynomial
  Taylor polynomials are not appropriate for interpolation, because the approximations are needed only at points close to x_0. For ordinary computational purposes it is more efficient to use methods that include information at various points. To generalize the concept of linear interpolation, consider construction of a polynomial of degree at most n that passes through $n+1$ points

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

to approximate that $n+1$ points, we will use the well-established Lagrange Interpolating Polynomial.

Theorem 11.1: nth Lagrange Interpolating Polynomial

If $x_0, x_1, \dots, x_n$ are $n+1$ distinct numbers and f is a function whose values are given at these numbers, then a unique polynomial $P(x)$ of degreen at most n exists with

$$f(x_k) = P(x_k)$$

for each $k = 0, 1, \dots, n$ This polynomial is given by

$$P(x) = f(x_0)L_{n,0}(x) + \cdots + f(x_n)L_{n,n}(x) = \sum_{k=0}^{n} f(x_k)L_{n,k}(x) \tag{11.3}$$

where, for each $k = 0, 1, \dots, n$,

$$\begin{aligned} L_{n,k}(x) &= \frac{(x-x_0)(x-x_1)\dots(x-x_{k-1})(x-x_{k+1})\dots(x-x_n)}{(x_k-x_0)(x_k-x_1)\dots(x_k-x_{k-1})(x_k-x_{k+1})(x_k-x_n)} \\ &= \prod_{i=0,\ i\neq k}^{n} \frac{(x-x_i)}{(x_k-x_i)} \end{aligned} \tag{11.4}$$

Theorem 11.2: Approximating a Function by an Interpolating Polynomial

Suppose $x_0, x_1, \dots, x_n$ are distinct numbers in the interval $[a, b]$ and $f \in C^{n+1}[a, b]$. Then, for each x in $[a, b]$, a number $\xi(x)$ in (a, b) exists with

$$f(x) = P(x) + \frac{f^{(n+1)(\xi(x))}}{(n+1)!}(x - x_0)(x - x_1)\dots(x - x_n) \tag{11.5}$$

where $P(x)$ is the interpolating polynomial given in Eq. (11.3).

- **Example:**
The example is taken from example 1, chapter 3.1 of [5].
Use the numbers (or nodes) $x_0 = 2, x_1 = 2.5$, and $x_2 = 4$ to find the second interpolating polynomial for $f(x) = \frac{1}{x}$.

Solution:
We need to determine the coefficient polynomials $L_0(x), L_1(x)$, and $L_2(x)$. In nested form they are

$$\begin{aligned} L_0(x) &= \frac{(x-2.5)(x-4)}{(2-2.5)(2-4)} = (x-6.5)x + 10 \\ L_1(x) &= \frac{(x-2)(x-4)}{(2.5-2)(2-4)} = \frac{(-4x+24)x - 32}{3} \\ L_2(x) &= \frac{(x-2)(x-2.5}{(4-2)(4-2.5)} = \frac{(x-4.5)x + 5}{3} \end{aligned}$$

Since

$$\begin{aligned} f(x_0) &= f(2) = 0.5 \\ f(x_1) &= f(2.5) = 0.4 \\ f(x_2) &= f(4) = 0.25 \end{aligned}$$

we will have

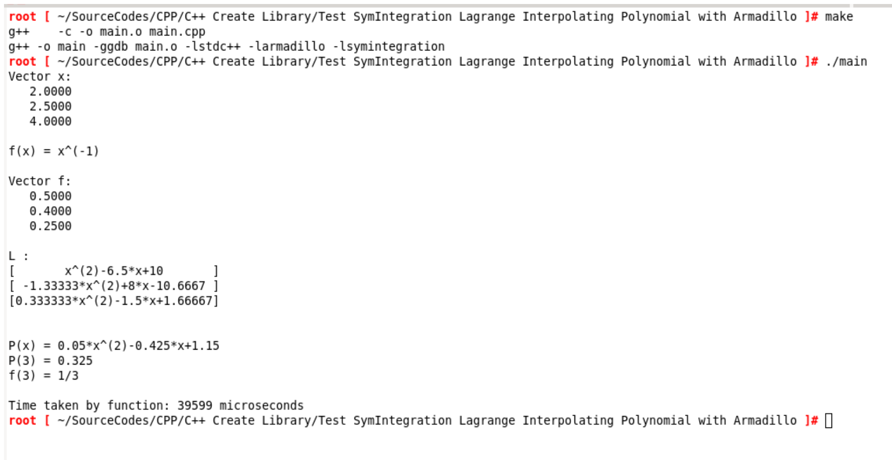
$$\begin{aligned} P(x) &= \sum_{k=0}^{2} f(x_k)L_k(x) \\ &= 0.5\left((x-6.5)x + 10\right) + 0.4\left(\frac{(-4x+24)x - 32}{3}\right) + 0.25\left(\frac{(x-4.5)x + 5}{3}\right) \\ &= (0.05x - 0.425)x + 1.15 \end{aligned}$$

We can now check with $x = 3$

$$\begin{aligned} f(3) &= \frac{1}{3} \approx 0.33333 \\ P(3) &= 0.325 \end{aligned}$$

it is quite a good approximation.

The C++ code is using SymIntegration and Armadillo to help us compute the polynomial Lagrange $P(x)$, we can input the initial nodes $x_0, x_1, \dots, x_N$ from textfile **vectorX.txt**, and then the function $f(x)$ can be inputted in the main source code symbolically.

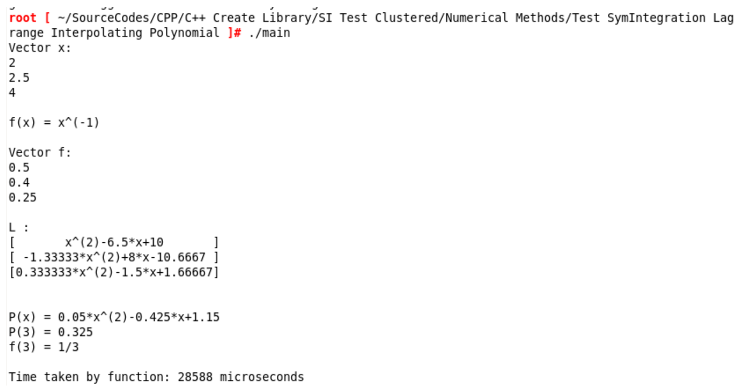


```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Lagrange Interpolating Polynomial with Armadillo ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -larmadillo -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Lagrange Interpolating Polynomial with Armadillo ]# ./main
Vector x:
   2.0000
   2.5000
   4.0000

f(x) = x^(-1)

Vector f:
   0.5000
   0.4000
   0.2500

L :
[       x^(2)-6.5*x+10       ]
[ -1.33333*x^(2)+8*x-10.6667 ]
[0.333333*x^(2)-1.5*x+1.66667]


P(x) = 0.05*x^(2)-0.425*x+1.15
P(3) = 0.325
f(3) = 1/3

Time taken by function: 39599 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Lagrange Interpolating Polynomial with Armadillo ]#
```

Figure 11.6: *The computation to approximate $f(x) = \frac{1}{x}$ with nodes $x_0 = 2, x_1 = 2.5$, and $x_2 = 4$ using Lagrange interpolating polynomial with SymIntegration and Armadillo (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Lagrange Interpolating Polynomial with Armadillo/main.cpp**).*

Figure 11.7: *The plot for Lagrange interpolating polynomial with Hamzstlab, SymIntegration and Armadillo for $f(x) = \frac{1}{x}$ with nodes $x_0 = 2, x_1 = 2.5$, and $x_2 = 4$. (**Hamzstlab-Mathematics/examples/Interpolation/main.cpp**).*

Another C++ code that we create is using SymIntegration only, after doing work on linear algebra we add more functions so it can have better matrix and vector operations. The result is amazing, the speed almost twice faster than when we are using Armadillo.



```
root [ ~/SourceCodes/CPP/C++ Create Library/SI Test Clustered/Numerical Methods/Test SymIntegration Lag
range Interpolating Polynomial ]# ./main
Vector x:
2
2.5
4

f(x) = x^(-1)

Vector f:
0.5
0.4
0.25

L :
[       x^(2)-6.5*x+10       ]
[ -1.33333*x^(2)+8*x-10.6667 ]
[0.333333*x^(2)-1.5*x+1.66667]


P(x) = 0.05*x^(2)-0.425*x+1.15
P(3) = 0.325
f(3) = 1/3

Time taken by function: 28588 microseconds
```

Figure 11.8: *The computation to approximate $f(x) = \frac{1}{x}$ with nodes $x_0 = 2, x_1 = 2.5$, and $x_2 = 4$ using Lagrange interpolating polynomial with SymIntegration only(**SymIntegration/Examples/Numerical Methods/Test SymIntegration Lagrange Interpolating Polynomial/main.cpp**).*

III. Numerical Methods for Optimization

- Minimizing a single-variable function

[H*] The numerical methods for minimizing a single-variable function can be broadly categorized into interval reduction methods and descent methods.

Interval reduction methods, like Golden Section search, iteratively narrow down an interval known to contain the minimum, while descent methods, such as gradient descent, use derivative information to iteratively move towards the minimum.

[H*] The interval reduction methods:

1. **Golden Section Search**
   This method is a bracketing method that works by iteratively narrowing down the interval containing the minimum. It uses the golden ratio to determine two points within the interval, evaluates the function at these points, and then discards a portion of the interval based on the function values. This process is repeated until the interval is sufficiently small.
2. **Fibonacci Search**
   Similar to Golden Section Search, Fibonacci Search uses the Fibonacci sequence to determine the points at which to evaluate the function. It offers optimal performance for a fixed number of function evaluations.

[H*] The descent methods:

1. **Gradient Descent**
   This method uses the gradient (derivative) of the function to determine the direction of steepest descent and iteratively moves towards the minimum. At each step, the algorithm moves in the opposite direction of the gradient.
2. **Newton's Method**
   This method uses both the first and second derivatives (gradient and Hessian) to find the minimum. It offers faster convergence than gradient descent but requires the calculation of the Hessian.

Many algorithms for geometry optimization are based on some variant of the Newton-Raphson (NR) scheme. In order to find the minima, the first derivative $f'(x) = 0$ (must be zero) and the second derivative $f''(x) > 0$ (must be positive). To find the maxima, the first derivative $f'(x) = 0$ (must be zero) and the second derivative $f''(x) < 0$ (must be negative).

[H*] The other methods:

1. **Brent's Method**
   A hybrid method that combines the bracketing approach of Golden Section Search with the local quadratic approximation of Newton's method, offering a good balance of robustness and efficiency.
2. **Downhill Simplex Method (Nelder-Mead)**
   A direct search method that does not require derivative information. It works by iteratively moving and transforming a simplex (a geometric shape with $n+1$ vertices in n-dimensional space) in the search space to find the minimum.

[H*] The best method for minimizing a single-variable function depends on the specific characteristics of the function and the desired level of accuracy. For well-behaved functions, gradient-based methods like Newton's method can offer fast convergence. For functions whose derivatives are difficult or impossible to compute, direct search methods like Brent's method or the downhill simplex method may be more suitable.

- Gradient Descent
  Gradient Descent (GD) is an iterative first-order optimisation algorithm, used to find a local maximum/minimum of a given function. This method is commonly used in machine learning (ML) and deep learning (DL) to minimize a cost/loss function (e.g. in a linear regression). Due to its importance and ease of implementation, this algorithm is usually taught at the beginning of almost all machine learning courses. It is also widely used in areas like control engineering (robotics, chemical), computer games, mechanical engineering.

[H*] This method was proposed long before the era of modern computers by Augustin-Louis Cauchy in 1847.

[H*] Gradient descent algorithm does not work for all functions. There are two specific requirements. A function has to be:

1. Differentiable
   Differentiable means a function has a derivative for each point in its domain. Typical non-differentiable functions have a step or a cusp or a discontinuity, such as $f(x) = \frac{1}{x}, f(x) = \sqrt{|x|}$
2. Convex
   For a univariate function, this means that the line segment connecting two function's points lays on or above its curve (it does not cross it). If it does, then it means that it has a local minimum which is not a global one.
   In mathematics notation, for two points x_1 and x_2 laying on the function's curve this condition is expressed as:

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2) \tag{11.6}$$

where λ denotes a point's location on a section line and its value has to be between 0 (left point) and 1 (right point), e.g. $\lambda = 0.5$ means a location in the middle.

Another way to check mathematically if a univariate function is convex is to calculate the second derivative and check if itsvalue is always bigger than 0.

$$\frac{d^2 f(x)}{dx^2} > 0$$

[H*] A gradient fo an n-dimensional function $f(x)$ at a given point p is defined as

$$\nabla f(p) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(p) \\ \frac{\partial f}{\partial x_2}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{bmatrix} \tag{11.7}$$

The upside-down triangle is a notation called nabla, or you can also call it del operator.

[H*] **Gradient Descent Algorithm**
Gradient Descent Algorithm iteratively calculates the next point using gradient at the current position, scales it (by a learning rate) and subtracts obtained value from the current position (makes a step). It subtracts the value because we want to minimize the function (to maximize it would be adding). This process can be written as:

$$p_{n+1} = p_n - \gamma \nabla f(p_n) \tag{11.8}$$

There's an important parameter γ (read: gamma) which scales the gradient and thus controls the step size. In machine learning, gamma is called the learning rate and have a strong influence on performance.

The smaller the learning rate, the longer the GD converges, or may reach maximum iteration before reaching the optimum point.

If the learning rate is too big the algorithm may not converge to the optimal point (jump around) or even diverge completely.

The Gradient Descent method's steps are:

1. Choose a starting point (initialisation).
2. Calculate gradient at this point.
3. Make a scaled step in the opposite direction to the gradient (objective: minimize).
4. Repeat steps number 2 and 3 until one of the criteria is met:
   - Maximum number of iterations reached.
   - Step size is smaller than the tolerance (due to scaling or a small gradient)

[H*] **Example:**
Find the global minimum of a function that is given by

$$f(x) = x^2 - x + 3$$

Solution:
We will compute the first and second derivative of $f(x)$.

$$\frac{df(x)}{dx} = 2x - 1$$
$$\frac{d^2 f(x)}{dx^2} = 2$$

Because the second derivative is always bigger than 0, $f(x)$ is strictly convex.

Now we examine the first derivative and equate it to zero.

$$\begin{aligned}\frac{df(x)}{dx} &= 0 \\ 2x - 1 &= 0 \\ x &= 0.5\end{aligned}$$

$x = 0.5$ is the critical point.

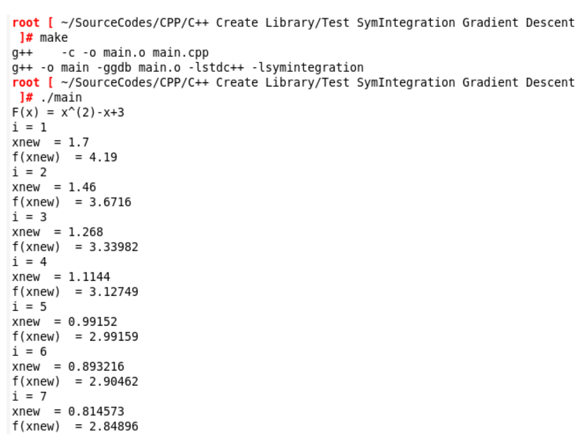
Now, we will check with the gradient descent method, first we will choose the initial value and the parameter γ (the learning rate):

$$x_{old} = 2, \quad \gamma = 0.1$$

then we will have

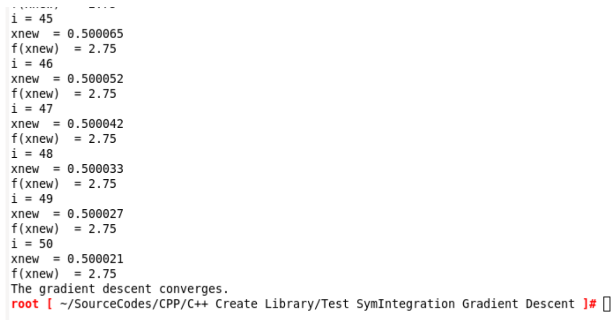
$$\begin{aligned}x_{new} &= x_{old} - \gamma \nabla f(x_{old}) \\ x_{new} &= 2 - 0.1(2(2) - 1) = 1.7 \quad (i = 1) \\ x_{new} &= 1.7 - 0.1(2(1.7) - 1) = 1.46 \quad (i = 2) \\ x_{new} &= 1.46 - 0.1(2(1.46) - 1) = 1.268 \quad (i = 3) \\ &\vdots \\ x_{new} &= 0.500027 - 0.1(2(0.500027) - 1) = 0.500021 \quad (i = 50)\end{aligned}$$

It is proven then with the gradient descent method that the critical point is at $x = 0.5$. The computation can be done with **SymIntegration**, as long as the function $f(x)$ is differentiable and the derivative can be computed then we can use gradient descent method. Polynomials without the combination of trigonometry and transcendentals functions can be differentiated easily, even the fraction polynomial.



```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Gradient Descent
 ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Gradient Descent
 ]# ./main
F(x) = x^(2)-x+3
i = 1
xnew  = 1.7
f(xnew)  = 4.19
i = 2
xnew  = 1.46
f(xnew)  = 3.6716
i = 3
xnew  = 1.268
f(xnew)  = 3.33982
i = 4
xnew  = 1.1144
f(xnew)  = 3.12749
i = 5
xnew  = 0.99152
f(xnew)  = 2.99159
i = 6
xnew  = 0.893216
f(xnew)  = 2.90462
i = 7
xnew  = 0.814573
f(xnew)  = 2.84896
```

Figure 11.9: *The computation for minimization problem for function $f(x) = x^2 - x + 3$ with gradient descent method. (**SymIntegration/Examples/Operations Research/Test SymIntegration Gradient Descent/main.cpp**).*



```
i = 45
xnew  = 0.500065
f(xnew)  = 2.75
i = 46
xnew  = 0.500052
f(xnew)  = 2.75
i = 47
xnew  = 0.500042
f(xnew)  = 2.75
i = 48
xnew  = 0.500033
f(xnew)  = 2.75
i = 49
xnew  = 0.500027
f(xnew)  = 2.75
i = 50
xnew  = 0.500021
f(xnew)  = 2.75
The gradient descent converges.
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Gradient Descent ]#
```

Figure 11.10: *The gradient descent method took 50 iteration to converges with $\gamma = 0.1$. (**SymIntegration/Examples/-Operations Research/Test SymIntegration Gradient Descent/main.cpp**).*

[H*] **Example:**
Find the global minimum of a function that is given by

$$f(x,y) = \frac{1}{2}x^2 + 2x + y^2 + y + 3$$

Solution:
In the case of a multivariate function, we need to compute the vector of derivatives in each main direction (along variable axes). Because we are interested only in a slope along one axis and we don't care about others, these derivatives are called partial derivatives.

We will have

$$\begin{aligned} z = f(x,y) &= \frac{1}{2}x^2 + 2x + y^2 + y + 3 \\ \frac{\partial f}{\partial x} &= x + 2 \\ \frac{\partial f}{\partial y} &= 2y + 1 \end{aligned}$$

then the nabla is

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} x + 2 \\ 2y + 1 \end{bmatrix}$$

at initialization we will input the initial value and the parameter γ

$$\begin{aligned} x_{old} &= -5 \\ y_{old} &= 21 \\ \gamma &= 0.1 \end{aligned}$$

thus the gradient descent method

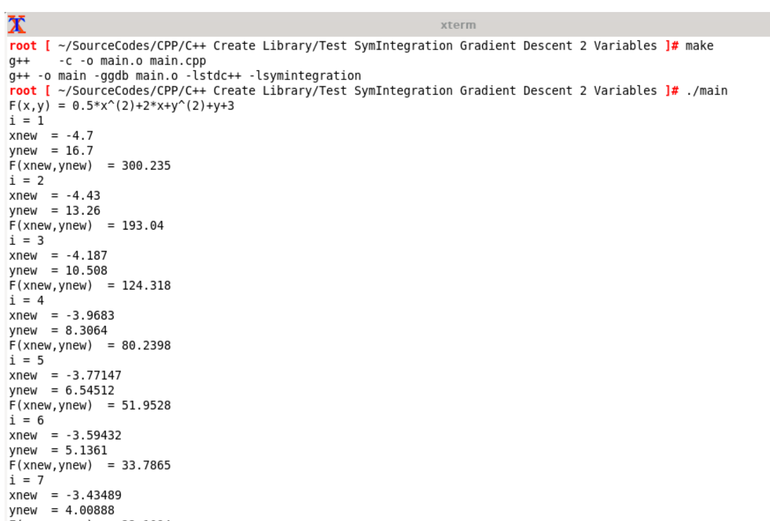
$$
\begin{aligned}
x_{new} &= x_{old} - \gamma \nabla f(x_{old}) \\
x_{new} &= -5 - 0.1((-5)+2) = -4.7 \quad (i=1) \\
x_{new} &= -4.7 - 0.1((-4.7)+2) = -4.43 \quad (i=2) \\
x_{new} &= -4.43 - 0.1((-4.43)+2) = -4.187 \quad (i=3) \\
&\vdots \\
x_{new} &= -2.01718 - 0.1((-2.01718)+2) = -2.01546 \quad (i=50)
\end{aligned}
$$

$$
\begin{aligned}
y_{new} &= y_{old} - \gamma \nabla f(y_{old}) \\
y_{new} &= 21 - 0.1(2(21)+1) = 16.7 \quad (i=1) \\
y_{new} &= 16.7 - 0.1(2(16.7)+1) = 13.26 \quad (i=2) \\
y_{new} &= 13.26 - 0.1(2(13.26)+1) = 10.508 \quad (i=3) \\
&\vdots \\
y_{new} &= -0.499616 - 0.1(2(-0.499616)+1) = -0.499693 \quad (i=50)
\end{aligned}
$$

and the value for function $f(x,y)$ at every iteration

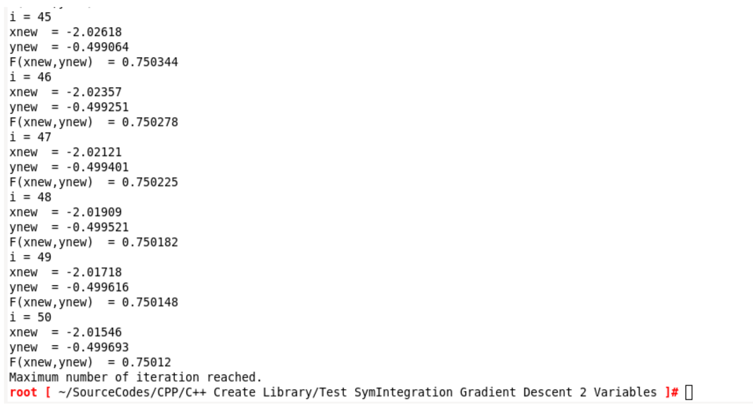
$$
\begin{aligned}
f(x_{new}, y_{new}) &= 300.235 \quad (i=1) \\
f(x_{new}, y_{new}) &= 193.04 \quad (i=2) \\
f(x_{new}, y_{new}) &= 124.318 \quad (i=3) \\
&\vdots \\
f(x_{new}, y_{new}) &= 0.75012 \quad (i=50)
\end{aligned}
$$

so the function $f(x,y) = \frac{1}{2}x^2 + 2x + y^2 + y + 3$ reaches the minimum of 0.75012 at $(-2.01546, -0.499693)$.



```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Gradient Descent 2 Variables ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Gradient Descent 2 Variables ]# ./main
F(x,y) = 0.5*x^(2)+2*x+y^(2)+y+3
i = 1
xnew  = -4.7
ynew  = 16.7
F(xnew,ynew)  = 300.235
i = 2
xnew  = -4.43
ynew  = 13.26
F(xnew,ynew)  = 193.04
i = 3
xnew  = -4.187
ynew  = 10.508
F(xnew,ynew)  = 124.318
i = 4
xnew  = -3.9683
ynew  = 8.3064
F(xnew,ynew)  = 80.2398
i = 5
xnew  = -3.77147
ynew  = 6.54512
F(xnew,ynew)  = 51.9528
i = 6
xnew  = -3.59432
ynew  = 5.1361
F(xnew,ynew)  = 33.7865
i = 7
xnew  = -3.43489
ynew  = 4.00888
F(xnew,ynew)  = 22.1094
```

Figure 11.11: *The computation for minimization problem for function $f(x,y) = \frac{1}{2}x^2 + 2x + y^2 + y + 3$ with gradient descent method. (**SymIntegration/Examples/Operations Research/Test SymIntegration Gradient Descent 2 Variables/main.cpp**).*



```
i = 45
xnew  = -2.02618
ynew  = -0.499064
F(xnew,ynew)  = 0.750344
i = 46
xnew  = -2.02357
ynew  = -0.499251
F(xnew,ynew)  = 0.750278
i = 47
xnew  = -2.02121
ynew  = -0.499401
F(xnew,ynew)  = 0.750225
i = 48
xnew  = -2.01909
ynew  = -0.499521
F(xnew,ynew)  = 0.750182
i = 49
xnew  = -2.01718
ynew  = -0.499616
F(xnew,ynew)  = 0.750148
i = 50
xnew  = -2.01546
ynew  = -0.499693
F(xnew,ynew)  = 0.75012
Maximum number of iteration reached.
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Gradient Descent 2 Variables ]#
```

Figure 11.12: *The gradient descent method took 50 iteration to converges with $\gamma = 0.1$. (**SymIntegration/Examples/-Operations Research/Test SymIntegration Gradient Descent 2 Variables/main.cpp**).*

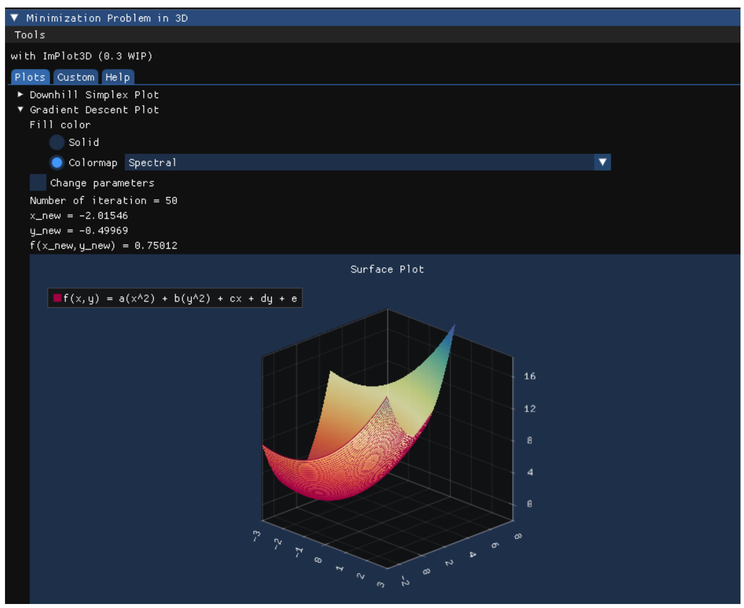


Figure 11.13: *The interactive surface plot for $f(x,y) = \frac{1}{2}x^2 + 2x + y^2 + y + 3$ with Hamzstlab Mathematics and SymIntegration combined so you can see the optimum / minimum value there. (**Hamzstlab-Mathematics/examples/3D Minimization/main.cpp**).*

- Downhill Simplex
  The downhill simplex, introduced by Nelder and Mead, also known as the Nelder-Mead' method [17], it is a method for solving multidimensional, unconstrained, numerical optimization problems. This type of problem is one of the most common in engineering and science, and this method is one of the simplest and fairly effective to solve the problem.

  [H*] We define a minimization problem as

$$\arg\ \min f(\boldsymbol{x}) \tag{11.9}$$

$$\boldsymbol{x} \in \mathbb{R}^N,\ f : \mathbb{R}^N \rightarrowtail \mathbb{R}^1$$

[H*] **Method and simplex transforms**

A simplex is a collection of connected points in $\mathbb{R}^N$. We call it a k-simplex where k is the degree that indicates how many points it contains. A k-simplex contains $k+1$ points, so a triangle which has 3 points is a 2-simplex,a tetrahedron which has 4 points is a 3-simplex and so on.

When optimizing a function with N parameters we will use a simplex with $N+1$ points. Meaning the degree k of the simplex is the same as the dimension of the parameter space. So for a problem with 2 parameters, $N = 2$, we use a 2-simplex, a triangle, that has three points. A 1 parameter problem would use a line segment in 1D and a 3 parameter problem would use a tetrahedron in 3D.

Each point on the simplex, denoted by $\boldsymbol{x}_i, 0 \leq i \leq N$, corresponds to a complete set of parameters and therefore to a value of the objective function $f_i = f(\boldsymbol{x}_i)$.

The core of the method is to keep the points sorted so that $f(\boldsymbol{x}_0)$ is the current lowest value, i.e. the best point, and $\boldsymbol{x}_N$ is the worst. We then calculate the center of mass of all points except $\boldsymbol{x}_N$:

$$\boldsymbol{c} = \frac{1}{N}\sum_{i=0}^{N-1} \boldsymbol{x}_i \tag{11.10}$$

Afterwards we attempt to replace the point $\boldsymbol{x}_N$ with one along the line formed by $\overrightarrow{\boldsymbol{x}_N\boldsymbol{c}}$ according to a set of transformations called reflection, expansion, and contraction. A shrink transform that moves all points closer to the current best can also be performed. We associate each transform with a parameter, α for reflection, β for contraction, γ for expansion and δ for shrink, that should satisfy the following constraints:

- $\alpha > 0$
- $0 < \beta < 1$
- $\gamma > \alpha, \gamma > 1$
- $0 < \delta < 1$

[H*] **Reflection**

The following illustrations are a 2D problem with a triangle simplex (can be used for function with two independent variable such as $f(x, y)$). The point $\boldsymbol{x}_0$ with blue color is the current best, the point $\boldsymbol{x}_1$ with green color is the second best, and the point $\boldsymbol{x}_2$ with light red color is the third best / the one being transformed except for when the simplex performs a shrink and all but the best are changed. The blue lines indicate the current simplex and the light red lines are the new one.

For every iteration the reflected point $\boldsymbol{x}_r$ is the one tried first

$$\boldsymbol{x}_r = \boldsymbol{c} + \alpha(\boldsymbol{c} - \boldsymbol{x}_2) \tag{11.11}$$

If this point is better than the second best but not better than the current best such that

$$f(\boldsymbol{x}_0) \leq f(\boldsymbol{x}_r) < f(\boldsymbol{x}_1)$$

we replace $\boldsymbol{x}_2$ with $\boldsymbol{x}_r$, otherwise we do either expansion, contraction or shrink.

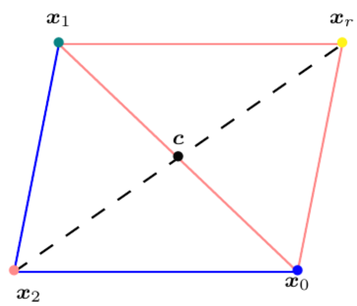


Figure 11.14: x_2 *is mirrored in* c *and creates a reflected point* x_r*.*

[H*] **Expansion**
If x_r is better than the current best,

$$f(x_r) < f(x_0)$$

we continue to explore and compute the expansion point x_e

$$x_e = c + \gamma(x_r - c) \tag{11.12}$$

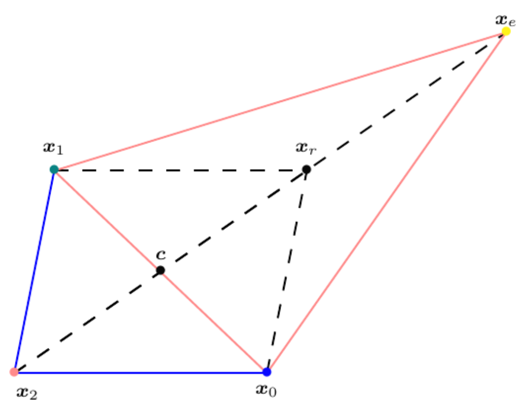


Figure 11.15: *The expansion point* x_e *is computed by moving further away from* c*.*

Then one of two approaches can be used:

- Greedy minimization
  in which the reflected and expanded points are compared and the better of the two is picked such that if

  $$f(x_e) < f(x_r) < f(x_0)$$

  we accept x_e ($x_2 = x_e$) and terminate the iteration.

  Otherwise, if

  $$f(x_r) \leq f(x_e)$$

  then x_r is accepted ($x_2 = x_r$).

- Greedy expansion
  we will accept $\boldsymbol{x}_e$ if
  $$f(\boldsymbol{x}_e) < f(\boldsymbol{x}_1)$$
  and
  $$f(\boldsymbol{x}_r) < f(\boldsymbol{x}_0)$$
  and skip comparing $f(\boldsymbol{x}_e)$ to $f(\boldsymbol{x}_r)$. This will keep the simplex as large as possible which can have some advantages for non-smooth functions.

[H*] **Contraction**
If $\boldsymbol{x}_r$ is worse than the current second worst point, in 2D case it is $\boldsymbol{x}_1$, which means

$$f(\boldsymbol{x}_r) \geq f(\boldsymbol{x}_1)$$

we compute a contraction point using $\boldsymbol{x}_2$ and $\boldsymbol{x}_r$.

It will be

- Outside
  if
  $$f(\boldsymbol{x}_1) \leq f(\boldsymbol{x}_r) < f(\boldsymbol{x}_2)$$
  We compute the contraction point as
  $$\boldsymbol{x}_c^{out} = \boldsymbol{c} + \beta(\boldsymbol{x}_r - \boldsymbol{c}) \tag{11.13}$$
  and if
  $$f(\boldsymbol{x}_c^{out}) \leq f(\boldsymbol{x}_r)$$
  then $\boldsymbol{x}_c^{out}$ is accepted ($\boldsymbol{x}_2 = \boldsymbol{x}_c^{out}$). Otherwise, a shrink of the simplex is performed.

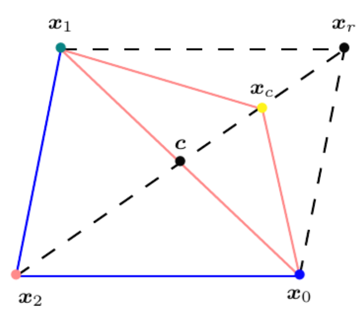


Figure 11.16: *The contraction point $\boldsymbol{x}_c^{out}$ is computed outside the old simplex.*

- Inside
  if
  $$f(\boldsymbol{x}_r) \geq f(\boldsymbol{x}_2)$$
  We compute the contraction point as
  $$\boldsymbol{x}_c^{in} = \boldsymbol{c} + \beta(\boldsymbol{x}_2 - \boldsymbol{c}) \tag{11.14}$$
  and if
  $$f(\boldsymbol{x}_c^{in}) \leq f(\boldsymbol{x}_2)$$

then x_c^{in} is accepted ($x_2 = x_c^{in}$). Otherwise, a shrink of the simplex is performed.

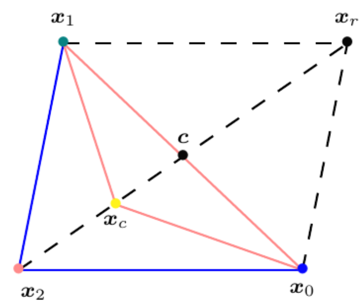


Figure 11.17: *The contraction point x_c^{in} is computed inside the old simplex.*

[H*] **Shrink**
the last resort that will be performed if none of the criteria so far have been met is to shrink the entire simplex by moving all points closer to the current best points.

$$s_{i-1} = x_0 + \delta(x_i - x_0), \quad i = \{1, 2, \ldots, N\} \tag{11.15}$$

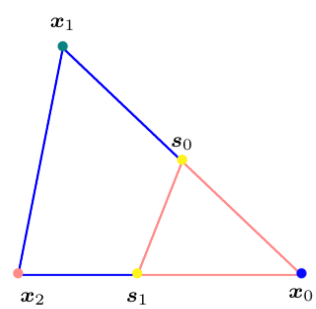


Figure 11.18: *Shrink the simplex by moving every point closer to the current best.*

[H*] **Initialization and termination**
When initializing the simplex, usually one point x_0 is to be given by the user (as the initial value / input), then the other points are created by placing them at certain distance μ from the initial point along the coordinate axes in the parameter space $\mathbb{R}^N$ of the objective function.

$$x_i = x_0 + \mu \hat{e}_i, \quad i = \{1, 2, \ldots, N\} \tag{11.16}$$

The algorithm terminates in one of three different ways. Either by having the simplex become sufficiently small, such that the active search space is thought to not have any considerable variation in function value. Another criteria is for all the function values of the points of the simplex to have a similar enough value. And finally the algorithm must be set to run for a maximum number of iterations or function evaluations.

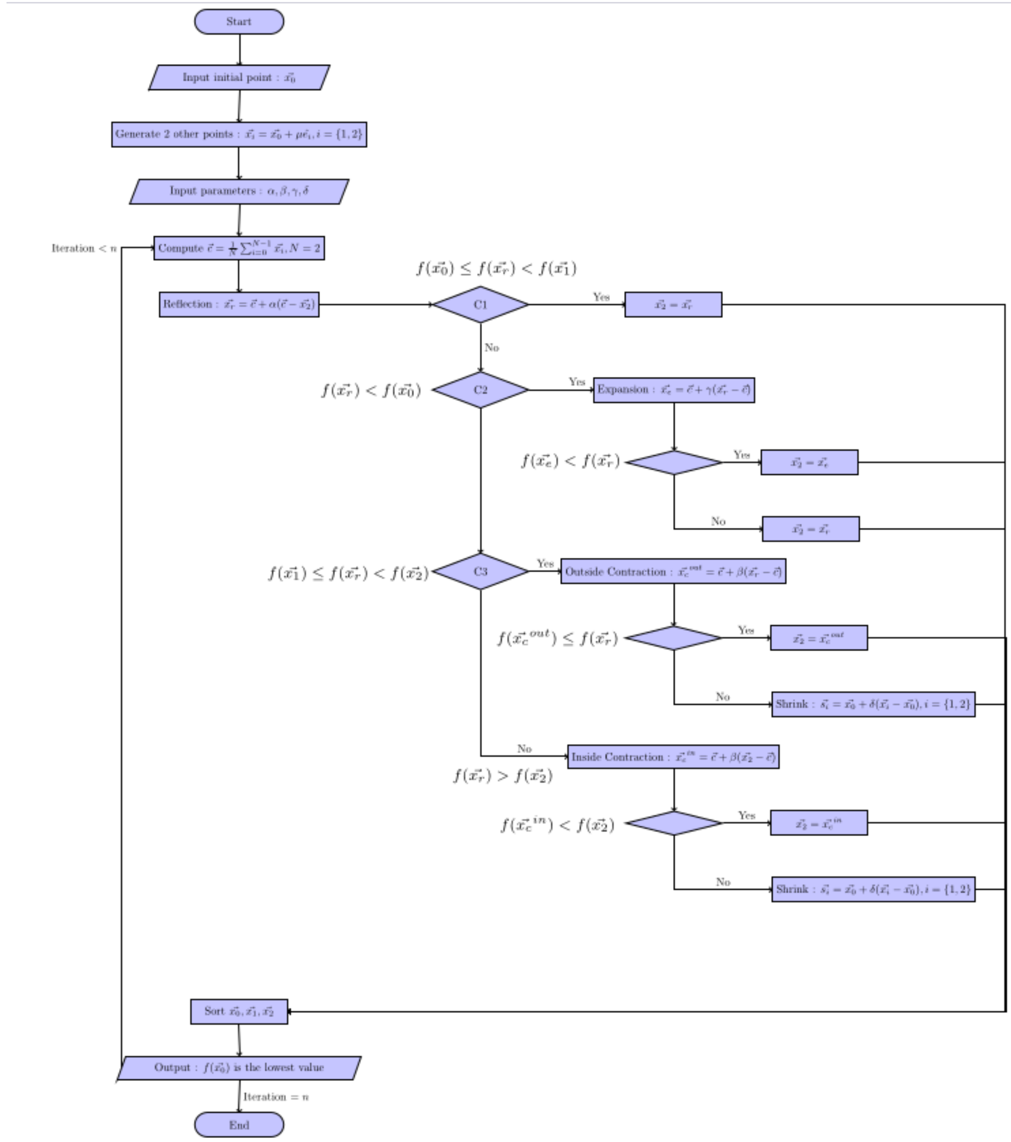


Figure 11.19: *The flowchart for downhill simplex algorithm that is used to make the C++ code for computing the solution.*

[H*] The downhill simplex method has some pros and cons. Its major strength is that it does not require the gradient / derivative computation of f. the gradient of the objective function is often not known as an analytical expression and can therefore be very expensive to evaluate. If so it has to be done using some form of finite difference scheme. This computational cost increases with the dimensionality of the problem and is also subject to potential numerical stability prolems if some care is not taken. Since downhill simplex does not require to compute the gradient of f, this makes this method applicable to non-smooth functions where the gradient may not exist on the whole domain.

This method has a memory footprint on the order of about $(N + 2) \times N$ since we always have to store $N+2$ sets of N parameters. Since we don't have to compute any gradient, then this method converges slowly compared to gradient based methods, generally because it requires more function evaluations.

[H*] **Example:**
Find the value x and y that minimizes

$$f(\boldsymbol{x}) = f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2, \quad s = 0.75,\ t = 1 \tag{11.17}$$

Solution:
First, we choose the initial point:

$$\boldsymbol{x}_0 = \begin{bmatrix} 0.1 \\ 0.5 \end{bmatrix}$$

The other two points can be generated from $\boldsymbol{x}_0$ by placing a certain distance $\mu = 0.05$ that we choose, thus

$$\boldsymbol{x}_1 = \boldsymbol{x}_0 + \mu\hat{e}_1 = \begin{bmatrix} 0.1 \\ 0.5 \end{bmatrix} + 0.05\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.15 \\ 0.5 \end{bmatrix}$$

$$\boldsymbol{x}_2 = \boldsymbol{x}_0 + \mu\hat{e}_2 = \begin{bmatrix} 0.1 \\ 0.5 \end{bmatrix} + 0.05\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.55 \end{bmatrix}$$

We will have

$$\begin{aligned} f(\boldsymbol{x}_0) &= -0.058442 \\ f(\boldsymbol{x}_1) &= -0.0604927 \\ f(\boldsymbol{x}_2) &= -0.066630 \end{aligned}$$

(It is unsorted, but we try to compute with unsorted vector first) From here we can compute $\boldsymbol{c}$

$$\boldsymbol{c} = \frac{1}{2}\sum_{i=0}^{1}\boldsymbol{x}_i = \begin{bmatrix} 0.125 \\ 0.5 \end{bmatrix}$$

Now, we need to input the parameters as

$$\alpha = 1, \quad \beta = 0.5, \quad \gamma = 2, \quad \delta = 0.5$$

Then we will perform the reflection operation first,

$$\boldsymbol{x}_r = \boldsymbol{c} + \alpha(\boldsymbol{c} - \boldsymbol{x}_2) = \begin{bmatrix} 0.15 \\ 0.45 \end{bmatrix}$$

From here we will compare the value of $f(\boldsymbol{x}_0), f(\boldsymbol{x}_1), f(\boldsymbol{x}_2)$ and $f(\boldsymbol{x}_r)$.

If $f(\boldsymbol{x}_0) < f(\boldsymbol{x}_r) < f(\boldsymbol{x}_1)$ then

$$\boldsymbol{x}_2 = \boldsymbol{x}_r$$

and we go to iteration 2.

If $f(\boldsymbol{x}_r) < f(\boldsymbol{x}_0)$, then go to the expansion step.

If $f(\boldsymbol{x}_r) \geq f(\boldsymbol{x}_1)$, then go to the contraction step.

In this case we have

$$\begin{aligned} f(\boldsymbol{x}_0) &= -0.058442 \\ f(\boldsymbol{x}_1) &= -0.0604927 \\ f(\boldsymbol{x}_2) &= -0.0666302 \\ f(\boldsymbol{x}_r) &= -0.0523666 \end{aligned}$$

$f(\boldsymbol{x}_r) \geq f(\boldsymbol{x}_1)$ so we go and compute the contraction point, since $f(\boldsymbol{x}_r) \geq f(\boldsymbol{x}_2)$ we compute the inside contraction point

$$\boldsymbol{x}_c^{in} = \boldsymbol{c} + \beta(\boldsymbol{x}_0 - \boldsymbol{c}) = \begin{bmatrix} 0.1125 \\ 0.5 \end{bmatrix}$$

then we will have

$$\begin{aligned} f(\boldsymbol{x}_0) &= -0.058442 \\ f(\boldsymbol{x}_1) &= -0.0604927 \\ f(\boldsymbol{x}_2) &= -0.0666302 \\ f(\boldsymbol{x}_c) &= -0.0588848 \end{aligned}$$

since $f(\boldsymbol{x}_c) > f(\boldsymbol{x}_2)$ then we perform shrink that will replace $\boldsymbol{x}_1$ with $\boldsymbol{s}_0$ and $\boldsymbol{x}_2$ with $\boldsymbol{s}_1$

$$\boldsymbol{s}_0 = \boldsymbol{x}_0 + \delta(\boldsymbol{x}_1 - \boldsymbol{x}_0) = \begin{bmatrix} 0.1250 \\ 0.5250 \end{bmatrix}$$

$$\boldsymbol{s}_1 = \boldsymbol{x}_0 + \delta(\boldsymbol{x}_2 - \boldsymbol{x}_0) = \begin{bmatrix} 0.1375 \\ 0.5625 \end{bmatrix}$$

So we have reached the end of the first iteration, which is using the shrink and we get these values at the end of this iteration:

$$\begin{aligned} f(\boldsymbol{x}_0) &= -0.058442 \\ f(\boldsymbol{x}_1) &= -0.0604927 \\ f(\boldsymbol{x}_2) &= -0.0666302 \end{aligned}$$

Continue till the last number of iteration. We use 6 iterations and we use C++ to help us in computing, thus we obtain the best vector $\boldsymbol{x}_0$ that minimizes the function $f(x, y)$ is

$$\boldsymbol{x}_0 = \begin{bmatrix} 0.1916 \\ 0.5 \end{bmatrix}$$

with $f(\boldsymbol{x}_0) = -0.063016$.

Since it is only using 6 iterations, thus the real optimal solution can be different if we add more iterations, we also sort the vectors $\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2$ at the end of iteration 1, if it is moved before iteration 1 then we will get faster convergence. We try the code with sorted vector before iteration 1 and obtain $f(\boldsymbol{x}_0) = -0.086404$ at the end of iteration 6.

We create the computation with **SymIntegration** and **Armadillo** and another one is with **SymIntegration** and **Eigen** for comparison, both produce the same answer but the one with **Eigen** has faster compiling time, we use 6 iterations with unsorted initial vectors then we sort the initial vectors and use 37 iterations, and then for the plotting in 3D we use Hamzstlab Mathematics with backend **ImGui** and **ImPlot3D** so we can change the value of s and t and the surface plot will change interactively as well.

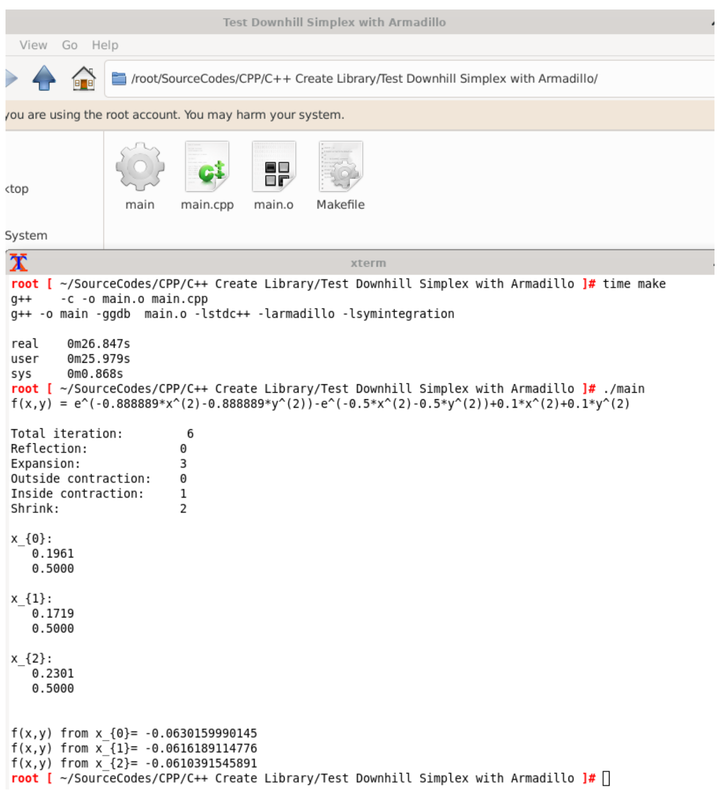


Figure 11.20: *The computation of downhill simplex for* $f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2,\ s = 0.75,\ t = 1$ *with SymIntegration and Armadillo (**SymIntegration/Examples/Operations Research/Test Downhill Simplex with Armadillo/main.cpp**).*

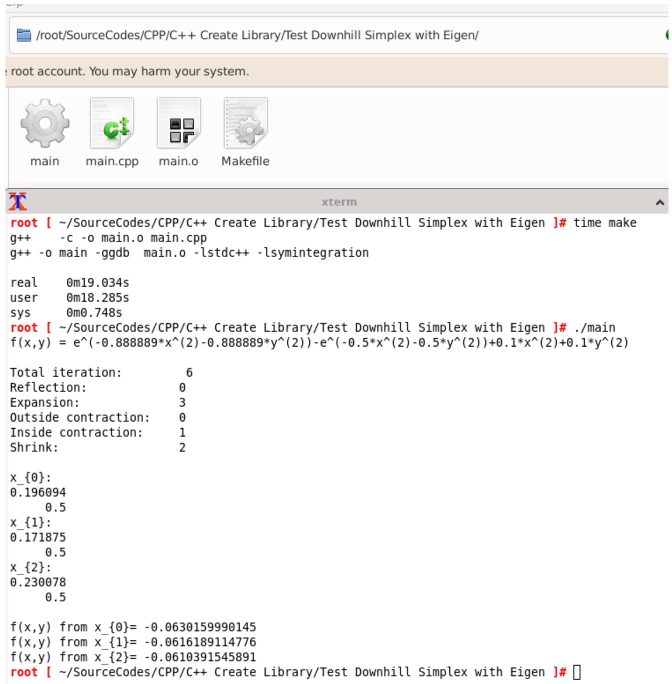


Figure 11.21: *The computation of downhill simplex for* $f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2,\ s = 0.75,\ t = 1$ *with SymIntegration and Eigen (**SymIntegration/Examples/Operations Research/Test Downhill Simplex with Eigen/main.cpp**).*

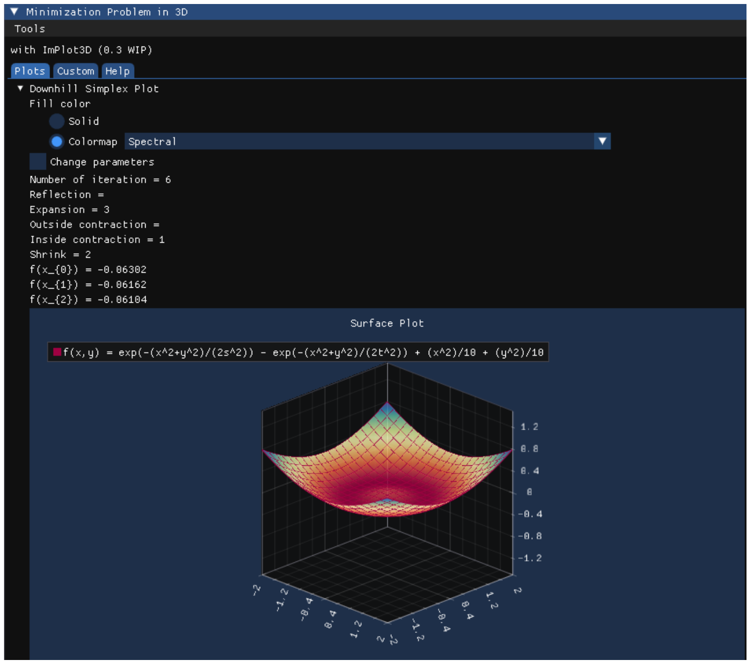


Figure 11.22: *The interactive surface plot for* $f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2$, $s = 0.75$, $t = 1$ *with Hamzstlab Mathematics and SymIntegration combined so you can see the optimum / minimum value there. (**Hamzstlab-Mathematics/examples/3D Minimization/main.cpp**).*

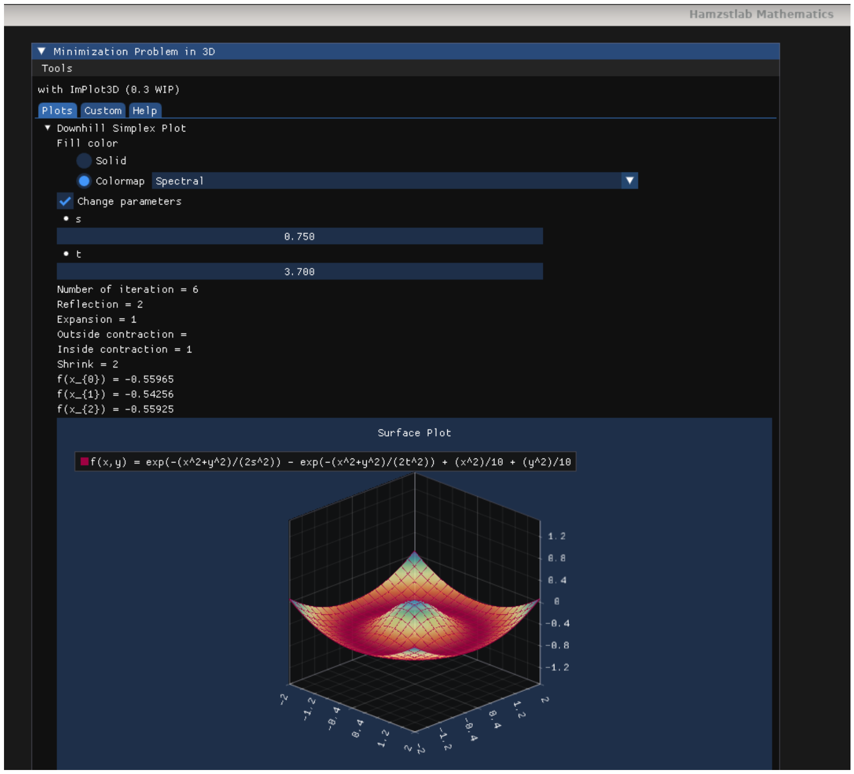


Figure 11.23: *The interactive surface plot for* $f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2$, $s = 0.75$ *when you change the value of* t *the computation and the plot change in an instant adapting to the parameter that has been changed. (**Hamzstlab-Mathematics/examples/3D Minimization/main.cpp**).*

```
g++ -o main -ggdb  main.o -lstdc++ -larmadillo -lsymintegration

real    0m29.108s
user    0m28.112s
sys     0m0.893s
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Armadillo ]# ./main
Vector x_{0}:
   0.1000
   0.5000

Vector x_{1}:
   0.1500
   0.5000

Vector x_{2}:
   0.1000
   0.5500

After sorting,
x_{0}:
   0.1000
   0.5500

x_{1}:
   0.1500
   0.5000

x_{2}:
   0.1000
   0.5000

f(x,y) = e^(-0.888889*x^(2)-0.888889*y^(2))-e^(-0.5*x^(2)-0.5*y^(2))+0.1*x^(2)+0.1*y^(2)

Total iteration:         37
Reflection:             5
Expansion:              14
Outside contraction:    0
Inside contraction:     18
Shrink:                 0

x_{0}:
   0.0973
   0.8914

x_{1}:
   0.0852
   0.8891

x_{2}:
   0.0629
   0.8977


f(x,y) from x_{0}= -0.0992255745363
f(x,y) from x_{1}= -0.0992180473087
f(x,y) from x_{2}= -0.0992086382243
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Armadillo ]# 
```

Figure 11.24: *The computation of downhill simplex for* $f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2,\ s = 0.75,\ t = 1$ *with 37 iterations (**SymIntegration/Examples/Operations Research/Test Downhill Simplex with Armadillo/main.cpp**).*

```
troot [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Eigen ]#
time make
g++    -c -o main.o main.cpp
g++ -o main -ggdb  main.o -lstdc++ -lsymintegration

real    0m19.921s
user    0m19.108s
sys     0m0.760s
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Eigen ]# ./main
Vector x_{0}:
0.1
0.5
Vector x_{1}:
0.15
 0.5
Vector x_{2}:
 0.1
0.55
After sorting,
x_{0}:
 0.1
0.55
x_{1}:
0.15
 0.5
x_{2}:
0.1
0.5
f(x,y) = e^(-0.888889*x^(2)-0.888889*y^(2))-e^(-0.5*x^(2)-0.5*y^(2))+0.1*x^(2)+0.1*y^(2)

Total iteration:         37
Reflection:             5
Expansion:              14
Outside contraction:    0
Inside contraction:     18
Shrink:                 0

x_{0}:
0.0972656
 0.891406
x_{1}:
0.0851562
 0.889063
x_{2}:
0.0628906
 0.897656

f(x,y) from x_{0}= -0.0992255745363
f(x,y) from x_{1}= -0.0992180473087
f(x,y) from x_{2}= -0.0992086382243
root [ ~/SourceCodes/CPP/C++ Create Library/Test Downhill Simplex with Eigen ]# 
```

Figure 11.25: *The computation of downhill simplex for* $f(x,y) = e^{-\frac{x^2+y^2}{2s^2}} - e^{-\frac{x^2+y^2}{2t^2}} + \frac{1}{10}x^2 + \frac{1}{10}y^2$, $s = 0.75$, $t = 1$ *with 37 iterations (**SymIntegration/Examples/Operations Research/Test Downhill Simplex with Eigen/main.cpp**).*

* the C++ codes are already updated so the initial vectors will be sorted before the **for** looping.

IV. Numerical Differentiation

[H*] The derivative of the function f at x_0 is

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \tag{11.18}$$

This formula gives an obvious way to generate an approximation to $f'(x)$; simply compute

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

for small values of h. Although this may be obvious, it is not very successful, die to roundoff error.

To approximate $f'(x_0)$, suppose first that $x_0 \in (a,b)$ where $f \in C^2[a,b]$, and that $x_1 = x_0 + h$ for some $h \neq 0$ that is sufficiently small to ensure that $x_1 \in [a,b]$. We construct the first

Lagrange polynomial $P_{0,1}(x)$ for f determined by x_0 and x_1, with its error term:

$$\begin{aligned} f(x) &= P_{0,1}(x) + \frac{(x-x_0)(x-x_1)}{2!} f''(\xi(x)) \\ &= \frac{f(x_0)(x-x_0-h)}{-h} + \frac{f(x_0+h)(x-x_0)}{h} + \frac{(x-x_0)(x-x_0-h)}{2} f''(\xi(x)) \end{aligned}$$

for some $\xi(x)$ in $[a,b]$. Differentiating gives

$$\begin{aligned} f'(x) &= \frac{f(x_0+h)-f(x_0)}{h} + D_x\left[\frac{(x-x_0)(x-x_0-h)}{2} f''(\xi(x))\right] \\ &= \frac{f(x_0+h)-f(x_0)}{h} + \frac{2(x-x_0)-h}{2} f''(\xi(x)) + \frac{(x-x_0)(x-x_0-h)}{2} D_x(f''(\xi(x))) \end{aligned}$$

so

$$f'(x) \approx \frac{f(x_0+h)-f(x_0)}{h}$$

One difficulty with this formula is that we have no information about $D_x f''(\xi(x))$, so the truncation error cannot be estimated. When x is x_0, however, the coefficent of $D_x f''(\xi(x))$ is 0, and the formula simplies to

$$f'(x_0) = \frac{f(x_0+h)-f(x-0)}{h} - \frac{h}{2} f''(\xi) \tag{11.19}$$

for small values of h, the difference quotient $\frac{f(x_0+h)-f(x-0)}{h}$ can be used to approximate $f'(x_0)$ with an error bounded by $M\frac{|h|}{2}$, where M is a bound on $|f''(\xi)|$ for $x \in [a,b]$. This formula is known as the forward-difference formula if $h > 0$ and the backward-difference formula if $h < 0$.

[H*] To obtain general derivative approximation formulas, suppose that $\{x_0, x_1, \ldots, x_n\}$ are $(n+1)$ distinct numbers in some interval I and that $f \in C^{n+1}(I)$. From Theorem 11.2 we will have,

$$f(x) = \sum_{k=0}^{n} f(x_k) L_k(x) + \frac{(x-x_0)\ldots(x-x_n)}{(n+1)!} f^{(n+1)}(\xi(x))$$

for some $\xi(x)$ in I, where $L_k(x)$ denotes the kth Lagrange coefficient polynomial for f at $x_0, x_1, \ldots, x_n$. Differentiating this expression gives

$$\begin{aligned} f'(x) = \sum_{k=0}^{n} f(x_k) L_k'(x) &+ D_x\left[\frac{(x-x_0)\ldots(x-x_n)}{(n+1)!}\right] f^{(n+1)}(\xi(x)) \\ &+ \frac{(x-x_0)\ldots(x-x_n)}{(n+1)!} D_x[f^{(n+1)}(\xi(x))] \end{aligned}$$

Again, we have a problem estimating the truncation error unless x is one of the numbers x_j. In this case, the term multiplying $D_x[f^{(n+1)}(\xi(x))]$ is 0, and the formula becomes

$$f'(x_j) = \sum_{k=0}^{n} f(x_k) L_k'(x_j) + \frac{f^{(n+1)}(\xi(x_j))}{(n+1)!} \prod_{k=0,\ k\neq j}^{n} (x_j - x_k) \tag{11.20}$$

which is called an $(n+1)$-point formula to approximate $f'(x_j)$.
In general, using more evaluation points in Eq. (11.20) produces greater accuracy, although

the number of functional evaluations and growth of roundoff error discourages this somewhat. The most common formulas are those involving three and five evaluation points.

We first derive some useful three-point formulas and consider aspects of their errors. Since

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

we have

$$L_0'(x) = \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}$$

Similarly,

$$L_1'(x) = \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}$$
$$L_2'(x) = \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}$$

Hence, from Eq. (11.20),

$$\begin{aligned} f'(x_j) = f(x_0)\left[\frac{2x_j - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}\right] + f(x_1)\left[\frac{2x_j - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}\right] \\ + f(x_2)\left[\frac{2x_j - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}\right] + \frac{1}{6} f^{(3)}(\xi_j) \prod_{k=0, k\neq j}^{2} (x_j - x_k) \end{aligned} \tag{11.21}$$

for each $j = 0, 1, 2$, where the notation ξ_j indicates that this point depends on x_j.

The three formulas from Eq. (11.21) become especially useful if the nodes are equally spaced, that is, when

$$x_1 = x_0 + h$$
$$x_2 = x_0 + 2h$$

for some $h \neq 0$.

Using Eq. (11.21) with $x_j = x_0,\ x_1 = x_0 + h$, and $x_2 = x_0 + 2h$ gives

$$f'(x_0) = \frac{1}{h}\left[-\frac{3}{2} f(x_0) + 2f(x_1) - \frac{1}{2} f(x_2)\right] + \frac{h^2}{3} f^{(3)}(\xi_0)$$

Doing the same for $x_j = x_1$ gives

$$f'(x_1) = \frac{1}{h}\left[-\frac{1}{2} f(x_0) + \frac{1}{2} f(x_2)\right] - \frac{h^2}{6} f^{(3)}(\xi_1)$$

and for $x_j = x_2$,

$$f'(x_2) = \frac{1}{h}\left[\frac{1}{2} f(x_0) - 2f(x_1) + \frac{3}{2} f(x_2)\right] + \frac{h^2}{3} f^{(3)}(\xi_2)$$

Since $x_1 = x_0 + h$ and $x_2 = x_0 + 2h$, these formulas can also be expressed as

$$\begin{aligned}
f'(x_0) &= \frac{1}{h}\left[-\frac{3}{2}f(x_0) + 2f(x_0+h) - \frac{1}{2}f(x_0+2h)\right] + \frac{h^2}{3}f^{(3)}(\xi_0)\\
f'(x_0+h) &= \frac{1}{h}\left[-\frac{1}{2}f(x_0) + \frac{1}{2}f(x_0+2h)\right] - \frac{h^2}{6}f^{(3)}(\xi_1)\\
f'(x_0+2h) &= \frac{1}{h}\left[\frac{1}{2}f(x_0) - 2f(x_0+h) + \frac{3}{2}f(x_0+2h)\right] + \frac{h^2}{3}f^{(3)}(\xi_2)
\end{aligned}$$

As a matter of convenience, the variable substitution x_0 for $x_0 + h$ is used in the middle equation to change this formula to an approximation for $f'(x_0)$. A similar change, x_0 for $x_0 + 2h$, is used in the last equation. This gives three formulas for approximating $f'(x_0)$:

$$\begin{aligned}
f'(x_0) &= \frac{1}{2h}\left[-3f(x_0) + 4f(x_0+h) - f(x_0+2h)\right] + \frac{h^2}{3}f^{(3)}(\xi_0)\\
f'(x_0) &= \frac{1}{2h}\left[-f(x_0-h) + f(x_0+h)\right] - \frac{h^2}{6}f^{(3)}(\xi_1)\\
f'(x_0) &= \frac{1}{2h}\left[f(x_0-2h) - 4f(x_0-h) + 3f(x_0)\right] + \frac{h^2}{3}f^{(3)}(\xi_2)
\end{aligned}$$

Finally, note that since the last of these equations can be obtained from the first by simply replacing h with $-h$, there are actually only two formulas:

$$f'(x_0) = \frac{1}{2h}\left[-3f(x_0) + 4f(x_0+h) - f(x_0+2h)\right] + \frac{h^2}{3}f^{(3)}(\xi_0) \tag{11.22}$$

where ξ_0 lies between x_0 and $x_0 + 2h$, and

$$f'(x_0) = \frac{1}{2h}\left[f(x_0+h) - f(x_0-h)\right] - \frac{h^2}{6}f^{(3)}(\xi_1) \tag{11.23}$$

where ξ_1 lies between $(x_0 - h)$ and $(x_0 + h)$.

Although the error in both Eqs. (11.22) and (11.23) are $O(h^2)$, the error in Eq. (11.23) are approximately half the error in Eq. (11.22). This is because Eq. (11.23) uses data on both sides of x_0 and Eq. (11.22) uses data on only one side. Note also that f needs to be evaluated at only two points in Eq. (11.23), whereas in Eq. (11.22) three evaluations are needed.

Similarly, the five-point formulas that involve evaluating the function at two more points is

$$f'(x_0) = \frac{1}{12h}[f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)] + \frac{h^4}{30}f^{(5)}(\xi) \tag{11.24}$$

where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.

The other five-point formula is useful for end-point approximations, particularly with regard to the clamped cubic spline interpolation. It is

$$\begin{aligned}
f'(x_0) = \frac{1}{12h}[&-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h)\\
&+ 16f(x_0+3h) - 3f(x_0+4h)] + \frac{h^4}{5}f^{(5)}(\xi)
\end{aligned} \tag{11.25}$$

where ξ lies between x_0 and $x_0 + 4h$. Left-endpoint approximations are found using this formula with $h > 0$ and right-endpoint approximations with $h < 0$.

[H*] When we use three-point formula that use both sides of point, the optimal choice for h theoretically can be found when we find the minimum of function $e(h)$ (the roundoff errors) and solve it for h [5]

$$e(h) = \frac{\epsilon}{h} + \frac{h^2}{6}M \tag{11.26}$$

where

$$M = \max_{x \in [x_0 - a, x_0 + a]} |f'''(x)| \tag{11.27}$$

with a is a small number such as 0.1. With this formula we can determine the optimal choice of h only for certain cases. In practice, we cannot compute an optimal h to use in approximating the derivative, since we have no knowledge of the third derivative of the function or it will cost more.

GlanzFreya' Guide 11.1: Approximating Second Order Derivative

Methods can also be derived to find approximations to higher derivatives of a function using only tabulated values of the function at various points. The derivation is algebraically tedious, however, so only a representative procedure will be presented. Expand a function f in a third Taylor polynomial about a point x_0 and evaluate at $x_0 + h$ and $x_0 - h$. Then

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 + \frac{1}{6}f'''(x_0)h^3 + \frac{1}{24}f^{(4)}(\xi_1)h^4$$

and

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{1}{2}f''(x_0)h^2 - \frac{1}{6}f'''(x_0)h^3 + \frac{1}{24}f^{(4)}(\xi_{-1})h^4$$

where $x_0 - h < \xi_{-1} < x_0 < \xi_1 < x_0 + h$.

If we add these equations, the term involving $f'(x_0)$ cancels and we obtain

$$f(x_0+h) + f(x_0-h) = 2f(x_0) + f''(x_0)h^2 + \frac{1}{24}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})]h^4$$

Solving this equaiton for $f''(x_0)$ gives

$$f''(x_0) = \frac{1}{h^2}[f(x_0-h) - 2f(x_0) + f(x_0+h)] - \frac{h^2}{24}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})] \quad (11.28)$$

Suppose $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$. Since $\frac{1}{2}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})]$ is between $f^{(4)}(\xi_1)$ and $f^{(4)}(\xi_{-1})$, the Intermediate Value Theorem implies that a number ξ exists between ξ_1 and ξ_{-1}, and hence in $(x_0 - h, x_0 + h)$, with

$$f^{(4)}(\xi) = \frac{1}{2}[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})]$$

This permits us to rewrite the approximation equation Eq. (11.28) as

$$f''(x_0) = \frac{1}{h^2}[f(x_0-h) - 2f(x_0) + f(x_0+h)] - \frac{h^2}{12}f^{(4)}(\xi) \quad (11.29)$$

for some ξ, where $x_0 - h < \xi < x_0 + h$.

[H*] **Example:**
Let $f(x) = \ln(x)$ and $x_0 = 1.8$, use forward-difference formula to approximate $f'(1.8)$ with $h = 0.1, 0.01, 0.001$.

Then, let $f_2(x) = xe^x$ and $x_0 = 2.0$, use three-point formulas and five-point formulas to approximate $f'(2.0)$ with $h = 0.1$ and $h = -0.1$.

Solution:
Since $f'(x) = \frac{1}{x}$, the exact value of $f'(1.8)$ is 0.55555.

For f_2, since $f_2'(x) = (x+1)e^x$, the exact value of $f_2'(x)$ is 22.167168.

In SymIntegration we can approximate the numerical differentiation problem with forward-difference formula:
numericaldifferentiation(const Symbolic &f, const Symbolic &x, double x_0, double h)
where f is the function of x, x_0 is the value that we want to approximate, h is the step size.

For three-point formulas, the functions in SymIntegration are:
numericaldifferentiation3pointoneside(const Symbolic &f, const Symbolic &x, double x_0, double h)
numericaldifferentiation3pointbothsides(const Symbolic &f, const Symbolic &x, double x_0, double h)

For five-point formulas, the functions in SymIntegration are:
numericaldifferentiation5pointoneside(const Symbolic &f, const Symbolic &x, double x_0, double h)
numericaldifferentiation5pointbothsides(const Symbolic &f, const Symbolic &x, double x_0, double h)

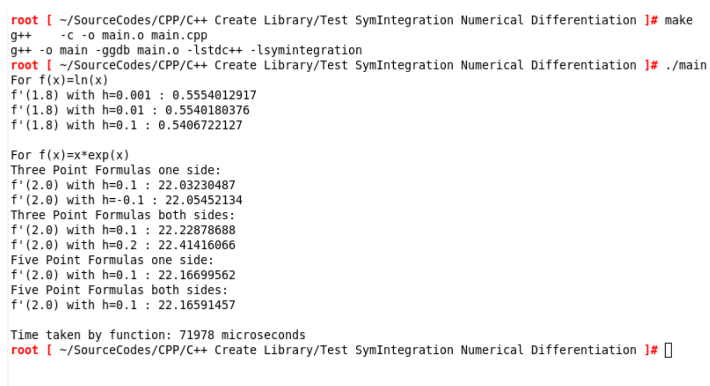

```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Numerical Differentiation ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Numerical Differentiation ]# ./main
For f(x)=ln(x)
f'(1.8) with h=0.001 : 0.5554012917
f'(1.8) with h=0.01 : 0.5540180376
f'(1.8) with h=0.1 : 0.5406722127

For f(x)=x*exp(x)
Three Point Formulas one side:
f'(2.0) with h=0.1 : 22.03230487
f'(2.0) with h=-0.1 : 22.05452134
Three Point Formulas both sides:
f'(2.0) with h=0.1 : 22.22878688
f'(2.0) with h=0.2 : 22.41416066
Five Point Formulas one side:
f'(2.0) with h=0.1 : 22.16699562
Five Point Formulas both sides:
f'(2.0) with h=0.1 : 22.16591457

Time taken by function: 71978 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Numerical Differentiation ]#
```

Figure 11.26: *The computation of numerical differentiation for $f(x) = \ln(x)$ with forward-difference formula and for $f(x) = xe^x$ with three-point and five-point formulas (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Numerical Differentiation//main.cpp**).*

[H*] **Richardson's Extrapolation**
Richardson's extrapolation is used to generate high-accuracy results while using low-order formulas. Extrapolation can be applied whenever it is known that an approximation technique has an error term with a predictable form, one that depends on a parameter, usually the step size h.

Suppose that for each number $h \neq 0$ we have a formula $N(h)$ that approximates an unknown value M and that the truncation error involved with the approximation has the form

$$M - N(h) = K_1 h + K_2 h^2 + K_3 h^3 + \ldots,$$

for some collection of unknown constants $K_1, K_2, K_3, \ldots$.

Since the truncation error is $O(h)$, we would expect, for example, that

$$M - N(0.1) \approx 0.1K_1$$
$$M - N(0.01) \approx 0.01K_1$$

and, in general, $M - N(h) \approx K_1 h$, unless there was a large variation in magnitude among the constants $K_1, K_2, K_3, \ldots$.

The object of extrapolation is to find an easy way to combine the rather inaccurate $O(h)$ approximations in an appropriate way to produce formulas with a higher-order truncation error. Suppose, for example, we could combine the $N(h)$ formulas so as to produce an $O(h^2)$ approximation formula, $\hat{N}(h)$, for M with

$$M - \hat{N}(h) = \hat{K}_2 h^2 + \hat{K}_3 h^3 + \ldots,$$

for some, again unknown, collection of constants $\hat{K}_1, \hat{K}_2, \ldots$. Then we would have

$$M - \hat{N}(0.1) \approx 0.01\hat{K}_2$$
$$M - \hat{N}(0.01) \approx 0.0001\hat{K}_2$$

and so on. If the constants K_1 and $\hat{K}_2$ are roughly of the same magnitude, then the $\hat{N}(h)$ approximations would be much better than the corresponding $N(h)$ approximations. The extrapolation continues by combining the $\hat{N}(h)$ approximations in a manner that produces formulas with $O(h^3)$ truncation error, and so on.
To see specifically how we can generate these higher-order formulas, let us consider the formula for approximating M of the form

$$M = N(h) + K_1 h + K_2 h^2 + K_3 h^3 + \ldots \tag{11.30}$$

Since the formula is assumed to hold for all positive h, consider the result when we replace the parameter h by half its value. Then we have the formula

$$M = N\left(\frac{h}{2}\right) + K_1 \frac{h}{2} + K_2 \frac{h^2}{4} + K_3 \frac{h^3}{8} + \ldots$$

Subtracting Eq. (11.30) from twice this equation eliminates the term involving K_1 and gives

$$M = \left[N\left(\frac{h}{2}\right) + \left(N\left(\frac{h}{2}\right) - N(h)\right)\right] + K_2\left(\frac{h^2}{2} - h^2\right) + K_3\left(\frac{h^3}{4} - h^3\right) + \ldots$$

to facilitate this discussion, we define $N_1(h) \equiv N(h)$ and

$$N_2(h) = N_1\left(\frac{h}{2}\right) + \left[N_1\left(\frac{h}{2}\right) - N_1(h)\right]$$

Then we have the $O(h^2)$ approximation formula for M:

$$M = N_2(h) - \frac{K_2}{2}h^2 - \frac{3K_3}{4}h^3 - \ldots \tag{11.31}$$

If we now replace h by $\frac{h}{2}$ in this formula, we have

$$M = N_2\left(\frac{h}{2}\right) - \frac{K_2}{8}h^2 - \frac{3K_3}{32}h^3 - \ldots \tag{11.32}$$

this can be combined with Eq. (11.31) to eliminate the h^2 term. Specifically, subtracting Eq. (11.31) from 4 times Eq. (11.32) gives

$$3M = 4N_2\left(\frac{h}{2}\right) - N_2(h) + \frac{3K_3}{8}h^3 + \dots$$

and dividing by 3 gives an $O(h^3)$ formula for approximating M:

$$M = \left[N_2\left(\frac{h}{2}\right) + \frac{N_2\left(\frac{h}{2}\right) - N_2(h)}{3}\right] + \frac{K_3}{8}h^3 + \dots$$

By defining

$$N_3(h) \equiv N_2\left(\frac{h}{2}\right) + \frac{N_2\left(\frac{h}{2}\right) - N_2(h)}{3}$$

we have the $O(h^3)$ formula:

$$M = N_3(h) + \frac{K_3}{8}h^3 + \dots$$

The process is continued by constructing an $O(h^4)$ approximation

$$N_4(h) = N_3\left(\frac{h}{2}\right) + \frac{N_3\left(\frac{h}{2}\right) - N_3(h)}{7}$$

an $O(h^5)$ approximation

$$N_5(h) = N_4\left(\frac{h}{2}\right) + \frac{N_4\left(\frac{h}{2}\right) - N_4(h)}{15}$$

and so on. In general, if M can be written in the form

$$M = N(h) + \sum_{j=1}^{m-1} K_j h^j + O(h^m) \tag{11.33}$$

then for each $j = 2, 3, \dots, m$, we have an $O(h^j)$ approximation of the form

$$N_j(h) = N_{j-1}\left(\frac{h}{2}\right) + \frac{N_{j-1}\left(\frac{h}{2}\right) - N_{j-1}(h)}{2^{j-1} - 1} \tag{11.34}$$

These approximations are generated by rows in the order indicated by the numbered entries. This is done to take best advantage of the highest-order formulas. The extrapolation can be applied whenever the truncation error for a formula has the form

$$\sum_{j=1}^{m-1} K_j h^{\alpha_j} + O(h^{\alpha_m})$$

for a collection of constants K_j and when $\alpha_1 < \alpha_2 < \alpha_3 < \cdots < \alpha_m$.

$O(h)$	$O(h^2)$	$O(h^3)$	$O(h^4)$
1 : $N_1(h) \equiv N(h)$			
2 : $N_1\left(\frac{h}{2}\right) \equiv N\left(\frac{h}{2}\right)$	**3 :** $N_2(h)$		
4 : $N_1\left(\frac{h}{4}\right) \equiv N\left(\frac{h}{4}\right)$	**5 :** $N_2\left(\frac{h}{2}\right)$	**6 :** $N_3(h)$	
7 : $N_1\left(\frac{h}{8}\right) \equiv N\left(\frac{h}{8}\right)$	**8 :** $N_2\left(\frac{h}{4}\right)$	**9 :** $N_3\left(\frac{h}{4}\right)$	**10 :** $N_4(h)$

Table 11.1: *The extrapolation table*

[H*] The centered difference formula in Eq. (11.23) to approximate $f'(x_0)$ can be expressed with an error formula

$$f'(x_0) = \frac{1}{2h}[f(x_0+h) - f(x_0-h)] - \frac{h^2}{6}f'''(x_0) - \frac{h^4}{120}f^{(5)}(x_0) - \dots$$

Since this error formula contains only even powers of h, extrapolation is more effective than as outlined in the opening discussin. In this case, we have the $O(h^2)$ approximation

$$f'(x_0) = N_1(h) - \frac{h^2}{6}f'''(x_0) - \frac{h^4}{120}f^{(5)}(x_0) - \dots \tag{11.35}$$

where

$$N_1(h) \equiv N(h) = \frac{1}{2h}[f(x_0+h) - f(x_0-h)]$$

Replacing h by $\frac{h}{2}$ in this formula gives the approximation

$$f'(x_0) = N_1\left(\frac{h}{2}\right) - \frac{h^2}{24}f'''(x_0) - \frac{h^4}{1920}f^{(5)}(x_0) - \dots$$

Subtracting Eq. (11.35) from 4 times this equation eliminates the $O(h^2)$ term that involves $f'''(x_0)$ and gives

$$3f'(x_0) = 4N_1\left(\frac{h}{2}\right) - N_1(h) + \frac{h^4}{160}f^{(5)}(x_0) + \dots$$

Dividing by 3 provides an $O(h^4)$ formula

$$f'(x_0) = N_2(h) + \frac{h^4}{480}f^{(5)}(x_0) + \dots$$

where

$$N_2(h) = N_1\left(\frac{h}{2}\right) + \frac{N_1\left(\frac{h}{2}\right) - N_1(h)}{3}$$

Continuing this procedure gives, for each $j = 2, 3, \dots$, and $O(h^{2j})$ approximation

$$N_j(h) = N_{j-1}\left(\frac{h}{2}\right) + \frac{N_{j-1}\left(\frac{h}{2}\right) - N_{j-1}(h)}{4^{j-1} - 1} \tag{11.36}$$

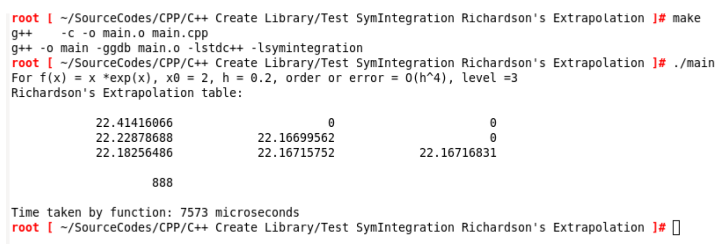
Notice that the denominator of the quotient is $4^{j-1} - 1$ instead of $2^{j-1} - 1$ because we are now eliminating powers of h^2 instead of powers of h. Since $\left(\frac{h}{2}\right)^2 = \frac{h^2}{4}$, the multipliers used to eliminate the powers of h^2 are powers of 4 instead of 2.

[H*] **Example:**
Suppose that $x_0 = 2.0, h = 0.2$, and $f(x = xe^x)$, create the extrapolation table with Richardson's extrapolation.

Solution:

$$\begin{aligned} N_1(0.2) &= N(0.2) = \frac{1}{0.4}[f(2.2) - 1.8] = 22.414160 \\ N_1(0.1) &= N(0.1) = 22.228786 \\ N_1(0.05) &= N(0.05) = 22.182564 \end{aligned}$$

The exact value of $f'(x) = xe^x + e^x$ at $x_0 = 2.0$ to six decimal places is 22.167168. We see that $N_1(0.05)$ has only one decimal place of accuracy, while $N_3(0.2)$ are exact up to 6 decimal places.



```
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Richardson's Extrapolation ]# make
g++    -c -o main.o main.cpp
g++ -o main -ggdb main.o -lstdc++ -lsymintegration
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Richardson's Extrapolation ]# ./main
For f(x) = x *exp(x), x0 = 2, h = 0.2, order or error = O(h^4), level =3
Richardson's Extrapolation table:

           22.41416066                   0                   0
           22.22878688         22.16699562                   0
           22.18256486         22.16715752         22.16716831

                   888

Time taken by function: 7573 microseconds
root [ ~/SourceCodes/CPP/C++ Create Library/Test SymIntegration Richardson's Extrapolation ]# 
```

Figure 11.27: *The computation of Richardson's extrapolation for $x_0 = 2.0, h = 0.2$, and $f(x = xe^x)$ (**SymIntegration/Examples/Numerical Methods/Test SymIntegration Richardson's Extrapolation//main.cpp**).*

We use the Richardson's extrapolation formula of this (with centered difference formula to approximate $f'(x_0)$):

$$N_j(h) = N_{j-1}\left(\frac{h}{2}\right) + \frac{N_{j-1}\left(\frac{h}{2}\right) - N_{j-1}(h)}{m^{j-1} - 1} \tag{11.37}$$

with m as the order of error as in $O(h^m)$.

In SymIntegration we can use this function to obtain the extrapolation table that is using the Eq. (11.37):
richardsonextrapolation(const Symbolic &f, const Symbolic &x, double x_0, double h, int l, int m)
where f is the function of x, x_0 is the value that we want to approximate, h is the step size, l is the level / size of the square matrix, and m is the order of error.

V. Numerical Integration

[H*] The need often arises for evaluating the definite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain. The basic method involved in approximating $\int_a^b f(x)\,dx$ is called numerical quadrature. It uses a sum

$$\sum_{i=0}^{n} a_i f(x_i)$$

to approximate $\int_a^b f(x)\,dx$.

The methods of quadrature are based on the interpolation polynomials. We first select a set of distinct nodes $\{x_0, x_1, \ldots, x_n\}$ from the interval $[a, b]$. Then we integrate the Lagrange interpolating polynomial

$$P_n(x) = \sum_{i=0}^{n} f(x_i) L_i(x)$$

and its truncation error term over $[a, b]$ to obtain

$$\begin{aligned}\int_a^b f(x)\,dx &= \int_a^b \sum_{i=0}^{n} f(x_i) L_i(x)\,dx + \int_a^b \prod_{i=0}^{n}(x - x_i)\frac{f^{(n+1)(\xi(x))}}{(n+1)!}\,dx \\ &= \sum_{i=0}^{n} a_i f(x_i) + \frac{1}{(n+1)!}\int_a^b \prod_{i=0} n(x - x_i) f^{(n+1)}(\xi(x))\,dx\end{aligned}$$

where $\xi(x)$ is in $[a, b]$ for each x and

$$a_i = \int_a^b L_i(x)\,dx$$

for each $i = 0, 1, 2, \ldots, n$.

The quadrature formula is, therefore,

$$\int_a^b f(x)\,dx \approx \sum_{i=0}^{n} a_i f(x_i)$$

with error given by

$$E(f) = \frac{1}{(n+1)!}\int_a^b \prod_{i=0}^{n}(x - x_i) f^{(n+1)}(\xi(x))\,dx$$

[H*] To derive the Trapezoidal rule for approximating $\int_a^b f(x)\,dx$. let $x_0 = a,\ x_1 = b,\ h = b - a$ and use the linear Lagrange polynomial:

$$P_1(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1)$$

Then,

$$\begin{aligned}\int_a^b f(x)\,dx = \int_{x_0}^{x_1}\left[\frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1)\right]dx \\ + \frac{1}{2}\int_{x_0}^{x_1} f''(\xi(x))(x - x_0)(x - x_1)\,dx\end{aligned} \tag{11.38}$$

Since $(x - x_0)(x - x_1)$ does not change sign on $[x_0, x_1]$, the Weighted Mean Value Theorem for Integrals can be applied to the error term to give, for some ξ in (x_0, x_1).

$$\begin{aligned}
\int_{x_0}^{x_1} f''(\xi(x))(x - x_0)(x - x_1)\, dx &= f''(\xi(x)) \int_{x_0}^{x_1} (x - x_0)(x - x_1)\, dx \\
&= f''(\xi(x)) \left[\frac{x^3}{3} - \frac{(x_1 + x_0)}{2} x^2 + x_0 x_1 x \right]_{x_0}^{x_1} \\
&= -\frac{h^3}{6} f''(\xi)
\end{aligned}$$

Consequently, Eq. (11.38) implies that

$$\begin{aligned}
\int_a^b f(x)\, dx &= \left[\frac{(x - x_1)^2}{2(x_0 - x_1)} f(x_0) + \frac{(x - x_0)^2}{2(x_1 - x_0)} f(x_1) \right] - \frac{h^3}{12} f''(\xi) \\
&= \frac{(x_1 - x_0)}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi)
\end{aligned}$$

Since $h = x_1 - x_0$, we have the following rule:

$$\int_a^b f(x)\, dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi) \tag{11.39}$$

this is called the Trapezoidal rule because when f is a function with positive values, $\int_a^b f(x)\, dx$ is approximated by the area in a trapezoid.

Since the error term for the Trapezoidal rule involves f'', the rule gives the exact result when applied to any function whose second derivative is identically zero, that is, any polynomial of degree one or less.

[H*] Simpson's rule results from integrating over $[a, b]$ the second Lagrange polynomial with nodes $x_0 = a,\ x_2 = b$, and $x_1 = a + h$, where $h = \frac{b-a}{2}$.

Therefore,

$$\begin{aligned}
\int_a^b f(x)\, dx = \int_{x_0}^{x_2} \Big[&\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \\
&+ \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2) \Big]\, dx \\
&+ \int_{x_0}^{x_2} \frac{(x - x_0)(x - x_1)(x - x_2)}{6} f^{(3)}(\xi(x))\, dx
\end{aligned}$$

Deriving Simpson's rule in this manner, however, provides only an $O(h^4)$ error term involving $f^{(3)}$. By approaching this problem in another way, a higher-order term involving $f^{(4)}$ can be derived.

To illustrate this alternative formula, suppose that f is expanded in the third Taylor polynomial about x_1. Then for each x in $[x_0, x_2]$, a number $\xi(x)$ in (x_0, x_2) exists with

$$f(x) = f(x_1) + f'(x_1)(x - x_1) + \frac{f''(x_1)}{2}(x - x_1)^2 + \frac{f'''(x_1)}{6}(x - x_1)^3 + \frac{f^{(4)}(\xi(x))}{24}(x - x_1)^4$$

and

$$\int_{x_0}^{x_2} f(x)\,dx = \Big[f(x_1)(x-x_0) + \frac{f'(x_1)}{2}(x-x_1)^2 + \frac{f''(x_1)}{6}(x-x_1)^3 + \frac{f'''(x_1)}{24}(x-x_1)^4\Big]_{x_0}^{x_2} + \frac{1}{24}\int_{x_0}^{x_2} f^{(4)}(\xi(x))(x-x_1)^4\,dx \tag{11.40}$$

since $(x-x_1)^4$ is never negative on $[x_0, x_2]$, the Weighted Mean Value Theorem for Integrals implies that

$$\frac{1}{24}\int_{x_0}^{x_2} f^{(4)}(\xi(x))(x-x_1)^4\,dx = \frac{f^{(4)}(\xi_1)}{24}\int_{x_0}^{x_2}(x-x_1)^4\,dx = \frac{f^{(4)}(\xi_1)}{120}(x-x_1)^5\Big]_{x_0}^{x_2}$$

for some number ξ_1 in (x_0, x_2).

However, $h = x_2 - x_1 = x_1 - x_0$, so

$$(x_2-x_1)^2 - (x_0-x_1)^2 = (x_2-x_1)^4 - (x_0-x_1)^4 = 0$$

whereas

$$(x_2-x_1)^3 - (x_0-x_1)^3 = 2h^3$$
$$(x_2-x_1)^5 - (x_0-x_1)^5 = 2h^5$$

Consequently, Eq. (11.40) can be rewritten as

$$\int_{x_0}^{x_2} f(x)\,dx = 2hf(x_1) + \frac{h^3}{3}f''(x_1) + \frac{f^{(4)}(\xi_1)}{60}h^5$$

If we now replace $f'''(x_1)$ by the approximation given in Eq. (11.29), we have

$$\begin{aligned}\int_{x_0}^{x_2} f(x)\,dx &= 2hf(x_1) + \frac{h^3}{3}\left[\frac{1}{h^2}[f(x_0) - 2f(x_1) + f(x_2)] - \frac{h^2}{12}f^{(4)}(\xi_2)\right] + \frac{f^{(4)}(\xi_1)}{60}h^5 \\ &= \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{12}\left[\frac{1}{3}f^{(4)}(\xi_2) - \frac{1}{5}f^{(4)}(\xi_1)\right]\end{aligned}$$

it can be shown by alternative methods that the values ξ_1 and ξ_2 in this expression can be replaced by a common value ξ in (x_0, x_2). This gives Simpson's rule:

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90}f^{(4)}(\xi) \tag{11.41}$$

Since the error term involves the fourth derivative of f, Simpson's rule gives exact results when applied to any polynomial of degree three or less.

[H*] **Example:**
Compute the definite integral for $x^2, x^4, \frac{1}{x+1}, \sqrt{1+x^2}, \sin\, x, e^x$ using the Trapezoidal rule and Simpson's rule on the interval $[0, 2]$.

Solution:

VI. Boundary-Value Problems for Ordinary Differential Equations

[H*]

VII. Numerical Solutions to Partial Differential Equations

[H*]

Chapter 12

Complex Numbers

"Tobikata Wo Wasureta Tori No You Ni (A Little bird who forgot how to fly)." - ***Star Ocean 3 OST - MISIA***

I. Preliminaries

[H*] Complex analysis, traditionally known as the theory of functions of a complex variable, is the branch of mathematical analysis that investigates functions of complex numbers. It is helpful in many branches of mathematics, including algebraic geometry, number theory, analytic combinatorics, applied mathematics; as well as in physics, including the branches of hydrodynamics, thermodynamics, and particularly quantum mechanics. By extension, use of complex analysis also has applications in engineering fields such as nuclear, aerospace, mechanical and electrical engineering.

As a differentiable function of a complex variable is equal to its Taylor series (that is, it is analytic), complex analysis is particularly concerned with analytic functions of a complex variable (that is, holomorphic functions).

[H*] Complex number arises when no one on this planet up until 2024 A.D. able to compute this:

$$\sqrt{-1}$$

That number is called imaginary number, the notation is $i = \sqrt{-1}$.

[H*] Let $x, y \in \mathbb{R}$, the complex number $z \in \mathbb{C}$ can be defined as

$$z = x + iy \tag{12.1}$$

Some important properties of complex numbers:

- $\text{Re}(z) = x$ and $\text{Im}(z) = y$ are called the real part of z and the imaginary part of z, respectively.
- $|z| = \sqrt{x^2 + y^2}$ is called the modulus (or absolute value) of z.
- $\overline{z} = x - yi$ is called the complex conjugate of z.
- $z\overline{z} = x^2 + y^2 = |z|^2$
- The angle θ is called an argument of z.
- $\text{Re}(z) = |z| \cos\ \theta$
- $\text{Im}(z) = |z| \sin\ \theta$

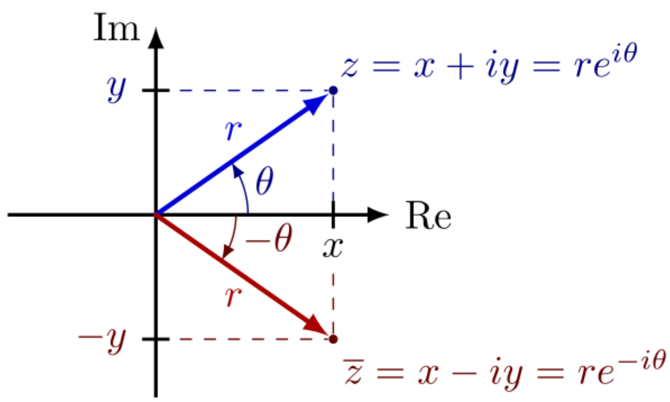


Figure 12.1: *The complex number can be represented as $z = x + iy = re^{i\theta}$, with $r = |z|$ and the angle θ is the argument of z.*

- $z = |z|(\cos\ \theta + i\ \sin\ \theta)$ is called the polar form of z.

[H*] We try to apply Newton-Raphson method to compute the root of $x^2 + 1 = 0$ and it cannot even compute i.

- General Theory of nth Order Linear Equations

  [H*] The first
  [H*]

- The Method of Variation of Parameters

  [H*] The first
  [H*]

i. C++ Plot: 2D Plot of Complex Function

Suppose we have a complex function

$$e^{ix}$$

and we want to plot it.

Note that

$$e^{ix} = \cos\ x + i\ \sin\ x$$

With **gnuplot** we can plot the function as separate functions, the first function is the real part, the second function is the imaginary part.

Thus, the real part of e^{ix} is $\cos\ x$ and the imaginary part of e^{ix} is $\sin\ x$.

```
#include <vector>
#include <cmath>
#include <utility>
#include <boost/tuple/tuple.hpp>
```

```
f(x) = 1+x^2

f'(x) = 2*x

p_{0} = 0.7853982

   n        p_{n}               p_{n} in Degree
   0        0.7853982           45.0
   1        -0.24392074         -13.975629
   2        1.9278858           110.45972
   3        0.7045915           40.37012
   4        -0.35733533         -20.473806
   5        1.2205782           69.93398
   6        0.20064712          11.496233
   7        -2.3916135          -137.02936
   8        -0.98674273         -56.536194
   9        0.013346314         0.7646875
  10        -37.456852          -2146.1196
  11        -18.715078          -1072.295
  12        -9.330823           -534.61676
  13        -4.6118255          -264.23813
  14        -2.1974957          -125.907234
  15        -0.8712162          -49.91701
  16        0.13830233          7.9241395
  17        -3.5461168          -203.17754
  18        -1.6320591          -93.5101
  19        -0.509668           -29.201826
  20        0.72619677          41.60801
  21        -0.32542026         -18.645208
  22        1.3737646           78.710915
  23        0.3229189           18.50189
  24        -1.3869171          -79.4645
  25        -0.33294678         -19.076445
  26        1.3352681           76.505226
  27        0.2931775           16.797832
  28        -1.5588628          -89.31626
  29        -0.4586848          -26.280704
  30        0.8607309           49.316246
  31        -0.15053618         -8.625088
  32        3.2461925           185.99313
  33        1.4690697           84.17149
  34        0.3941834           22.585045
  35        -1.0713533          -61.384026
  36        -0.06897712         -3.9520977
  37        7.2142925           413.3485
  38        3.5378394           202.70326
  39        1.6275904           93.25407
  40        0.50659263          29.02562
  41        -0.73369            -42.037342
```

Figure 12.2: *The Newton-Raphson method to compute the root of* $x^2 + 1 = 0$ *cannot converge.*

```
#include "gnuplot-iostream.h"

int main() {
	Gnuplot gp;

	// Don't forget to put "\n" at the end of each line!
	gp << "set xrange [-10:5]\nset yrange [-3:3]\n";
	// '-' means read from stdin. The send1d() function sends data to
	    gnuplot's stdin.
	gp << "i = {0,1}\n";
	gp << "plot real(exp(i*x)), imag(exp(i*x))\n";

	// To do parametric plot uncomment below and comment the plot line
	    above
	//gp << "set parametric\n";
	//gp << "plot real(exp(i*t)), imag(exp(i*t))\n";

	return 0;
}
```

Code 25: *main.cpp "2D plot of complex function"*

To compile it, type:
g++ -o main main.cpp -lboost_iostreams
./main

or with Makefile, type:
make
./main
s
Explanation for the codes:

- In **gnuplot** complex constants are expressed as $\{< \text{real} >, < \text{imag} >\}$, where $< \text{real} >$ and $< \text{imag} >$ must be numerical constants. For instance:

$$a + bi \rightarrow \{a, b\}$$

$$i \rightarrow \{0, 1\}$$

$$5 - 2i \rightarrow \{5, -2\}$$

  Gnuplot can plot only real quantities. Thus we can plot the real part of e^{ix} which is $\cos\ x$ and the imaginary part of e^{ix} which is $\sin\ x$, but first we need to define the variable i as $\{0, 1\}$.

```
	gp << "i = {0,1}\n";
	gp << "plot real(exp(i*x)), imag(exp(i*x))\n";
```

- If you want to do parametric plot with

$$(\text{real}(e^{ix}), \text{imag}(e^{ix}))$$

  you can uncomment the two lines at the bottom and comment the plot line above it

```
//gp << "plot real(exp(i*x)), imag(exp(i*x))\n";

// To do parametric plot uncomment below and comment the plot
    line above
gp << "set parametric\n";
gp << "plot real(exp(i*t)), imag(exp(i*t))\n";
```

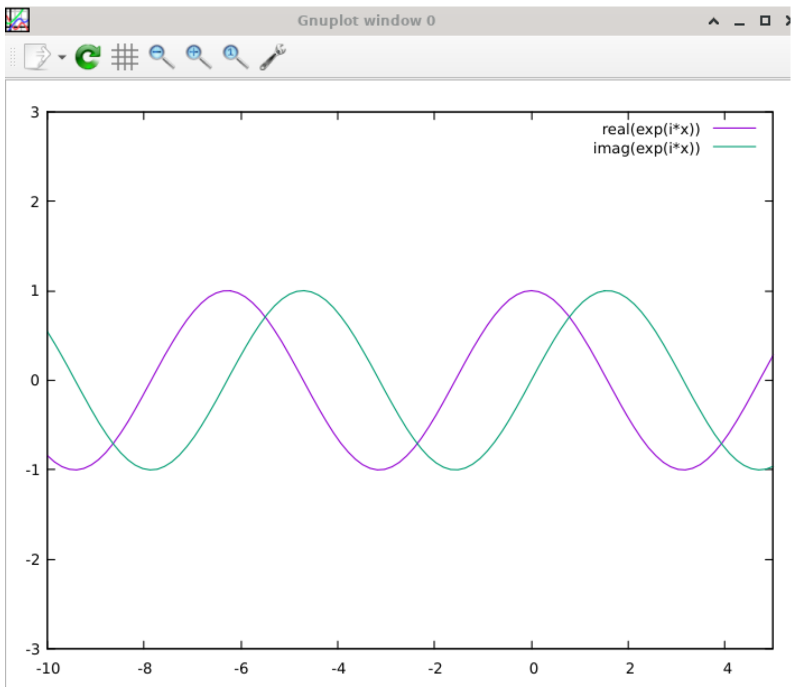


Figure 12.3: *The 2D plot of the real part of e^{ix} and the imaginary part of e^{xi} (**DFSimulatorC/Source Codes/C++/C+++ Gnuplot SymbolicC++/ch23-Numerical Linear Algebra/2D Plot Complex Function/main.cpp**).*

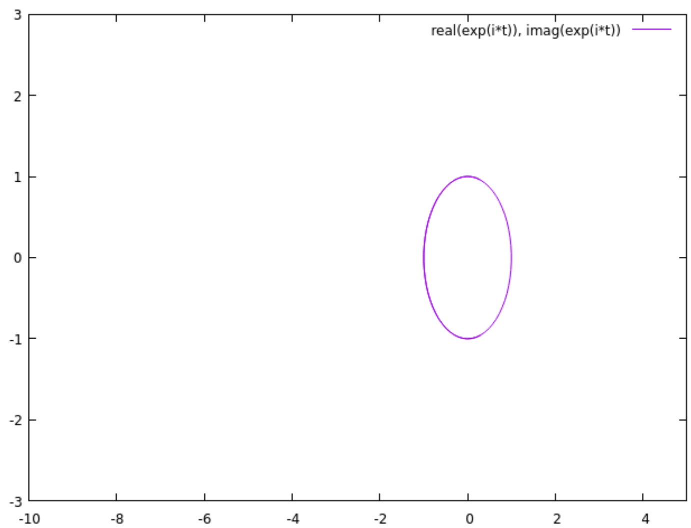


Figure 12.4: *The 2D parametric plot of* $(real(e^{ix}), imag(e^{ix}))$ *(**DFSimulatorC/Source Codes/C++/C+++ Gnuplot SymbolicC++/ch23-Numerical Linear Algebra/2D Plot Complex Function/main.cpp**).*

Chapter 13

Dynamical System

"Chaos would suit me just fine." - ***Yuber (Suikoden III)***

I. Double Pendulum

[H*] In physics and mathematics, a double pendulum, also known as a chaotic pendulum, is a pendulum with another pendulum attached to its end, forming a complex physical system that exhibits rich dynamics behavior with a strong sensitivity to initial conditions. The motion of a double pendulum is governed by a pair of coupled ordinary differential equations and is chaotic.

[H*] There are several variants of the double pendulum; the two limbs may be of equal or unequal length and masses, they may be simple pendulums or compound pendulums (complex pendulums) and the motion may be in three dimensions or restricted to one vertical plane.

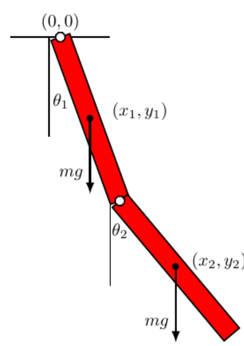


Figure 13.1: *The illustration of double compound pendulum .*

In the following analysis the limbs are taken to be identical compound pendulums of length l and mass m, and the motion is restricted to two dimensions.

In a compound pendulum, the mass is distributed along its length. If the double pendulum mass is evenly distributed, then the center of mass of each limb is at its midpoint, and the limb has a moment of inertia of

$$I = \frac{1}{12}ml^2$$

about that point.

Although it is possible to derive the equations of a double pendulum with Newtonian mechanics, it is considered to be cumbersome to work with as it would require resolving vectors with respect to constraint forces. So it is more convenient to use the angles between each limb and the vertical as the generalized coordinates defining the configuration of the system. These angles are denoted θ_1 and θ_2. The position of the center of mass of each rod may be written in terms of these two coordinates. If the origin of the Cartesian coordinate system is taken to be at the point of suspension of the first pendulum, then the center of mass of this pendulum is at:

$$x_1 = \frac{1}{2} l \sin\theta_1$$
$$y_1 = -\frac{1}{2} l \cos\theta_1$$

and the center of mass of the second pendulum is at

$$x_2 = l\left(\sin\theta_1 + \frac{1}{2}\sin\theta_2\right)$$
$$y_2 = -l\left(\cos\theta_1 + \frac{1}{2}\cos\theta_2\right)$$

This is enough information to write out the Lagrangian.

[H*] **Lagrangian for Double Compound Pendulum**

The Lagrangian is given by

$$\begin{aligned} L &= \text{kinetic energy} - \text{potential energy} \\ &= \frac{1}{2}m(v_1^2 + v_2^2) + \frac{1}{2}I(\dot\theta_1^2 + \dot\theta_2^2) - mg(y_1 + y_2) \\ &= \frac{1}{2}m(\dot x_1^2 + \dot y_1^2 + \dot x_2^2 + \dot y_2^2) + \frac{1}{2}I(\dot\theta_1^2 + \dot\theta_2^2) - mg(y_1 + y_2) \end{aligned} \tag{13.1}$$

The first term is the linear kinetic energy of the center of mass of the bodies and the second term is the rotational kinetic energy around the center of mass of each rod. The last term is the potential energy of the bodies in a uniform gravitational field. The dot-notation indicates the time derivative of the variable in question.

Using the values of x_1 and y_1 defined above, we have

$$\dot x_1 = \dot\theta_1\left(\frac{1}{2} l \cos\theta_1\right)$$
$$\dot y_1 = \dot\theta_1\left(\frac{1}{2} l \sin\theta_1\right)$$

which leads to

$$v_1^2 = \dot x_1^2 + \dot y_1^2 = \frac{1}{4}\dot\theta_1^2 l^2\left(\cos^2\theta_1 + \sin^2\theta_1\right) = \frac{1}{4}l^2\dot\theta_1^2$$

Similarly, for x_2 and y_2 we have

$$\dot x_2 = l\left(\dot\theta_1\cos\theta_1 + \frac{1}{2}\dot\theta_2\cos\theta_2\right)$$
$$\dot y_2 = l\left(\dot\theta_1\sin\theta_1 + \frac{1}{2}\dot\theta_2\sin\theta_2\right)$$

and therefore

$$\begin{aligned}
v_2^2 &= \dot{x}_2^2 + \dot{y}_2^2 \\
&= l^2 \left(\dot{\theta}_1^2 \cos^2\theta_1 + \dot{\theta}_1^2 \sin^2\theta_1 + \frac{1}{4}\dot{\theta}_2^2 \cos^2\theta_2 + \frac{1}{4}\dot{\theta}_2^2 \sin^2\theta_2 + \dot{\theta}_1\dot{\theta}_2 \cos\theta_1 \cos\theta_2 + \dot{\theta}_1\dot{\theta}_2 \sin\theta_1 \sin\theta_2 \right) \\
&= l^2 \left(\dot{\theta}_1^2 + \frac{1}{4}\dot{\theta}_2^2 + \dot{\theta}_1\dot{\theta}_2 \cos(\theta_1 - \theta_2) \right)
\end{aligned}$$

Substituting the coordinates above into the definition of the Lagrangian, and rearranging the equation, gives

$$\begin{aligned}
L &= \frac{1}{2}m(v_1^2 + v_2^2) + \frac{1}{2}I(\dot{\theta}_1^2 + \dot{\theta}_2^2) - mg(y_1 + y_2) \\
&= \frac{1}{2}ml^2 \left(\frac{1}{4}\dot{\theta}_1^2 + \dot{\theta}_1^2 + \frac{1}{4}\dot{\theta}_2^2 + \dot{\theta}_1\dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) + \frac{1}{24}ml^2 \left(\dot{\theta}_1^2 + \dot{\theta}_2^2 \right) \\
&\quad - mgl \left(-\frac{1}{2}\cos\theta_1 - \cos\theta_1 - \frac{1}{2}\cos\theta_2 \right) \\
&= \frac{1}{6}ml^2 \left(\dot{\theta}_2^2 + 4\dot{\theta}_1^2 + 3\dot{\theta}_1\dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) + \frac{1}{2}mgl(3\cos\theta_1 + \cos\theta_2)
\end{aligned}$$

The equations of motion can now be derived using the Euler-Lagrange equations, which are given by

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i} = 0, \quad i = 1, 2. \tag{13.2}$$

Equations of motion describe how an object's position, velocity, and acceleration change over time. We begin with the equation of motion for θ_1. The derivatives of the Lagrangian are given by

$$\frac{\partial L}{\partial \theta_1} = -\frac{1}{2}ml^2\dot{\theta}_1\dot{\theta}_2 \sin(\theta_1 - \theta_2) - \frac{3}{2}mgl\sin\theta_1$$

and

$$\frac{\partial L}{\partial \dot{\theta}_1} = \frac{4}{3}ml^2\dot{\theta}_1 + \frac{1}{2}ml^2\dot{\theta}_2 \cos(\theta_1 - \theta_2)$$

Thus

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_i} = \frac{4}{3}ml^2\ddot{\theta}_1 + \frac{1}{2}ml^2\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \frac{1}{2}ml^2\dot{\theta}_2(\dot{\theta}_1 - \dot{\theta}_2)\sin(\theta_1 - \theta_2)$$

Combining these results and simplifying yields the first equation of motion,

$$\frac{4}{3}l\ddot{\theta}_1 + \frac{1}{2}l\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \frac{1}{2}l\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + \frac{3}{2}g\sin\theta_1 = 0 \tag{13.3}$$

Similarly, the derivatives of the Lagrangian with respect to θ_2 and $\dot{\theta}_2$ are given by

$$\frac{\partial L}{\partial \theta_2} = -\frac{1}{2}ml^2\dot{\theta}_1\dot{\theta}_2 \sin(\theta_1 - \theta_2) - \frac{1}{2}mgl\sin\theta_2$$

and

$$\frac{\partial L}{\partial \dot{\theta}_2} = \frac{1}{3}ml^2\dot{\theta}_2 + \frac{1}{2}ml^2\dot{\theta}_1 \cos(\theta_1 - \theta_2)$$

Thus

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_2} = \frac{1}{3}ml^2\ddot{\theta}_2 + \frac{1}{2}ml^2\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \frac{1}{2}ml^2\dot{\theta}_1(\dot{\theta}_1 - \dot{\theta}_2)\sin(\theta_1 - \theta_2)$$

Plugging these results into the Euler-Lagrange equation and simplifying yields the second equation of motion,

$$\frac{1}{3}l\ddot{\theta}_2 + \frac{1}{2}l\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \frac{1}{2}l\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + \frac{1}{2}g \sin\theta_2 = 0 \tag{13.4}$$

No closed form solutions for θ_1 and θ_2 as functions of time are known, therefore the system can only be solved numerically, using the Runge Kutta method or similar techniques.

[H*] **Lagrangian for Double Simple Pendulum**
The knowledge of this writing is taken from Wolfram Mathematica webpage written by Eric W. Weisstein.

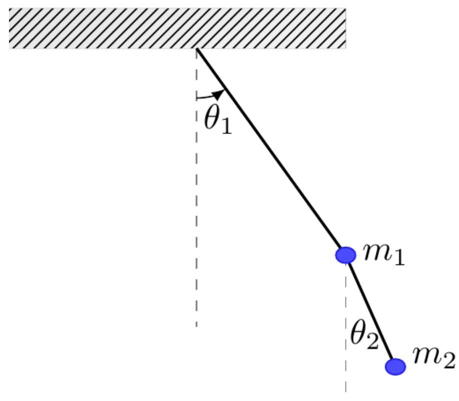


Figure 13.2: *The illustration of double simple pendulum .*

Consider a double bob pendulum with masses m_1 and m_2 attached by rigid massless wires of lengths l_1 and l_2. Further, let the angles of the two wires with the vertical be denoted by θ_1 and θ_2. Finally, let gravity be given by g. Then the positions of the bobs are given by

$$\begin{aligned} x_1 &= l_1 \sin\theta_1 \\ y_1 &= -l_1 \cos\theta_1 \\ x_2 &= l_1 \sin\theta_1 + l_2 \sin\theta_2 \\ y_2 &= -l_1 \cos\theta_1 - l_2 \cos\theta_2 \end{aligned}$$

the potential energy of the system is given by

$$\begin{aligned} V &= m_1 g y_1 + m_2 g y_2 \\ &= -(m_1 + m_2) g l_1 \cos\theta_1 - m_2 g l_2 \cos\theta_2 \end{aligned}$$

the potential energy is negative because it is going downward, and the first potential energy for m_1 is the sum of $m_1 + m_2$ because the first bob is carrying the mass of the second bob pendulum.

The kinetic energy is given by

$$\begin{aligned} T &= \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 \\ &= \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 \left[l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right] \end{aligned}$$

The Lagrangian is then

$$\begin{aligned}
L &= T - V \\
&= \frac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2 l_2^2\dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) - (-(m_1 + m_2)gl_1\cos\theta_1 - m_2 g l_2\cos\theta_2) \\
&= \frac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2 l_2^2\dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) + ((m_1 + m_2)gl_1\cos\theta_1 + m_2 g l_2\cos\theta_2)
\end{aligned}$$

Therefore, the equations of motion can now be derived using the Euler-Lagrange equations, for $\dot{\theta}_1$

$$\begin{aligned}
\frac{\partial L}{\partial \dot{\theta}_1} &= m_1 l_1^2\dot{\theta}_1 + m_2 l_1^2\dot{\theta}_1 + m_2 l_1 l_2\dot{\theta}_2\cos(\theta_1 - \theta_2) \\
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) &= (m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) - m_2 l_1 l_2\dot{\theta}_2\sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)
\end{aligned}$$

and for θ_1

$$\frac{\partial L}{\partial \theta_1} = -l_1 g(m_1 + m_2)\sin\theta_1 - m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2)$$

so the Euler-Lagrange differential equation for θ_1 / the first equation of motion becomes

$$\begin{aligned}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) - \frac{\partial L}{\partial \theta_1} &= 0 \\
(m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) + m_2 l_1 l_2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + l_1 g(m_1 + m_2)\sin\theta_1 &= 0
\end{aligned} \tag{13.5}$$

Similarly, the equations of motion for $\dot{\theta}_2$

$$\begin{aligned}
\frac{\partial L}{\partial \dot{\theta}_2} &= m_2 l_2^2\dot{\theta}_2 + m_2 l_1 l_2\dot{\theta}_1\cos(\theta_1 - \theta_2) \\
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) &= m_2 l_2^2\ddot{\theta}_2 + m_2 l_1 l_2\ddot{\theta}_1\cos(\theta_1 - \theta_2) - m_2 l_1 l_2\dot{\theta}_1\sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)
\end{aligned}$$

and for θ_2

$$\frac{\partial L}{\partial \theta_2} = m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - l_2 m_2 g\sin\theta_2$$

so the Euler-Lagrange differential equation for θ_2 / the second equation of motion becomes

$$\begin{aligned}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) - \frac{\partial L}{\partial \theta_2} &= 0 \\
m_2 l_2^2\ddot{\theta}_2 + m_2 l_1 l_2\ddot{\theta}_1\cos(\theta_1 - \theta_2) - m_2 l_1 l_2\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + l_2 m_2 g\sin\theta_2 &= 0
\end{aligned} \tag{13.6}$$

[H*] **Chaotic Motion**
The double pendulum undergoes chaotic motion, and clearly shows a sensitive dependence on initial conditions.

i. Double Simple Pendulum Computation and Simulation

In SymIntegration there is a function to compute the lagrangian and the equations of motion, **lagrangian(**$x_1, y_1, x_2, y_2, \theta_1, \dot{\theta}_1, \ddot{\theta}_1, \theta_2, \dot{\theta}_2, \ddot{\theta}_2$**)**

We input them all as symbolic to obtain the equations of motion formula symbolically, and then if we want to plot or compute numerically at certain time t we can substitute the symbolic, e.g. θ_1 into numeric, including defining / substituting the mass for both bob pendulum (m_1, m_2), the length of the wire from the ceiling to the first bob pendulum (l_1), the length of the wire between the first and the second bob pendulum (l_2) and the gravity (g).

```
#include <iostream>
#include "symintegrationc++.h"
#include <bits/stdc++.h>
#include <cmath>

#include <chrono>

using namespace std::chrono;
using namespace std;
using namespace SymbolicConstant;

double division(double x, double y)
{
      return x/y;
}

int main(void)
{
      // Get starting timepoint
      auto start = high_resolution_clock::now();

      Symbolic l1("l1"), l2("l2"), t1("t1"), t2("t2"), dt1("t1'"), dt2("t2'"),
          ddt1("t1''"),ddt2("t2''");
      Symbolic x1, y1, x2, y2;

      x1 = l1*sin(t1);
      y1 = -l1*cos(t1);
      x2 = l1*sin(t1) + l2*sin(t2);
      y2 = -l1*cos(t1) - l2*cos(t2);

      cout << "x1 = " << x1 << endl;
      cout << "y1 = " << y1 << endl;
      cout << "x2 = " << x2 << endl;
      cout << "y2 = " << y2 << endl;

      cout << "Here we go..." << lagrangian(x1,y1,x2,y2,t1, dt1, dt1, t2, dt2,
          ddt2) << endl;
```

```
    // Get ending timepoint
    auto stop = high_resolution_clock::now();
    auto duration = duration_cast<microseconds>(stop - start);

    cout << "\nTime taken by function: " << duration.count() << " microseconds"
         << endl;

    return 0;
}
```

Code 26: *SymIntegration/Examples/Dynamical System/Test SymIntegration Double Pendulum Problem with Lagrangian Function/main.cpp*

Figure 13.3: *The computation to obtain the Lagrangian and the equations of motion for double simple pendulum (**Dynamical System/Test SymIntegration Double Pendulum Problem with Lagrangian Function/-main.cpp**).*

We are using Hamzstlab Physics 2D to create a double simple pendulum simulation, we can vary the length from the wall above to the first body and the length from the first body to the second body.

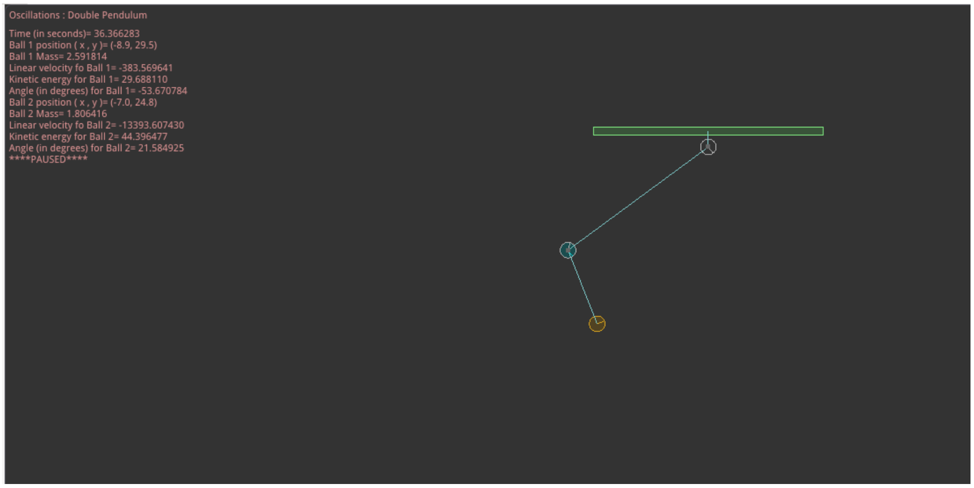


Figure 13.4: *The 2D simulation of a double simple pendulum (**Hamzstlab-Physics2D/tests/double_pendulum.cpp**).*

Box2D has created a 2D circle that can provide us with the angle of the rotation of the circle, in this pendulum case when the body is swinging to the right then the angle will be positive (counterclockwise rotation), and if the body is swinging to the left, the angle will be negative (clockwise rotation). The angle of the rotation is the same as the angle between the vertical axis and the rope that is connected to the circle / the body / bob.

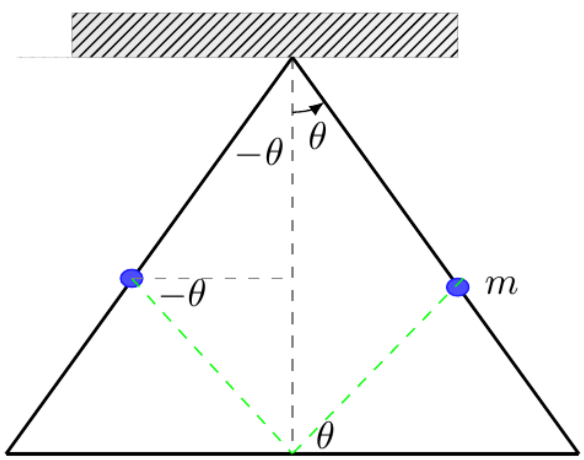


Figure 13.5: *The angle for the pendulum can be obtained easily from the rotation of the circle.*

Bibliography

[1] Banerjee, Sandi. (2022) Mathematical Modeling 2nd Edition: Models, Analysis, and Applications, CRC Press, Boca Raton, Florida, United States.

[2] Bazaraa, Mokhtar S., Jarvis, John J., Sherali, Hanif D. (1990) Linear Programming and Network Flows, John Wiley & Sons, New Jersey, United States.

[3] Borse, G.J. (1997) Numerical Methods with MATLAB: A Resouce for Scientists and Engineers, PWS Publishing Company, Worcester, United Kingdom.

[4] Boyce, William E., DiPrima, Richard C. (2010) Elementary Differential Equations and Boundary Value Problems 9th Edition, John Wiley & Sons, Hoboken, New Jersey, United States.

[5] Burden, Richard L., Faires, J. Douglas (2001) Numerical Analysis 7th Edition, Brooks/Cole, Pacific Grove, California, United States.

[6] Meyrick, Caroline, Dwamena, Kwame (2013) Mathematical Studies: Standard Level for the IB Diploma, Cambridge University Press, Cambridge, United Kingdom.

[7] Colley, Susan Jane (2012) Vector Calculus 4th Edition, Pearson, Boston, Massachussets, United States.

[8] Dorf, Richard C., Bishop, Robert H. (2010) Modern Control Systems 12th Edition, Pearson, New Jersey, United States.

[9] Durran, Dale R. (2010) Numerical Methods for Fluid Dynamics, Springer, New York, United States.

[10] Fabozzi, Frank J. (1999) Bond Markets, Analysis and Strategies 4th Edition, Prentice Hall, New Jersey, United States.

[11] Ferziger, Joel H., Peric, Milovan, Street, Robert L. (2020) Computational Methods for Fluid Dynamics 4th Edition, Springer Nature, Switzerland.

[12] Freya, Hamzst, Glanzsche, DS, DianFreya Math Physics Simulator with C++,Berlin-Sentinel Academy of Science, AliceGard.

[13] Gezerlis, Alex (2020) Numerical Methods in Physics with Python, Cambridge University Press, Cambridge, United Kingdom.

[14] Gockenbach, Mark S. (2002) Partial Differential Equations Analytical and Numerical Methods, Society for Industrial and Applied Mathematics, 3600 University City Science Center, Philadelphia, United States.

[15] Klok, Hayden, Nazarathy, Yoni (2021) Statistics with Julia, Springer, Cham, Switzerland.

[16] Lasthrim, Glanzsche, DS, Freya (2025) Lasthrim Projection 1st Edition, Berlin-Sentinel Academy of Science, AliceGard.

[17] Nelder J.A, Mead R. (1965). A simplex method for function minimization. The computer journal, 7(4), 308-313.

[18] Press, W.H., Teukolsky, S.A., Vetterling, W.T. Flannery, B.P. (2007) Numerical Recipes 3rd Edition, Cambridge University Press, Cambridge, United Kingdom.

[19] Purcell, Rigdon, Varberg (2007) Calculus 9th Edition, Pearson, Upper Saddle River, New Jersey, United States.

[20] Rao, SS (2006) Optimization Theory and Applications 2nd Edition, Wiley Eastern Limited, New Delhi, India.

[21] Rosen, Kenneth H. (2006) Discrete Mathematics and its Applications 6th Edition, McGraw-Hill, Boston, United States.

[22] Anton, Rorres (2006) Elementary Linear Algebra with Supplemental Applications 10th Edition, Wiley, Boston, United States.

[23] Anton, Howard, Rorres, Chris, Elementary Linear Algebra with Supplemental Applications 12th edition, Wiley, Hoboken, New Jersey, United States.

[24] Simmons, George F. (2016) Differential Equations with Applications and Historical Notes 3rd Edition, CRC Press, Boca Raton, Florida, United States.

[25] Strauss, Walter A. (2008) Partial Differential Equations: An Introduction 2nd Edition, John Wiley & Sons, New Jersey, United States.